

CHAPTER - 2

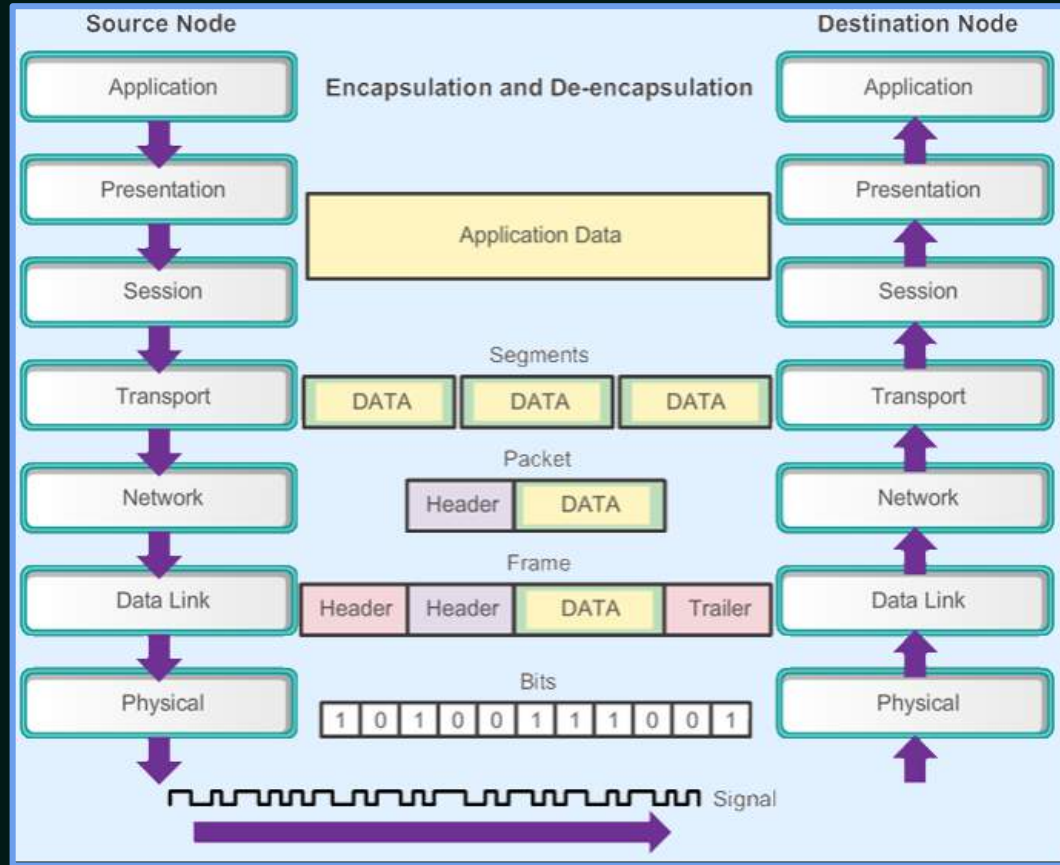
Data Link Layer

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the fundamental principles of physical layer.
- ★ Know about data and signals.

FUNDAMENTAL PRINCIPLES OF PHYSICAL LAYER



PHYSICAL LAYER – DATA AND SIGNALS

Data and Signals.

Source/Destination → Data → Signals.

FUNDAMENTAL PRINCIPLES OF PHYSICAL LAYER

- ★ One of the major functions of the physical layer is to move data in the form of electromagnetic signals across a transmission medium.
- ★ The data usable to a person or an application are not in a form that can be transmitted over a network.
- ★ For example, an image must first be changed to a form that transmission media can accept.
- ★ To be transmitted, data must be transformed to electromagnetic signals.

OUTCOMES

Upon the completion of this session, the learner will be able to

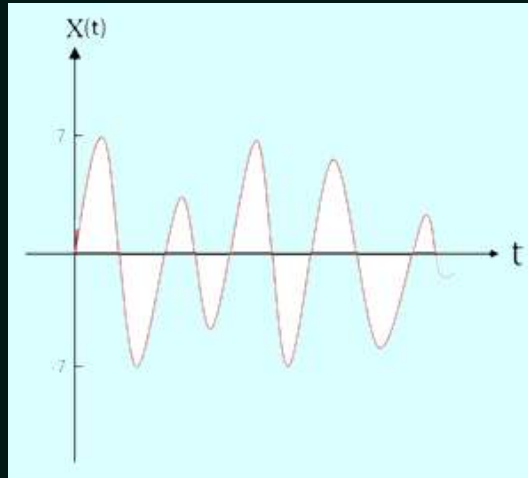
- ★ Understand analog and digital signals.
- ★ Know various physical layer media.
- ★ Compare various physical media.

SIGNAL

- ★ It is a function that represents the variation of a physical quantity with respect to time.
- ★ Example: Variation in temperature of a city in one day i.e. 24 hours.
- ★ Analog Signal and Digital Signal.

ANALOG SIGNAL

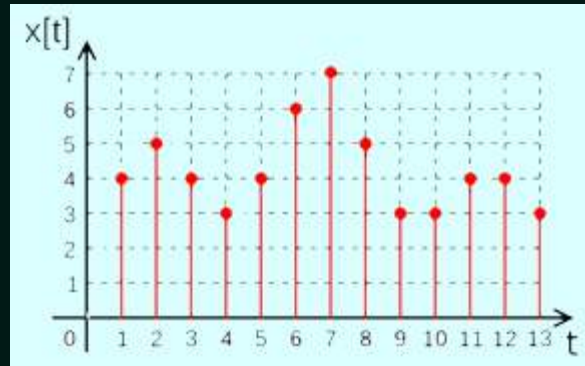
- ★ It is the signal that can take any value in the defined range.
- ★ All real-life signals are analog in nature.



The signal $x(t)$ can take any value between -7 to +7.

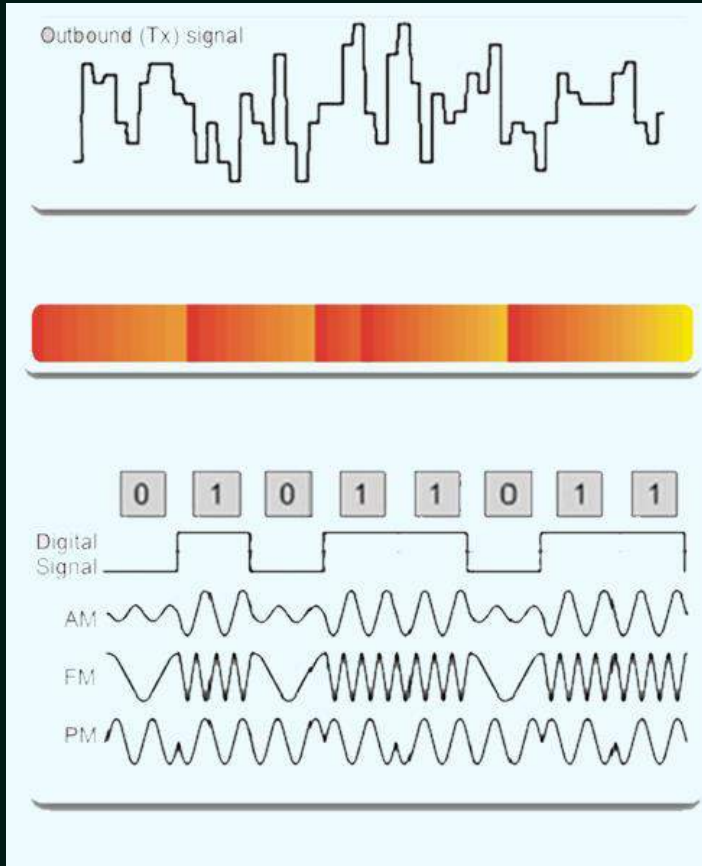
DIGITAL SIGNAL

- ★ It is the signal that can take on of the finite values at any given time.
- ★ In case of digital signals, we discretize both time and magnitude.



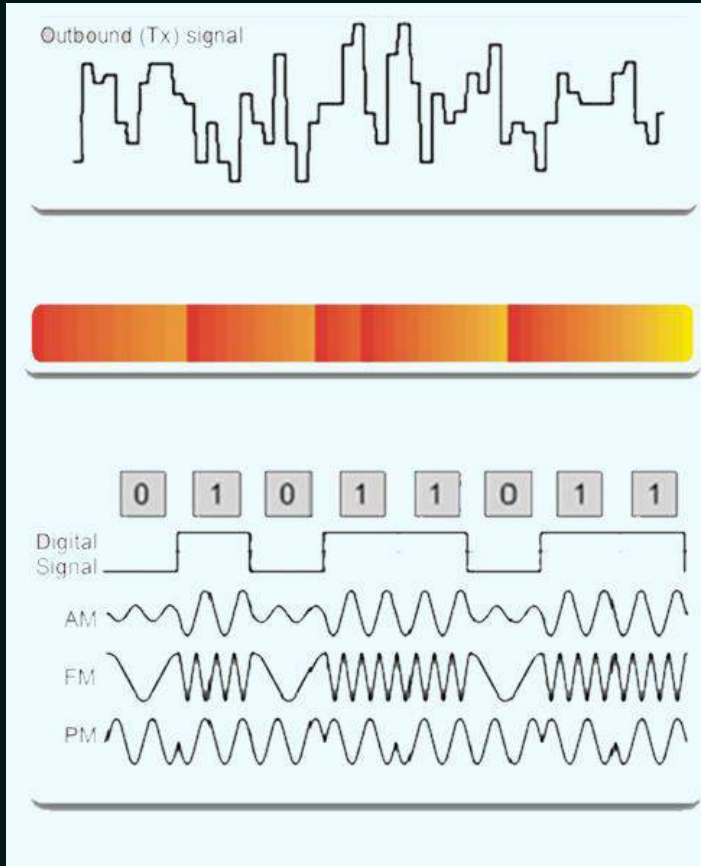
The signal $x[t]$ can only take one value out of 0, 1, 2, 3, 4, 5, 6, and 7 for any discrete value of time.

PHYSICAL LAYER MEDIA

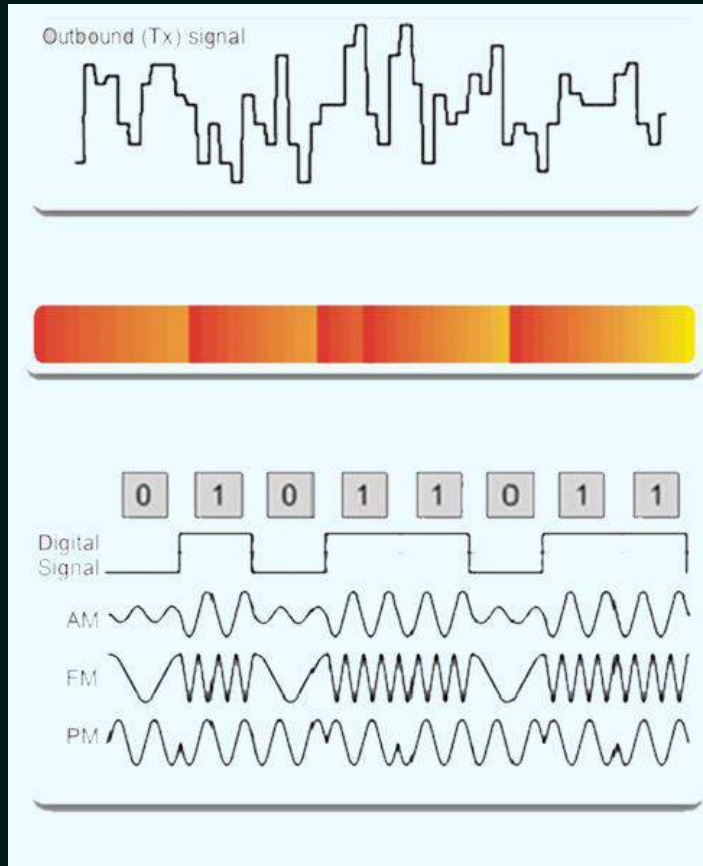


Electrical Signals
Copper cable

PHYSICAL LAYER MEDIA



PHYSICAL LAYER MEDIA



← Microwave signals
← Wireless

COMPARISON OF VARIOUS PHYSICAL MEDIA

Media	Physical Components	Signal
Copper Cable (Wired)	·UTP/STP ·Coaxial ·Connectors ·NICs ·Ports/·Interfaces	Electromagnetic Signal
Fiber Optic Cable (Wired)	·Single-mode Fiber ·Multimode Fiber ·Connectors ·NICs and Interfaces ·Lasers and LEDs	·A light pulse equals 1. ·No light pulse is 0.
Wireless Media	·Access Points ·NICs ·Radio ·Antennae	·Radio waves

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand wired and wireless media.
- ★ Understand UTP and STP.
- ★ Compare copper media and fiber optic media.
- ★ Know the popular wireless technologies.

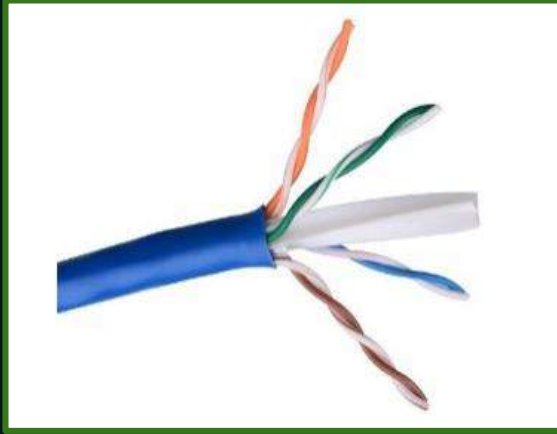
WIRED MEDIA

- ★ Copper cable (Ethernet cable)
 - Unshielded Twisted Pair (UTP).
 - Shielded Twisted Pair (STP).
- ★ Copper coaxial cable.
- ★ Fiber optic cable.

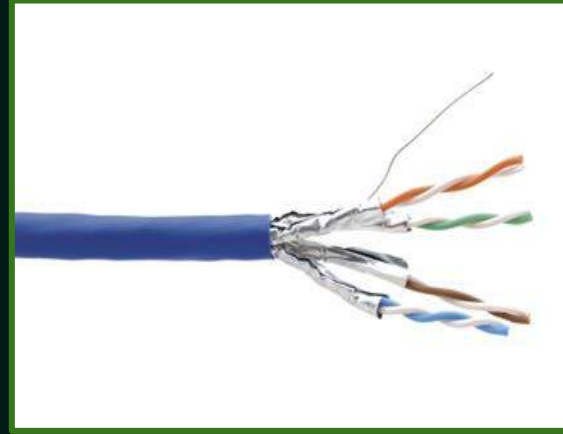
COPPER MEDIA - ETHERNET



COPPER MEDIA - ETHERNET



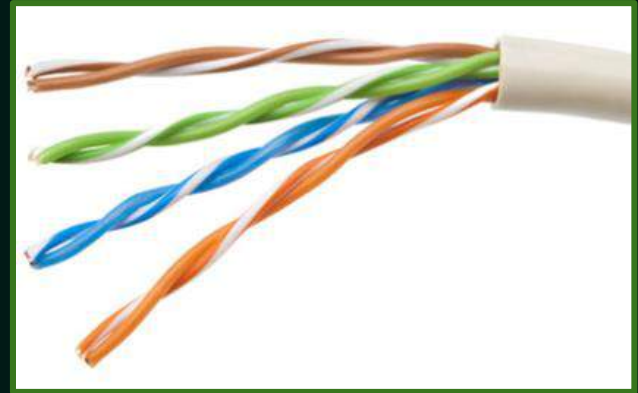
Unshielded Twisted Pair (UTP)
Ethernet Cable



Shielded Twisted Pair (STP)
Ethernet cable

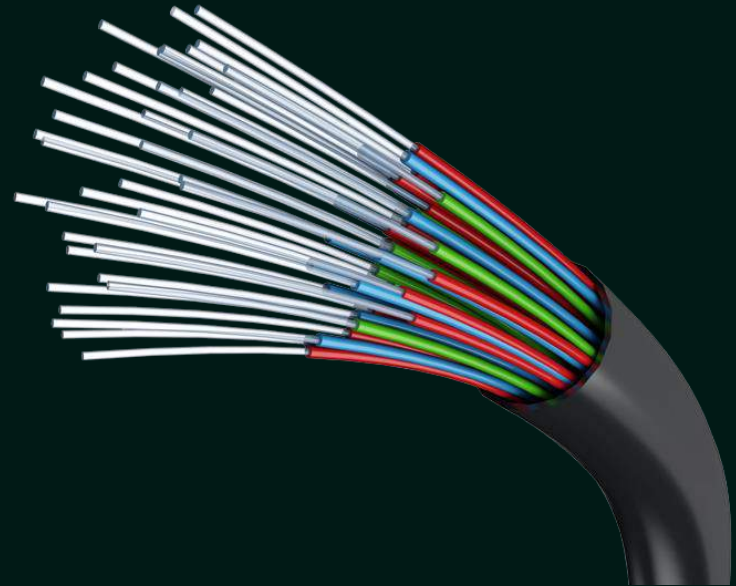
CROSSTALK

- ★ UTP cable does not use shielding to counter the effects of EMI and RFI.
- ★ The negative effect of crosstalk can be limited by varying the number of twists per wire pair.



FIBER OPTIC MEDIA

- ★ Light waves.
- ★ High Speed transmission.



COPPER VS FIBER OPTIC MEDIA

Implementation Points	Copper	Fiber Optic
Bandwidth Supported	10 Mbps – 10 Gbps	10 Mbps – 100 Gbps
Range	Relatively short (upto 100 meters)	Relatively High (upto 100,000 meters)
Immunity To EMI And RFI	Low	High (Completely immune)
Immunity To Electrical Hazards	Low	High (Completely immune)
Media And Connector Costs	Lowest	Highest
Installation Skills Required	Lowest	Highest
Safety	Lowest	Highest

WIRELESS MEDIA

Wireless media has the following areas of concern:

- ★ Coverage area
- ★ Interference
- ★ Security

WIRELESS TECHNOLOGIES



- IEEE 802.11 standards
- Commonly referred to as Wi-Fi.
- Uses CSMA/CA
- Variations include:
 - 802.11a: 54 Mbps, 5 GHz
 - 802.11b: 11 Mbps, 2.4 GHz
 - 802.11g: 54 Mbps, 2.4 GHz
 - 802.11n: 600 Mbps, 2.4 and 5 GHz
 - 802.11ac: 1 Gbps, 5 GHz
 - 802.11ad: 7 Gbps, 2.4 GHz, 5 GHz, and 60 GHz



- IEEE 802.15 standard
- Supports speeds up to 3 Mb/s
- Provides device pairing over distances from 1 to 100 meters.



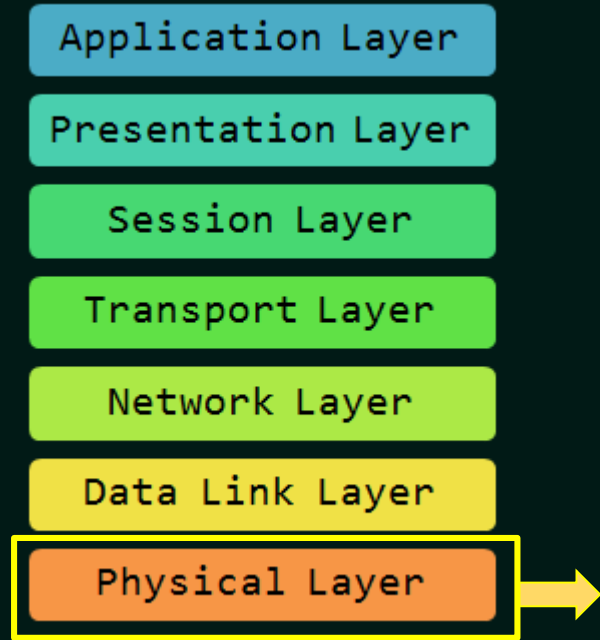
- IEEE 802.16 standard
- Provides speeds up to 1 Gbps
- Uses a point-to-multipoint topology to provide wireless broadband access.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Recapture the services offered by physical layer of the OSI model.
- ★ Understand the line configuration.
- ★ Understand the types of line configuration.

Neso Academy



It is responsible for transmitting bits over a medium.

SERVICES PROVIDED BY PHYSICAL LAYER

- ★ Physical characteristics of the media.
- ★ Representation of bits.
- ★ Data rate.
- ★ Synchronization of bits.
- ★ Line configuration.
- ★ Physical topology.
- ★ Transmission mode.

LINE CONFIGURATION

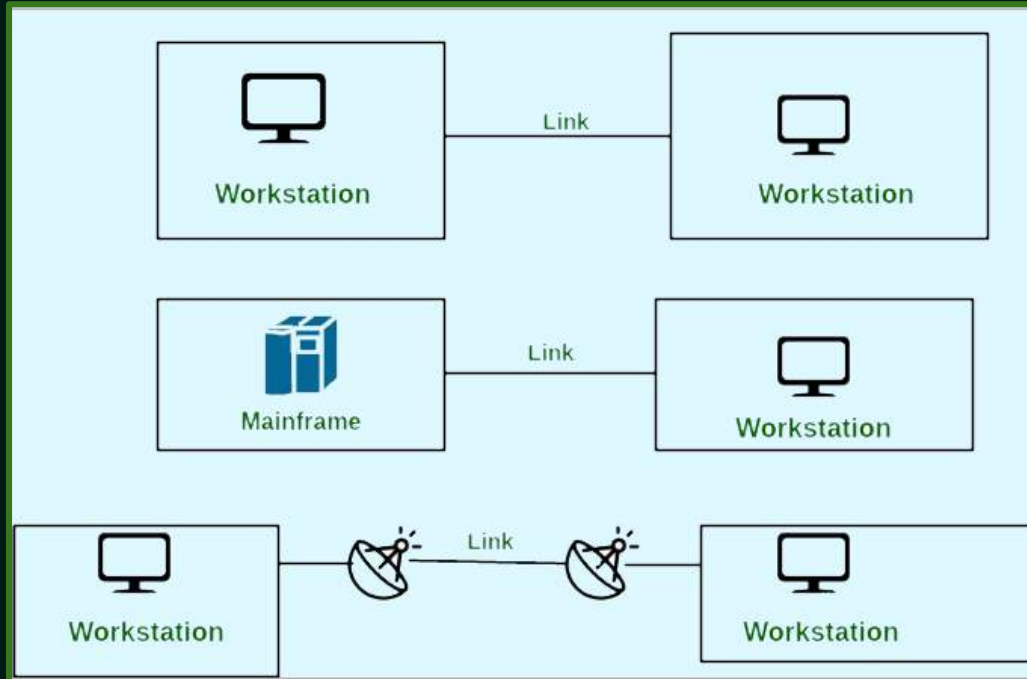
- ★ In a network, two or more nodes are connected by a communication link.
- ★ The communication link can be wired or wireless.
- ★ For visualization purpose, links are imagined as a line drawn between two points.
- ★ For communication to happen, two nodes must be connected to the same link at the same time.
- ★ This is called as line configuration or connection.

TYPES OF LINE CONFIGURATION

- ★ Point-to-Point Connection
- ★ Multipoint Connection

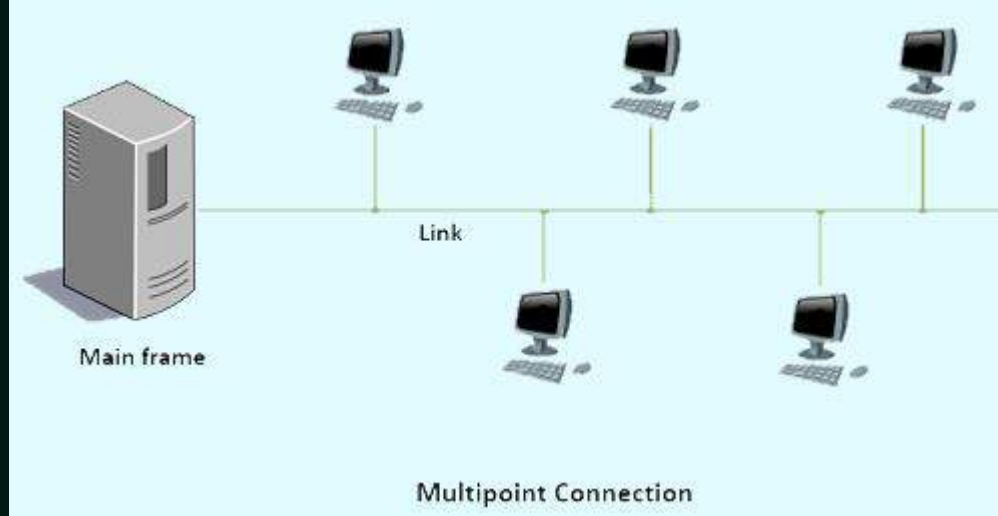
POINT-TO-POINT CONNECTION

- ★ A point-to-point connection provides a dedicated link between two devices.
- ★ The entire capacity of the link is reserved for transmission between those two devices.



MULTIPOINT CONNECTION

- ★ It is also called Multidrop configuration or Broadcast connection.
- ★ In this connection, two or more devices share a single link.
- ★ Since more than two devices share the link, the capacity of the channel is shared now.
- ★ It can be spatial or temporal.

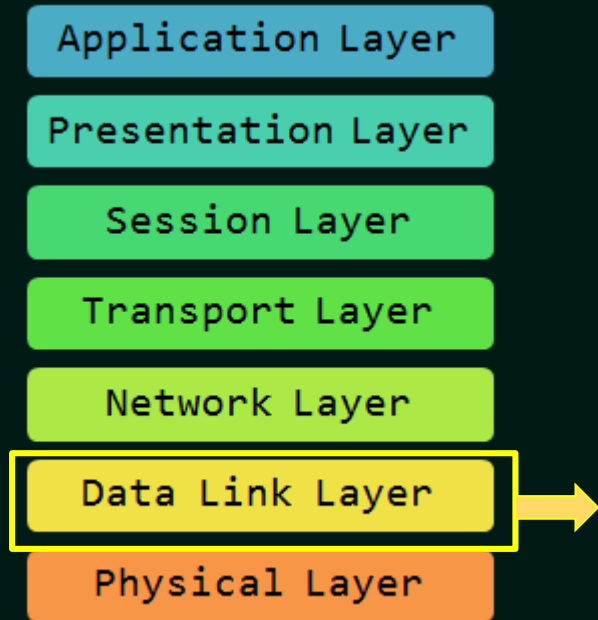


OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various services offered by data link layer.
- ★ Understand the services offered by data link layer.

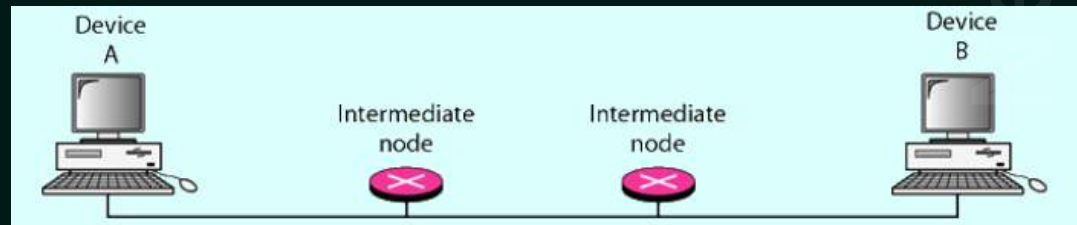
DATA LINK LAYER



It is responsible for moving data(frames) from one node to another node.

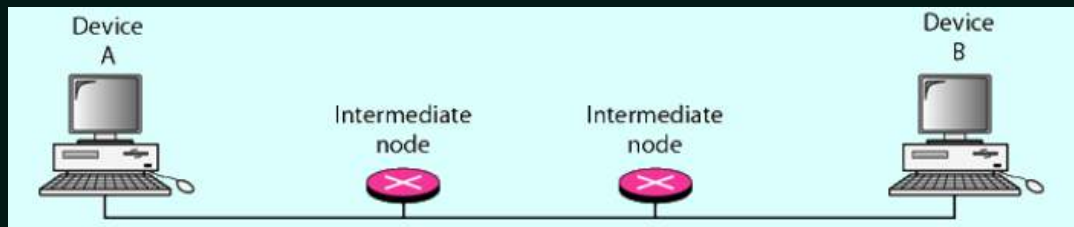
SERVICES PROVIDED BY DATA LINK LAYER

- ★ Framing.
- ★ Physical Addressing.
- ★ Flow Control.
- ★ Error Control.
- ★ Access Control.



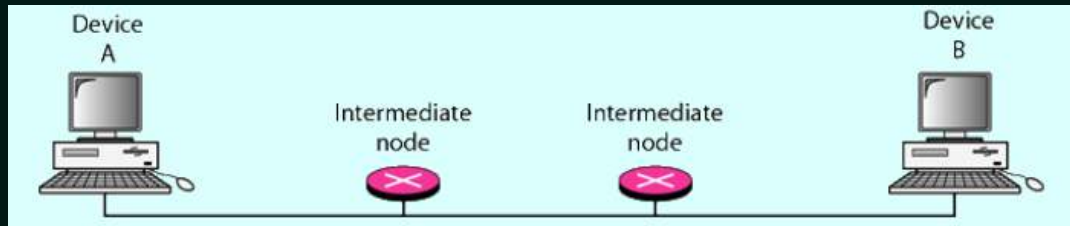
FRAMING

- ★ The data link layer needs to pack bits into frames, so that each frame is distinguishable from another.
- ★ Our postal system practices a type of framing.
- ★ The simple act of inserting a letter into an envelope separates one piece of information from another; the envelope serves as the delimiter.



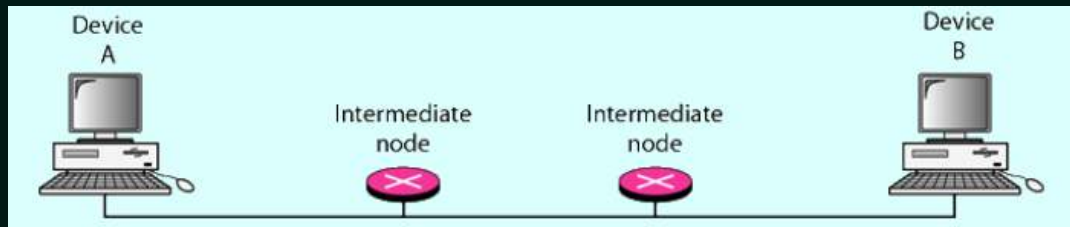
PHYSICAL ADDRESSING

- ★ A Frame is the encapsulation of the header and trailer information with the packet.
- ★ In the header, the source and the destination MAC address are dealt.



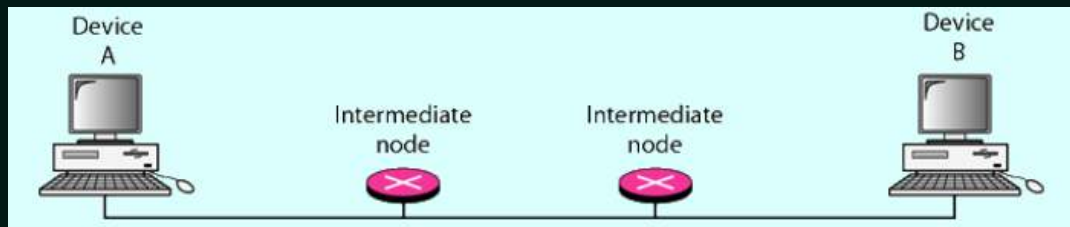
FLOW CONTROL

- ★ Flow Control is one of the duties of data link control sublayer.
- ★ The flow control in data link layer is end to end flow control.
- ★ Speed matching mechanism.
- ★ Flow control coordinates the amount of data that can be sent before receiving an acknowledgment.



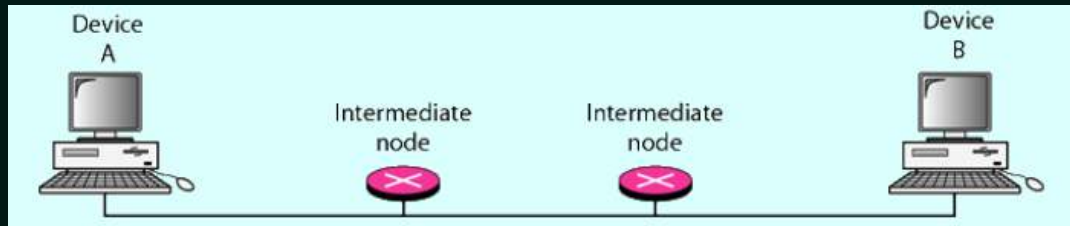
ACCESS CONTROL

- ★ Media Access control.



ERROR CONTROL

- ★ Error Detection.
- ★ Error Correction.

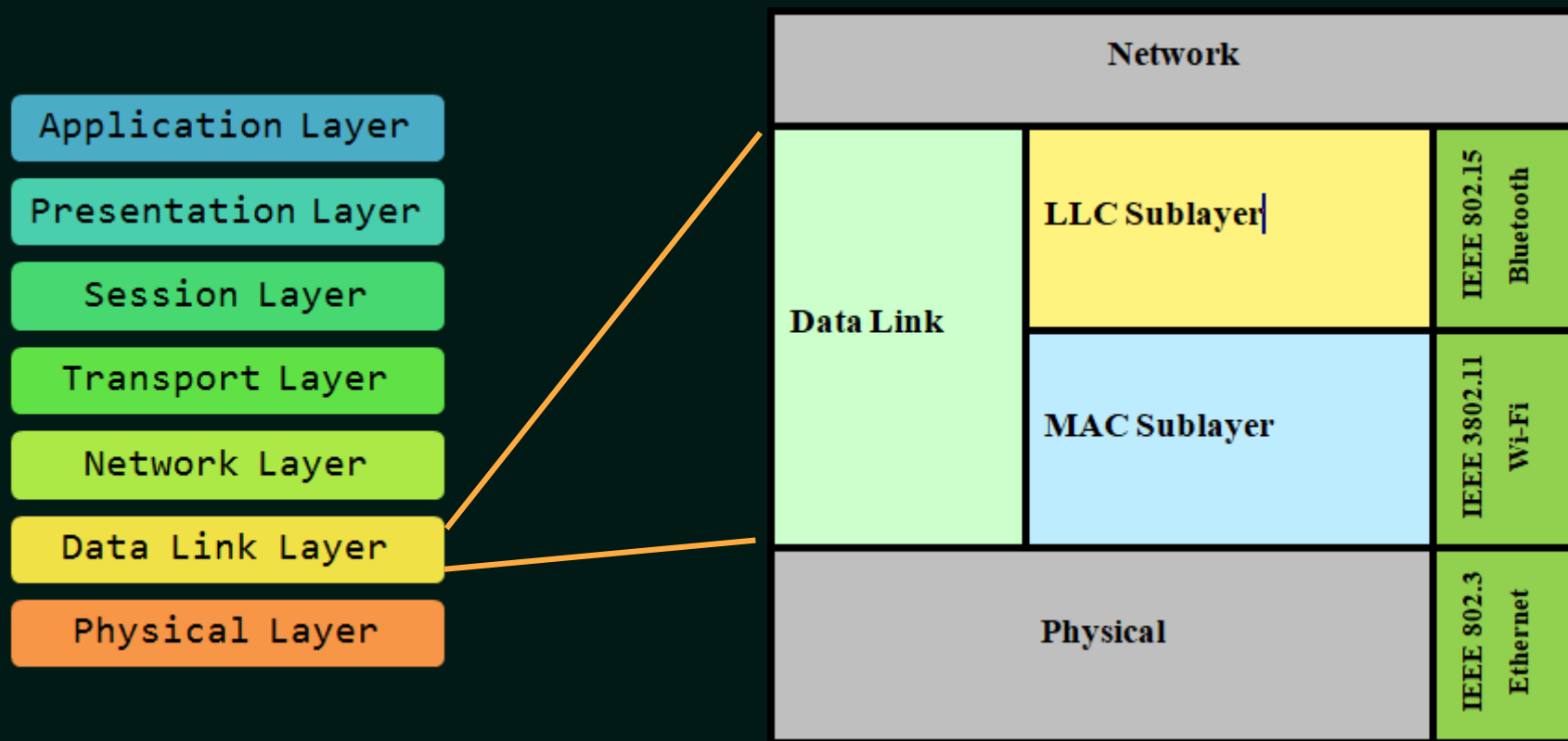


OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the sublayers of data link layer.
- ★ Know the responsibilities of LLC or DLC.
- ★ Know the responsibilities of MAC sublayer.

DATA LINK LAYER



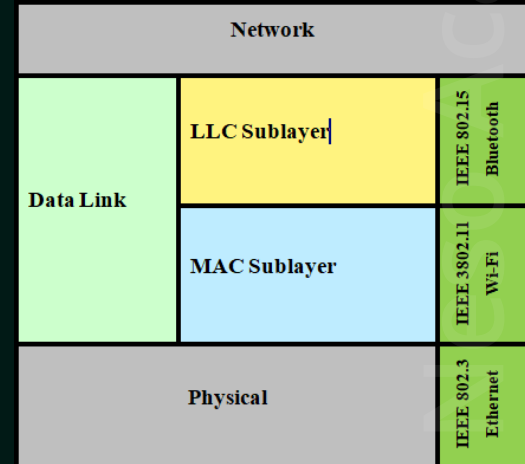
DATA LINK SUBLAYERS

Logical Link Control (LLC) or Data Link Control (DLC) Sublayer

- ★ Handles communication between upper and lower layers.
- ★ Takes the network protocol data and adds control information to help deliver the packet to the destination. (Flow control)

MAC Sublayer

- ★ Constitutes the lower sublayer of the data link layer.
- ★ Implemented by hardware, typically in the computer NIC.
- ★ Two primary responsibilities:
 - ★ Data encapsulation
 - ★ Media access control



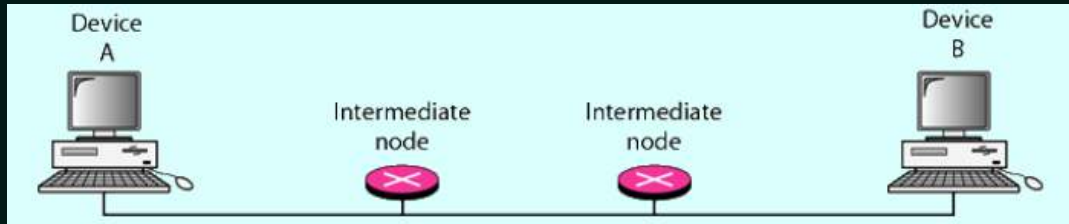
MAC SUBLAYER

Data encapsulation

- ★ Frame assembly before transmission and frame disassembly upon reception of a frame.
- ★ MAC layer adds a header and trailer to the network layer PDU.

Provides three primary functions:

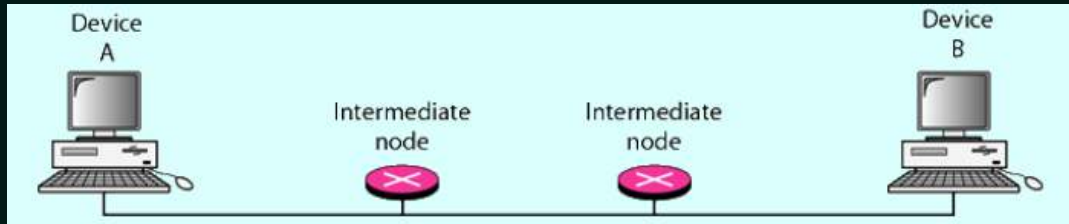
- ★ Framing.
- ★ Physical Addressing or MAC Addressing.
- ★ Error control.



Network		
Data Link	LLC Sublayer	IEEE 802.15 Bluetooth
	MAC Sublayer	IEEE 802.11 Wi-Fi
Physical		IEEE 802.3 Ethernet

MAC SUBLAYER

- ★ Responsible for the placement of frames on the media and the removal of frames from the media
- ★ Communicates directly with the physical layer.



Network		
Data Link	LLC Sublayer	IEEE 802.15 Bluetooth
	MAC Sublayer	IEEE 802.11 Wi-Fi
Physical		IEEE 802.3 Ethernet

ACTIVITY TIME

Mention the sublayer that is responsible for the service shown in the table.

Service	Sublayer
Flow Control	LLC or DLC
Framing	MAC
Physical Addressing	
Error Control	
Access Control	

OUTCOMES

Upon the completion of this session, the learner will be able to

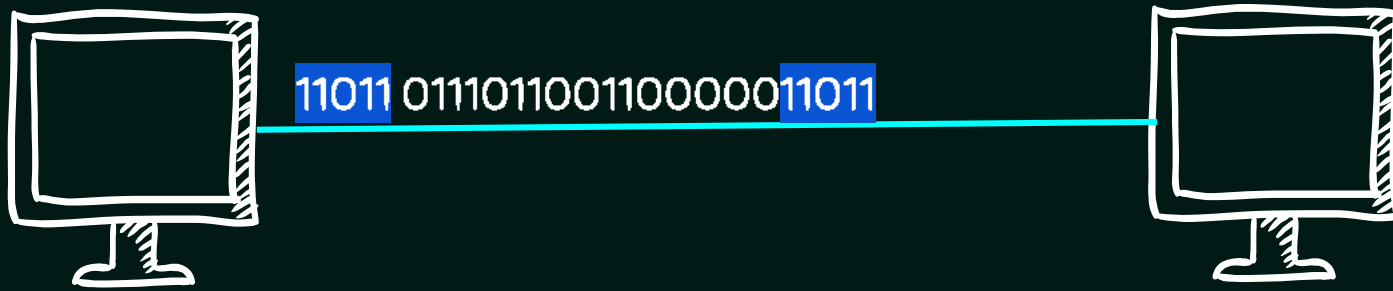
- ★ Understand framing.
- ★ Understand framing errors.

FRAMING



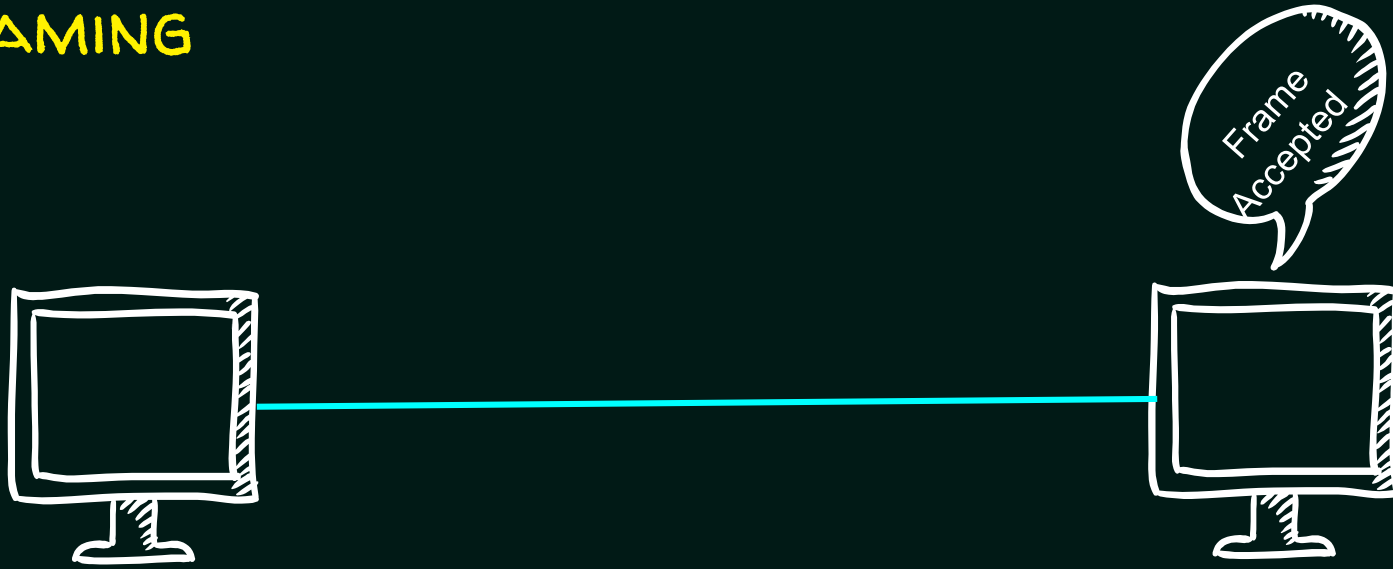
Bits flow between adaptors, frames between hosts

FRAMING



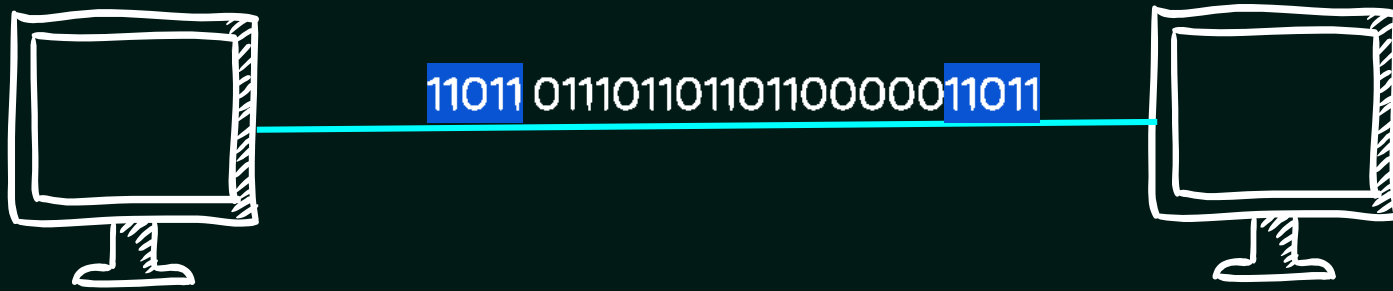
Let the start of frame and end of frame be `11011`

FRAMING



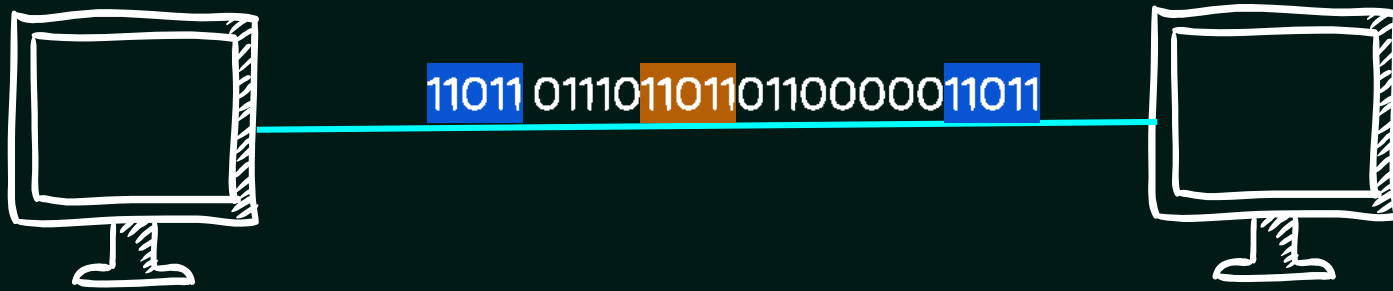
Let the start of frame and end of frame be 11011

ANY PROBLEM HERE...



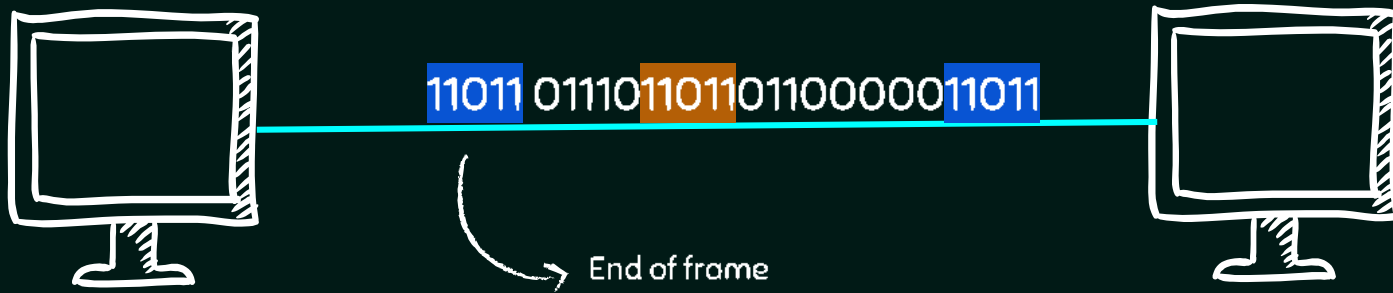
Let the start of frame and end of frame be `11011`

ANY PROBLEM HERE...



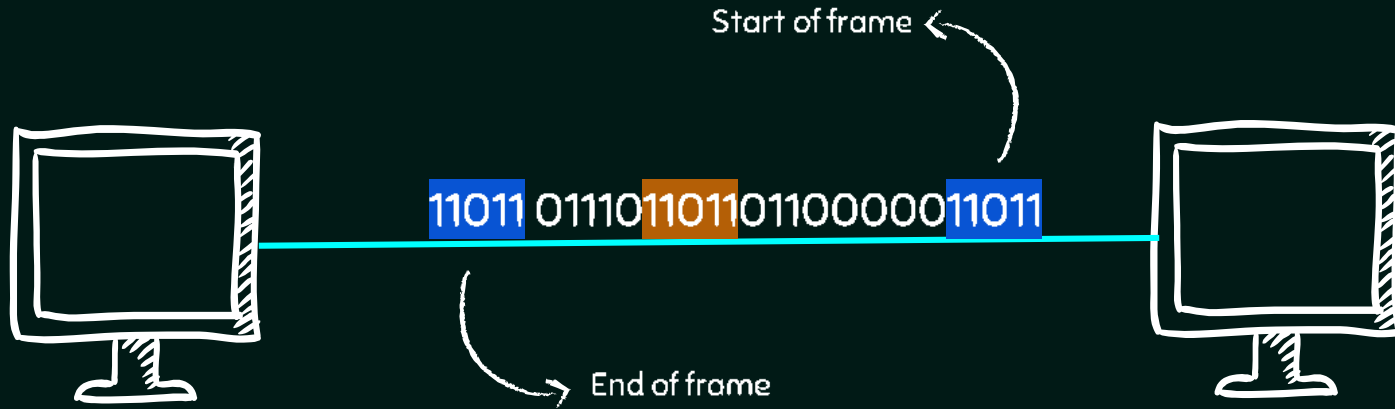
Let the start of frame and end of frame be 11011

FRAMING ERROR



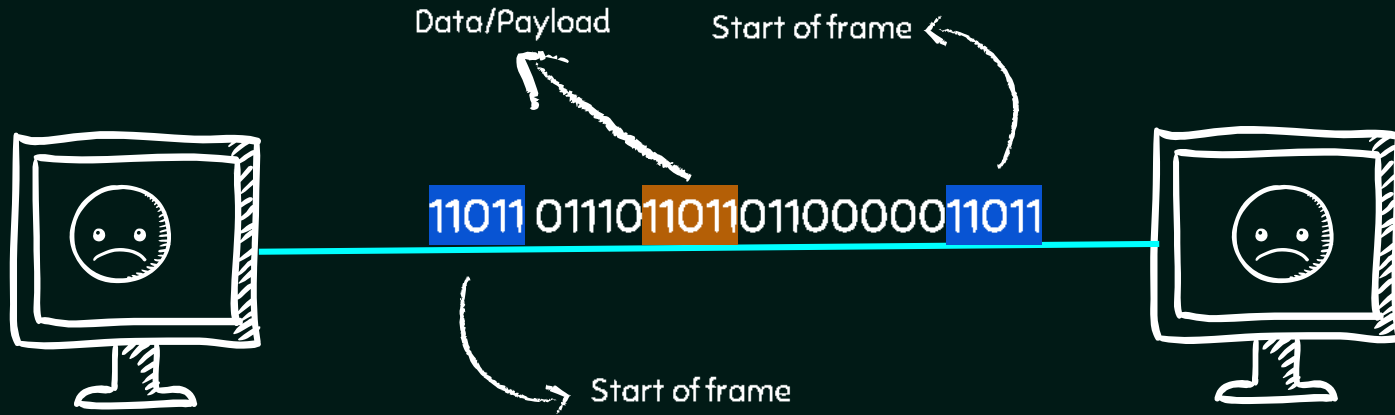
Let the start of frame and end of frame be 11011

FRAMING ERROR



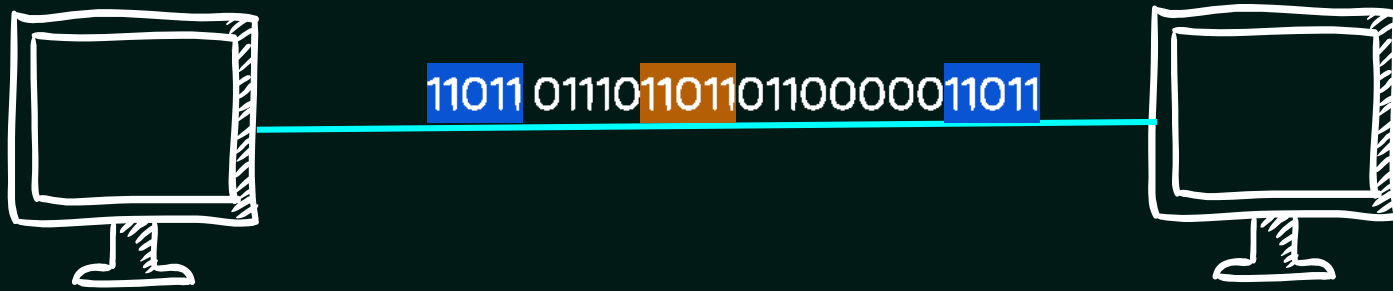
Let the start of frame and end of frame be 11011

FRAMING ERROR



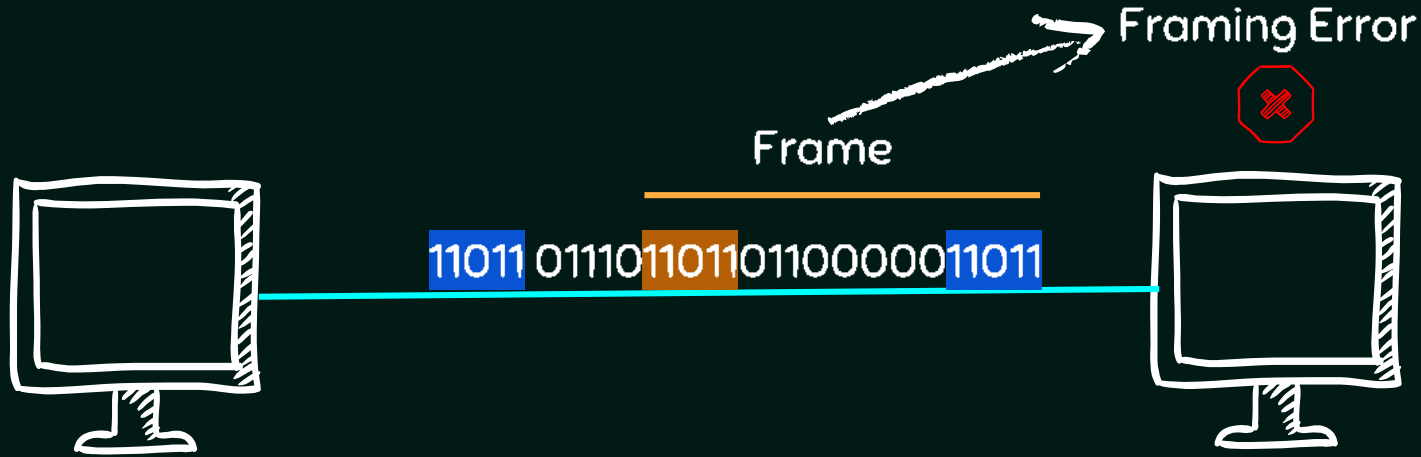
Let the start of frame and end of frame be 11011
Data = Start of Frame or End of Frame

FRAMING ERROR



Let the start of frame and end of frame be 11011

FRAMING ERROR



Let the start of frame and end of frame be `11011`

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the basics of framing.
- ★ Understand the types of framing.
- ★ Know various framing approaches.

FRAMING

- ★ Framing in the data link layer separates a frame distinguishable from another frame.
- ★ Frame = Header + Network Layer PDU + Trailer.
- ★ In packet switched networks, the block of data called frames are exchanged between nodes, not bits streams.



Bits flow between adaptors, frames between hosts

FRAMING

- ★ When node A wishes to transmit a frame to node B, it tells its adaptor to transmit a frame from the node's memory.
- ★ This results in a sequence of bits being sent over the link.
- ★ The adaptor on node B then collects together the sequence of bits arriving on the link and deposits the corresponding frame in B's memory.
- ★ Challenge: What set of bits constitute a frame?



Bits flow between adaptors, frames between hosts

TYPES OF FRAMING

1. Fixed-size framing.

- ★ Here the size of the frame is fixed and so the frame length acts as delimiter of the frame.
- ★ Consequently, it does not require additional boundary bits to identify the start and end of the frame.

2. Variable-size framing.

- ★ Here, the size of each frame to be transmitted may be different.
- ★ So additional mechanisms are kept to mark the end of one frame and the beginning of the next frame.

VARIOUS FRAMING APPROACHES



OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know bit oriented approach and bit oriented protocols.
- ★ Know byte oriented approach and byte oriented protocols.
- ★ Know clock based framing.

VARIOUS FRAMING APPROACHES



BIT ORIENTED APPROACH

- ★ It simply views the frame as a collection of bits.
- ★ These bits might come from
 - Some character set, such as ASCII

BIT ORIENTED APPROACH

- ★ It simply views the frame as a collection of bits.
- ★ In bit-oriented framing, data is transmitted as a sequence of bits that can be interpreted in the upper layers both as text as well as multimedia data.

BIT ORIENTED PROTOCOL

- ★ HDLC $\leftrightarrow$ High-Level Data Link Control

BYTE ORIENTED APPROACH

- ★ One of the oldest approaches to framing.
- ★ Here each frame is viewed as a collection of bytes (characters) rather than bits.
- ★ a.k.a Character Oriented Approach.

BYTE ORIENTED PROTOCOLS

- ★ BISYNC $\leftrightarrow$ Binary Synchronous Communication Protocol.
- ★ DDCMP $\leftrightarrow$ Digital Data Communication Message Protocol.
- ★ PPP $\leftrightarrow$ Point-to-Point Protocol

CLOCK BASED FRAMING

- ★ The third approach to framing is the clock based framing.
- ★ Example: SONET $\leftrightarrow$ Synchronous Optical Network.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand bit-oriented protocol.
- ★ Know the frame format of HDLC.
- ★ Know the types of HDLC frames.

BIT ORIENTED APPROACH

- ★ It simply views the frame as a collection of bits.

Bit Oriented Protocol

HDLC $\leftrightarrow$ High-Level Data Link Control

HDLC

- ★ The Synchronous Data Link Control (SDLC) protocol developed by IBM is an example of a bit-oriented protocol.
- ★ SDLC was later standardized by the ISO as the High-Level Data Link Control (HDLC) protocol.
- ★ Bit Oriented Protocol.

HDLC – FRAME FORMAT

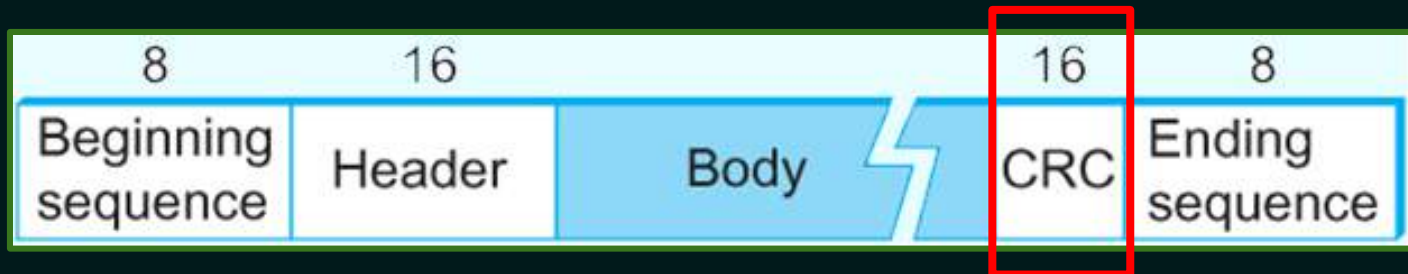
Beginning and Ending Sequences: 01111110

This sequence is also transmitted during any times that the link is idle so that the sender and receiver can keep their clocks synchronized.

Header: Address and Control Field.

Body: Payload (Variable size)

CRC: Cyclic Redundancy check – Error Detection



TYPES OF HDLC FRAMES

The type of frame is determined by the control field.

I-Frame: Information Frame.

S-Frame: Supervisory Frame.

U-Frame: Un-numbered Frame.

I-Frame	1st bit is 0
S-Frame	1st two bits is 10
U-Frame	1st two bits is 11

BIT ORIENTED APPROACH

- ★ It simply views the frame as a collection of bits.

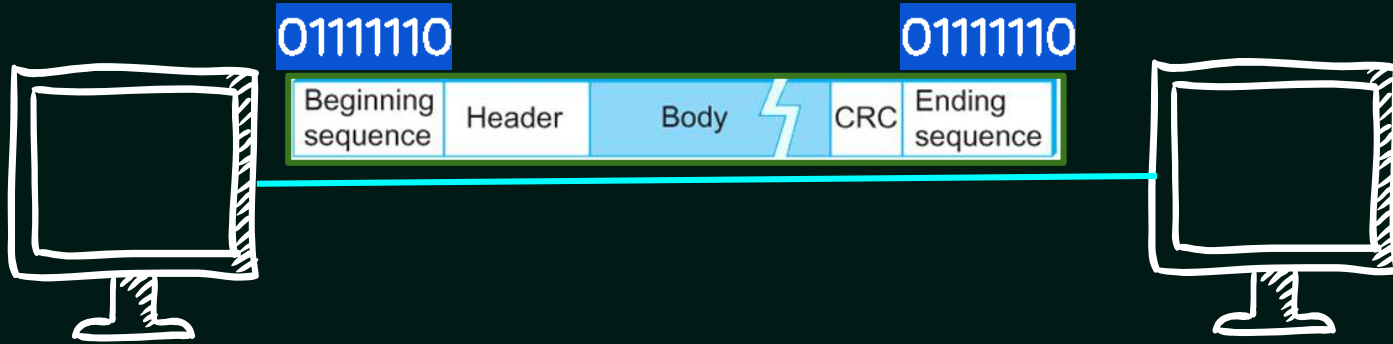
Bit Oriented Protocol

HDLC $\leftrightarrow$ High-Level Data Link Control

HDLC – FRAME FORMAT

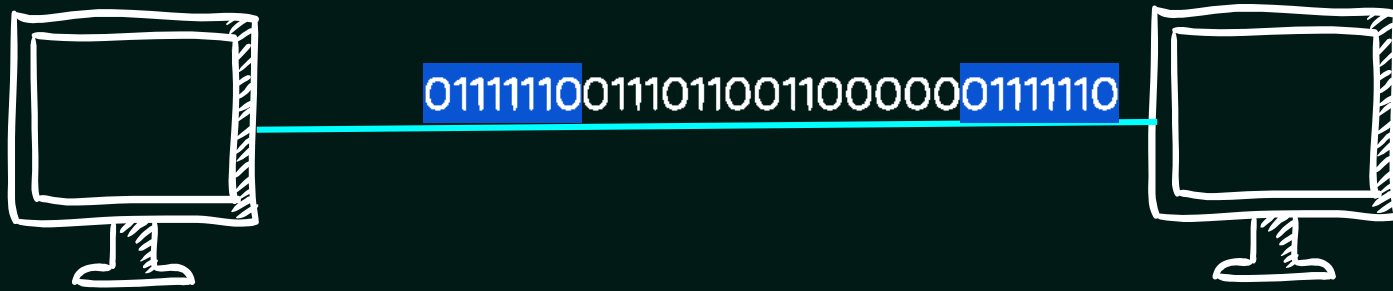


FRAMING



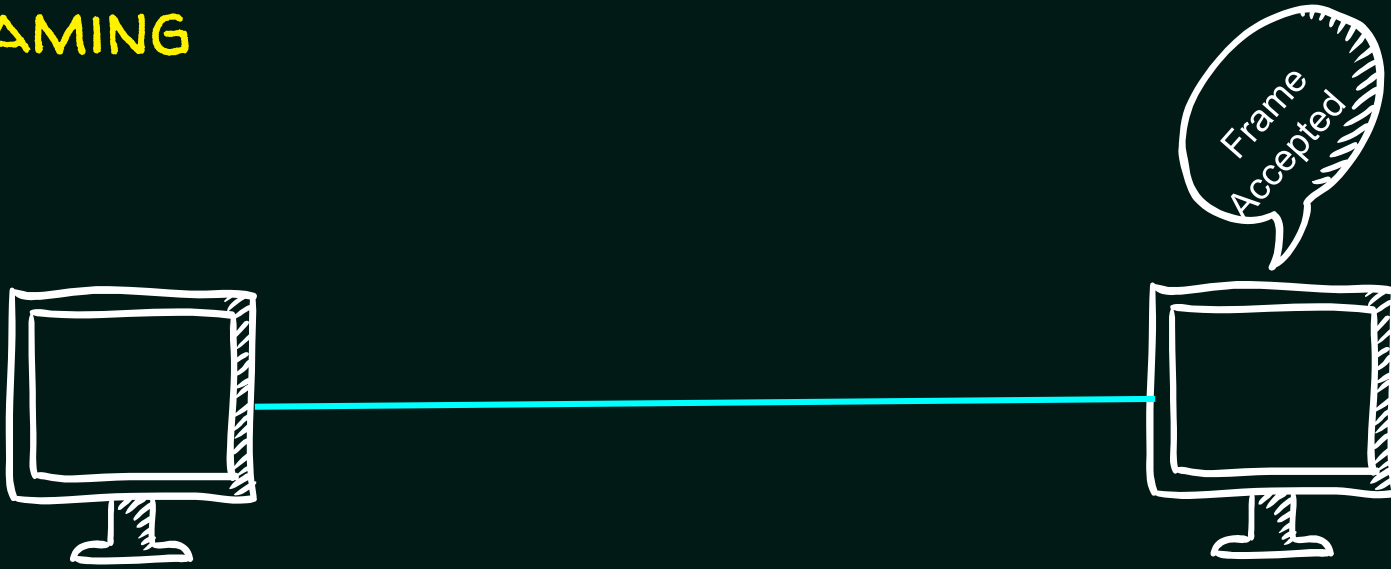
HDLC Protocol: Beginning and Ending Sequence is 01111110

FRAMING



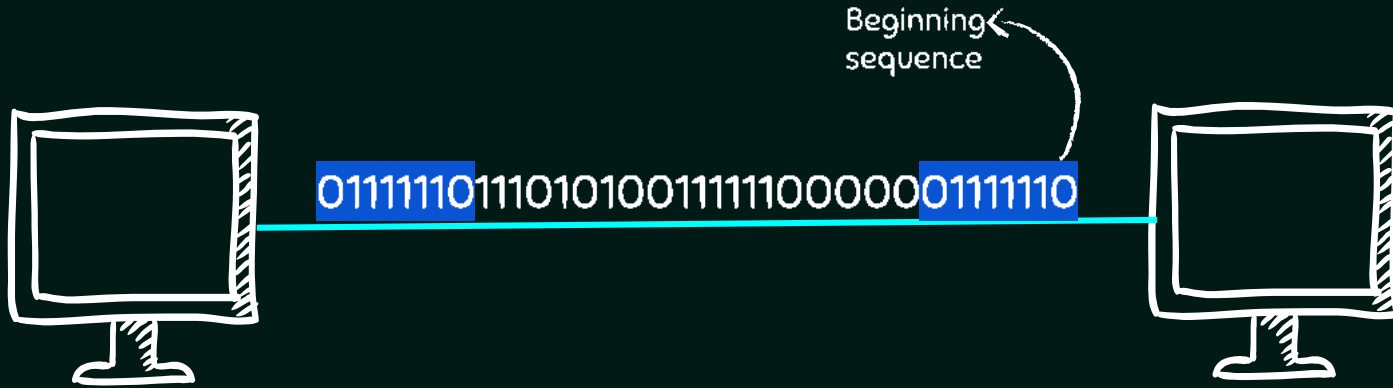
HDLC Protocol: Beginning and Ending Sequence is 01111110

FRAMING



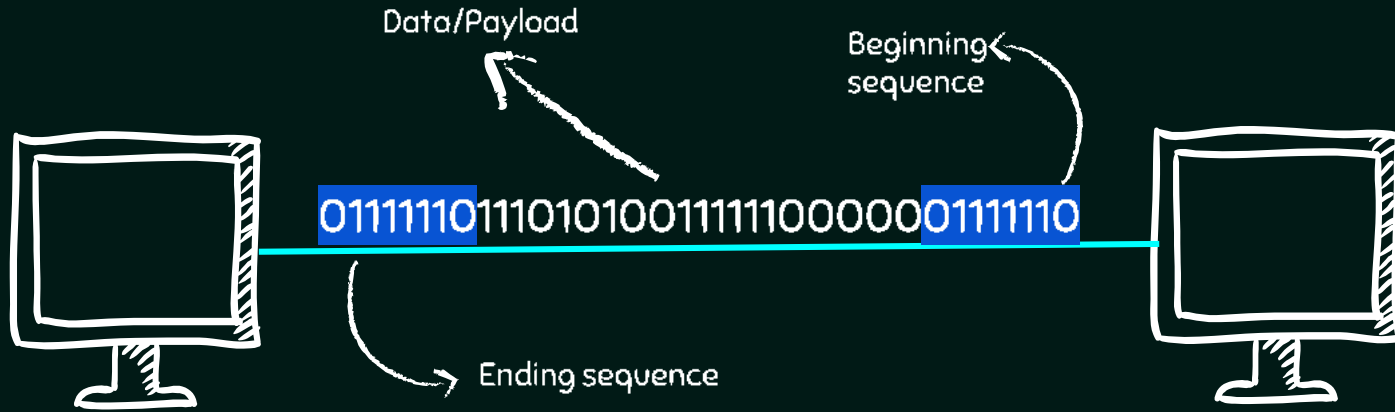
HDLC Protocol: Beginning and Ending Sequence is **01111110**

PROBLEM HERE...



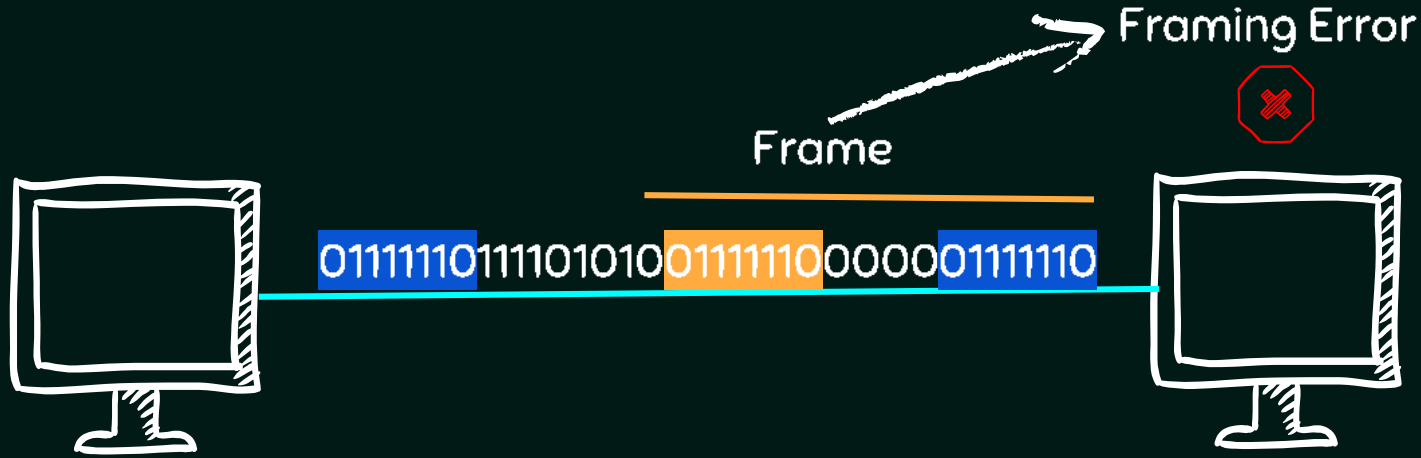
HDLC Protocol: Beginning and Ending Sequence is 01111110

PROBLEM HERE...



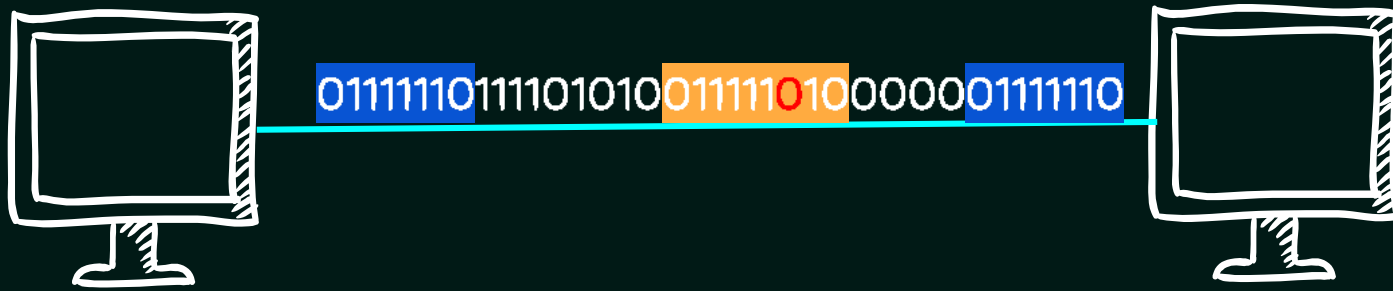
HDLC Protocol: Beginning and Ending Sequence is `01111110`

PROBLEM



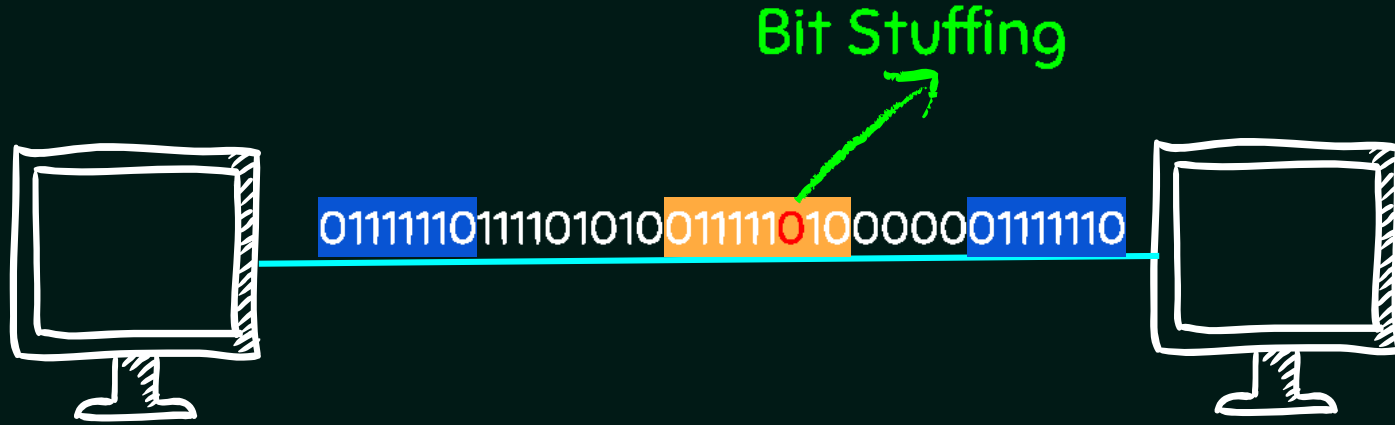
HDLC Protocol: Beginning and Ending Sequence is 01111110

SOLUTION



HDLC Protocol: Beginning and Ending Sequence is 01111110

BIT STUFFING



HDLC Protocol: Beginning and Ending Sequence is 01111110

HOMEWORK!

Correct? or Incorrect?

Bit sequence without stuffing: 1101011111010111110101111110

Bit sequence with bit stuffing: 110101111100101111101010111110110

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand byte-oriented approach
- ★ Know the byte-oriented protocol.
- ★ Know the frame format of BISYNC.
- ★ Understand character stuffing in BISYNC.

BYTE ORIENTED APPROACH

- ★ It simply views the frame as a collection of bytes or characters.

Byte Oriented Protocols

BISYNC $\leftrightarrow$ Binary Synchronous Communication Protocol

PPP $\leftrightarrow$ Point-to-Point Protocol

DDCMP $\leftrightarrow$ Digital Data Communication Message Protocol.

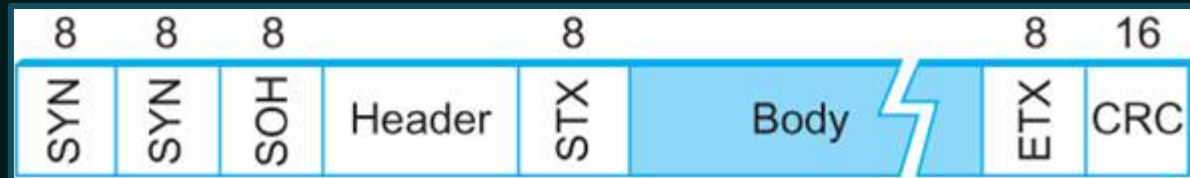
BiSYNC

- ★ It is a sentinel approach.
- ★ Developed by IBM.
- ★ Also preferred as BSC.
- ★ It is a byte-oriented protocol.

BISYNC – FRAME FORMAT

Frames transmitted beginning with leftmost field.

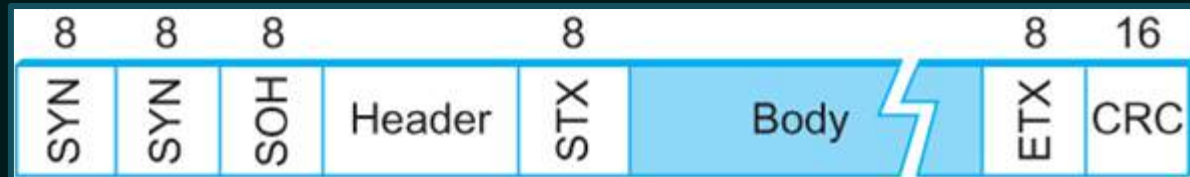
- ★ Beginning of a frame is denoted by sending a special SYN (synchronize) character.
- ★ Data portion of the frame is contained between special sentinel character STX (start of text) and ETX (end of text).
- ★ SOH : Start of Header.
- ★ DLE : Data Link Escape.
- ★ CRC: Cyclic Redundancy Check.



CHARACTER STUFFING

Byte stuffing or Character stuffing is the process of adding one extra byte whenever there is a flag or escape character in the text.

This is done by DLE in BISYNC protocol.



OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the byte-oriented protocol.
- ★ Know the frame format of PPP.
- ★ Understand character stuffing or byte stuffing in PPP.

BYTE ORIENTED APPROACH

- ★ It simply views the frame as a collection of bits.

Byte Oriented Protocols

BISYNC $\leftrightarrow$ Binary Synchronous Communication Protocol

PPP $\leftrightarrow$ Point-to-Point Protocol

DDCMP $\leftrightarrow$ Digital Data Communication Message Protocol.

PPP

- ★ PPP is a data link layer protocol.
- ★ PPP is a WAN protocol and which is commonly run over Internet links.
- ★ It is widely used in broadband communications having heavy loads and high speeds.
- ★ It is used to transmit multiprotocol data between two directly connected (point-to-point) computers.

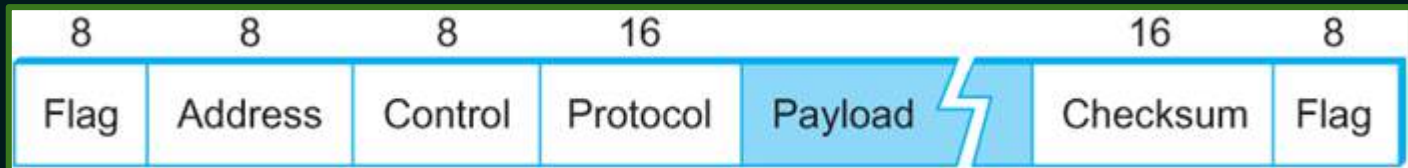
PPP – FRAME FORMAT

- ★ **Flag** – 1 byte that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
- ★ **Address** – 1 byte which is set to 11111111 in case of broadcast.
- ★ **Control** – 1 byte set to a constant value of 11000000.
- ★ **Protocol** – 1 or 2 bytes that define the type of data contained in the payload field.
- ★ **Payload** – This carries the data from the network layer. The maximum length of the payload field is 1500 bytes. However, this may be negotiated between the endpoints of communication.
- ★ **Checksum** – Error detection.



CHARACTER STUFFING

Byte stuffing or Character stuffing is the process of adding one extra byte whenever there is a flag sequence appear in the payload.



OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand DDCMP frame format.
- ★ Understand the framing error in DDCMP.

BYTE ORIENTED APPROACH

- ★ It simply views the frame as a collection of bytes or characters.

Byte Oriented Protocols

BISYNC $\leftrightarrow$ Binary Synchronous Communication Protocol

PPP $\leftrightarrow$ Point-to-Point Protocol

DDCMP $\leftrightarrow$ Digital Data Communication Message Protocol.

DDCMP

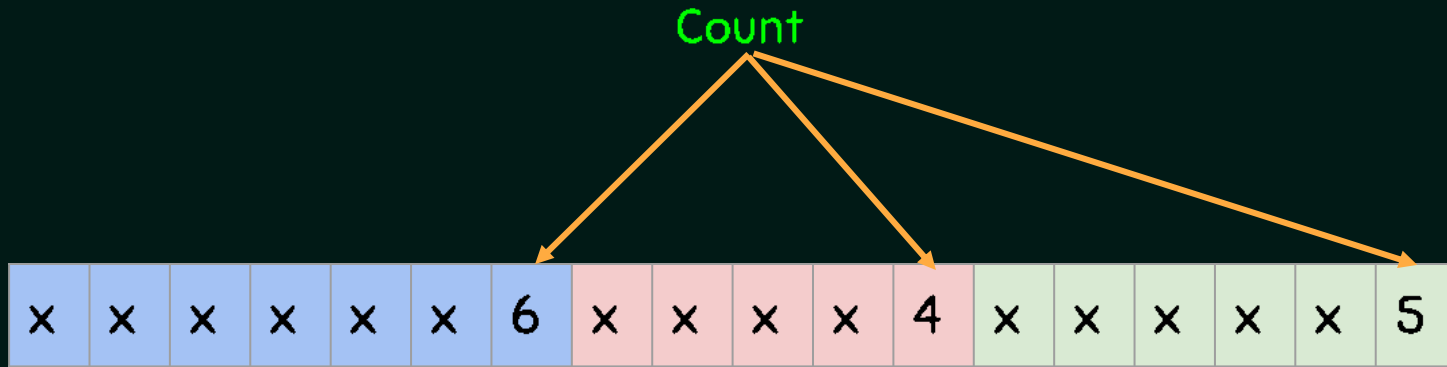
- ★ Byte-oriented communication protocol.
- ★ Devised by Digital Equipment Corporation.
- ★ It is a byte-counting approach.
- ★ Count field in the frame format.
- ★ **Count:** How many bytes are contained in the frame body?

DANGER WITH THE COUNT FIELD

One danger with this approach is that if transmission error could corrupt the count field then the end of the frame would not be correctly detected by the receiver.



DANGER WITH THE COUNT FIELD



DANGER WITH THE COUNT FIELD

Transmission Error



The diagram shows a horizontal sequence of 18 cells representing a data packet. The first 7 cells are blue and contain 'x', 'x', 'x', 'x', 'x', 'x', and '6'. The next 5 cells are pink and contain 'x', 'x', 'x', 'x', and '4'. A red arrow points from the text 'Transmission Error' to the pink cell containing '4'. The final 6 cells are green and contain 'x', 'x', 'x', 'x', 'x', and '5'.

x	x	x	x	x	x	6	x	x	x	x	4	x	x	x	x	x	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

DANGER WITH THE COUNT FIELD

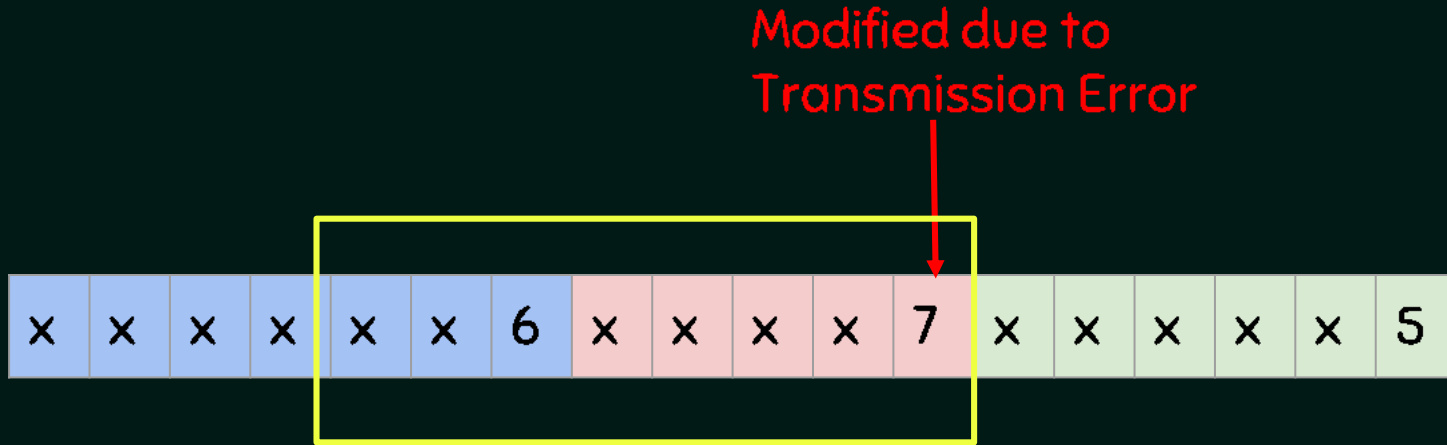
Modified due to
Transmission Error



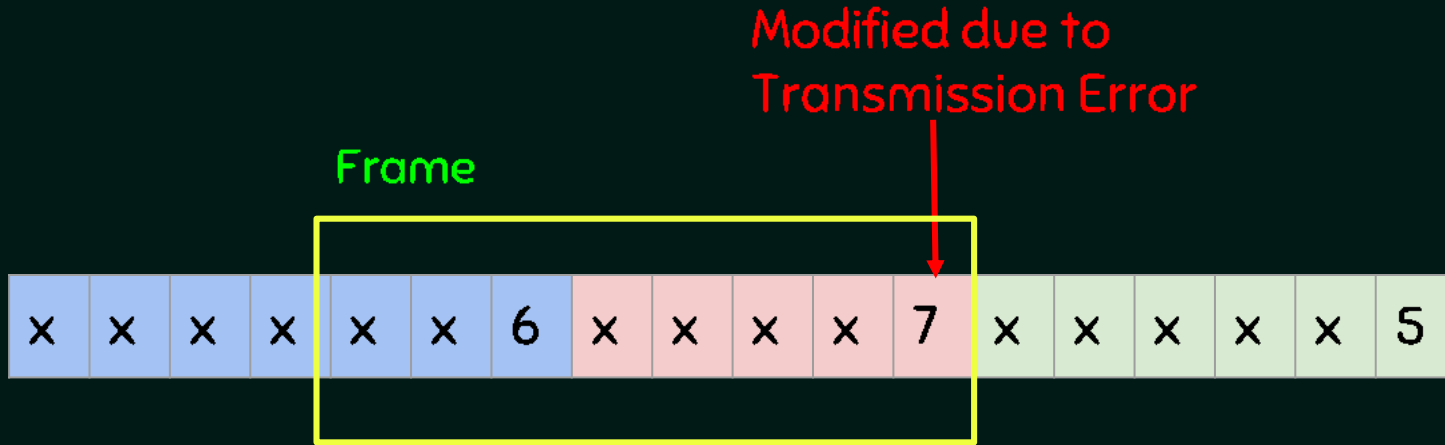
The diagram shows a horizontal sequence of 18 cells representing a data packet. The first 7 cells are light blue, the next 5 cells are light red, and the last 6 cells are light green. A red arrow points from the text 'Modified due to Transmission Error' to the 12th cell, which contains the number 7. The 7th cell contains the number 6, and the 18th cell contains the number 5. All other cells contain an 'x'.

x	x	x	x	x	x	6	x	x	x	x	7	x	x	x	x	x	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

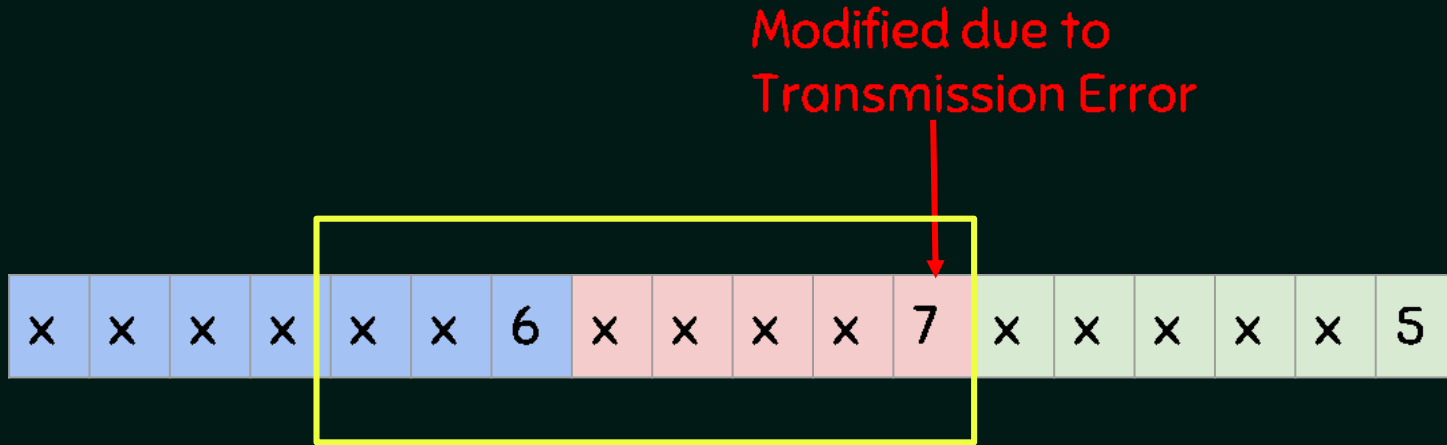
DANGER WITH THE COUNT FIELD



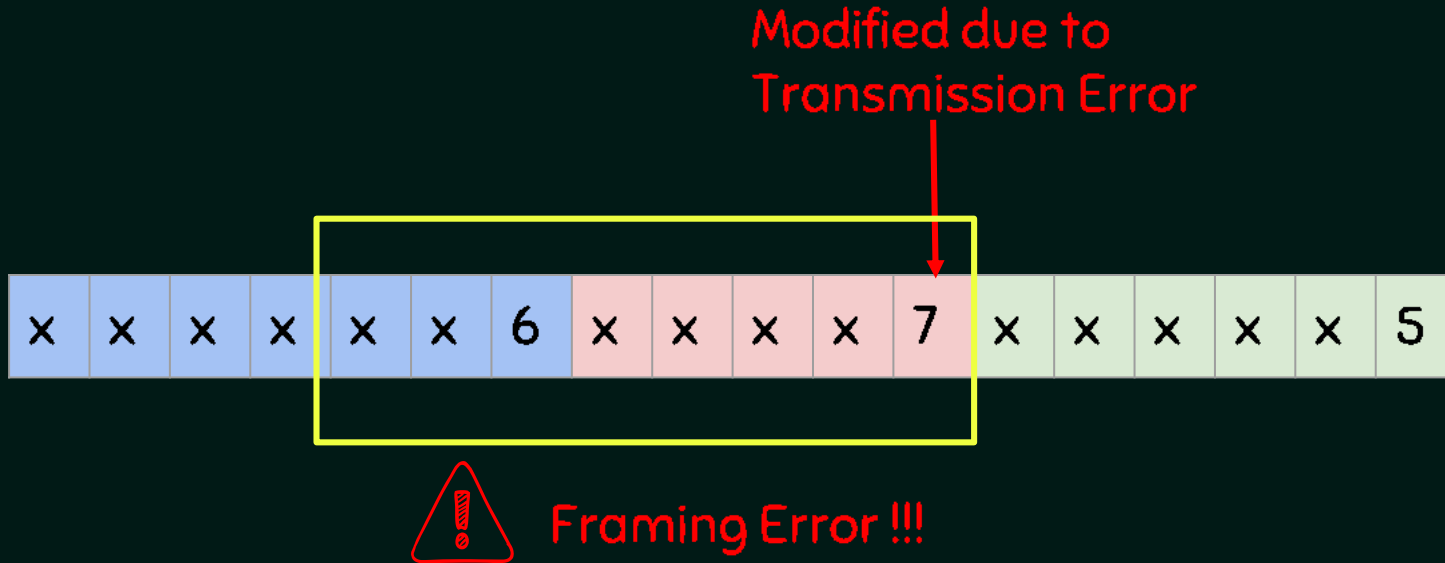
DANGER WITH THE COUNT FIELD



DANGER WITH THE COUNT FIELD



DANGER WITH THE COUNT FIELD



OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand transmission errors.
- ★ Know the types of errors.
- ★ Understand error detection and error correction.
- ★ Know the various error detection techniques.

ERROR

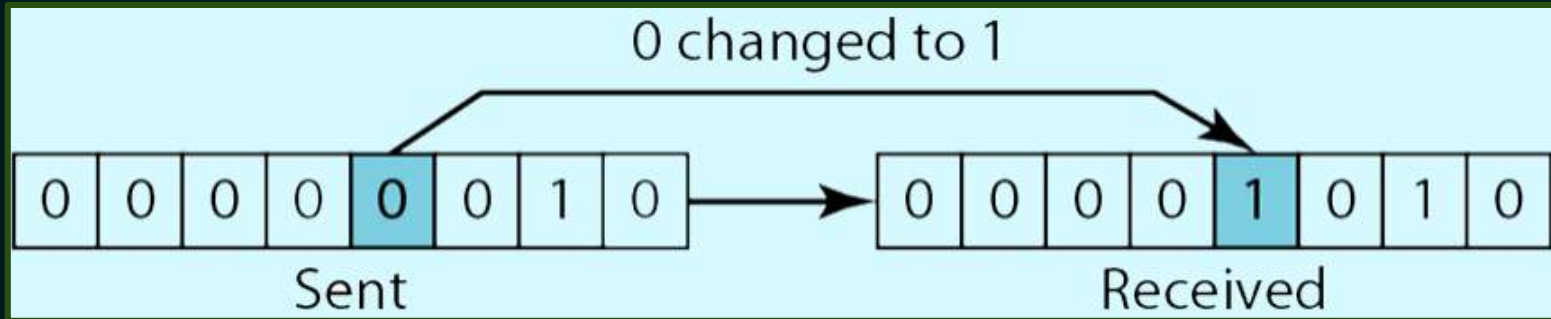
- ★ Data are transmitted in the network.
- ★ The data can be corrupted during transmission.
- ★ Transmission error.
- ★ For reliable communication, errors must be detected and corrected.
- ★ Error detection and correction are implemented either at the data link layer or the transport layer of the OSI model.

TYPES OF ERROR

1. Bit Error.
2. Burst Error.

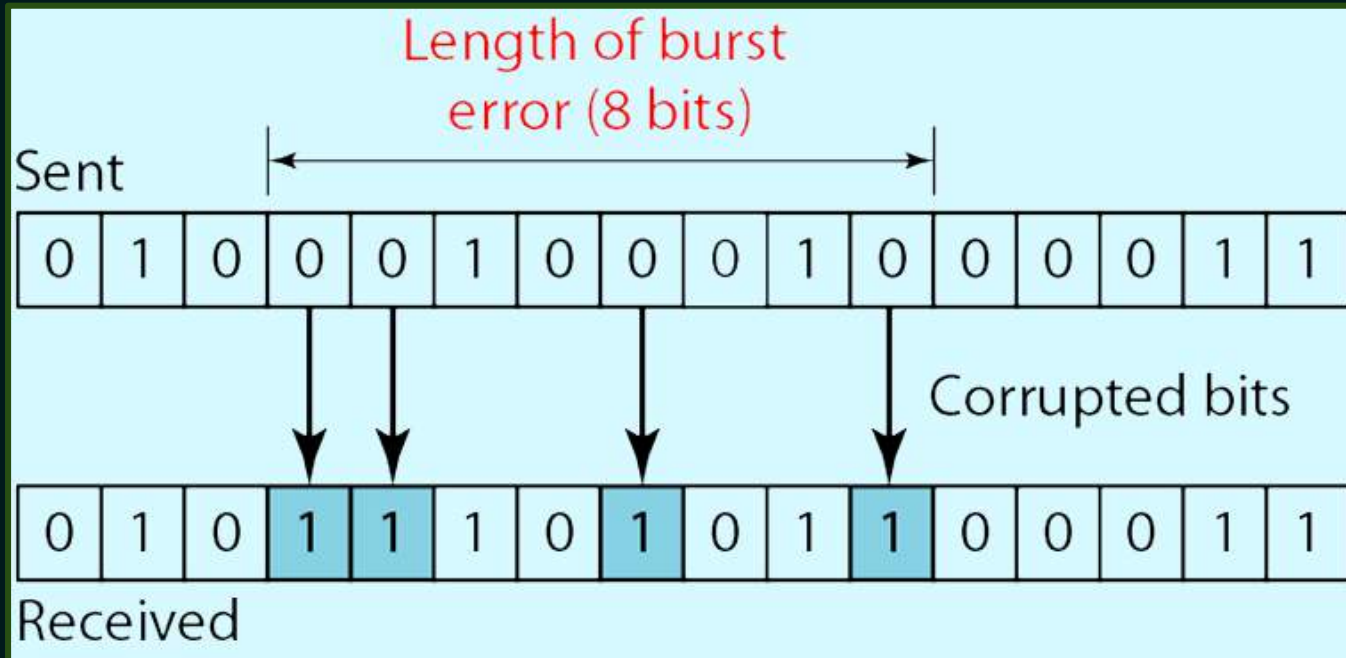
BIT ERROR

- ★ a.k.a single bit error.
- ★ In a single bit error, only 1 bit in the data unit has been changed.



BURST ERROR

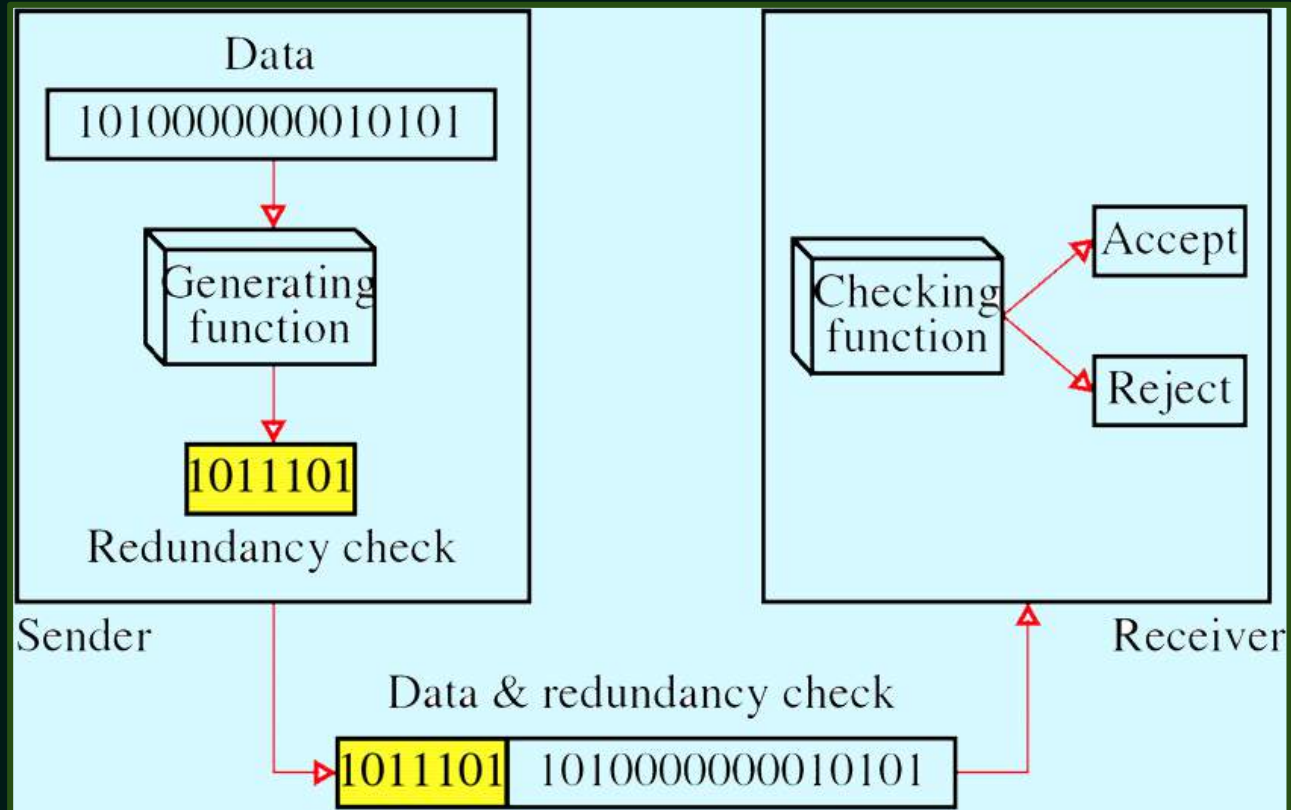
- ★ In burst error, 2 or more bits in the data unit have changed.



HOW TO DETECT THE ERRORS?

- ★ Error detection means to decide whether the received data is correct or not without having a copy of the original message.
- ★ To detect or correct errors, we need to send some extra bits with the data.
- ★ The extra bits are called as redundant bits.

REDUNDANCY

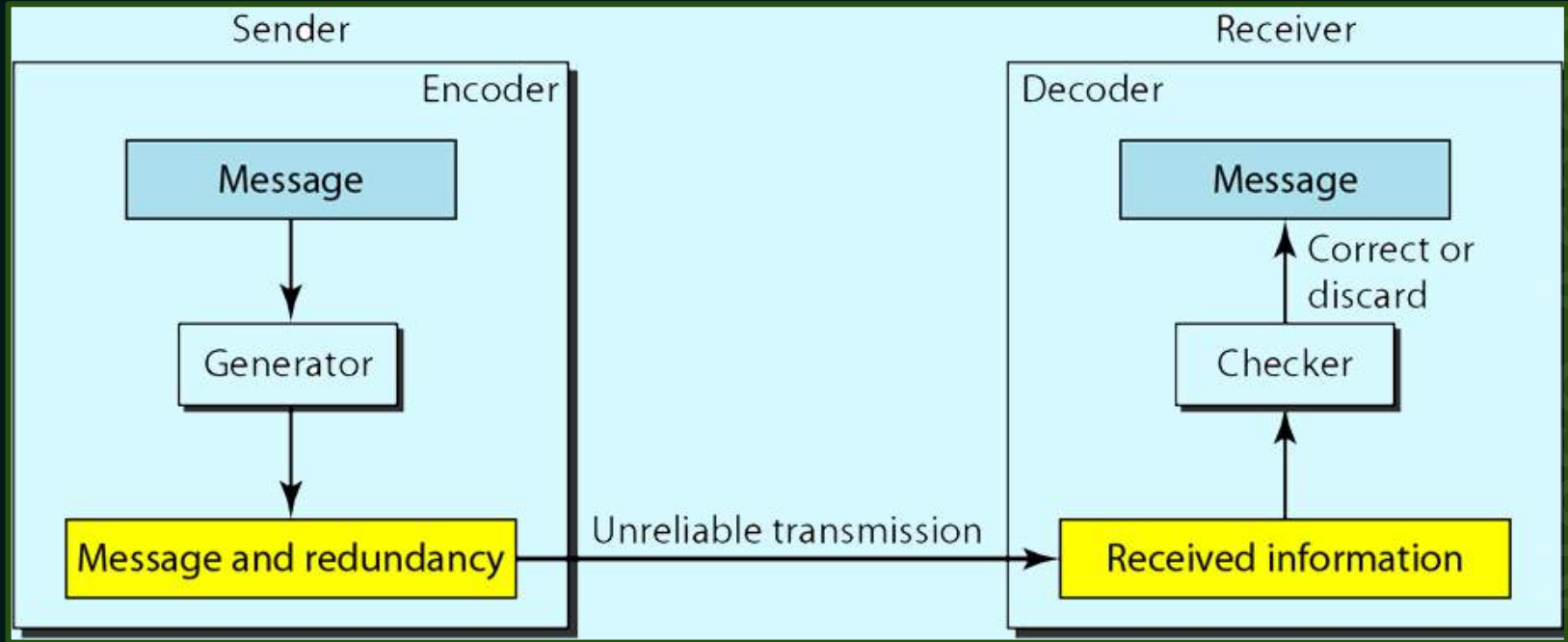


ERROR CORRECTION

It can be handled in two ways:

- 1) Receiver can have the sender retransmit the entire data unit.
- 2) The receiver can use an error-correcting code, which automatically corrects certain errors.

ERROR DETECTION/CORRECTION



ERROR DETECTION TECHNIQUES

Four types of redundancy checks are used in data communications. They are:

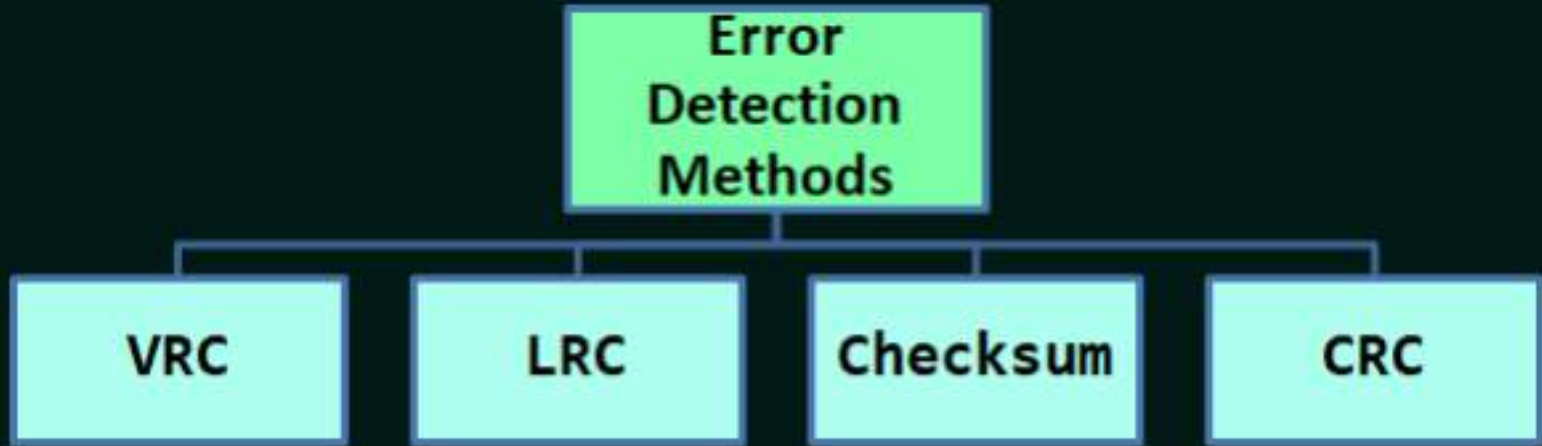
1. Vertical Redundancy Check (VRC)
2. Longitudinal Redundancy Check (LRC)
3. Checksum
4. Cyclic Redundancy Check (CRC)

OUTCOMES

Upon the completion of this session, the learner will be able to

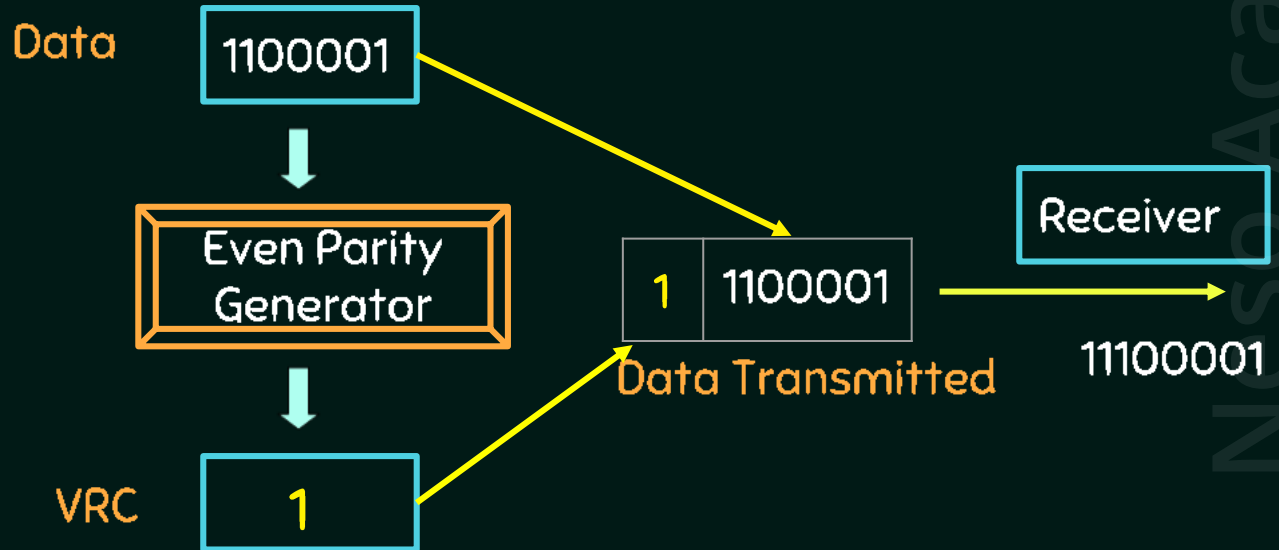
- ★ Know the types of error detection..
- ★ Understand VRC and its performance.
- ★ Understand LRC and its performance.

ERROR DETECTION



VRC

- ★ Vertical Redundancy check.
- ★ It is also called as parity check.



PERFORMANCE OF VRC

- ★ It can detect single bit error.
- ★ It can detect burst error only if the number of errors is odd.

Sender: 11100001 → Transmission Error 10100001 → Receiver rejects this data.

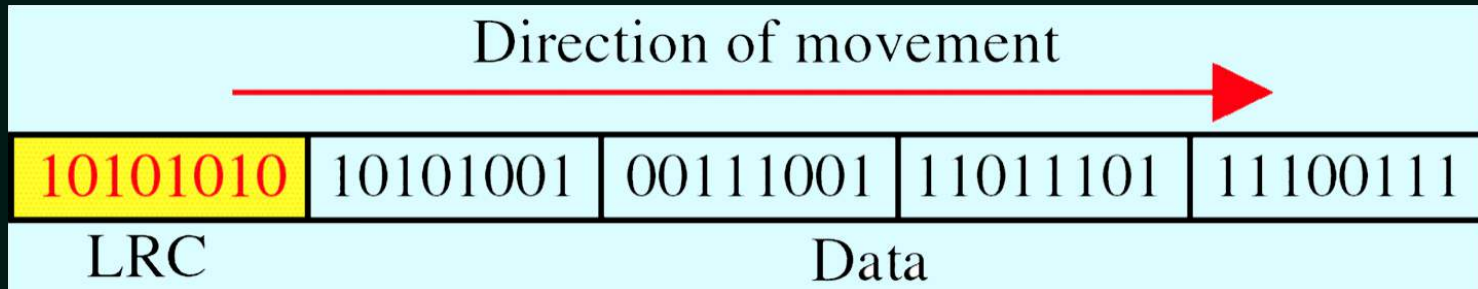
Sender: 11100001 → Transmission Error 10100101 → Receiver accepts this data.

LRC

- ★ Longitudinal Redundancy Check.
- ★ A block of bits is organized in rows and columns.
- ★ a.k.a Two Dimensional parity.
- ★ The parity bit is calculated for each column and sent along with the data.
- ★ The block of parity acts as the redundant bits.

LRC

	1	1	1	0	0	1	1	1
	1	1	0	1	1	1	0	1
	0	0	1	1	1	0	0	1
	1	0	1	0	1	0	0	1
LRC →	1	0	1	0	1	0	1	0



PERFORMANCE OF VRC

- ★ LCR increases the likelihood of detecting burst errors.
- ★ If two bits in one data units are damaged and two bits in exactly the same positions in another data unit are also damaged, the LRC checker will not detect an error.

HOME WORK – ANSWERS

1. Append the parity bit after each block shown below using VRC.

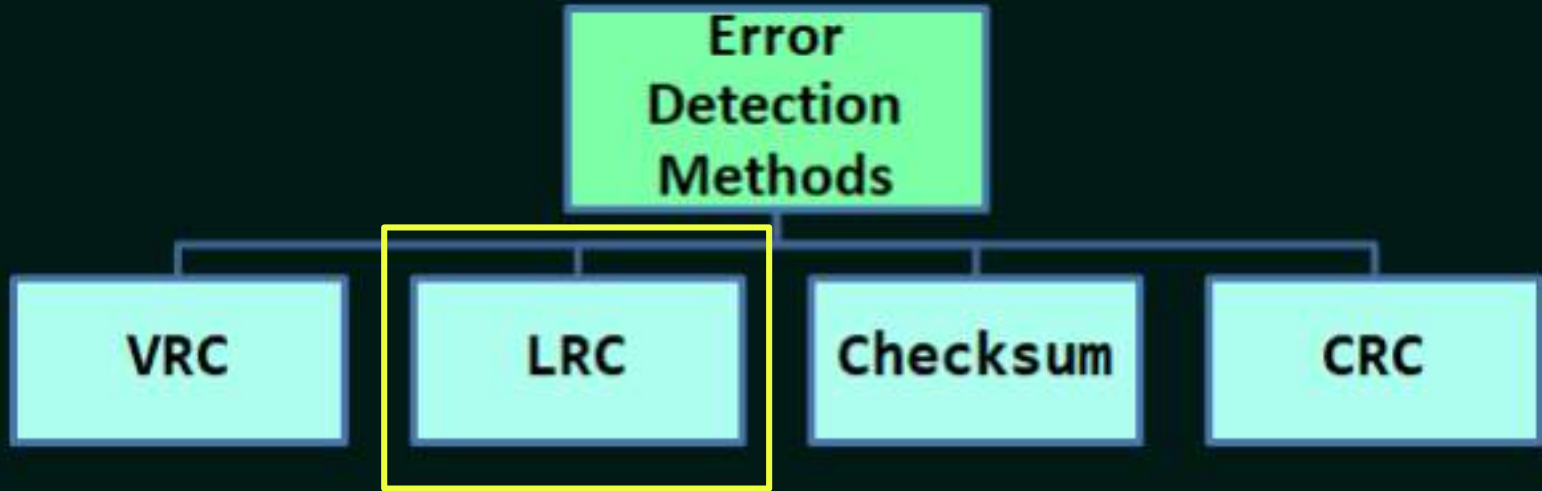
1110110 1101111 1110010

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the types of error detection.
- ★ Understand LRC and its performance.

ERROR DETECTION



LRC

- ★ Longitudinal Redundancy Check.
- ★ In LRC, a block of bits is organized in rows and columns.
- ★ a.k.a Two Dimensional parity.
- ★ The parity bit is calculated for each column and sent along with the data.
- ★ The block of parity acts as the redundant bits.

LRC – EXAMPLE

Find the LRC for the data blocks 11100111 11011101 00111001 10101001 and determine the data that is transmitted?

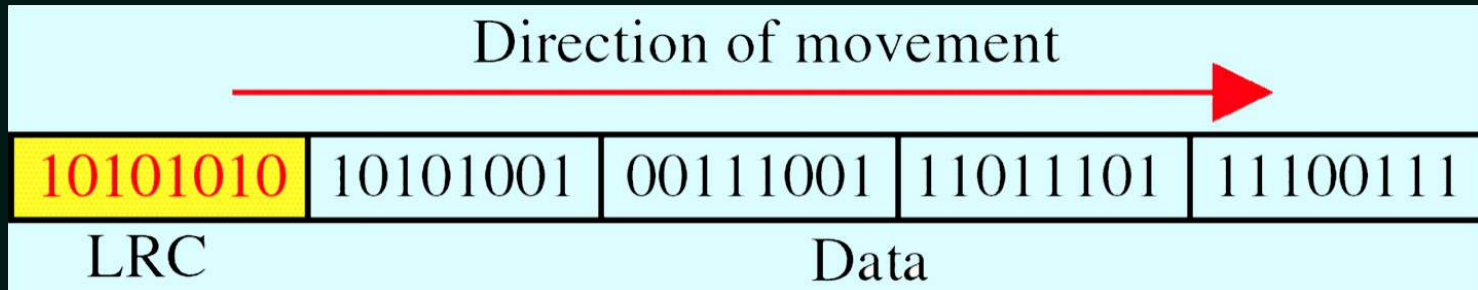
Odd no. of 1's	1
Even no. of 1's	0

LRC →

1	1	1	0	0	1	1	1
1	1	0	1	1	1	0	1
0	0	1	1	1	0	0	1
1	0	1	0	1	0	0	1
1	0	1	0	1	0	1	0

LRC - EXAMPLE

	1	1	1	0	0	1	1	1
	1	1	0	1	1	1	0	1
	0	0	1	1	1	0	0	1
	1	0	1	0	1	0	0	1
LRC →	1	0	1	0	1	0	1	0



PERFORMANCE OF LRC

- ★ LCR increases the likelihood of detecting burst errors.
- ★ If two bits in one data units are damaged and two bits in exactly the same positions in another data unit are also damaged, the LRC checker will not detect an error.

	1	1	1	0	0	1	1	1
	1	1	0	1	1	1	0	1
	0	0	1	1	1	0	0	1
	1	0	1	0	1	0	0	1
LRC →	1	0	1	0	1	0	1	0

HOMEWORK

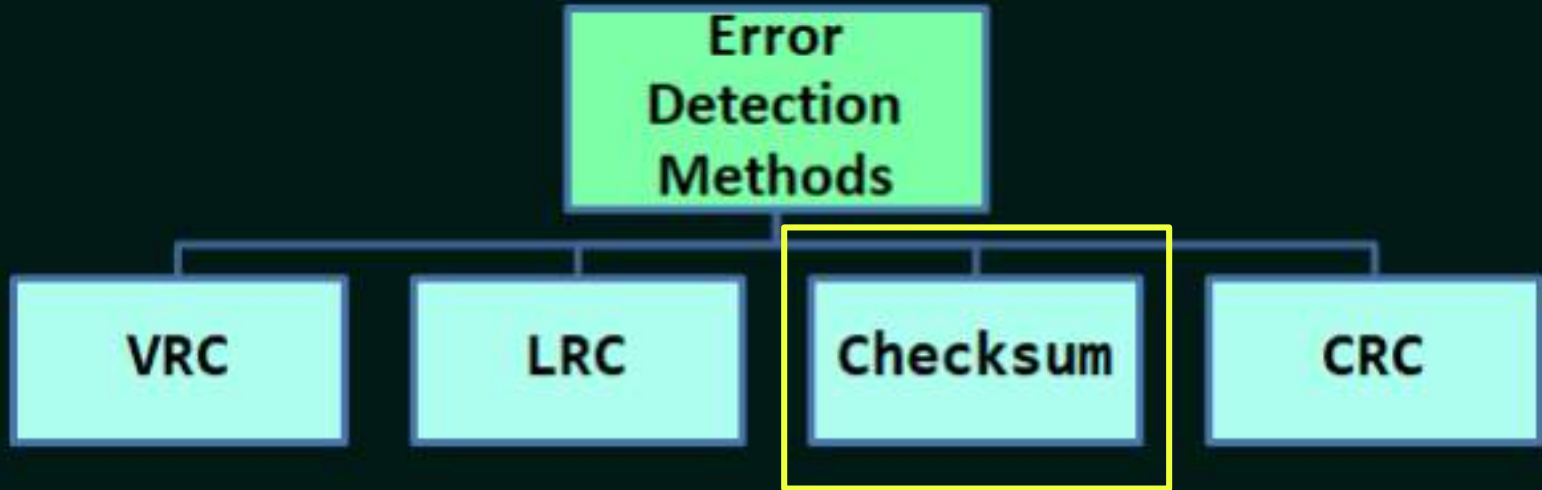
Find the LRC for the data blocks 01110111 10101001 01101001 10101010 and determine the data that is transmitted.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand checksum and its performance.

ERROR DETECTION



CHECKSUM

Checksum = Check + sum.

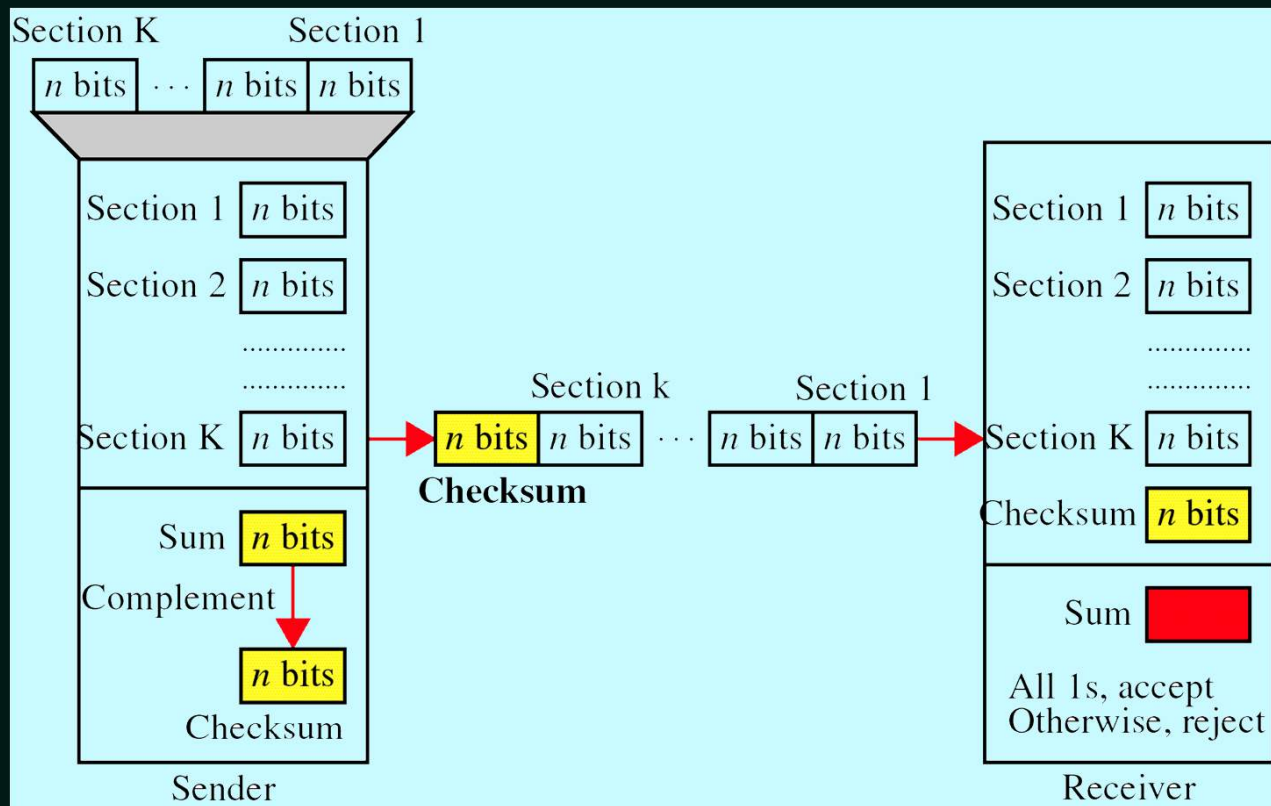
Sender side – Checksum Creation.

Receiver side – Checksum Validation.

CHECKSUM – OPERATION AT SENDER SIDE

1. Break the original message in to 'k' number of blocks with 'n' bits in each block.
2. Sum all the 'k' data blocks.
3. Add the carry to the sum, if any.
4. Do 1's complement to the sum = Checksum.

CHECKSUM



CHECKSUM – EXAMPLE

Consider the data unit to be transmitted is:

10011001111000100010010010000100

10011001	11100010	00100100	10000100
----------	----------	----------	----------

CHECKSUM - EXAMPLE



10011001

11100010

00100100

10000100

Carry

1 1 1 1 1

1 0 0 0 0 1 0 0

0 0 1 0 0 1 0 0

1 1 1 0 0 0 1 0

1 0 0 1 1 0 0 1

0 0 1 0 0 0 1 1

1 0

0 0 1 0 0 1 0 1

1's Complement

1 1 0 1 1 0 1 0

CHECKSUM - EXAMPLE



10011001

11100010

00100100

10000100

Carry

1 1 1 1 1

1 0 0 0 0 1 0 0

0 0 1 0 0 1 0 0

1 1 1 0 0 0 1 0

1 0 0 1 1 0 0 1

0 0 1 0 0 0 1 1

1 0

0 0 1 0 0 1 0 1

CHECKSUM

1 1 0 1 1 0 1 0

CHECKSUM - EXAMPLE



10011001

11100010

00100100

10000100

Carry

1

1

1

1

1

1

0

0

0

0

1

0

0

0

0

1

0

0

1

0

0

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CHECKSUM

1

1

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1

0

1

0

CHECKSUM - EXAMPLE

11011010

10011001

11100010

00100100

10000100

Carry

1

1

1

1

1

1

0

0

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1

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0

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1

0

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0

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0

1

0

0

1

0

0

0

1

1

1

0

0

0

1

0

0

1

0

1

CHECKSUM

1

1

0

1

1

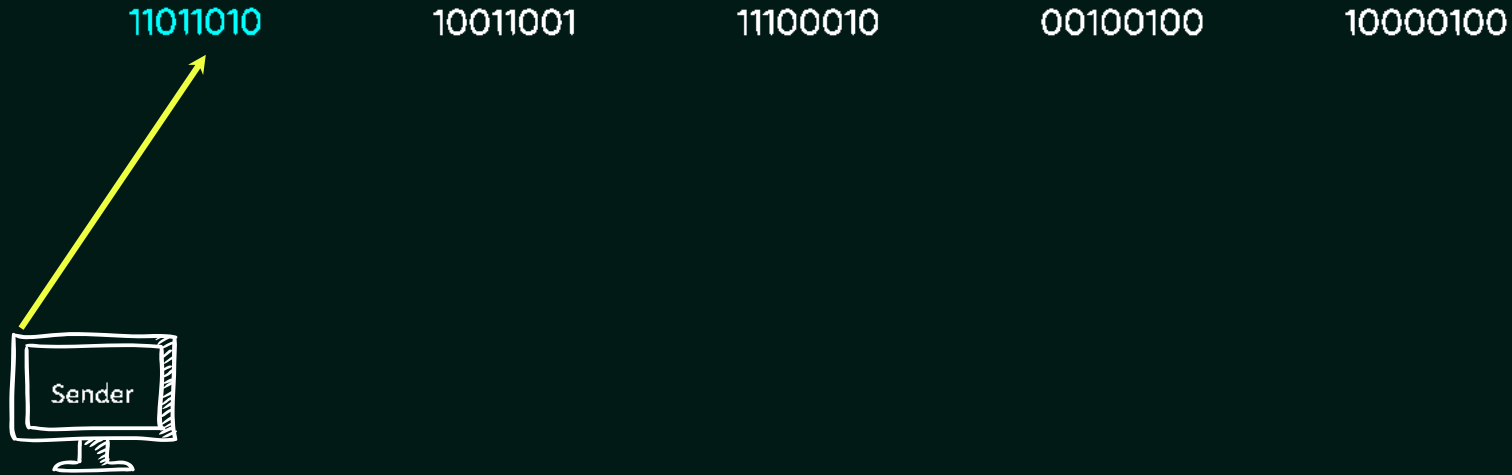
0

1

0



CHECKSUM - EXAMPLE



CHECKSUM – OPERATION AT RECEIVER SIDE

- ★ Collect all the data blocks including the checksum.
- ★ Sum all the data blocks and checksum
- ★ If the result is all 1's, ACCEPT; Else, REJECT.

CHECKSUM - EXAMPLE

11011010

10011001

11100010

00100100

10000100

Carry

1	1	1	1	1	1	1		
1	0	0	0	0	0	1	0	0
0	0	1	0	0	0	1	0	0
1	1	1	0	0	0	0	1	0
1	0	0	1	1	0	0	0	1
1	1	0	1	1	0	0	1	0
1	1	1	1	1	1	1	0	1
							1	0
1	1	1	1	1	1	1	1	1



PERFORMANCE OF CHECKSUM

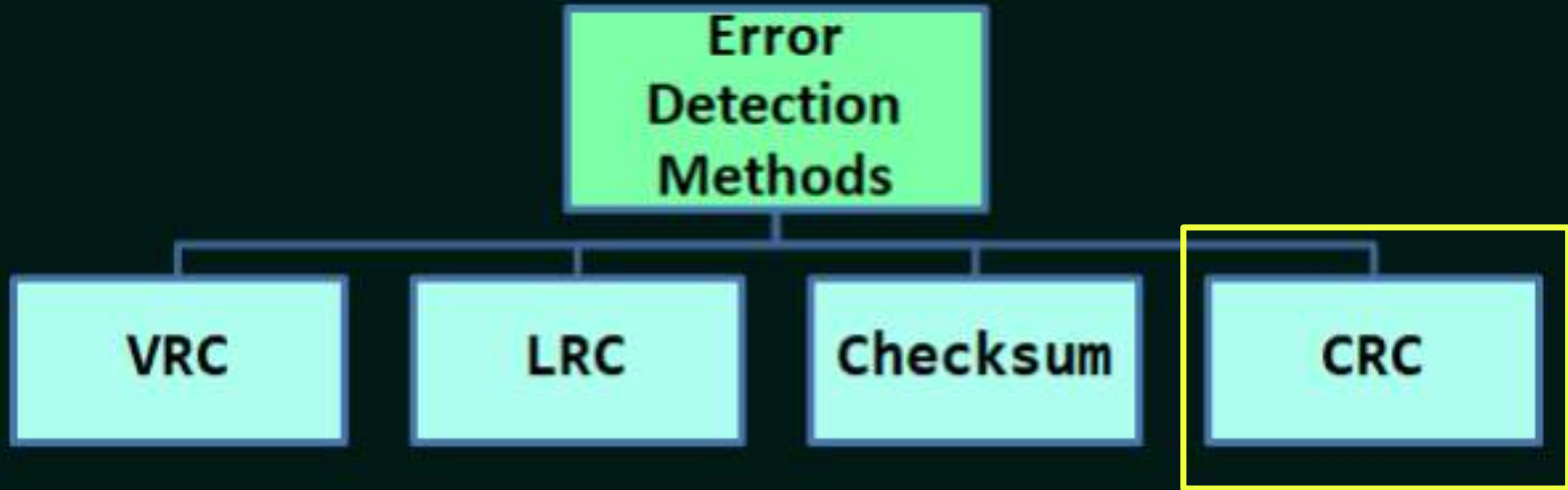
- ★ The checksum detects all errors involving an odd number of bits.
- ★ It detects most errors involving an even number of bits.
- ★ If one or more bits of a segment are damaged and the corresponding bit or bits of opposite value in a second segment are also damaged, the sums of those columns will not change and the receiver will not detect the error(s).

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the types of error detection.
- ★ Understand how to generate CRC.

ERROR DETECTION



CRC GENERATION AT SENDER SIDE

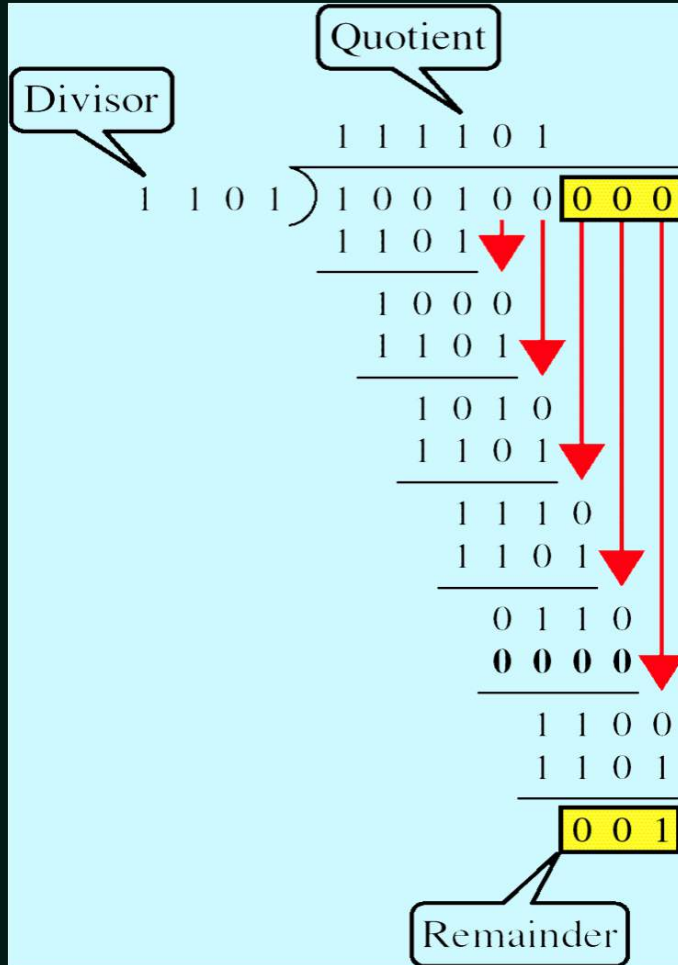
1. Find the length of the divisor 'L'.
2. Append 'L-1' bits to the original message.
3. Perform binary division operation.
4. Remainder of the division = CRC.

Note:

The CRC must be of L-1 bits.

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

CRC

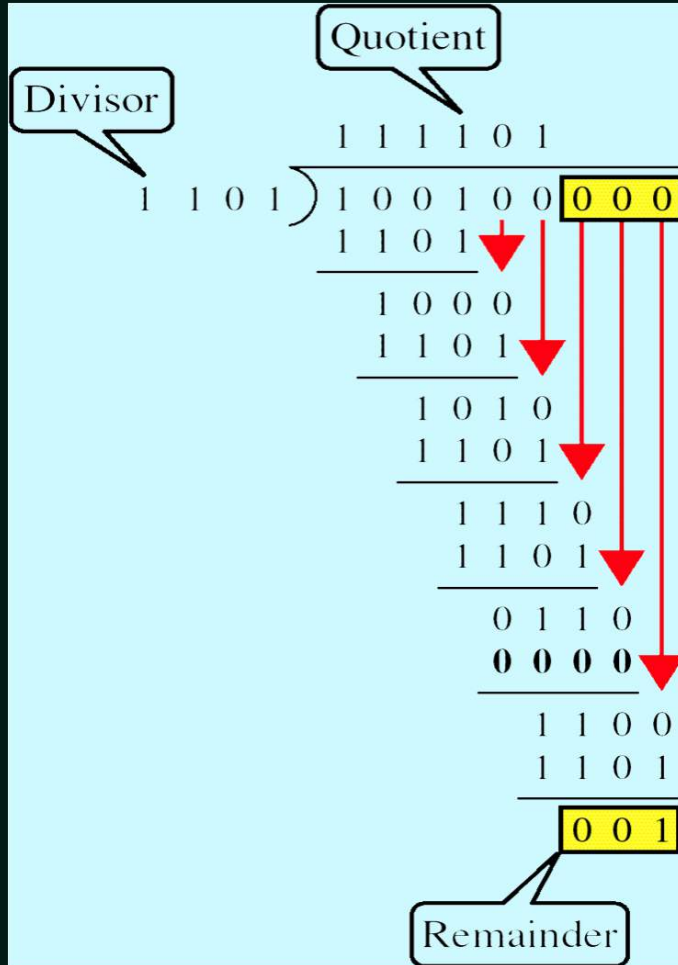


CRC: 001

Data Transmitted: 100100001

Data

CRC



CRC: 001

Data Transmitted: 100100001

CRC

HOMEWORK

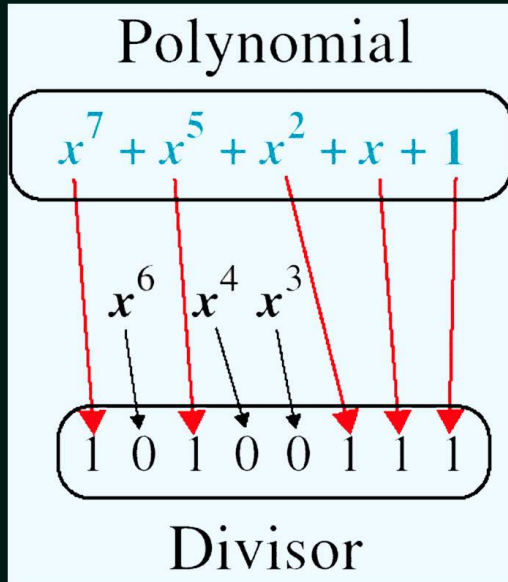
Find the CRC for 1110010101 with the divisor x^3+x^2+1 ?

Hint:

HOMEWORK

Find the CRC for 1110010101 with the divisor x^3+x^2+1 ?

Hint:



OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand how to detect error using CRC technique.

CRC

Find the CRC for the data block 100100 with the divisor 1101?

CRC

				1	1	1	1	0	1			
1	1	0	1	1	0	0	1	0	0	0	0	1
				1	1	0	1	↓	↓	↓	↓	↓
				1	0	0	0					
				1	1	0	1	↓				
				1	0	1	0					
				1	1	0	1	↓				
					1	1	1	0				
					1	1	0	1	↓	↓		
						0	1	1	0	1		
							1	1	0	1		
								0	0	0		



Data
Accepted

HOMEWORK

Suppose we want to transmit the message 11001001 and protect it from errors using the CRC polynomial x^3+1 . Use polynomial long division to determine the message that should be transmitted. Corrupt the left-most third bit of the transmitted message and show that the error is detected by the receiver using CRC technique.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Generate CRC code.
- ★ Determine the message to be transmitted.

QUESTION

The message 11001001 is to be transmitted using CRC polynomial x^3+1 to protect it from errors. The message that should be transmitted is:

[GATE CS 2007]

- A. 11001001000
- B. 11001001011
- C. 11001010
- D. 110010010011

CRC GENERATION AT SENDER SIDE

1. Find the length of the divisor 'L'.
2. Append 'L-1' bits to the original message.
3. Perform binary division operation.
4. Remainder of the division = CRC.
5. Message to be transmitted = Message + CRC

Note:

The CRC must be of L-1 bits.

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

CRC GENERATION AT SENDER SIDE

1. The length of the divisor is 4. ($x^3+1 = 1x^3+0x^2+0x^1+1$) = 1001
2. Append '4-1' bits to the original message. (3 bits are appended)
3. Perform binary division operation.
4. Remainder of the division = CRC.
5. Message to be transmitted = Message + CRC

Note:

The CRC must be of $L-1$ i.e $4-1=3$ bits.

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

CRC GENERATION



MESSAGE TO BE TRANSMITTED

Message to be transmitted = Original Message + CRC

Message to be transmitted = 11001001 011

QUESTION

The message 11001001 is to be transmitted using CRC polynomial x^3+1 to protect it from errors. The message that should be transmitted is:

[GATE CS 2007]

A. 11001001000

B. 11001001011 

C. 11001010

D. 110010010011

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the importance of network performance.
- ★ Understand bandwidth.
- ★ Understand throughput.
- ★ Know latency (delay) and its components.

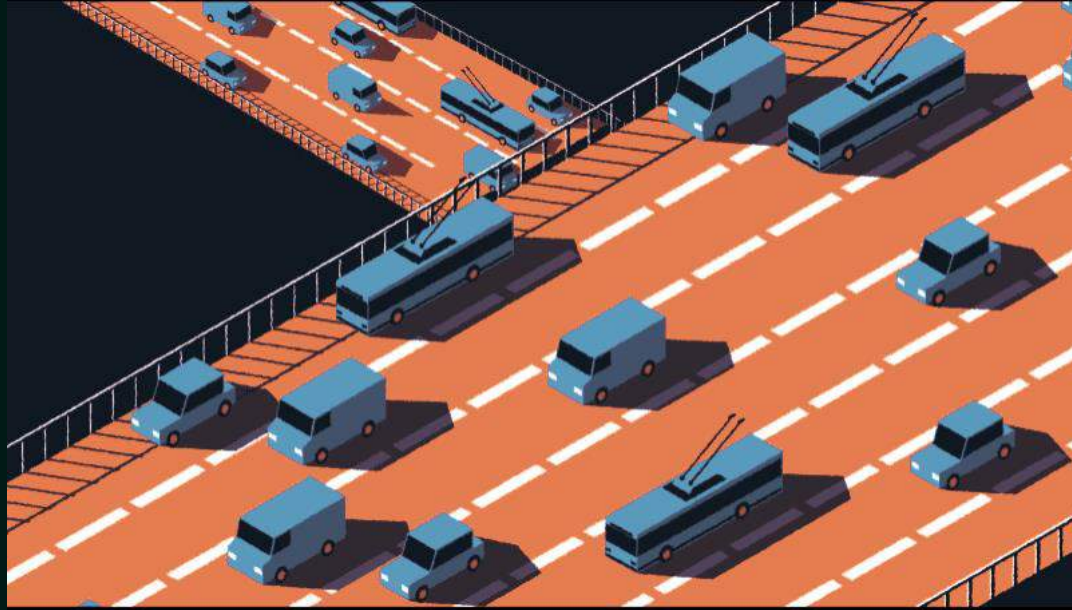
NETWORK PERFORMANCE

One important issue in networking is the performance of the network – how good is it?

Network performance is measured in following fundamental ways

- ★ Bandwidth
- ★ Throughput
- ★ Latency (Delay)

BANDWIDTH AND THROUGHPUT – ANALOGY



DERIVATIONS FROM ANALOGY

The bandwidth is 1000 cars per minute.

The throughput is 100 cars per minute.

BANDWIDTH

Informal: Maximum amount of data that can be transmitted per second.

Formal: The bandwidth of a network is given by the number of bits that can be transmitted over the network in a certain period of time.

Bandwidth in bps

Bandwidth = Capability.

Example: Gigabit Ethernet can provide a bandwidth of 1 Gbps.

Bandwidth in Hertz

A range of frequencies used to transmit signals which is measured in hertz.

THROUGHPUT

Informal: Actual amount of data that passes through the medium.

Formal: The throughput is a measure of how fast we can actually send data through a network.

Although bandwidth in bits per second and throughput seem the same, they are different.

A link may have a bandwidth of 'B' bps, but we can only send 'T' bps through this link with $T < B$ always.

THROUGHPUT

We may have a link with a bandwidth of 1 Mbps, but the devices connected to the end of the link may handle only 200 kbps. This means that we cannot send more than 200 kbps through this link.

LATENCY (DELAY)

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

COMPONENTS OF LATENCY (DELAY)

Latency is made of four components:

1. Transmission delay.
2. Propagation delay.
3. Queueing delay.
4. Processing delay.

Latency = Transmission delay + Propagation delay + Queueing delay + Processing delay.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the various components of latency.

LATENCY (DELAY)

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time, the first bit is sent out from the source.

COMPONENTS OF LATENCY (DELAY)

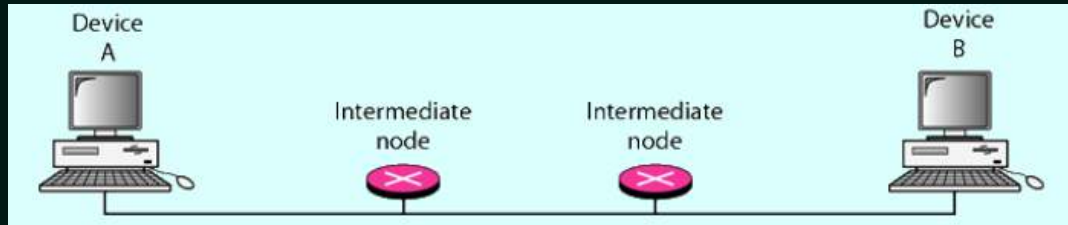
Latency is made of four components:

1. Transmission delay.
2. Propagation delay.
3. Queuing delay.
4. Processing delay.

Latency = Transmission delay + Propagation delay + Queuing delay + Processing delay.

TRANSMISSION DELAY

Time it takes to place the complete data packet on the transmission medium.



TRANSMISSION DELAY

Time it takes to place the complete data packet on the transmission medium.

$$\text{Transmission Time} = \frac{\text{Message size}}{\text{Bandwidth}}$$

PROPAGATION DELAY

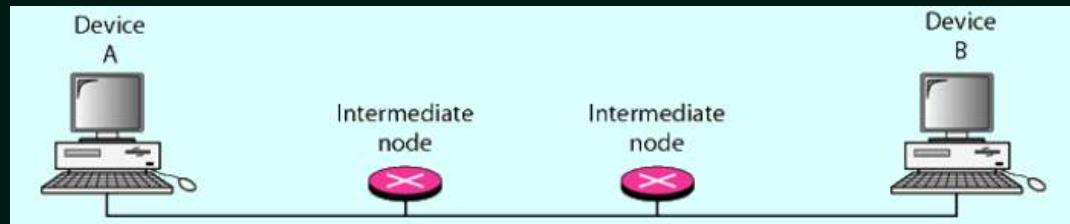
Time it takes for a bit to go from device A to device B.

The propagation time is calculated by dividing the distance by the propagation speed.

$$\textit{Propagation Time} = \frac{\textit{Distance}}{\textit{Propagation speed}}$$

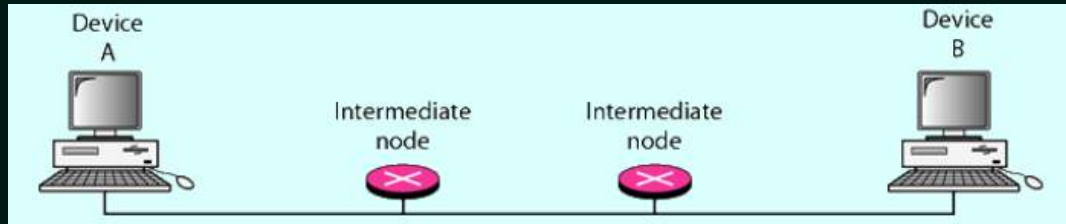
QUEUING DELAY

- ★ The third component in latency is the queuing time, the time needed for each intermediate or end device to hold the message before it can be processed.
- ★ The queuing time is not a fixed factor; it changes with the load imposed on the network.
- ★ When there is heavy traffic on the network, the queuing time increases.



PROCESSING DELAY

- ★ How much time the node takes to process the message?



OUTCOMES

Upon the completion of this session, the learner will be able to

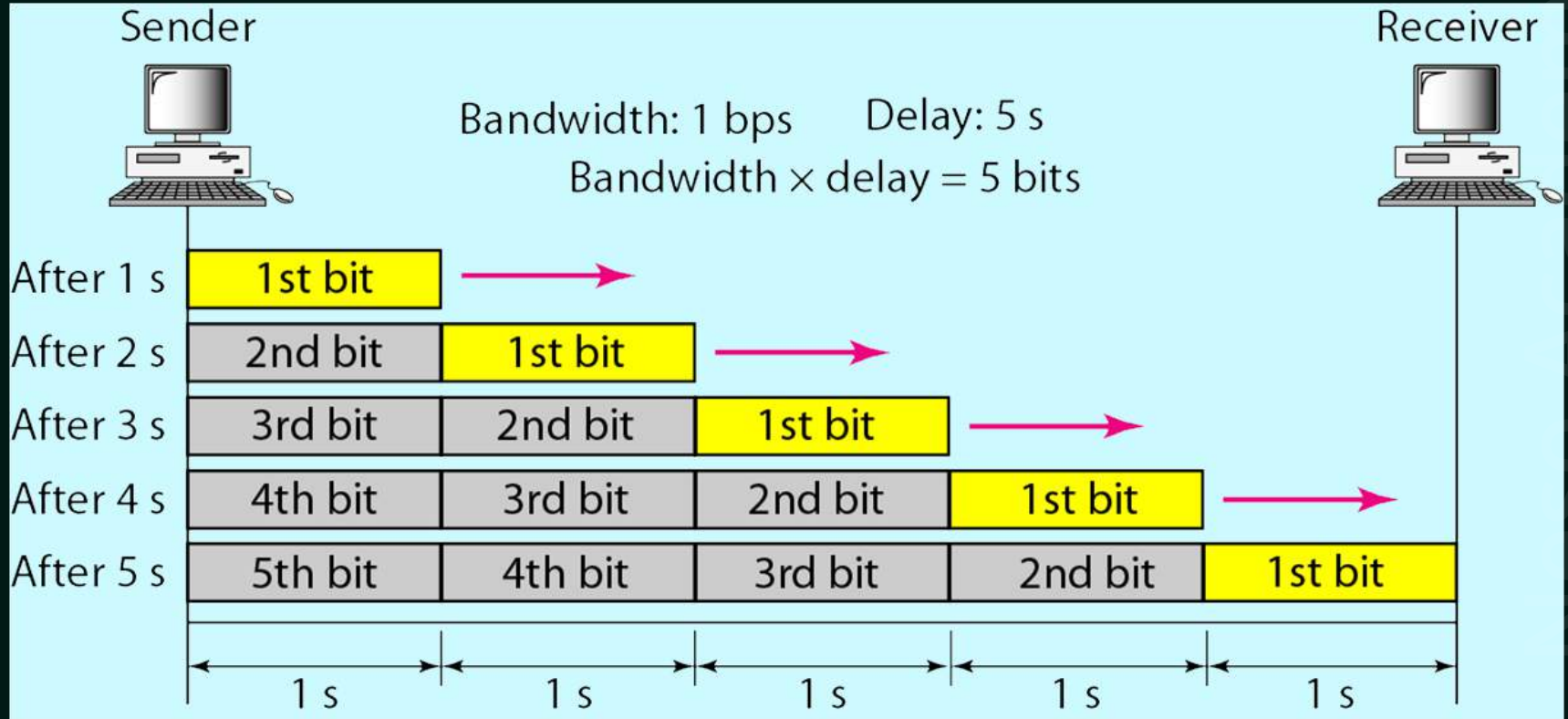
- ★ Understand bandwidth-delay product.

BANDWIDTH-DELAY PRODUCT

The bandwidth-delay product defines the number of bits that can fill the link.



BANDWIDTH-DELAY PRODUCT - EXAMPLE



BANDWIDTH-DELAY PRODUCT – SOLVED EXAMPLE

Consider that the link capacity of a channel is 512 Kbps and round – trip delay time is 1000ms.

Solution:

$$\begin{aligned}\text{The bandwidth delay product} &= 512 \text{ Kbps} \times 1000 \text{ ms} \\ &= 512 \times 1000 \text{ bits/sec} \times 1000 \times 10^{-3} \text{ sec} \\ &= 512,000 \text{ bits} \\ &= 64,000 \text{ bytes} \\ &= 62.5 \text{ KB}\end{aligned}$$

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Calculate propagation time.

QUESTION

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

Given

Distance = 12000 Km

Propagation speed = 2.4×10^8 m/s

To Find

Propagation Time

SOLUTION

$$\text{Propagation Time} = \frac{\text{Distance}}{\text{Propagation speed}}$$

$$\text{Propagation Time} = \frac{12000 \times 1000}{2.4 \times 10^8} \text{ sec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \text{ sec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \times \frac{10^{-3}}{10^{-3}} \text{ sec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \times \frac{1}{10^{-3}} \text{ millisec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \times \frac{10^3}{1} \text{ millisec}$$

$$\text{Propagation Time} = \frac{50 \times 100 \times 1000}{10^8} \times \frac{10^3}{1} \text{ millisec}$$

$$\text{Propagation Time} = 50 \text{ millisec OR } 50 \text{ ms}$$

OUTCOMES

Upon the completion of this session, the learner will be able to

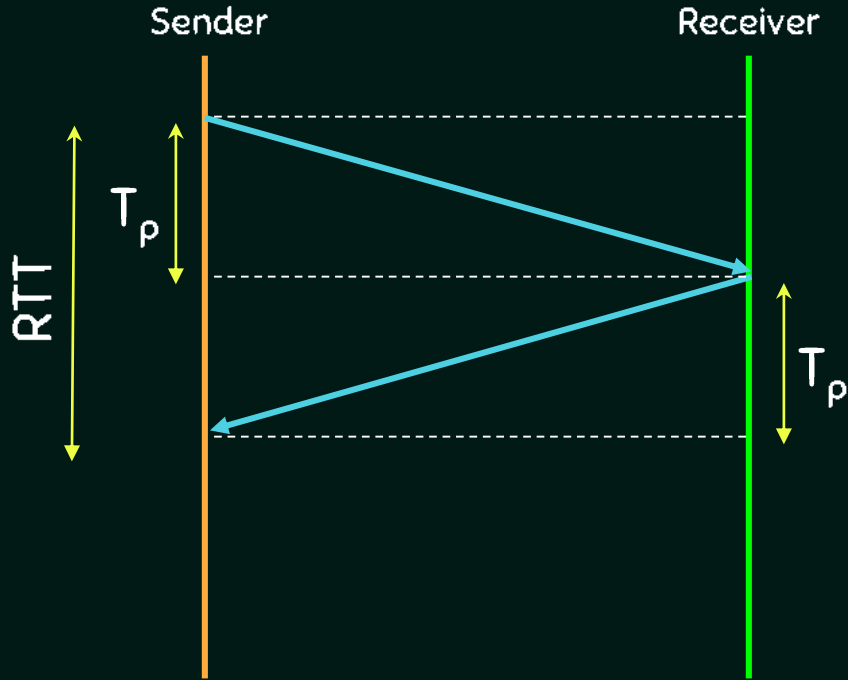
- ★ Recall propagation time.
- ★ Understand RTT.
- ★ Calculate RTT.

ROUND TRIP TIME (RTT)

- ★ a.k.a Round Trip Delay Time (RTD).
- ★ It is the length of time it takes for a signal to be sent plus the length of time it takes for an acknowledgement of that signal to be received.
- ★ This time therefore consists of the propagation times between the two point of signal.
- ★ If T_p is the Propagation time, then

$$RTT = 2 \times T_p$$

ROUND TRIP TIME (RTT)



$$RTT = T_p + T_p$$

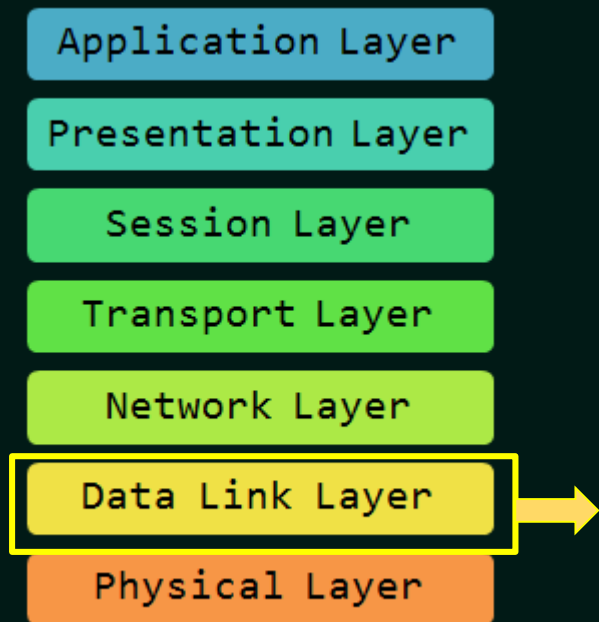
$$RTT \cong 2 \times T_p$$

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Recall link layer services.
- ★ Understand flow control.
- ★ Know the flow control protocols.

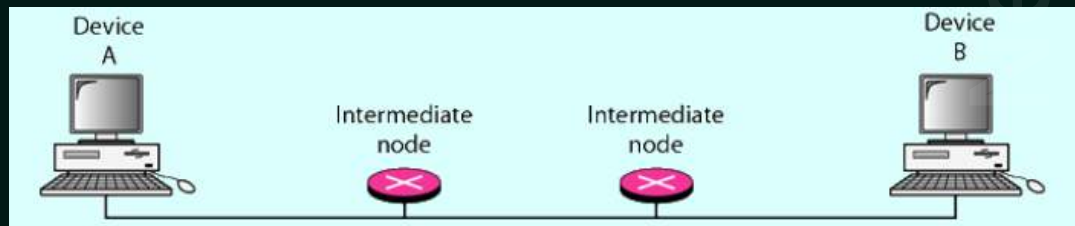
LINK LAYER SERVICES



It is responsible for moving data(frames) from one node to another node.

SERVICES PROVIDED BY DATA LINK LAYER

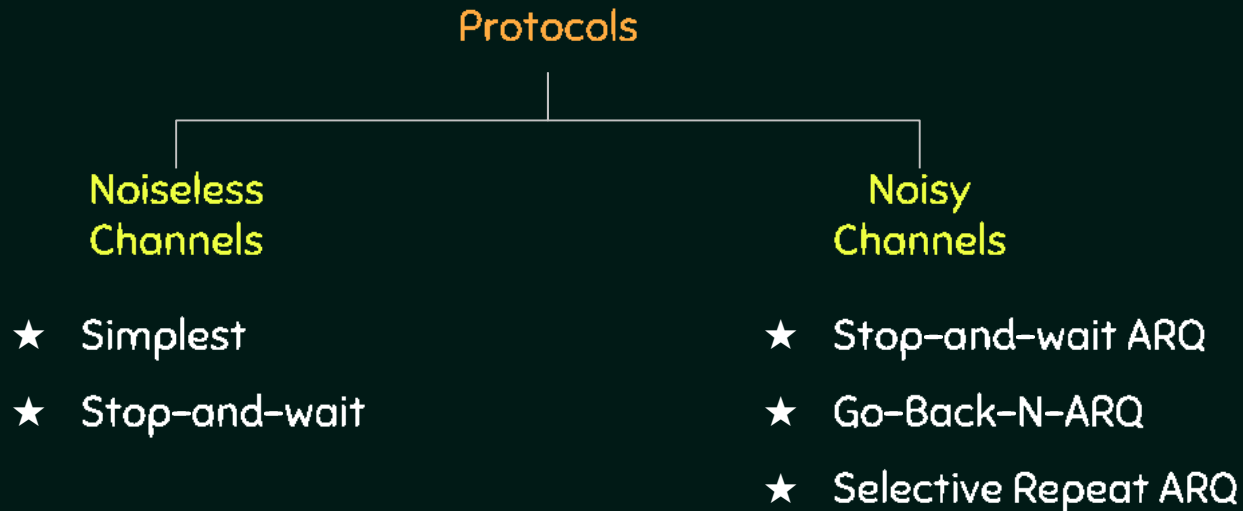
- ★ Framing.
- ★ Error Control.
- ★ Flow Control.
- ★ Physical Addressing.
- ★ Access Control.



FLOW CONTROL

- ★ Speed matching mechanism.
- ★ Flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- ★ Flow control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement from the receiver.
- ★ Receiver has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- ★ Receiver must inform the sender before the limits are reached and request that the transmitter to send fewer frames or stop temporarily.

FLOW CONTROL – PROTOCOLS

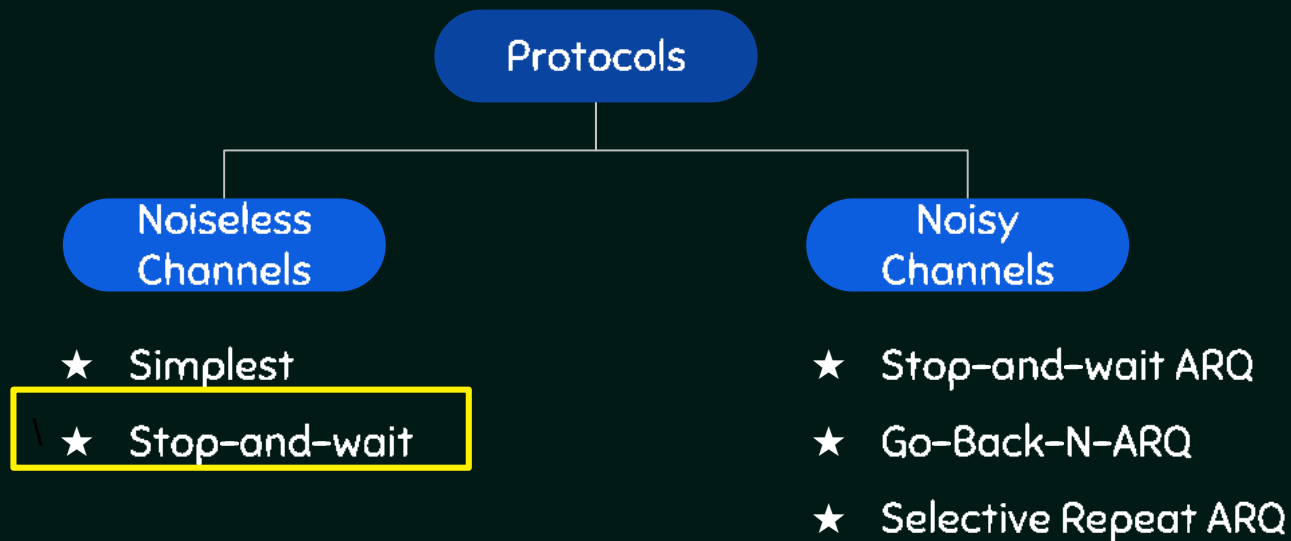


OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the working of simple stop-and-wait protocol.
- ★ Understand the problems with simple stop-and-wait protocol.

FLOW CONTROL



STOP-AND-WAIT PROTOCOL

- ★ Stop – and – Wait protocol is data link layer protocol for transmission of frames over noiseless channels.
- ★ It provides unidirectional data transmission with flow control facilities but without error control facilities.
- ★ The idea of stop-and-wait protocol is straightforward.
- ★ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.

PRIMITIVES OF STOP-AND-WAIT PROTOCOL

Sender side

Rule 1 : Send one data packet at a time.

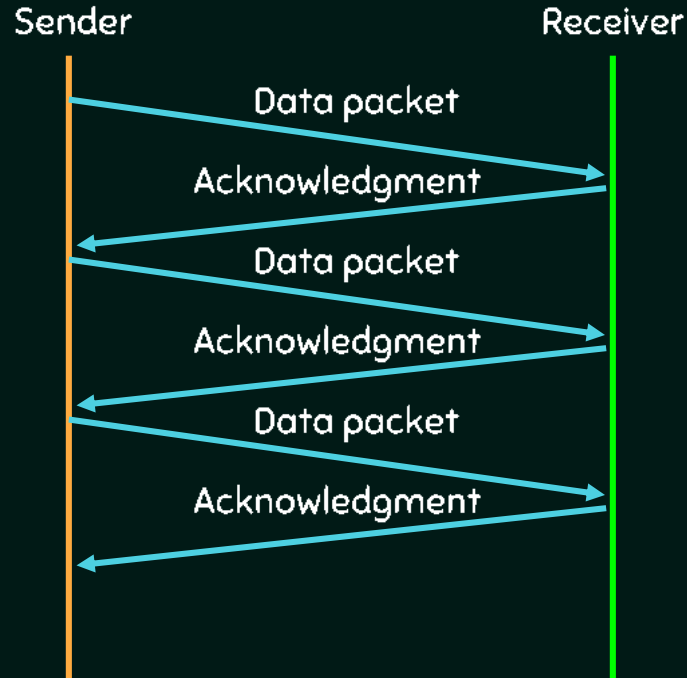
Rule 2 : Send the next packet only after receiving ACK for the previous.

Receiver side

Rule 1 : Receive and consume data packet.

Rule 2 : After consuming packet, ACK need to be sent (Flow Control).

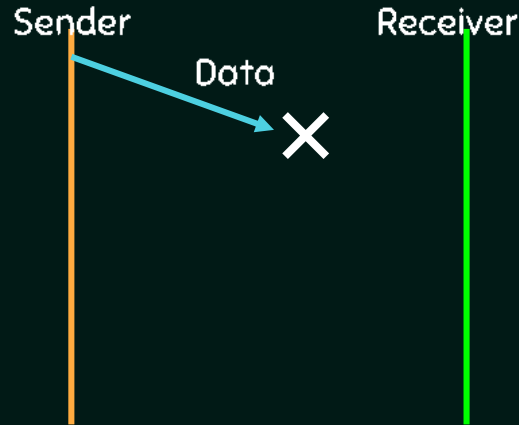
STOP-AND-WAIT PROTOCOL



PROBLEMS OF STOP-AND-WAIT PROTOCOL

1. Problems due to lost data.

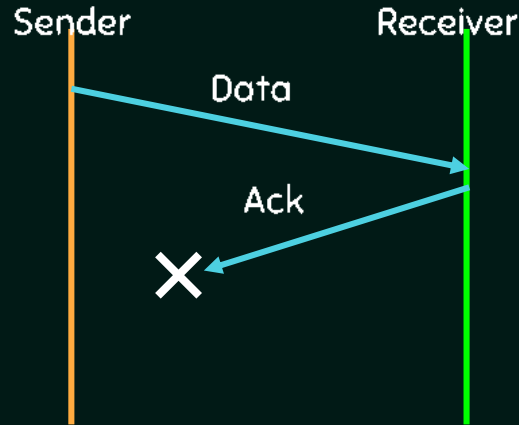
- ★ Sender waits for ack for an infinite amount of time.
- ★ Receiver waits for data an infinite amount of time.



PROBLEMS OF STOP-AND-WAIT PROTOCOL

2. Problems due to lost ACK.

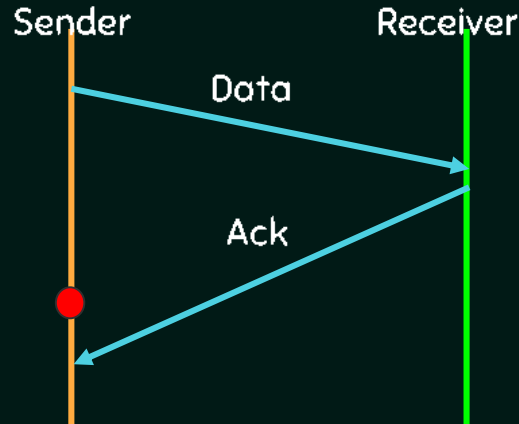
- ★ Sender waits for an infinite amount of time for ack.



PROBLEMS OF STOP-AND-WAIT PROTOCOL

3. Problems due to delayed ACK/data.

- ★ After timeout on sender side, a delayed ack might be wrongly considered as ack of some other data packet.



PROBLEMS OF STOP-AND-WAIT PROTOCOL

1. Problems due to lost data.

Sender waits for ack for an infinite amount of time.

Receiver waits for data an infinite amount of time.

1. Problems due to lost ACK.

Sender waits for an infinite amount of time for ack.

1. Problems due to delayed ACK/data.

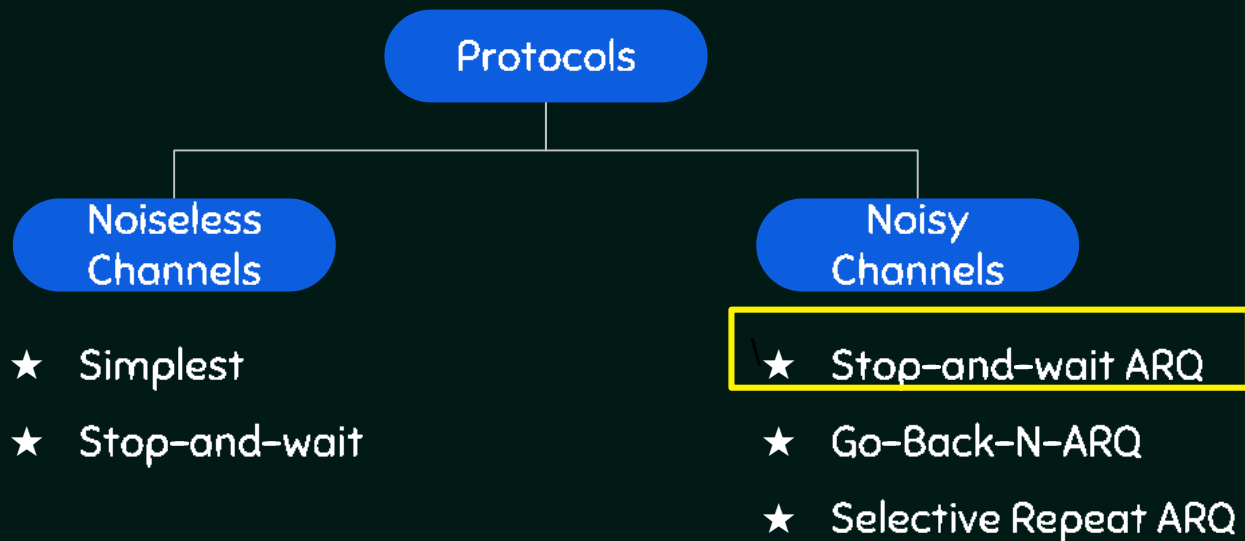
After timeout on sender side, a delayed ack might be wrongly considered as ack of some other data packet.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the working of stop-and-wait ARQ protocol.

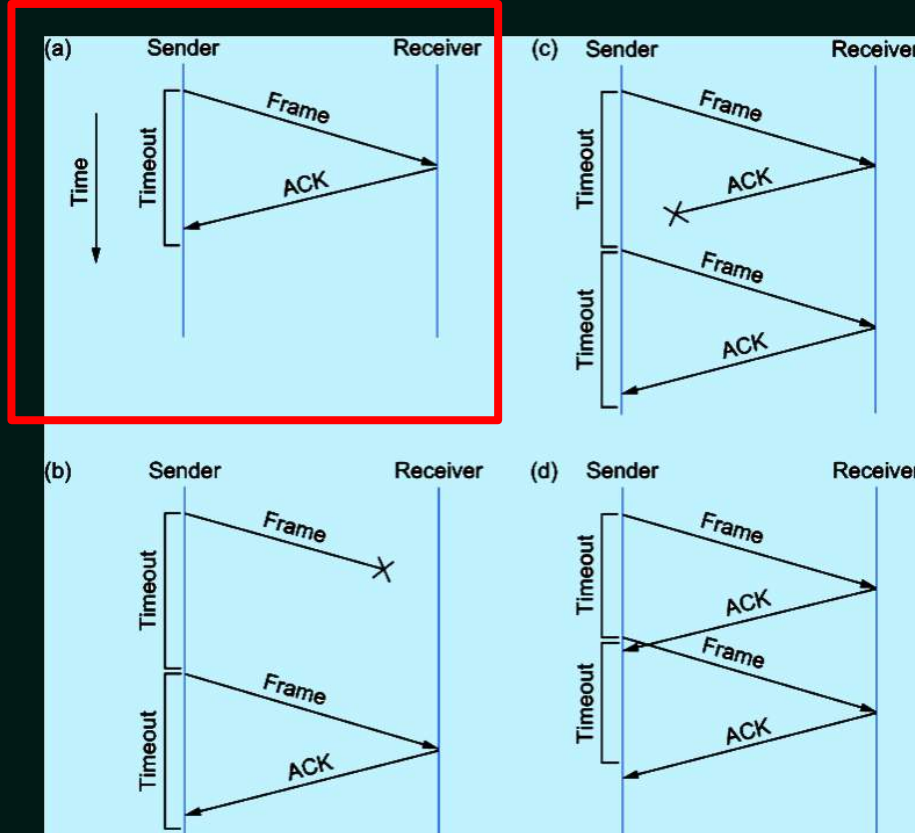
FLOW CONTROL



STOP-AND-WAIT ARQ PROTOCOL

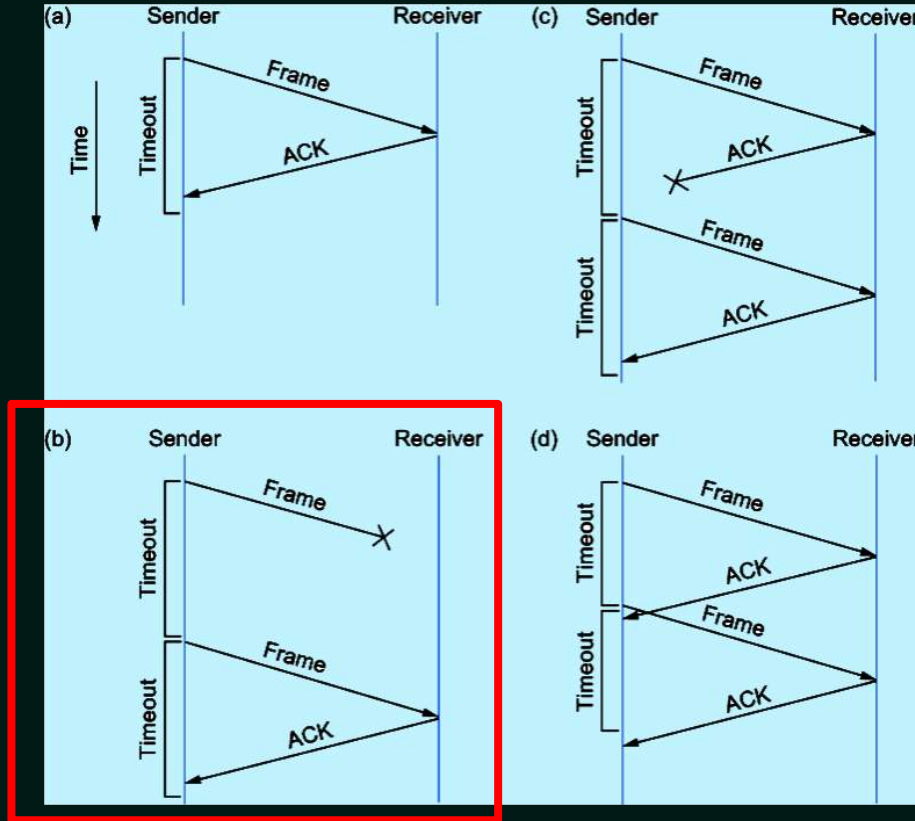
- ★ Idea of stop-and-wait protocol is straightforward.
- ★ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.
- ★ If the acknowledgement does not arrive after a certain period of time, the sender times out and retransmits the original frame.
- ★ Stop-and-Wait ARQ = Stop-and-Wait + Timeout Timer + Sequence number

STOP-AND-WAIT ARQ PROTOCOL



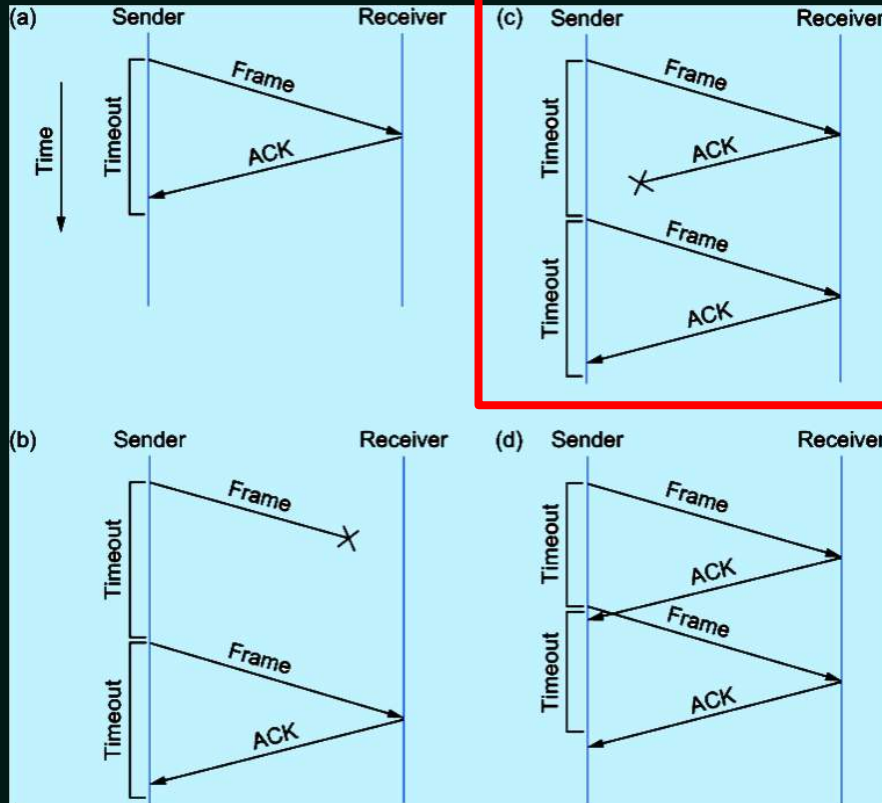
(a) The ACK is received before the timer expires;

STOP-AND-WAIT ARQ PROTOCOL



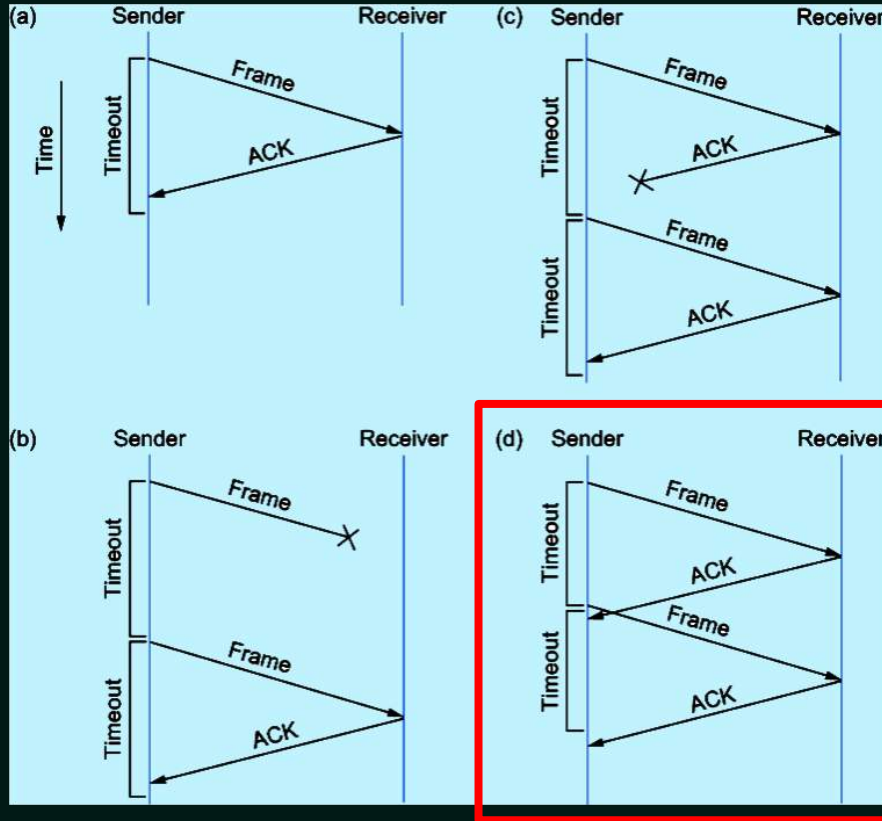
(b) The original frame is lost

STOP-AND-WAIT ARQ PROTOCOL



(c) The ACK is lost

STOP-AND-WAIT ARQ PROTOCOL



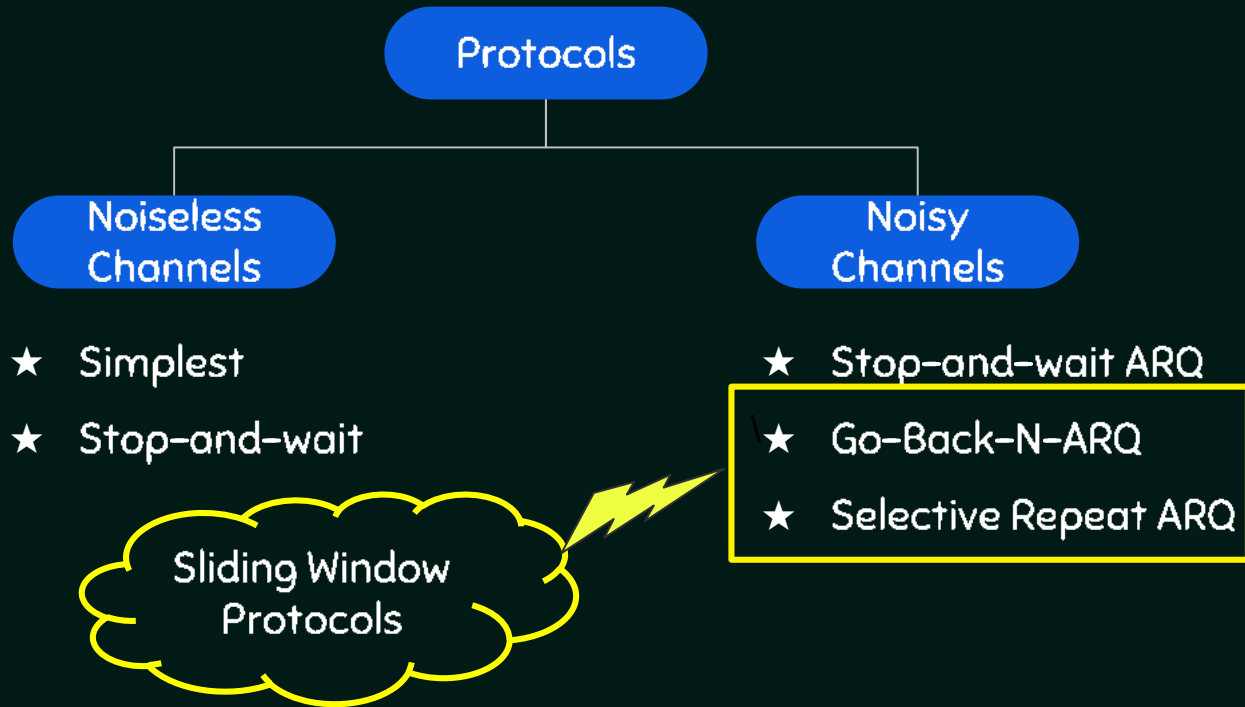
(d) The timeout fires too soon

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the drawbacks of stop-and-wait ARQ protocol.
- ★ Know the need for sliding window protocol.
- ★ Understand the working of sliding window protocol.

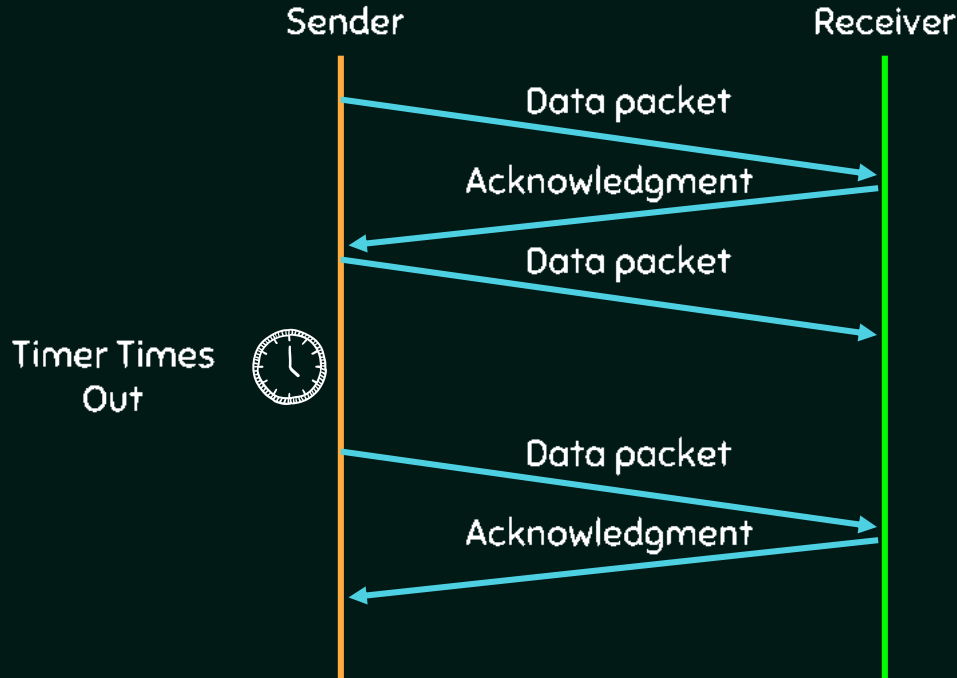
FLOW CONTROL



STOP-AND-WAIT ARQ PROTOCOL

- ★ Idea of stop-and-wait ARQ protocol is simple.
- ★ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.
- ★ If the acknowledgement does not arrive after a certain period of time, the sender times out and retransmits the original frame.

STOP-AND-WAIT ARQ PROTOCOL



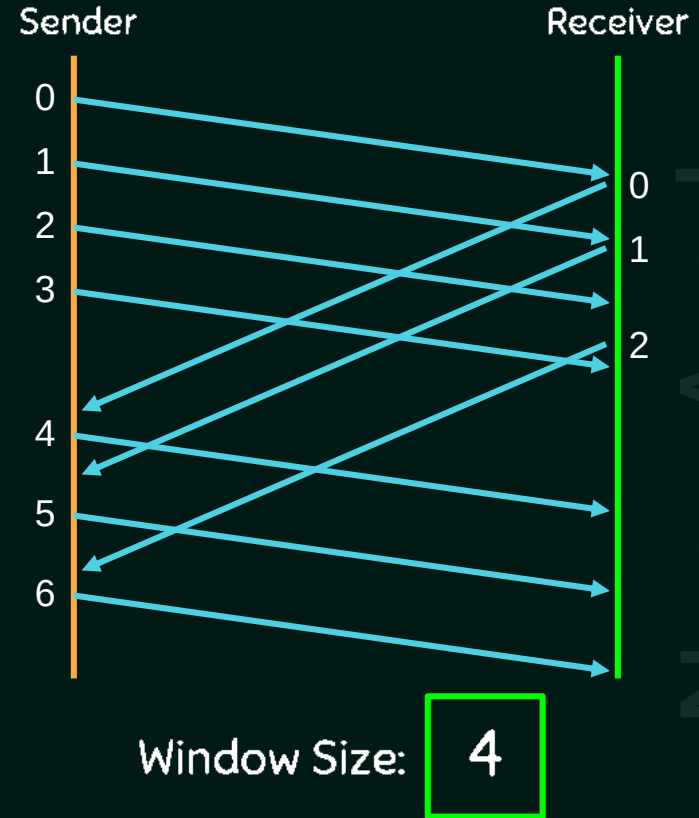
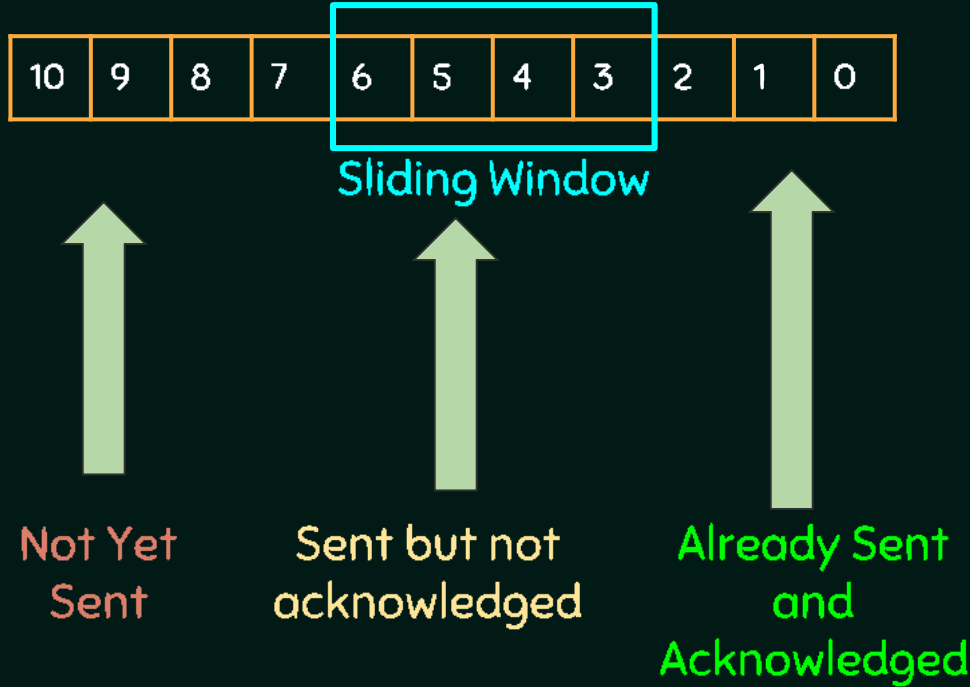
STOP-AND-WAIT ARQ – DRAWBACKS

- ★ One frame at a time.
- ★ Poor utilization of bandwidth.
- ★ Poor Performance

SLIDING WINDOW PROTOCOL

- ★ Send multiple frames at a time.
- ★ Number of frames to be sent is based on **Window size**.
- ★ Each frame is numbered → Sequence number.

WORKING OF SLIDING WINDOW PROTOCOL

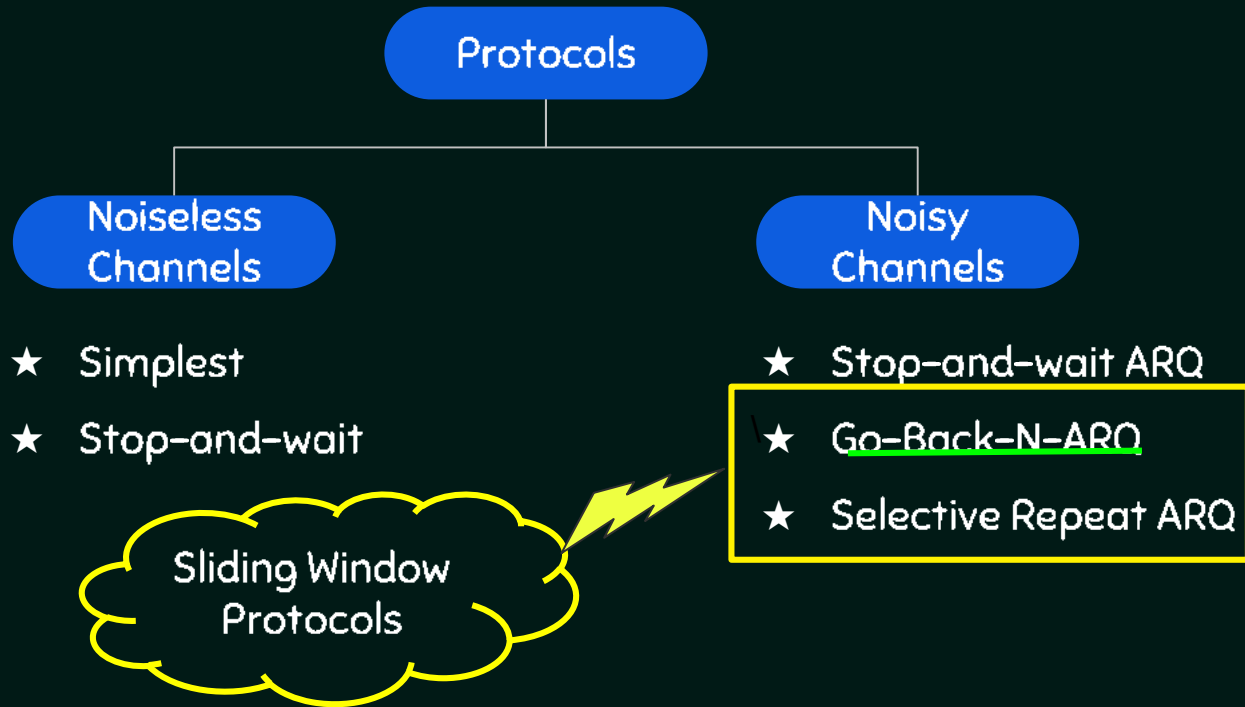


OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the working of Go-back-N ARQ.

SLIDING WINDOW PROTOCOLS



Go-Back-N ARQ



'N' is the sender window size.

Go-Back-N ARQ

- ★ Go - Back - N ARQ uses the concept of protocol pipelining i.e. the sender can send multiple frames before receiving the acknowledgment for the first frame.
- ★ There are finite number of frames and the frames are numbered in a sequential manner.
- ★ The number of frames that can be sent depends on the window size of the sender.
- ★ If the acknowledgment of a frame is not received within an agreed upon time period, all frames in the current window are transmitted.
- ★ The size of the sending window determines the sequence number of the outbound frames.

Go-Back-N ARQ

- ★ N – Sender's Window Size.
- ★ For example, if the sending window size is 4 (2^2), then the sequence numbers will be 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, and so on.
- ★ The number of bits in the sequence number is 2 to generate the binary sequence 00, 01, 10, 11.

WORKING OF GO-BACK-N ARQ

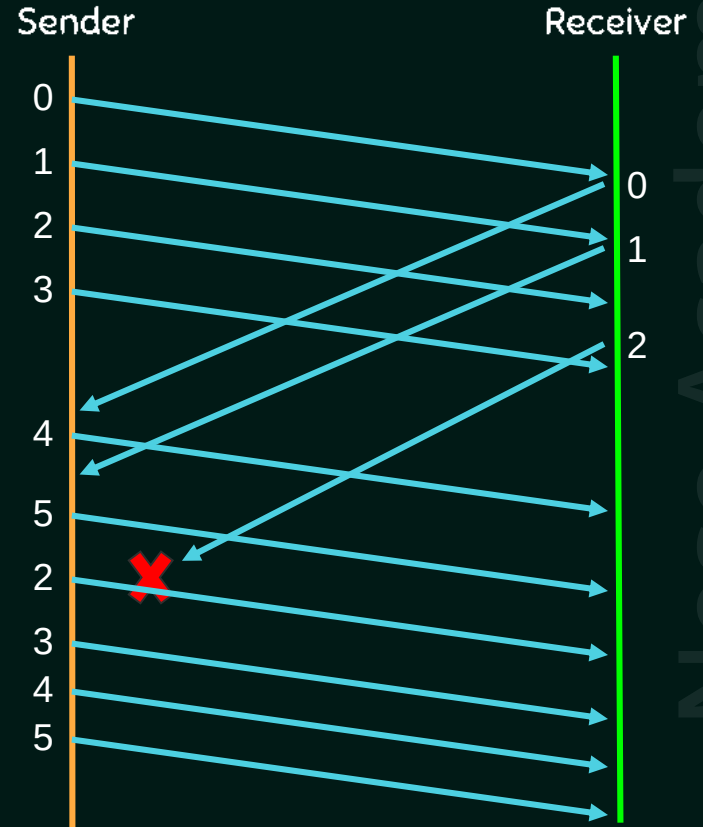


Sliding Window

Go-Back to 2

Window Size:

4

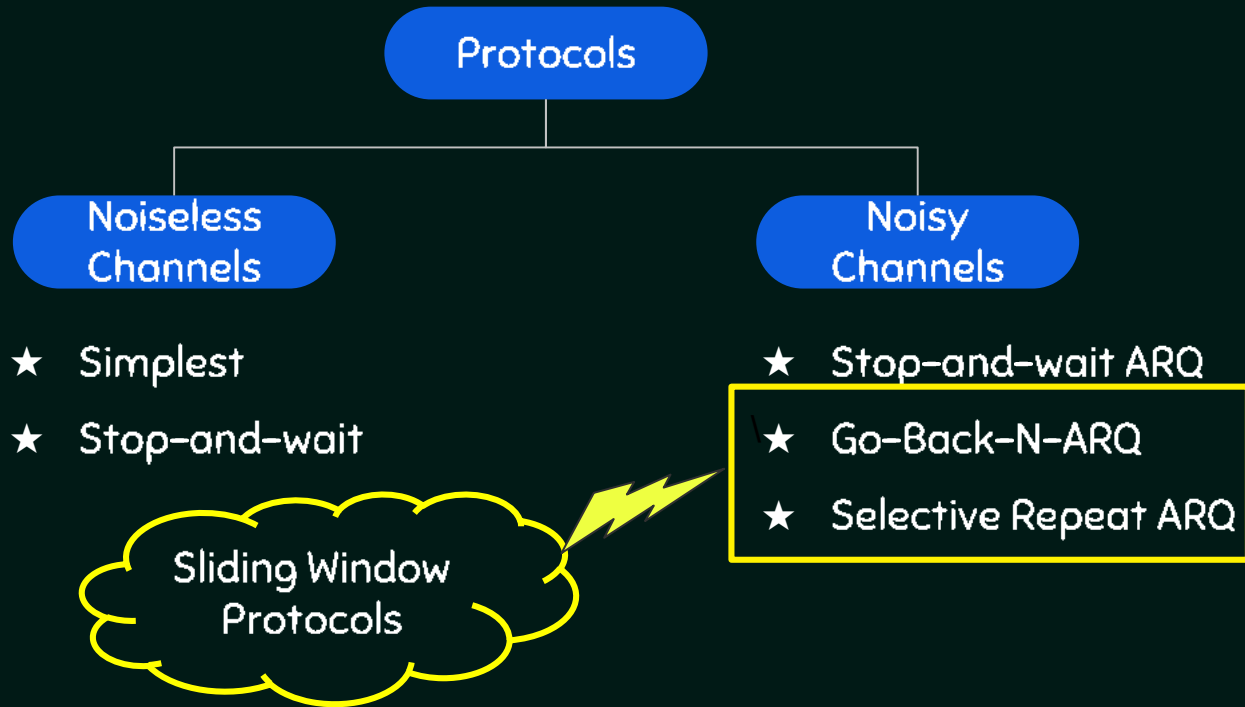


OUTCOMES

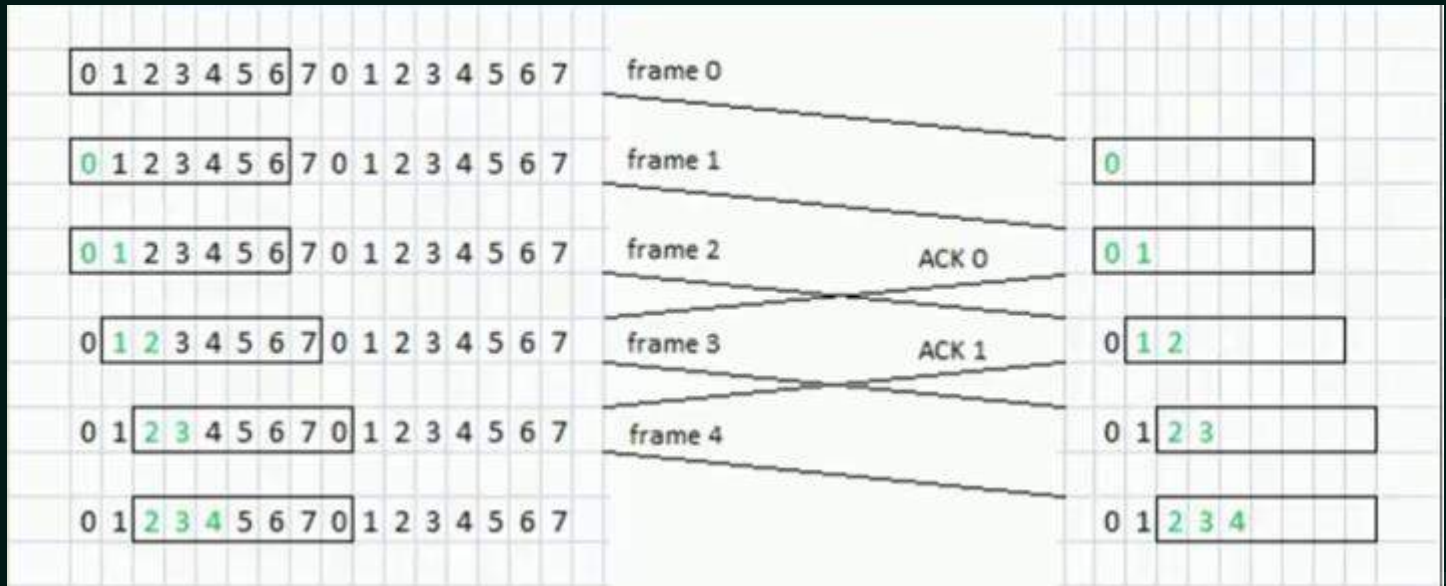
Upon the completion of this session, the learner will be able to

- ★ Calculate the sender and receiver's window size for 100% efficiency.
(GATE)
- ★ Know the importance of sequence numbers.
- ★ Find the minimum number of bits required to represent sequence numbers. (GATE)

SLIDING WINDOW PROTOCOLS



Sliding Window Protocol



Window Size for 100% Efficiency

$$\text{Window Size } (W_s) = 1 + \frac{2T_p}{T_t}$$

Where

T_p - Propagation Time

T_t - Transmission Time

Minimum number of bits in Sequence Number

$$\text{Min. No. of bits req} = \text{Ceil} \left(\log_2 \left(1 + \frac{2T_p}{T_t} \right) \right)$$

Where

T_p - Propagation Time

T_t - Transmission Time

OUTCOMES

Upon the completion of this session, the learner will be able to

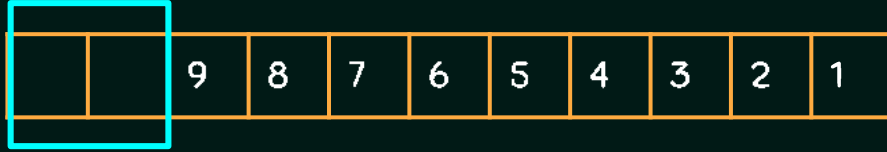
- ★ Understand the working of Go-back-N ARQ by solving a problem asked in GATE CS 2006.

QUESTION

Station A needs to send a message consisting of 9 packets to station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B? [GATE CS 2006]

- (A) 12
- (B) 14
- (C) 16
- (D) 18

SOLUTION

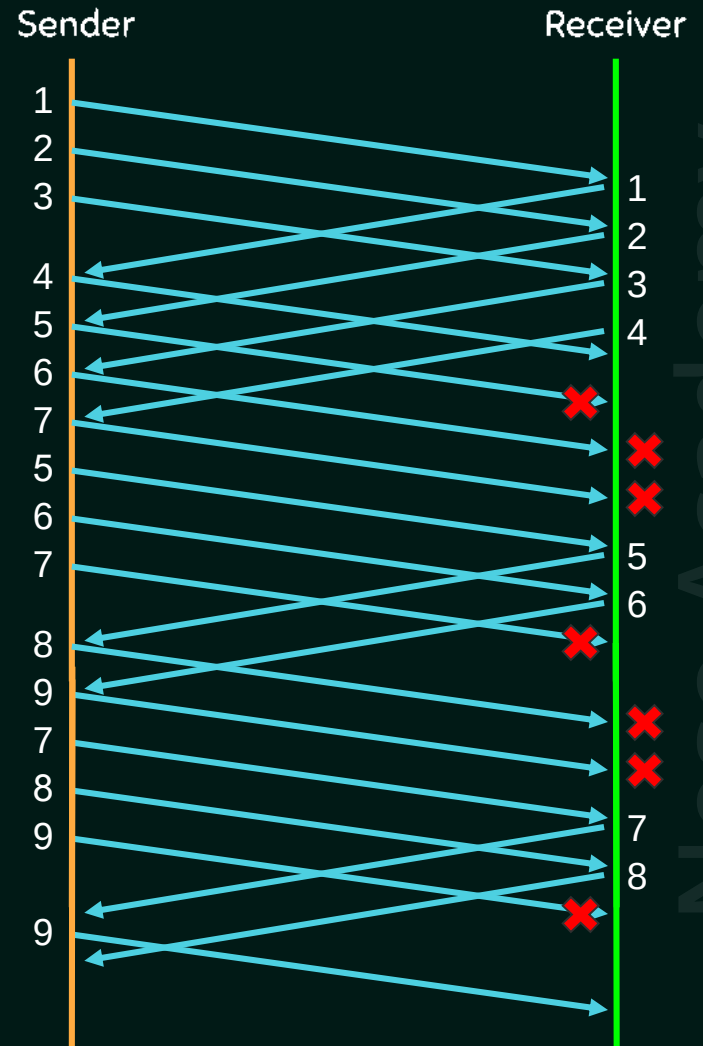


Window Size:

3

No. of packets transmitted by A (sender)

16



QUESTION

Station A needs to send a message consisting of 9 packets to station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

(A) 12

(B) 14

(C) 16 ✓

(D) 18

No. of packets transmitted by A (sender)

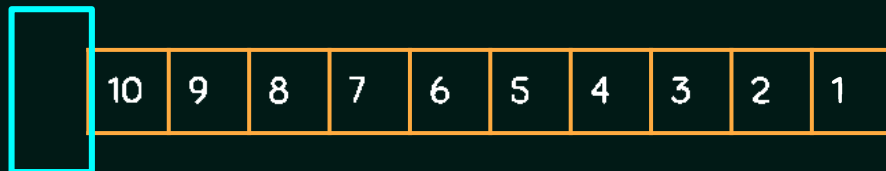
16

QUESTION

Host A wants to send 10 frames to Host B. The hosts agreed to go with Go-Back-4. How many number of frames are transmitted by Host A if every 6th frame that is transmitted by host A is either corrupted or lost?

SOLUTION

Sender Window



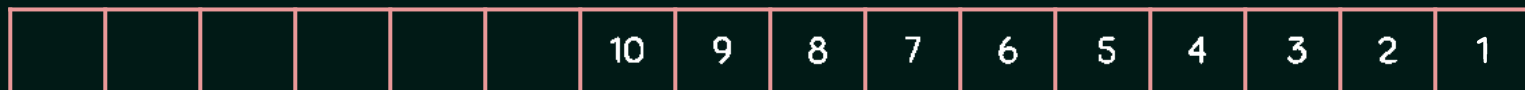
Window Size:

4

Frames transmitted by Host A



Frames Acknowledged by Host B



QUESTION

Host A wants to send 10 frames to Host B. The hosts agreed to go with Go-Back-4. How many number of frames are transmitted by Host A if every 6th frame that is transmitted by host A is either corrupted or lost?

ANSWER

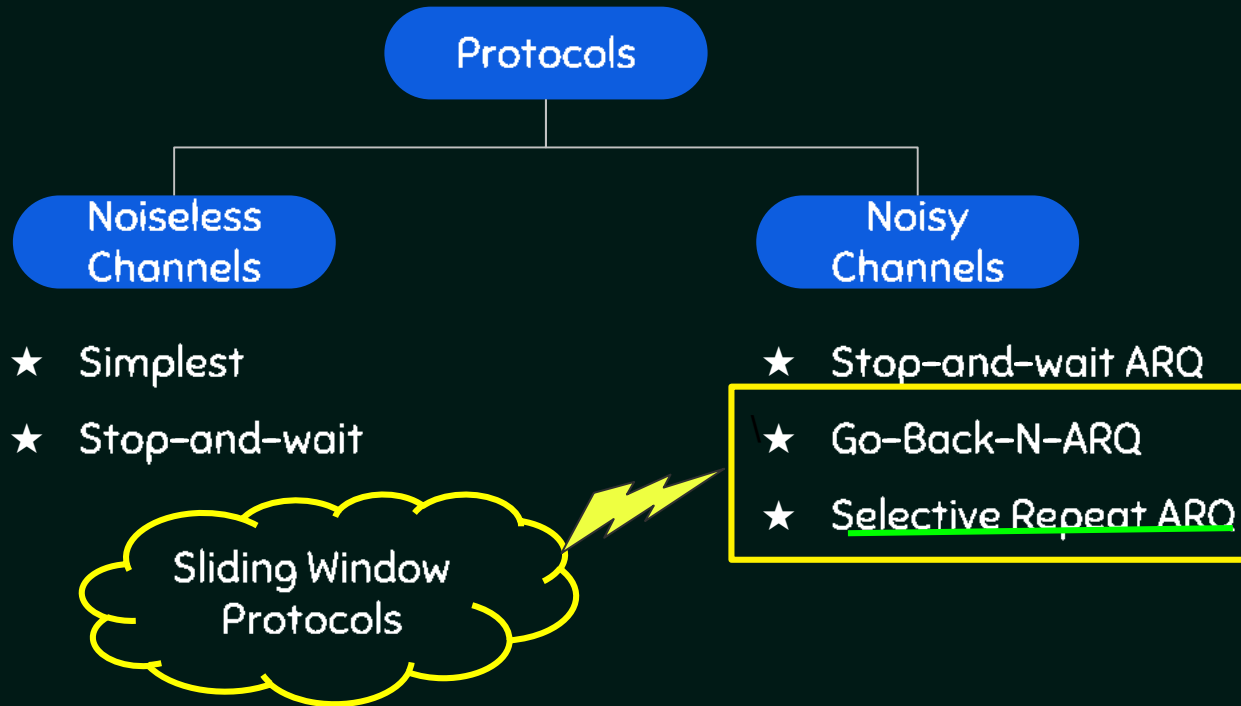
Number of frames transmitted by Host A: 17

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the working of Selective Repeat ARQ.

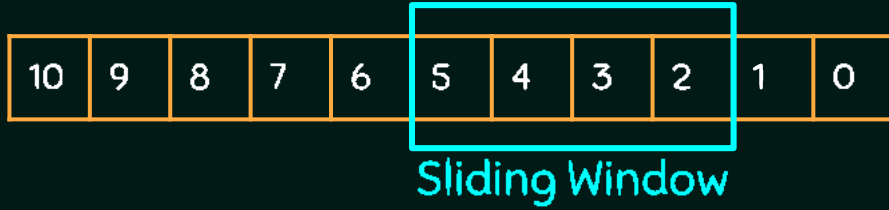
SLIDING WINDOW PROTOCOLS



SELECTIVE REPEAT ARQ

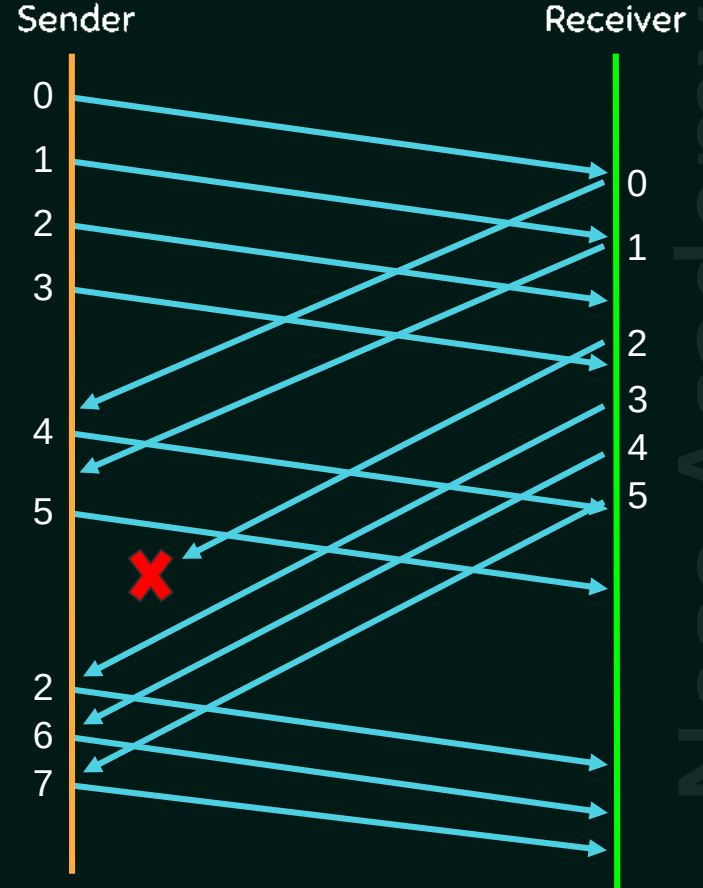
- ★ In Selective Repeat ARQ, only the erroneous or lost frames are retransmitted, while correct frames are received and buffered.
- ★ The receiver while keeping track of sequence numbers, buffers the frames in memory and sends NACK for only frame which is missing or damaged.
- ★ The sender will send/retransmit packet for which NACK is received.

WORKING OF SELECTIVE REPEAT



Window Size:

4

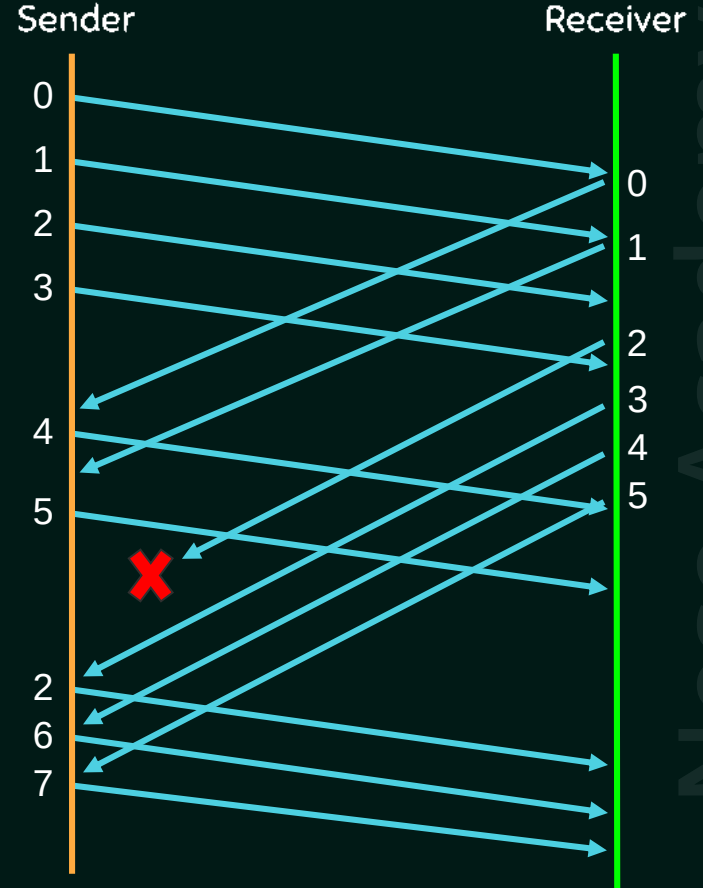


WORKING OF SELECTIVE REPEAT

In SR Protocol, only the required frame is retransmitted and not the entire window.

Window Size:

4

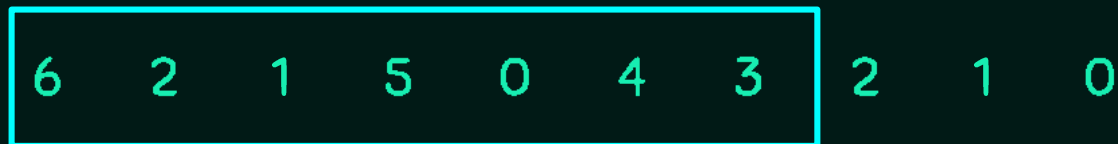


SOLUTION

In SR protocol, suppose frames through 0 to 4 have been transmitted. Now, imagine that 0 times out, 5 (a new frame) is transmitted, 1 times out, 2 times out and 6 (another new frame) is transmitted. At this point, what will be the outstanding packets in sender's window?

- a. 341526
- b. 3405126✓
- c. 0123456
- d. 654321
- e. None of the above

Sender's Window

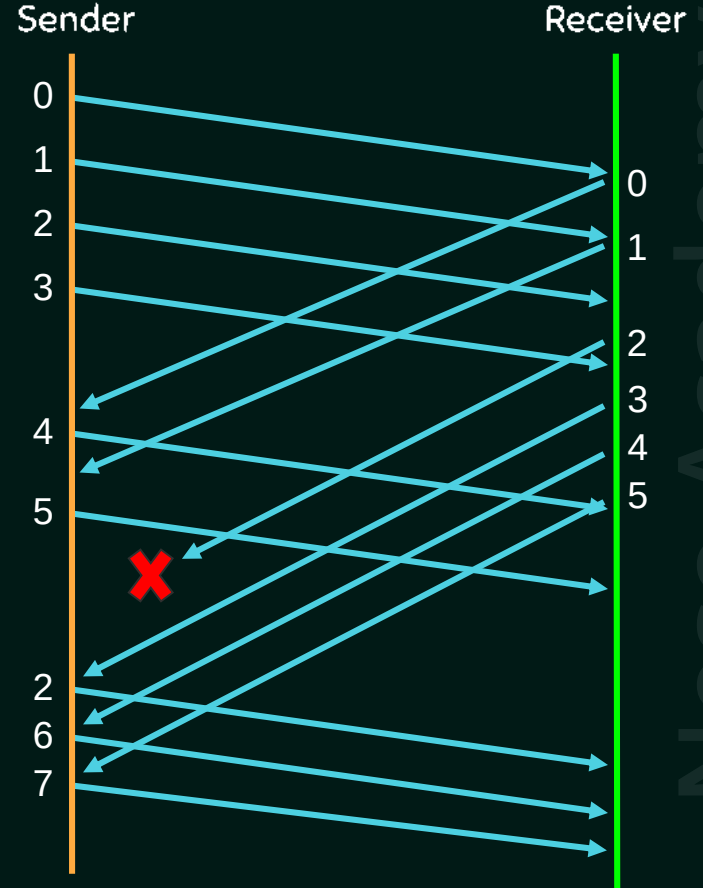


WORKING OF SELECTIVE REPEAT

In SR Protocol, only the required frame is retransmitted and not the entire window.

Window Size:

4



QUESTION

Host A wants to send 10 frames to Host B. The hosts agreed to go with SR ARQ. How many number of frames are transmitted by Host A if every 6th frame that is transmitted by host A is either corrupted or lost? Also compare the number of transmissions of SR ARQ with Go-Back-4 ARQ.

SOLUTION

Number of transmission with Go-Back-4 ARQ: 17

Refer previous lecture titled: Go-Back-N ARQ (Solved Problem 2)

SOLUTION

10	9	8	7	6	5	4	3	2	1
----	---	---	---	---	---	---	---	---	---

Frames transmitted by Host A

													10	9	6	8	7	6	5	4	3	2	1
--	--	--	--	--	--	--	--	--	--	--	--	--	----	---	---	---	---	---	---	---	---	---	---

Number of frames transmitted by Host A: 11

QUESTION

Host A wants to send 10 frames to Host B. The hosts agreed to go with SR ARQ. How many number of frames are transmitted by Host A if every 6th frame that is transmitted by host A is either corrupted or lost? Also compare the number of transmissions of SR ARQ with Go-Back-4 ARQ.

ANSWER

Go-Back-N: Number of frames transmitted by Host A: 17

Selective Repeat: Number of frames transmitted by Host A: 11

QUESTION

Station A uses 32 byte packets to transmit message to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

[GATE CS 2006]

- (A) 20
- (B) 40
- (C) 160
- (D) 320

SOLUTION

Refer Previous Lecture titled **Bandwidth – Delay Product**

Refer Previous Lecture titled **Sliding Window Protocol**

SOLUTION

Station A uses 32 byte packets to transmit message to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

Given Data

Delay = 80 milliseconds

Bandwidth = 128 kbps

Packet Size = 32 bytes

SOLUTION

Bandwidth – Delay Product = Bandwidth x Delay

Bandwidth – Delay Product = 128 kbps x 80 milliseconds

Bandwidth – Delay Product = 128 x 1024 x 80 milliseconds

Bandwidth – Delay Product = 128 x 1024 x 80 x 10⁻³ bits

Bandwidth – Delay Product = $\frac{128 \times 1024 \times 80 \times 10^{-3}}{8}$ bytes

Optimal Window Size = $\frac{128 \times 1024 \times 80 \times 10^{-3}}{8 \times 32}$ bytes

Optimal Window Size = 40

SOLUTION

Station A uses 32 byte packets to transmit message to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

- (A) 20
- (B) 40 ✓
- (C) 160
- (D) 320

QUESTION

The distance between two stations M and N is L kilometres. All frames are K bits long. The propagation time per kilometer is t seconds. Let R bits/second be the channel capacity. Assuming the processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is:

[GATE CS 2007]

A. $\log_2 \left(\frac{2LtR+2K}{K} \right)$

C. $\log_2 \left(\frac{2LtR+K}{K} \right)$

B. $\log_2 \left(\frac{2LtR}{K} \right)$

D. $\log_2 \left(\frac{2LtR+K}{2K} \right)$

SOLUTION

Let Propagation Delay = Lt sec

Round Trip Time = $2 \times$ Propagation Delay

Round Trip Time = $2 \times Lt$ sec

Round Trip Time = $2 Lt$ sec

No. of bits transmitted in round trip = $2 LtR$ bits.

SOLUTION

$$\text{No. of frames} = \left(\frac{2L+R}{K} \right)$$

Let the bits in the sequence numbers be b .

$$2^b = \left(\frac{2L+R}{K} \right)$$

Take log on both sides, we get

$$b = \text{Log} \left(\frac{2L+R}{K} \right)$$

QUESTION

The distance between two stations M and N is L kilometres. All frames are K bits long. The propagation time per kilometer is t seconds. Let R bits/second be the channel capacity. Assuming the processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is:

[GATE CS 2007]

A. $\log_2 \left(\frac{2LtR+2K}{K} \right)$

C. $\log_2 \left(\frac{2LtR+K}{K} \right)$

B. $\log_2 \left(\frac{2LtR}{K} \right)$ ✓

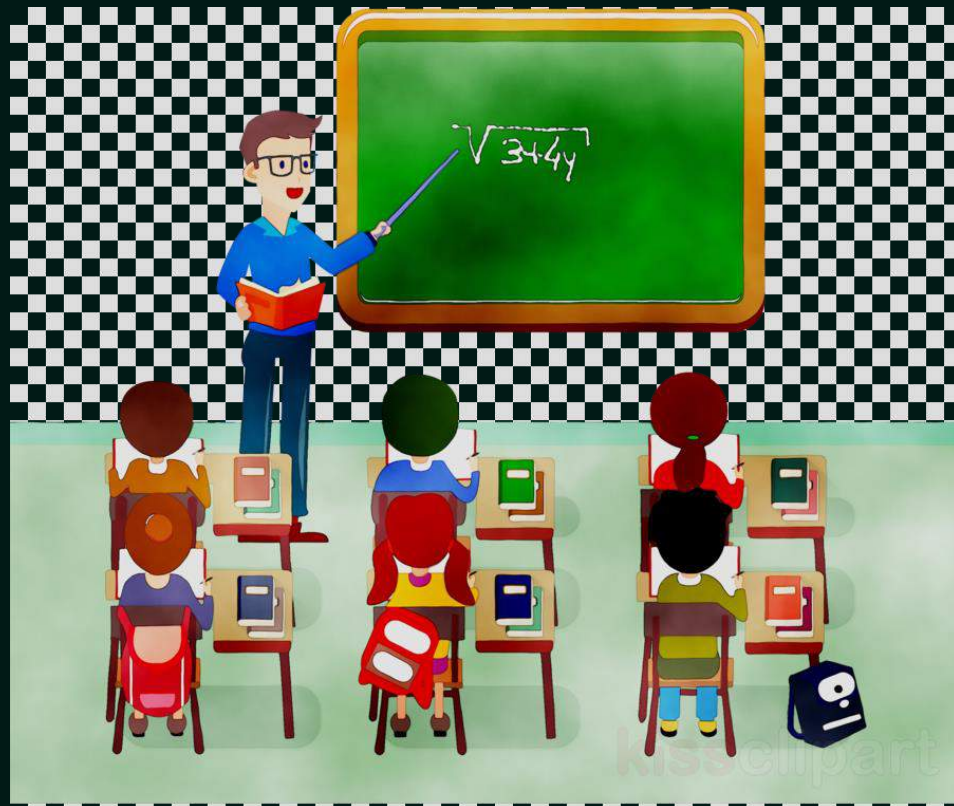
D. $\log_2 \left(\frac{2LtR+K}{2K} \right)$

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various multiple access protocols.

ANALOGY !

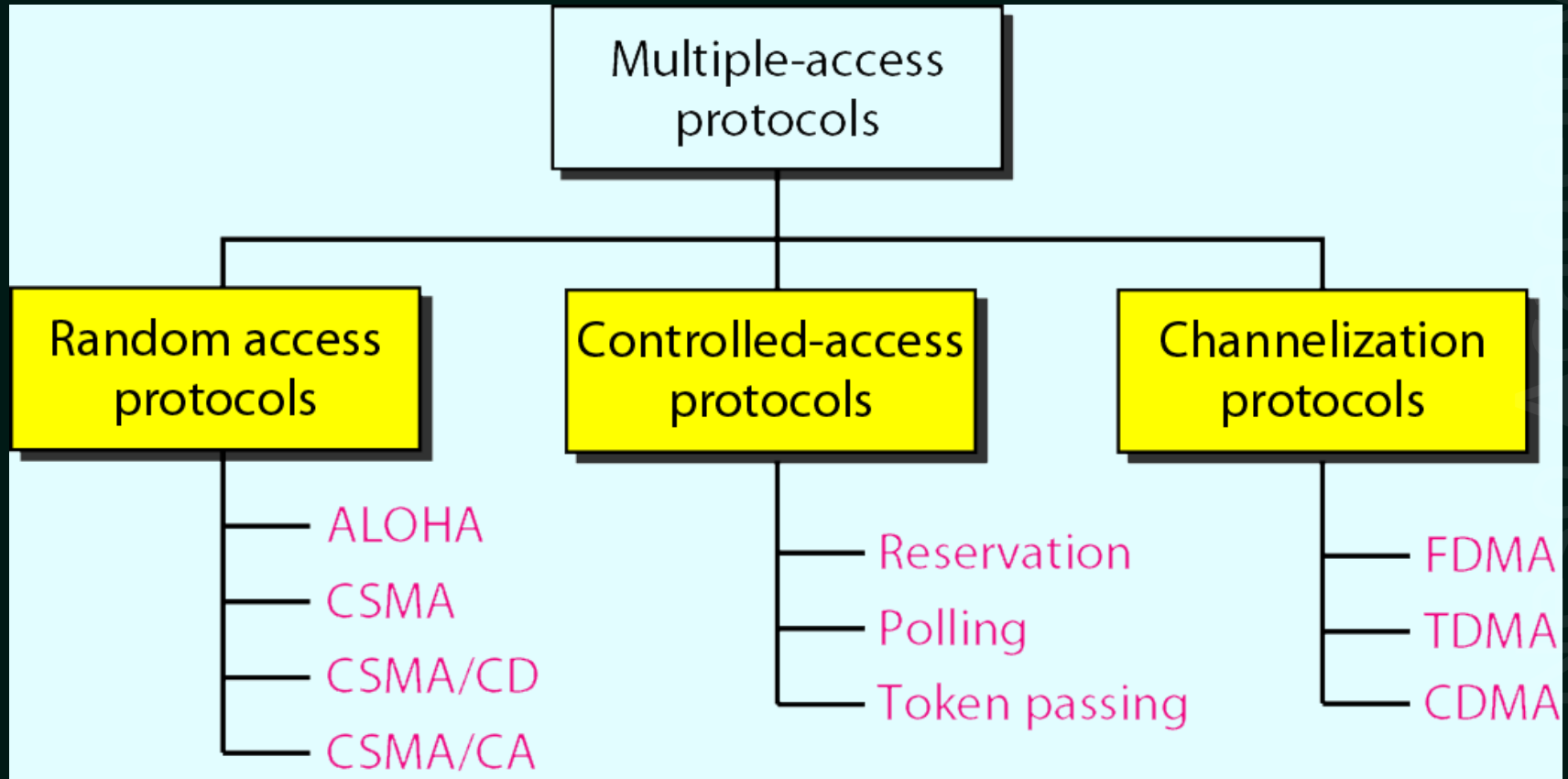


WHY MULTIPLE ACCESS PROTOCOLS?

If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously.

Hence multiple access protocols are required to decrease collision and avoid crosstalk.

MULTIPLE ACCESS PROTOCOLS



RANDOM ACCESS PROTOCOLS

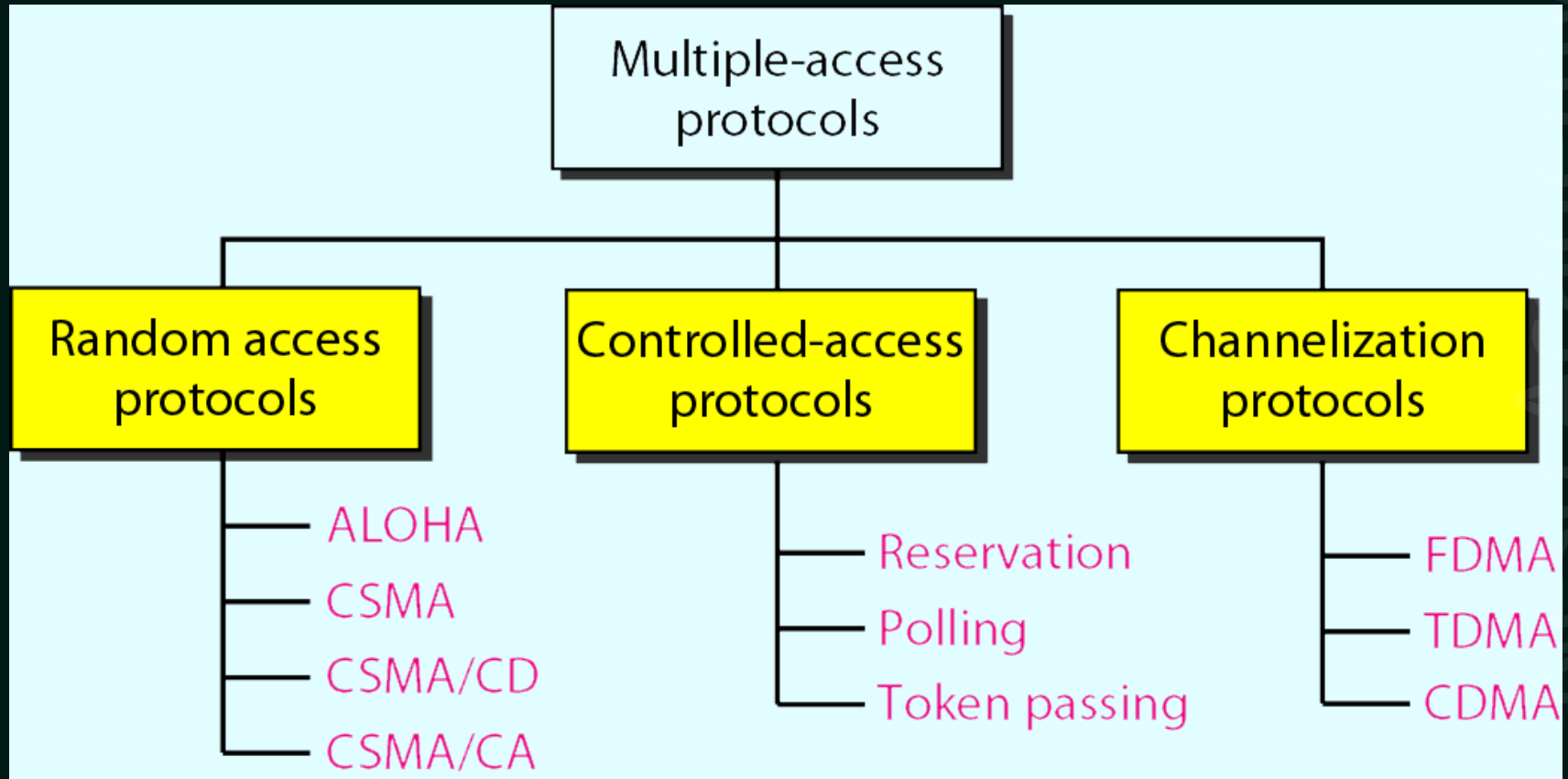
- ★ In this, all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state(idle or busy).
- ★ In a Random access method, each station has the right to the medium without being controlled by any other station.
- ★ If more than one station tries to send, there is an access conflict (COLLISION) and the frames will be either destroyed or modified.

RANDOM ACCESS PROTOCOLS

To avoid access conflict, each station follows a procedure.

- ★ When can the station access the medium ?
- ★ What can the station do if the medium is busy ?
- ★ How can the station determine the success or failure of the transmission ?
- ★ What can the station do if there is an access conflict ?

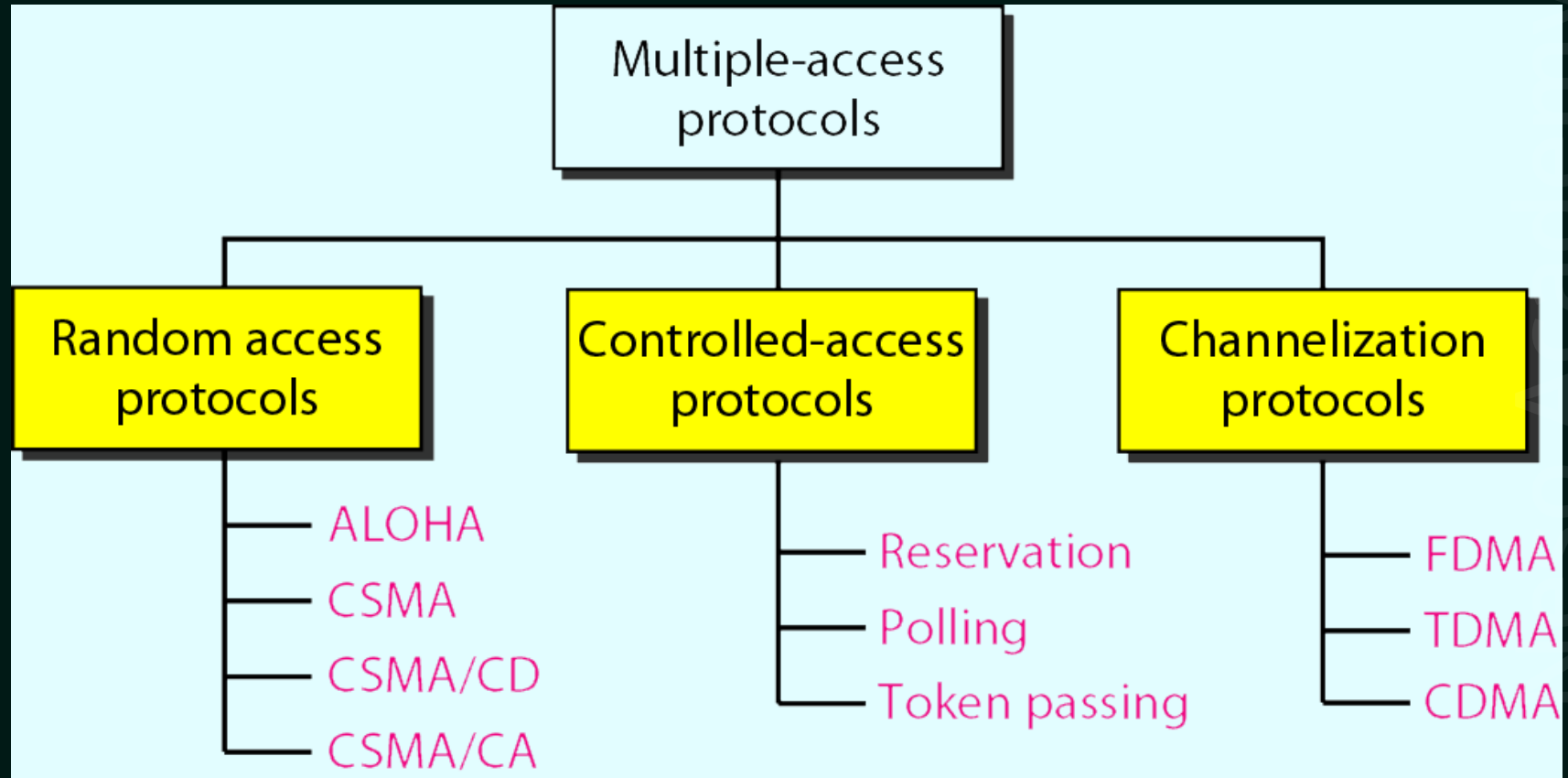
RANDOM ACCESS PROTOCOLS



CONTROLLED ACCESS PROTOCOLS

- ★ In controlled access, the stations consult one another to find which station has the right to send.
- ★ A station cannot send unless it has been authorized by other stations.

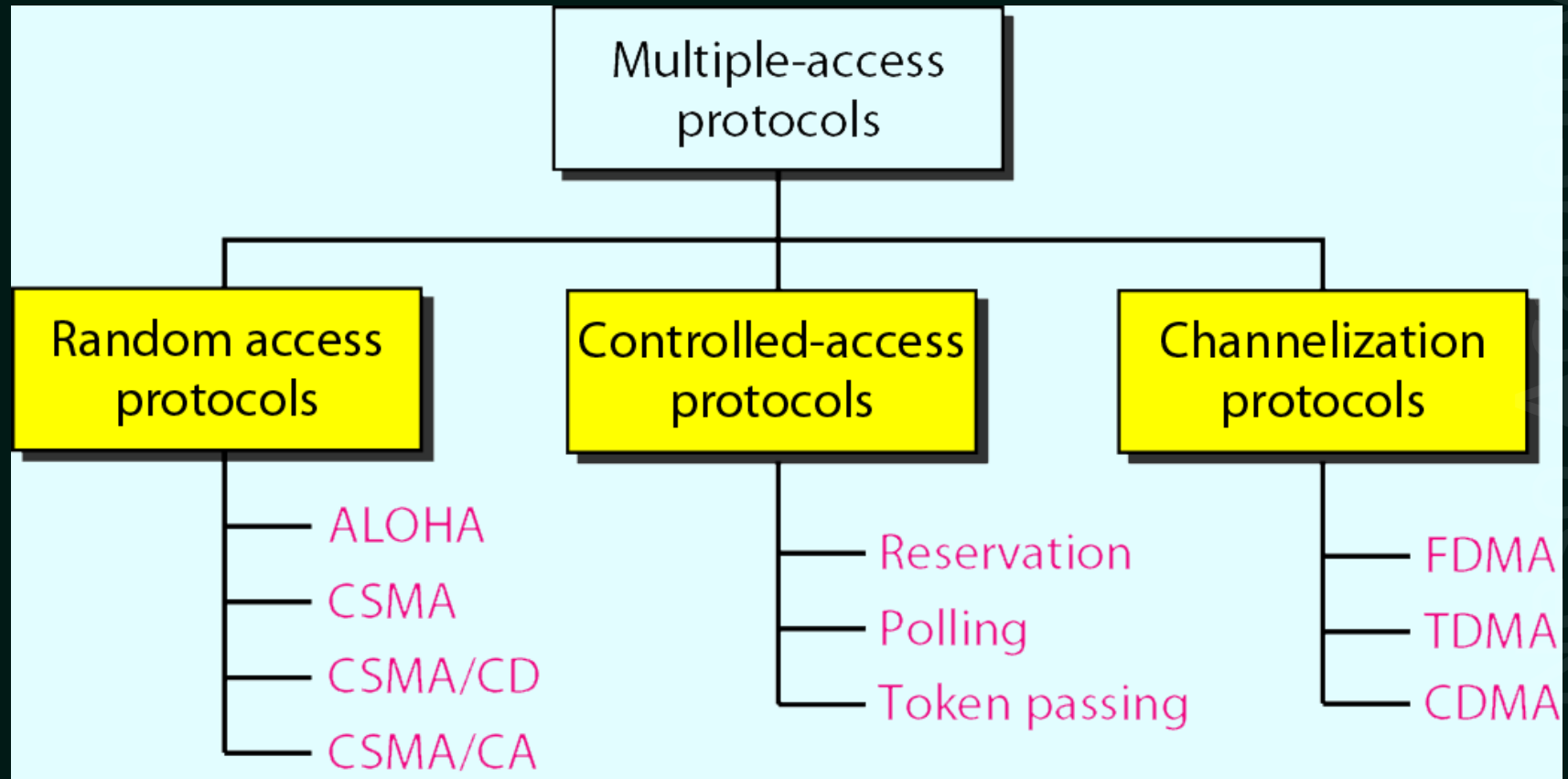
CONTROLLED ACCESS PROTOCOLS



CHANNELIZATION PROTOCOLS

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations.

CHANNELIZATION PROTOCOLS

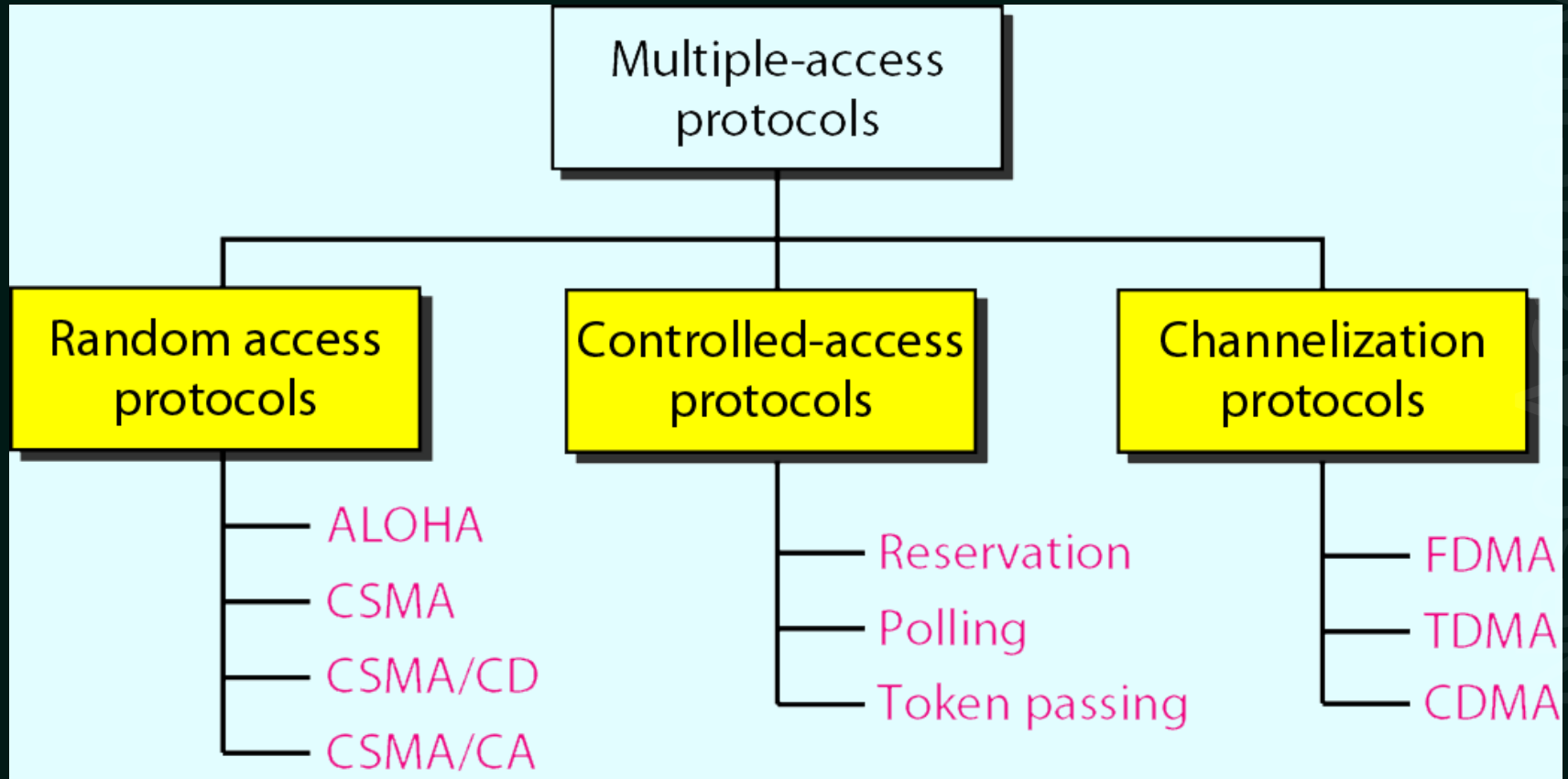


OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various multiple access protocols.
- ★ Understand collision.
- ★ Understand Pure Aloha.

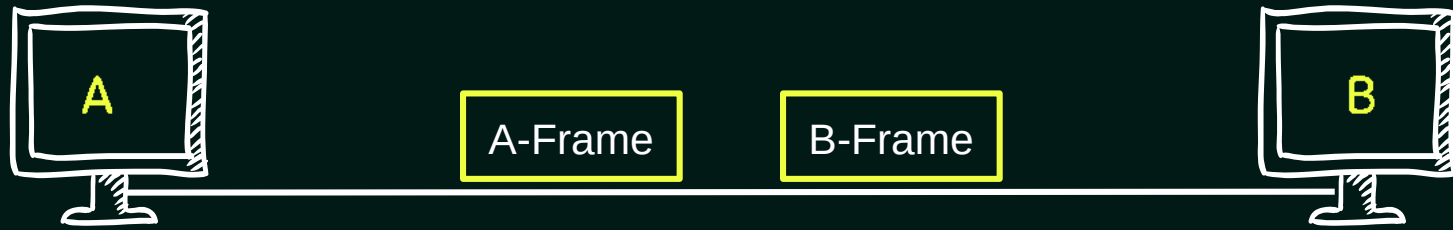
MULTIPLE ACCESS PROTOCOLS



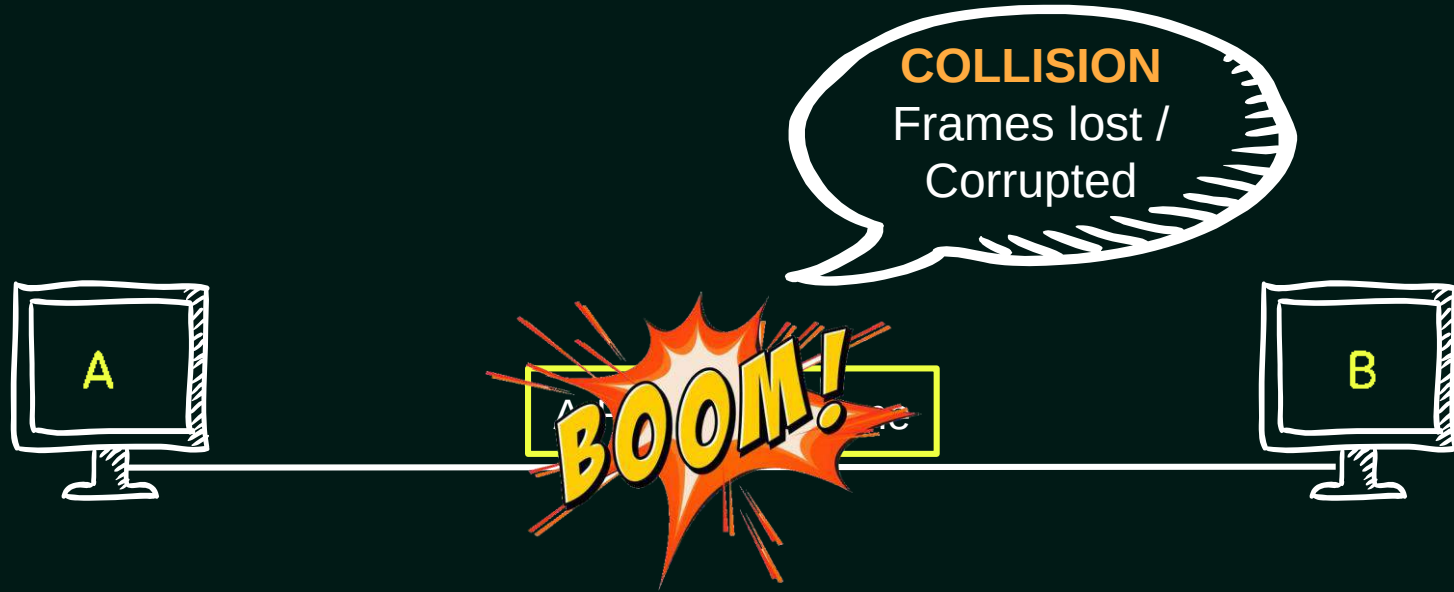
ALOHA

- ★ Aloha is a random access protocol.
- ★ It was actually designed for WLAN but it is also applicable for shared medium.
- ★ In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

COLLISION



COLLISION



ALOHA

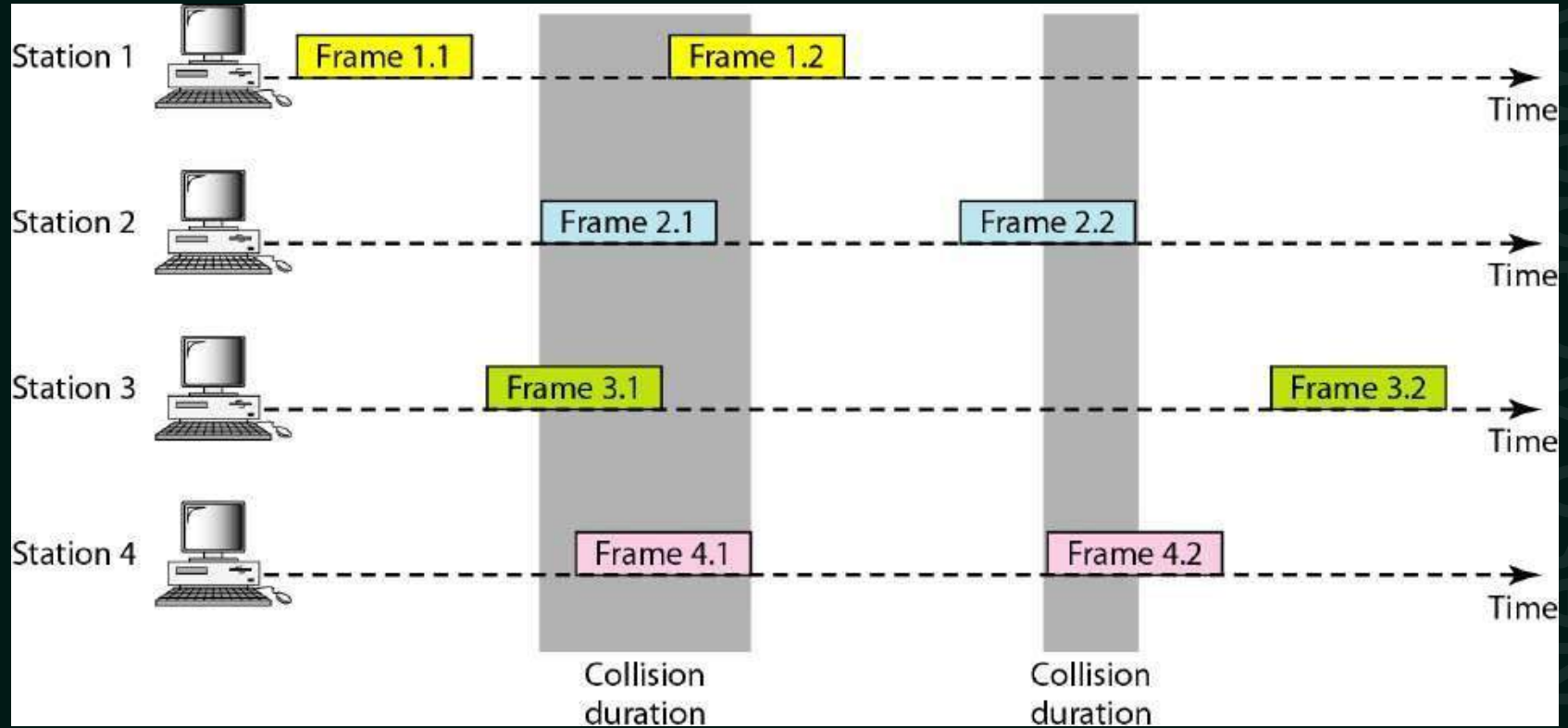
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- ★ It was actually designed for WLAN but it is also applicable for shared medium.
- ★ In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

Types:

- ★ Pure Aloha

- ★ Slotted Aloha

PURE ALOHA



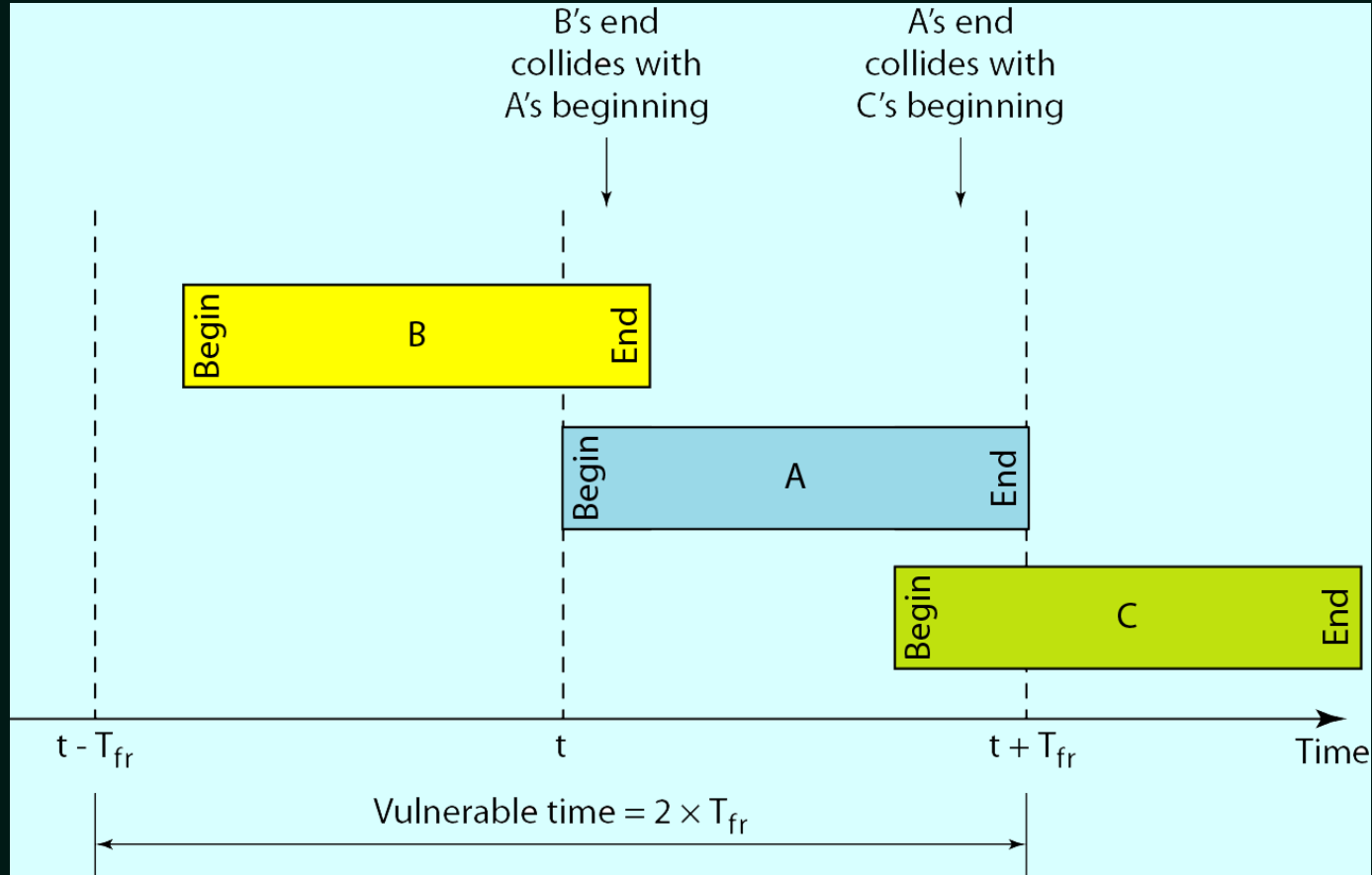
PURE ALOHA

- ★ Pure ALOHA allows stations to transmit whenever they have data to be sent.
- ★ When a station sends data it waits for an acknowledgement.
- ★ If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (T_b) and re-sends the data.
- ★ Since different stations wait for different amount of time, the probability of further collision decreases.
- ★ The throughput of pure aloha is maximized when frames are of uniform length.

PURE ALOHA

- ★ Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be garbled.
- ★ If the first bit of a new frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will have to be retransmitted later.

PURE ALOHA



PURE ALOHA

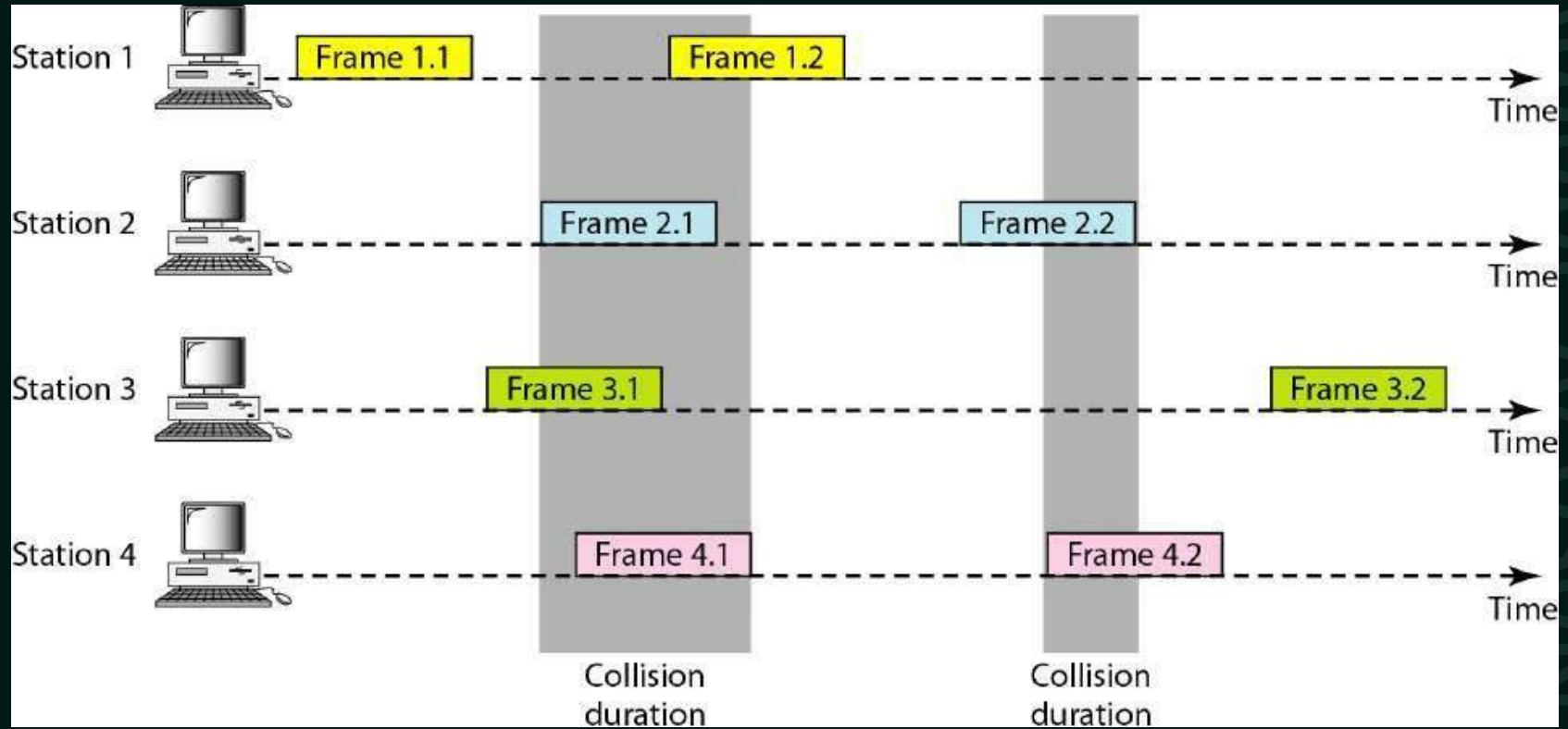
- ★ Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be garbled.
- ★ If the first bit of a new frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will have to be retransmitted later.

Vulnerable Time = $2 \times T_{fr}$

Throughput = $G \times e^{-2G}$; Where G is the number of stations wish to transmit in the same time.

Maximum throughput = 0.184 for $G=0.5$ ($1/2$)

PURE ALOHA



QUESTION

There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot?

[GATE CS 2007]

- A. $np(1-p)^{n-1}$
- B. $(1-p)^{n-1}$
- C. $p(1-p)^{n-1}$
- D. $1-(1-p)^{n-1}$

QUESTION

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

Given

Distance = 12000 Km

Propagation speed = 2.4×10^8 m/s

To Find

Propagation Time

SOLUTION

$$\text{Propagation Time} = \frac{\text{Distance}}{\text{Propagation speed}}$$

$$\text{Propagation Time} = \frac{12000 \times 1000}{2.4 \times 10^8} \text{ sec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \text{ sec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \times \frac{10^{-3}}{10^{-3}} \text{ sec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \times \frac{1}{10^{-3}} \text{ millisec}$$

$$\text{Propagation Time} = \frac{5000 \times 1000}{10^8} \times \frac{10^3}{1} \text{ millisec}$$

$$\text{Propagation Time} = \frac{50 \times 100 \times 1000}{10^8} \times \frac{10^3}{1} \text{ millisec}$$

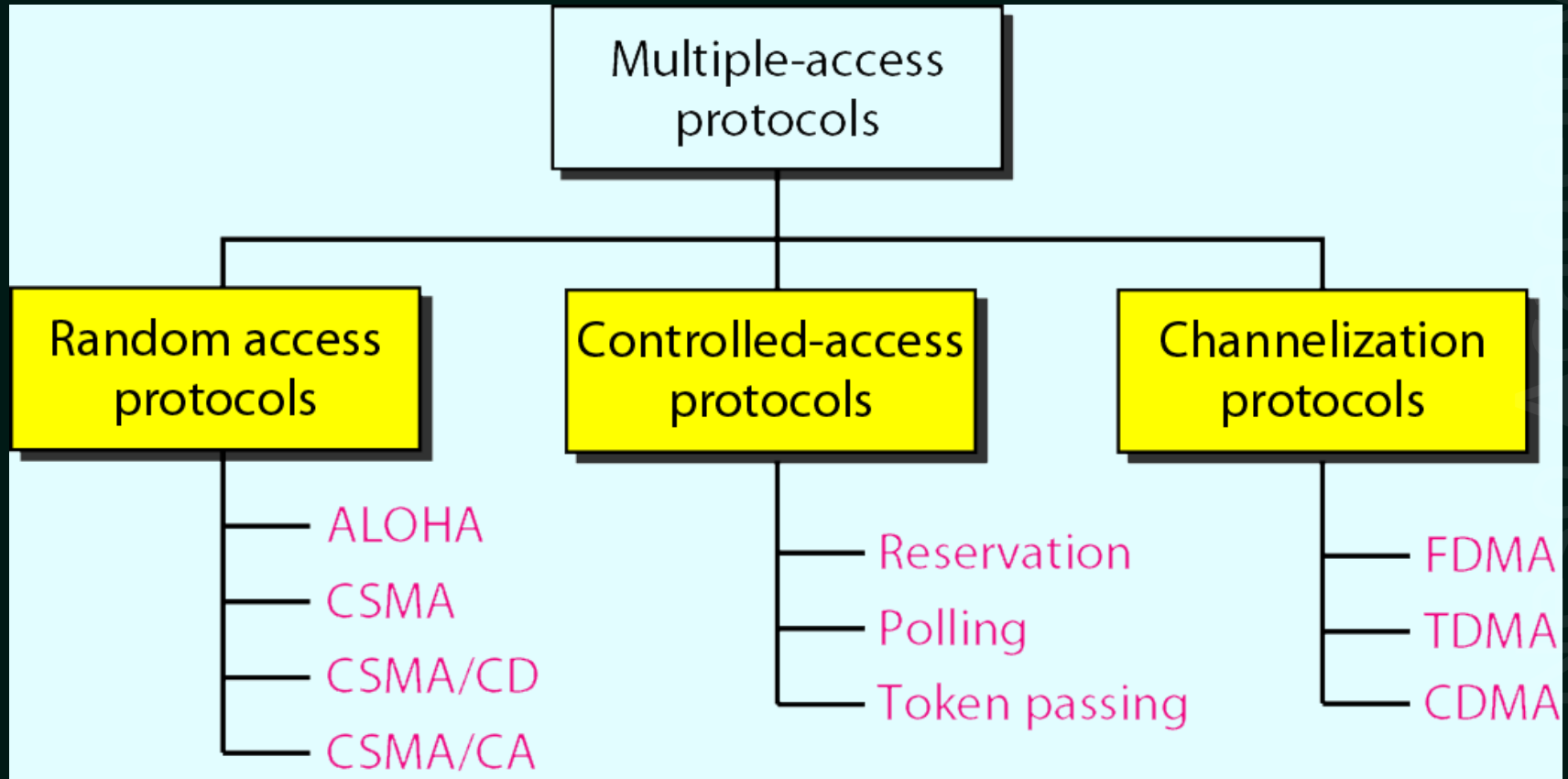
$$\text{Propagation Time} = 50 \text{ millisec OR } 50 \text{ ms}$$

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand Slotted Aloha.
- ★ Compare Pure Aloha and Slotted Aloha.

MULTIPLE ACCESS PROTOCOLS



ALOHA

- ★ Aloha is a random access protocol.
- ★ It was actually designed for WLAN but it is also applicable for shared medium.
- ★ In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

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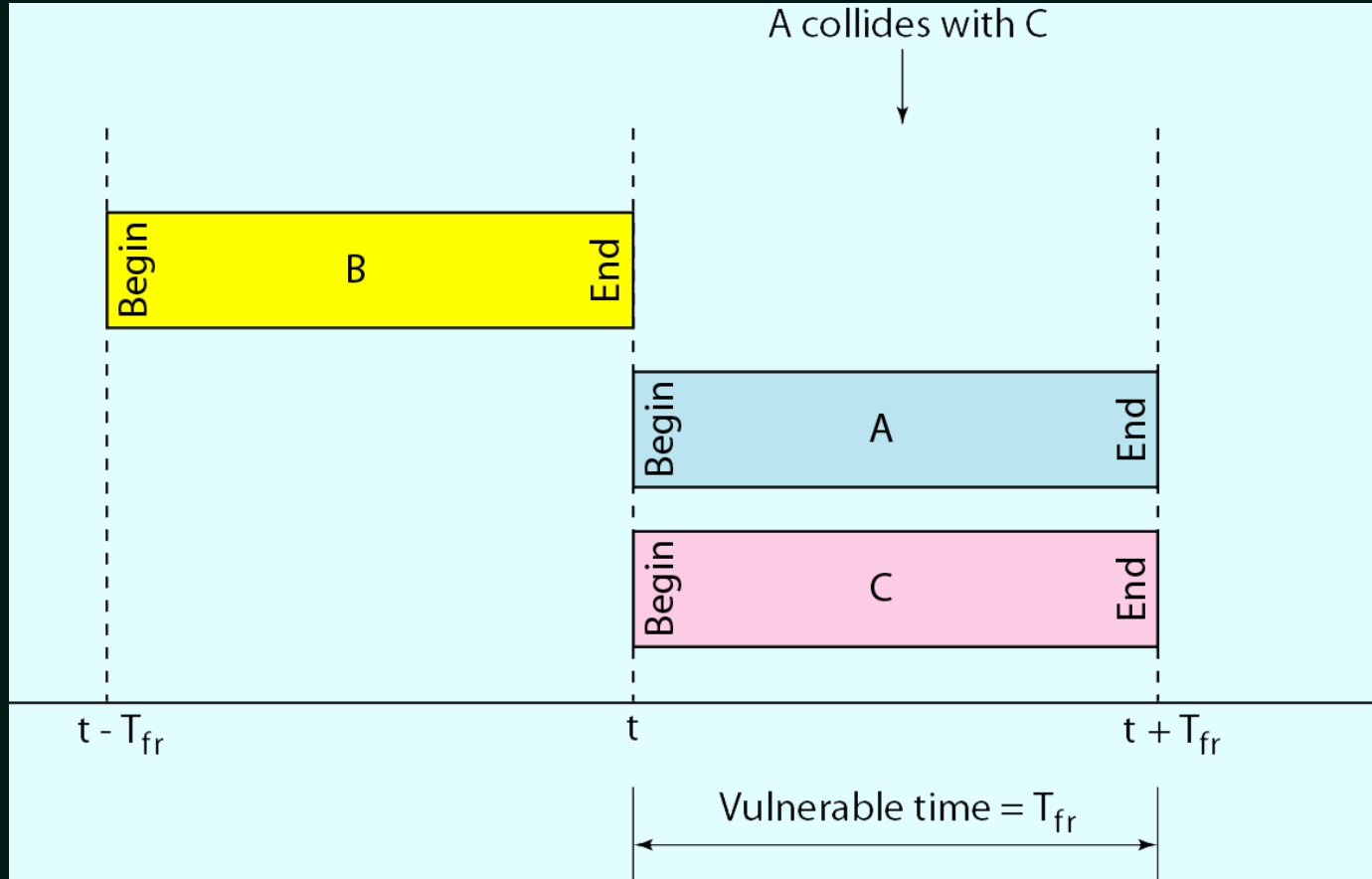
- ★ Pure Aloha

- ★ Slotted Aloha

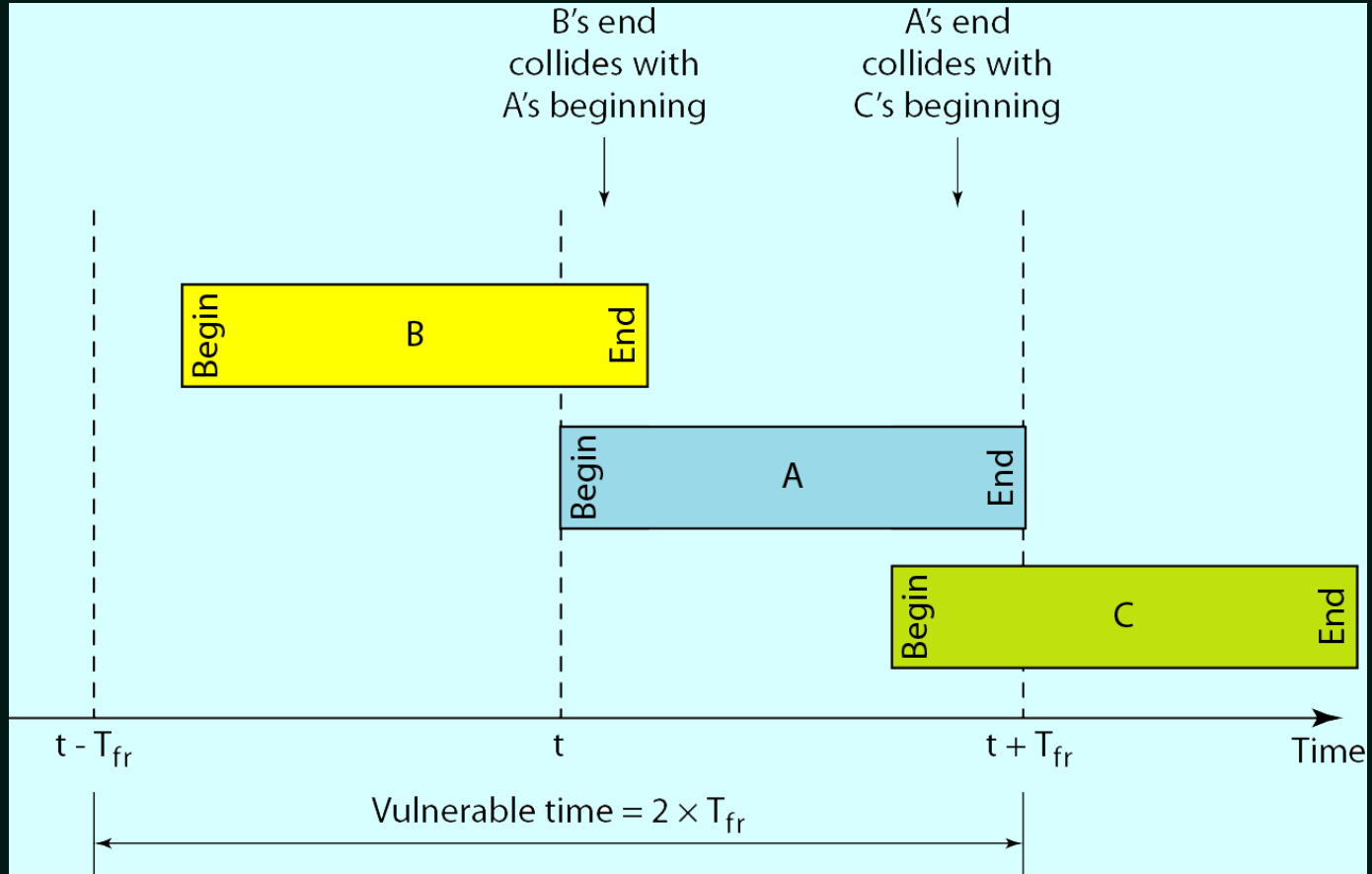
SLOTTED ALOHA

- ★ It was developed just to improve the efficiency of pure aloha as the chances for collision in pure aloha are high.
- ★ The time of the shared channel is divided into discrete time intervals called slots.
- ★ Sending of data is allowed only at the beginning of these slots.

SLOTTED ALOHA



PURE ALOHA



SLOTTED ALOHA

- ★ It was developed just to improve the efficiency of pure aloha as the chances for collision in pure aloha are high.
 - ★ The time of the shared channel is divided into discrete time intervals called slots.
 - ★ Sending of data is allowed only at the beginning of these slots.
 - ★ If a station misses out the allowed time, it must wait for the next slot.
- This reduces the probability of collision.

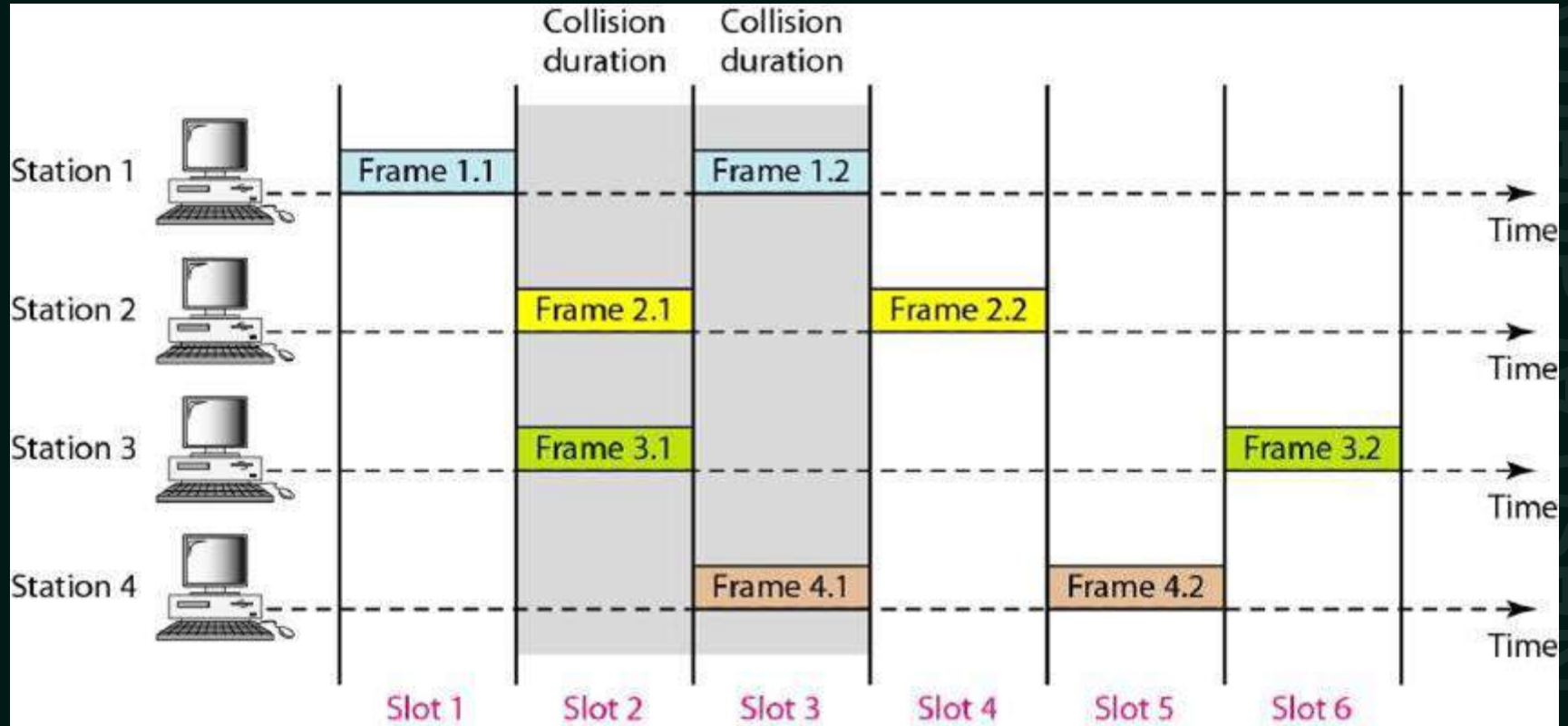
SLOTTED ALOHA

Vulnerable Time = Frame Transmission Time.

Throughput = $G \times e^{-G}$; Where G is the number of stations wish to transmit
in the same time.

Maximum throughput = 0.368 for $G=1$.

SLOTTED ALOHA



PURE ALOHA VS SLOTTED ALOHA

Pure Aloha	Slotted Aloha
Any station can transmit the data at any time.	Any station can transmit the data at the beginning of any time slot.
The time is continuous and not globally synchronized.	The time is discrete and globally synchronized.
Vulnerable time in which collision may occur $= 2 \times T_{Fr}$	Vulnerable time in which collision may occur $= T_{Fr}$
Probability of successful transmission of data packet $= G \times e^{-2G}$	Probability of successful transmission of data packet $= G \times e^{-G}$
Maximum efficiency = 18.4% (Occurs at $G = 1/2$)	Maximum efficiency = 36.8% (Occurs at $G = 1$)
Main advantage: Simplicity in implementation.	Main advantage: It reduces the number of collisions to half and doubles the efficiency of pure aloha.

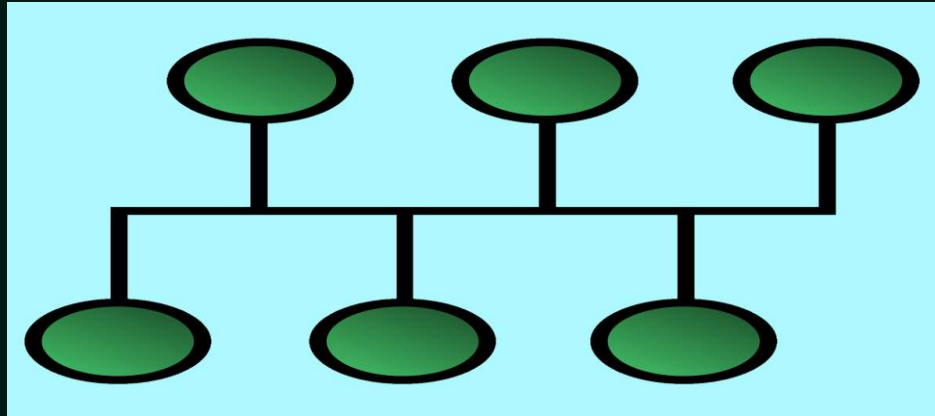
OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various multiple access protocols.
- ★ Understand CSMA protocol.
- ★ Types of CSMA.
- ★ Understand the behaviour of three persistent methods.

CSMA PROTOCOL

- ★ Carrier Sense Protocol.
- ★ To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed.
- ★ Principle of CSMA: “sense before transmit” or “listen before talk.”



CSMA PROTOCOL

- ★ Carrier Sense Protocol.
- ★ To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed.
- ★ Principle of CSMA: “sense before transmit” or “listen before talk.”
- ★ Carrier busy = Transmission is taking place.
- ★ Carrier idle = No transmission currently taking place.
- ★ The possibility of collision still exists because of propagation delay; a station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.

TYPES OF CSMA

1. 1-Persistent CSMA
2. P-Persistent CSMA
3. Non-Persistent CSMA
4. O-Persistent CSMA

CSMA/CD (CSMA with Collision Detection)

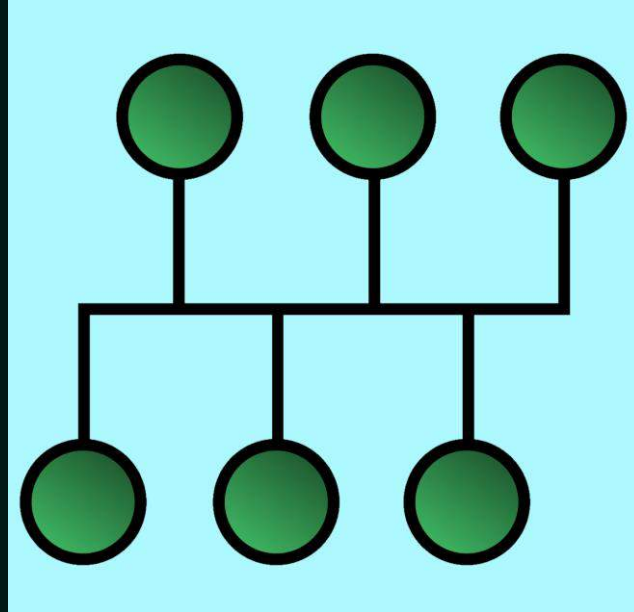
CSMA/CA (CSMA with Collision Avoidance)

1-PERSISTENT CSMA

- ★ Before sending the data, the station first listens to the channel to see if anyone else is transmitting the data at that moment.
- ★ If the channel is idle, the station transmits a frame.
- ★ If busy, then it senses the transmission medium continuously until it becomes idle.
- ★ Since the station transmits the frame with the probability of 1 when the carrier or channel is idle, this scheme of CSMA is called as 1-Persistent CSMA.
- ★ The propagation delay has an important effect on the performance of the protocol.
- ★ There is a small chance that just after a station begins sending, another station will become ready to send and sense the channel.

1-PERSISTENT CSMA

- ★ The longer the propagation delay, the more important this effect becomes, and the worse the performance of the protocol.
- ★ Even if the propagation delay is zero, there will still be collisions.



NON-PERSISTENT CSMA

- ★ Before sending, a station senses the channel. If no one else is sending, the station begins doing so itself.
- ★ However, if the channel is already in use, the station does not continually sense it for the purpose of seizing it immediately upon detecting the end of the previous transmission.
- ★ Instead, it waits a random period of time and then repeats the algorithm. Consequently, this algorithm leads to better channel utilization but longer delays than 1-persistent CSMA.

P-PERSISTENT CSMA

- ★ It applies to slotted channels.
- ★ When a station becomes ready to send, it senses the channel.
- ★ If it is idle, it transmits with a probability P .
- ★ With a probability $Q=1-P$, it defers until the next slot.
- ★ If that slot is also idle, it either transmits or defers again, with probabilities P and Q .
- ★ This process is repeated until either the frame has been transmitted or another station has begun transmitting.
- ★ In the latter case, the unlucky station acts as if there had been a collision (i.e., it waits a random time and starts again).
- ★ If the station initially senses the channel busy, it waits until the next slot and applies the above algorithm.

• Assume that N stations have a packet to send and the channel is busy.

- Then, due to the exponential number of nodes that will attempt to transmit once the medium becomes idle.
- If $P > 1$, then a collision is expected to occur. Therefore, the network must make sure that $0 \leq P \leq 1$ to avoid the collision when N is the maximum number of nodes that can be active on the line.

SNR:

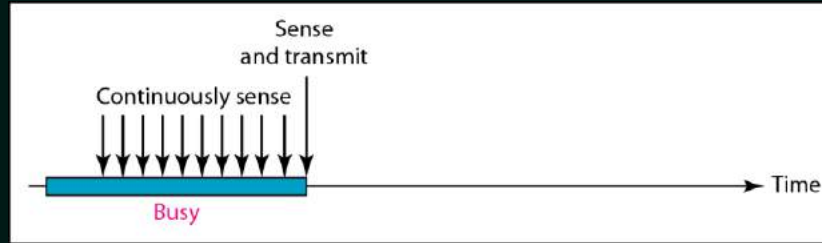
The logic used to check whether a station is able to transmit or not is very simple. First, the value of variable P is calculated and based on that value the decision is made.

- If $P < 0$, collision is expected.
- Use $P = 0$, if $P > 1$, if $P > 1$ then $P = 1$.

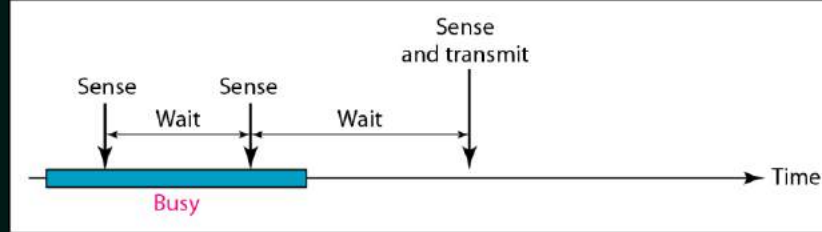
Now it comes the part of "how to calculate the variable P " using a very simple equation below, where N is the number of stations that are connected to the channel medium.

$$P_i = \left(1 - \frac{1}{N}\right)^{N-1}$$

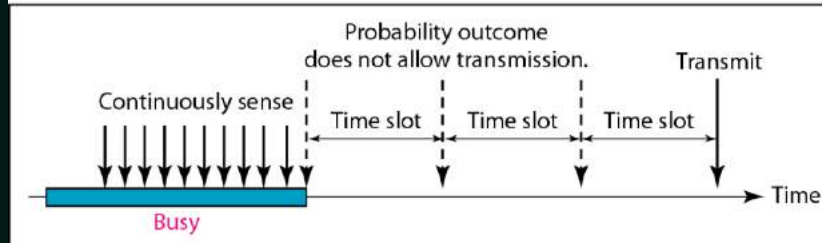
BEHAVIOUR OF THREE PERSISTENT METHODS



a. 1-persistent



b. Nonpersistent



c. p-persistent

O-PERSISTENT CSMA

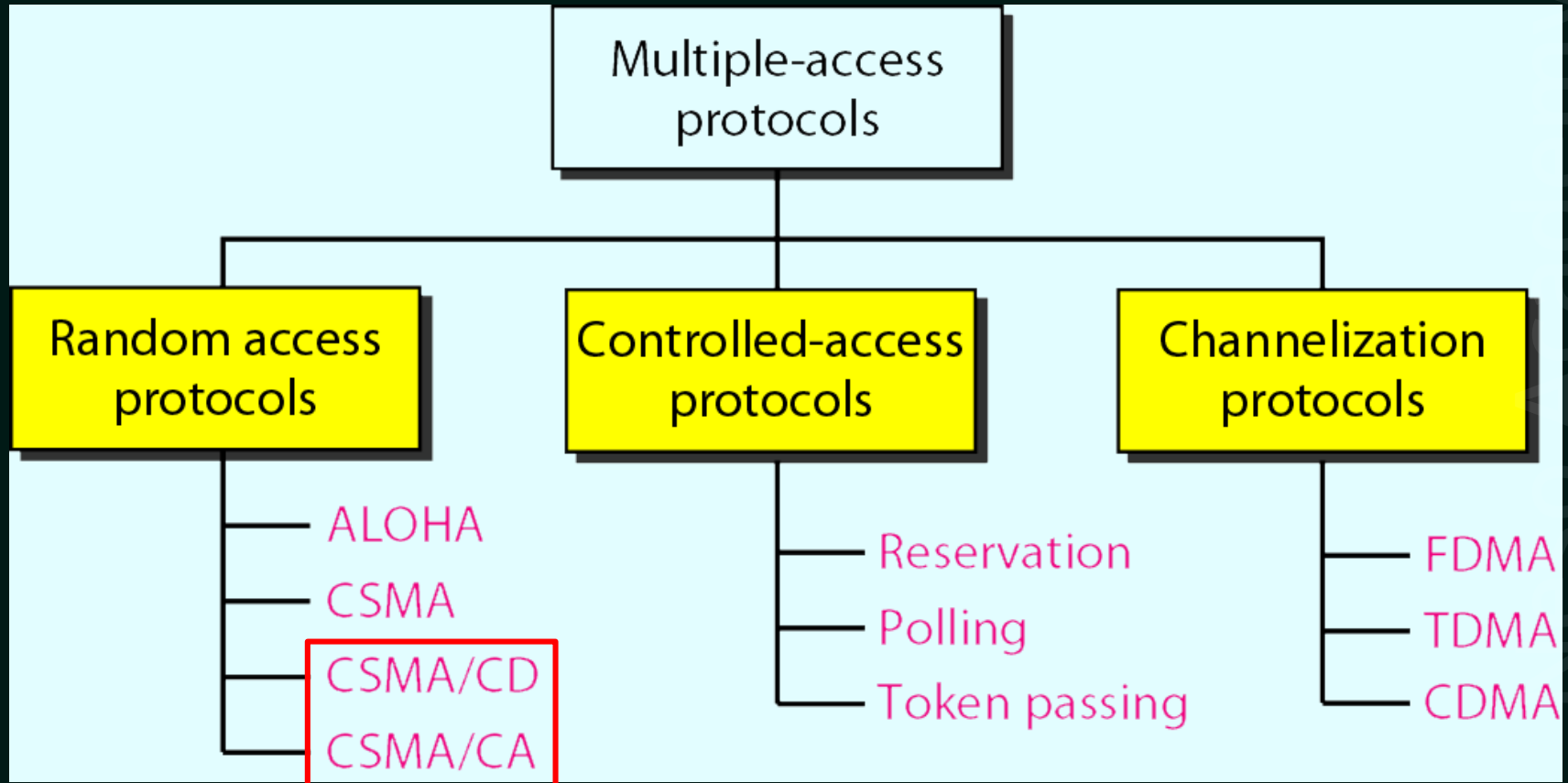
- ★ Each node is assigned a transmission order by a supervisory node.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various multiple access protocols.
- ★ Understand CSMA/CD.
- ★ Understand CSMA/CA.

MULTIPLE ACCESS PROTOCOLS



CSMA PROTOCOL

- ★ Principle of CSMA: “sense before transmit” or “listen before talk.”
- ★ Carrier busy = Transmission is taking place.
- ★ Carrier idle = No transmission currently taking place.

TYPES OF CSMA

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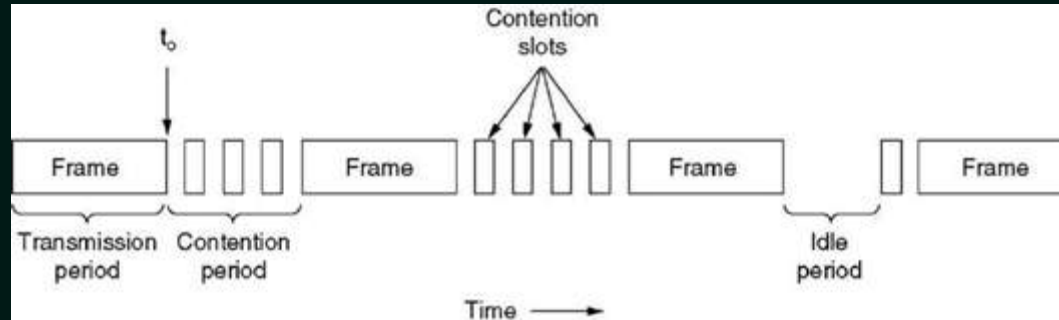
CSMA/CA (CSMA with Collision Avoidance)

CSMA/CD

- ★ If two stations sense the channel to be idle and begin transmitting simultaneously, they will both detect the collision almost immediately.
- ★ Rather than finish transmitting their frames, which are irretrievably garbled anyway, they should abruptly stop transmitting as soon as the collision is detected.
- ★ Quickly terminating damaged frames saves time and bandwidth.
- ★ This protocol, known as CSMA/CD (CSMA with Collision Detection) is widely used on LANs in the MAC sublayer.
- ★ Access method used by Ethernet: CSMA/CD.

CSMA/CD

- ★ At the point marked t_0 , a station has finished transmitting its frame.



CSMA/CD

- ★ At the point marked t_0 , a station has finished transmitting its frame.
- ★ Any other station having a frame to send may now attempt to do so. If two or more stations decide to transmit simultaneously, there will be a collision.
- ★ Collisions can be detected by looking at the power or pulse width of the received signal and comparing it to the transmitted signal.
- ★ After a station detects a collision, it aborts its transmission, waits a random period of time, and then tries again, assuming that no other station has started transmitting in the meantime.
- ★ Therefore, model for CSMA/CD will consist of alternating contention and transmission periods, with idle periods occurring when all stations are quiet.



IMPORTANT CONCEPT

CSMA/CD – For GATE aspirants

CSMA/CD – FOR GATE ASPIRANTS

$$\text{Efficiency} = \frac{1}{1 + 6.44 \times a}$$

$$a = \frac{T_p}{T_t}$$

- ★ If distance increases, efficiency of CSMA decreases.
- ★ CSMA is not suitable for long distance networks like WAN; but works optimally for LAN.
- ★ If length of packet is bigger, the efficiency of CSMA also increases; but maximum limit for length is 1500 Bytes.
- ★ Transmission Time $\geq$ Round Trip Time of 1 bit
- ★ Transmission Time $\geq 2 \times$ Propagation Time

CSMA/CA

- ★ Carrier-sense multiple access with collision avoidance (CSMA/CA) is a network multiple access method in which carrier sensing is used, but nodes attempt to avoid collisions by beginning transmission only after the channel is sensed to be "idle".
- ★ It is particularly important for wireless networks, where the collision detection of the alternative CSMA/CD is not possible due to wireless transmitters desensing their receivers during packet transmission.
- ★ CSMA/CA is unreliable due to the hidden node problem and exposed terminal problem.
- ★ Solution: RTS/CTS exchange.
- ★ CSMA/CA is a protocol that operates in the Data Link Layer (Layer 2) of the OSI model.

The 802.11 protocol allows transmissions to be protected against interference from other stations by using the RTS/CTS mechanism. Using RTS/CTS, the node wishing to transmit first sends an RTS (Request To Send) frame to the target node. The target node, if the channel is clear, responds with a CTS (Clear To Send) frame that not only informs the originator that it may transmit but also tells other stations to refrain from using the channel for the specified amount of time. RTS/CTS obviously adds some overhead to the transmission process, so it is normally used to protect longer frames which are more expensive to retransmit. RTS/CTS does not completely solve the hidden node problem, but it can significantly help under certain conditions. A slightly more in-depth coverage of RTS/CTS and challenges associated with it can be found e.g. [here](https://inet.omnetpp.org/docs/showcases/wireless/hiddennode/doc/index.html#:~:text=The%20hidden%20node%20problem%20occurs,when%20both%20nodes%20transmit%20simultaneously).

Go through the following link better understanding:

<https://inet.omnetpp.org/docs/showcases/wireless/hiddennode/doc/index.html#:~:text=The%20hidden%20node%20problem%20occurs,when%20both%20nodes%20transmit%20simultaneously>.

CSMA/CA

- ★ The Access method used by IEEE 802.11 Wi-Fi is CSMA/CA.

HOMEWORK!

Find the correct box.

Multiple Access Method used by

Ethernet : CSMA/CA

Wi-Fi : CSMA/CD

Multiple Access Method used by

Ethernet : CSMA/CD

Wi-Fi : CSMA/CA

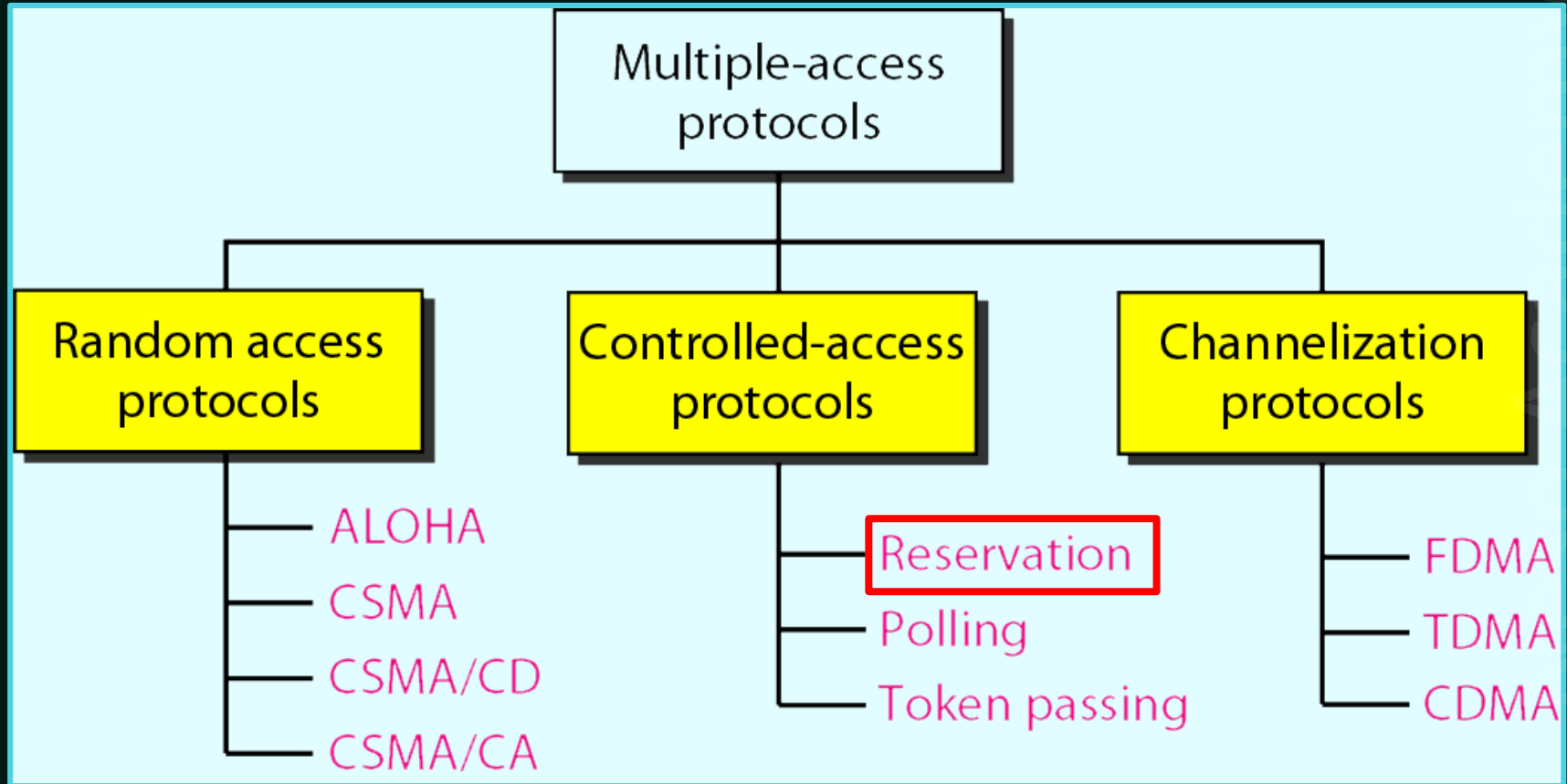
Please post your answer in the comments section

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various multiple access protocols.
- ★ Understand the Controlled Access Protocols – Reservation.

MULTIPLE ACCESS PROTOCOLS



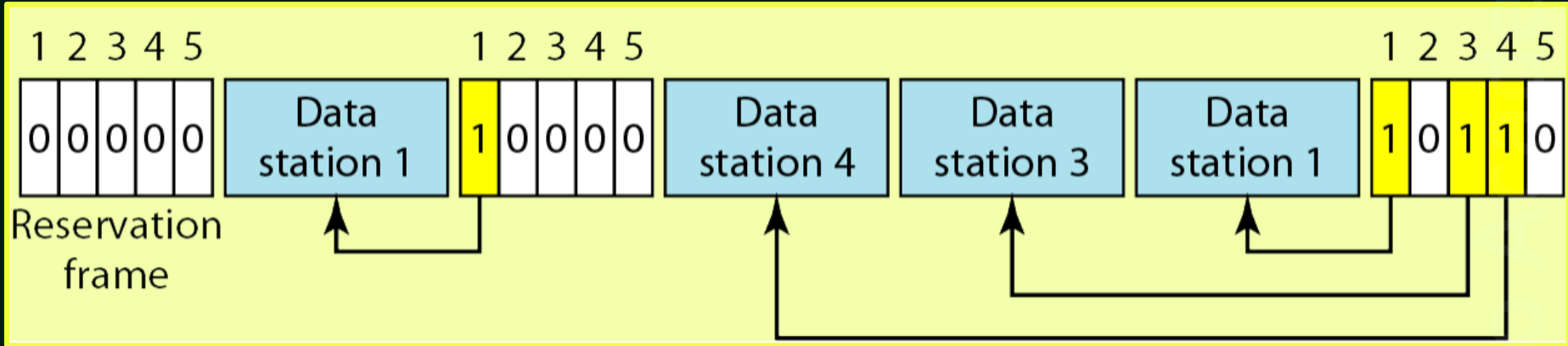
RESERVATION – ANALOGY



RESERVATION

- ★ A station need to make a reservation before sending data.
- ★ In each interval, a reservation frame precedes the data frames sent in that interval.
- ★ If there are N stations in the system, there are exactly N reservation minislots in the reservation frame.
- ★ Each minislot belongs to a station.
- ★ When a station needs to send a data frame, it makes a reservation in its own minislot.
- ★ The stations that have made reservations can send their data frames after the reservation frame.

RESERVATION

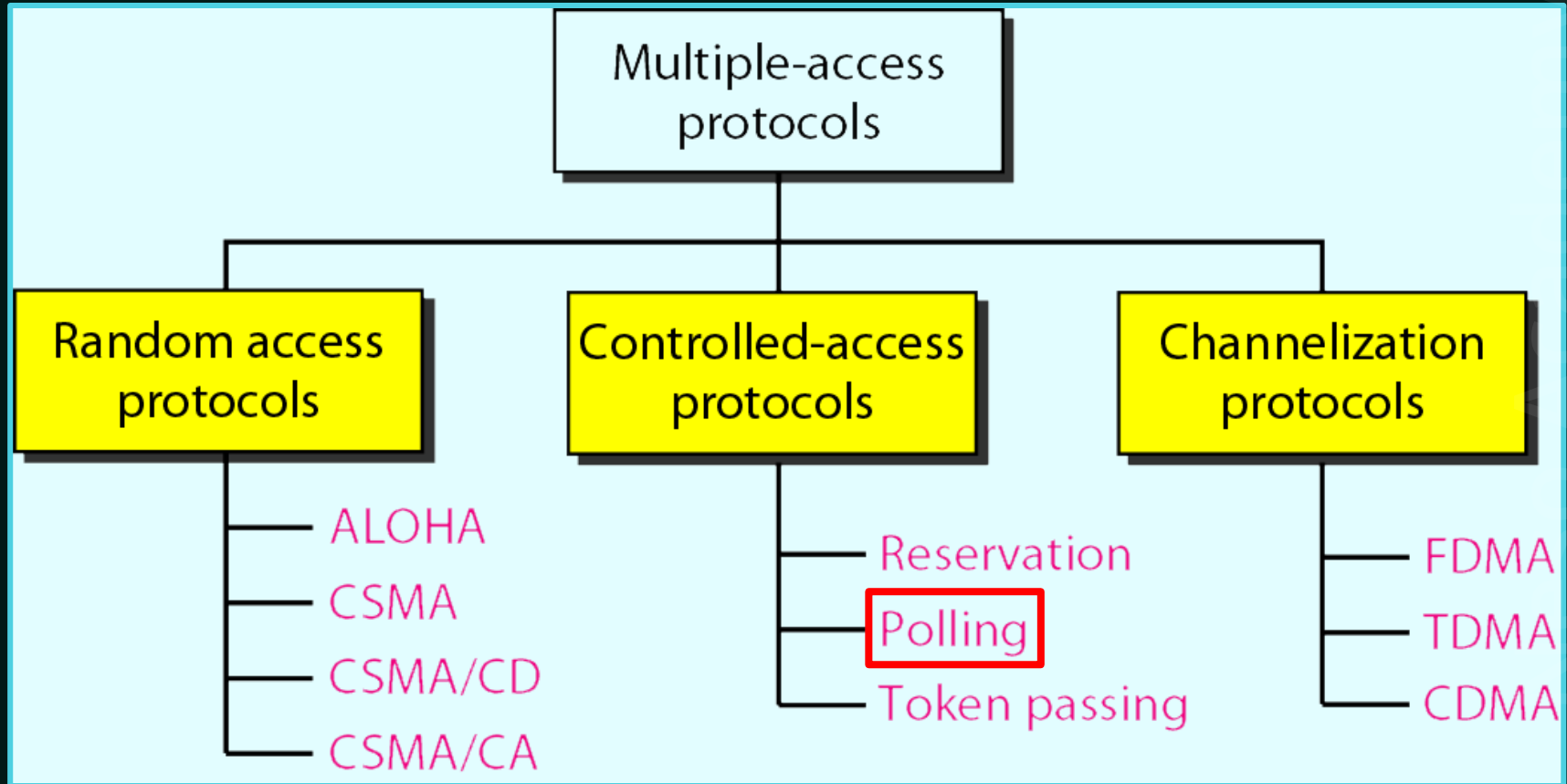


OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various multiple access protocols.
- ★ Understand the Controlled Access Protocol – Polling.
- ★ Know the efficiency of polling.

MULTIPLE ACCESS PROTOCOLS



POLLING - ANALOGY!



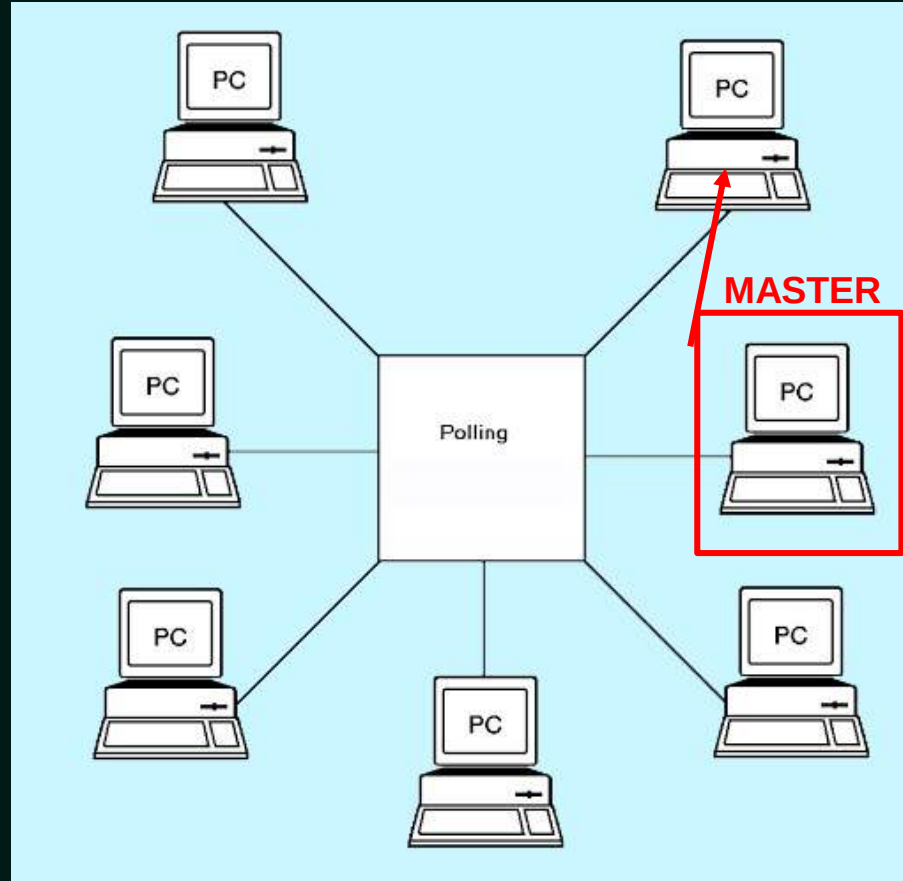
POLLING

- ★ The polling protocol requires one of the nodes to be designated as a Master node (Primary station).
- ★ The master node polls each of the nodes in a round-robin fashion.
- ★ In particular, the master node first sends a message to node 1, saying that it (node 1) can transmit up to some maximum number of frames.
- ★ After node 1 transmits some frames, the master node tells node 2 it (node 2) can transmit up to the maximum number of frames.
- ★ The master node can determine when a node has finished sending its frames by observing the lack of a signal on the channel.

POLLING

- ★ The procedure continues in this manner, with the master node polling each of the nodes in a cyclic manner.
- ★ The polling protocol eliminates the collision.
- ★ This allows polling to achieve a much higher efficiency.
- ★ The first drawback is that the protocol introduces a polling delay—the amount of time required to notify a node that it can transmit.
- ★ The second drawback, which is potentially more serious, is that if the master node fails, the entire channel becomes inoperative.

POLLING



POLLING – FUNCTIONS

- ★ **Poll function** : If the primary wants to receive data, it asks the secondaries if they have anything to send.
- ★ **Select function** : If the primary wants to send data, it tells the secondary to get ready to receive.

EFFICIENCY OF POLLING

Let T_{poll} be the time for polling and T_t be the time required for transmission of data. Then,

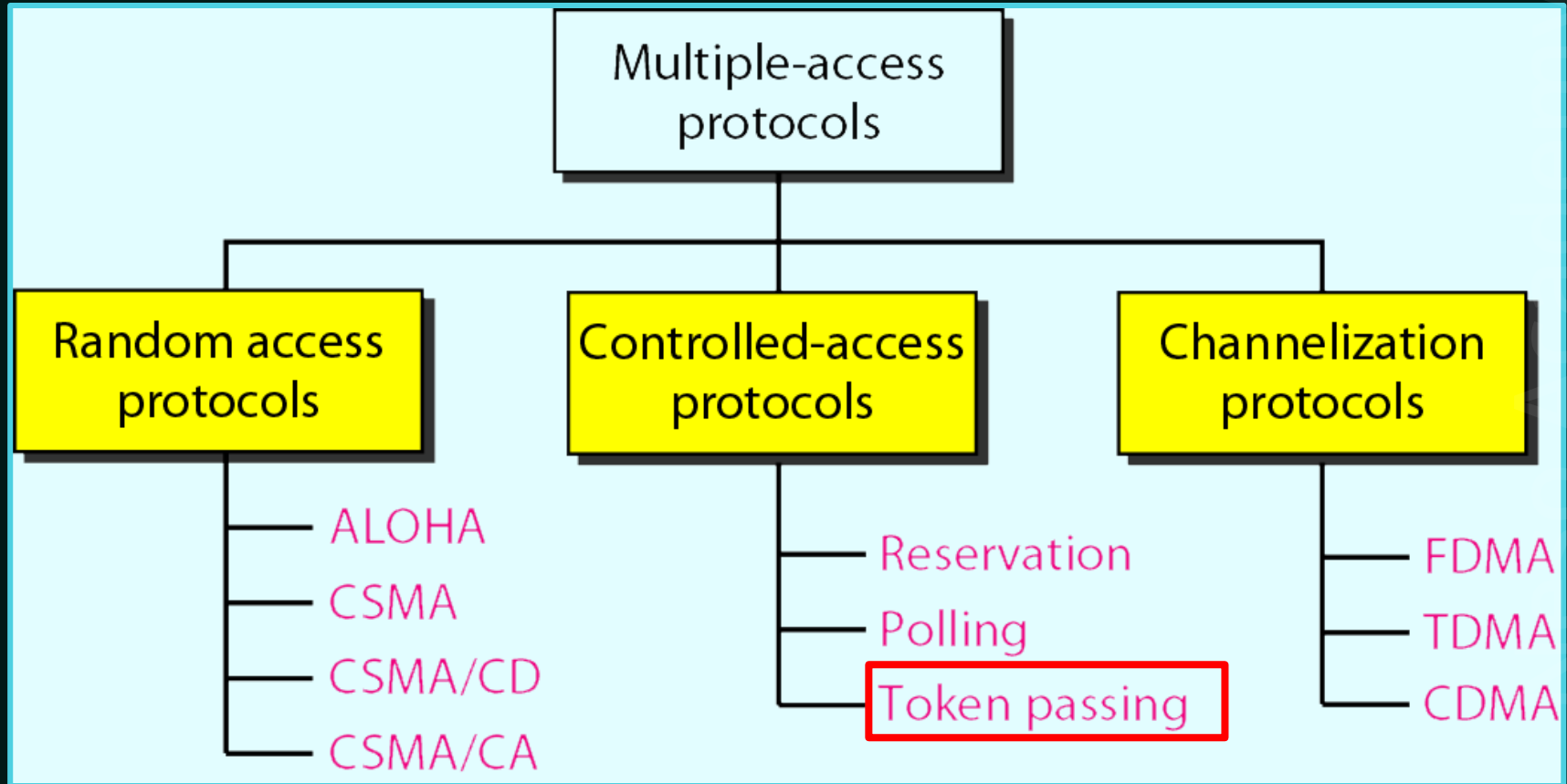
$$\text{Efficiency} = \frac{T_t}{T_t + T_{\text{poll}}}$$

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the various multiple access protocols.
- ★ Understand the Controlled Access Protocol – Token Passing.
- ★ Know the performance of Token Passing.

MULTIPLE ACCESS PROTOCOLS



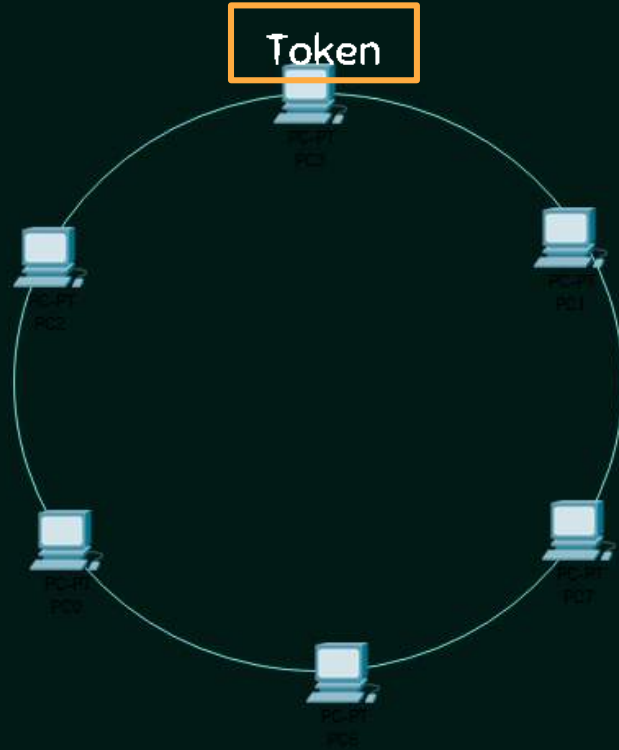
TOKEN PASSING

- ★ A station is authorized to send data when it receives a special frame called a token.
- ★ Here there is no master node.
- ★ A small, special-purpose frame known as a token is exchanged among the nodes in some fixed order.
- ★ When a node receives a token, it holds onto the token only if it has some frames to transmit; otherwise, it immediately forwards the token to the next node.

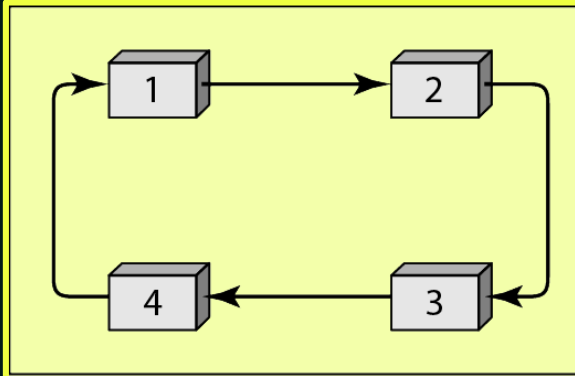
TOKEN PASSING

- ★ If a node does have frames to transmit when it receives the token, it sends up to a maximum number of frames and then forwards the token to the next node.
- ★ Token passing is decentralized and highly efficient. But it has problems as well.
- ★ For example, the failure of one node can crash the entire channel. Or if a node accidentally neglects to release the token, then some recovery procedure must be invoked to get the token back in circulation.

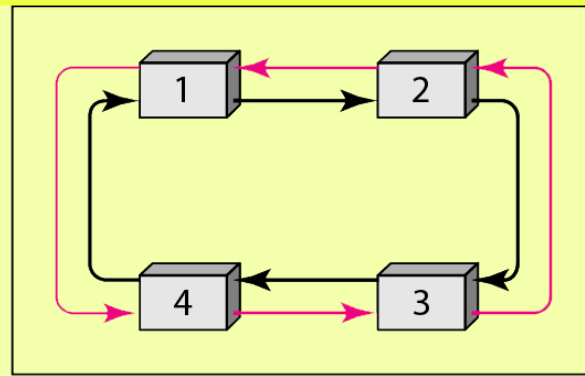
TOKEN PASSING



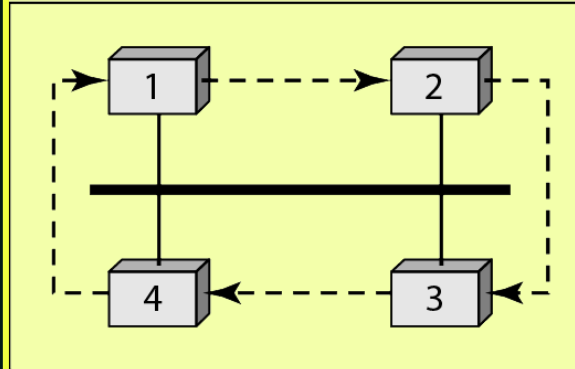
TOKEN PASSING



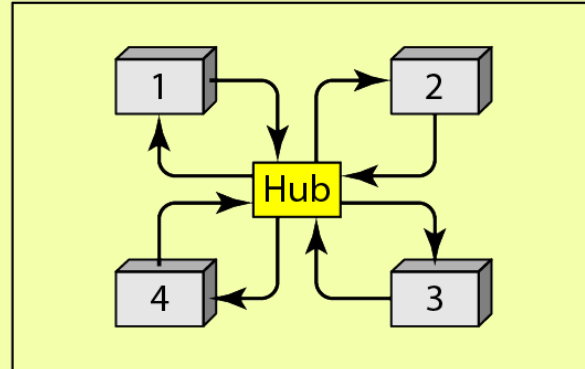
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

PERFORMANCE OF TOKEN PASSING

$$S = \frac{1}{1 + a/N} \quad ; \text{for } a < 1$$

$$S = \frac{1}{a(1 + 1/N)} \quad ; \text{for } a > 1$$

$$a = \frac{T_p}{T_t}$$

S = Throughput

N = number of stations

T_p = Propagation delay

T_t = Transmission delay

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the Ethernet protocol.
- ★ Know the evolution of Ethernet.
- ★ Understand the frame format of Ethernet.
- ★ Know the minimum and maximum length of Ethernet frames.
- ★ Understand the Ethernet address.

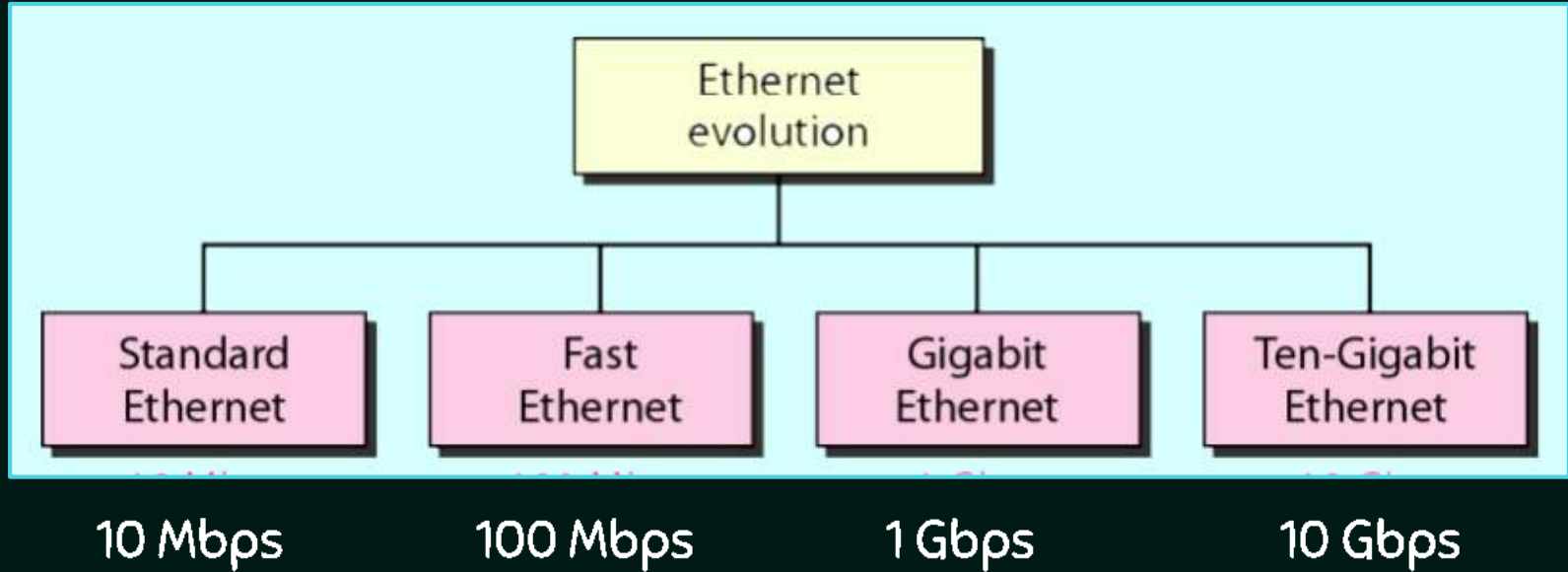
ETHERNET

- ★ One of the most widely used Wired LAN technologies.
- ★ Operates in the data link layer and the physical layer.
- ★ Family of networking technologies that are defined in the IEEE 802.2 and 802.3 standards.
- ★ Supports data bandwidths of 10, 100, 1000, 10,000, 40,000, and 100,000 Mbps (100 Gbps).

Ethernet Standards

- ★ Define Layer 2 protocols and Layer 1 technologies
- ★ Two separate sublayers of the data link layer to operate – Logical link control (LLC) and the MAC sublayers.

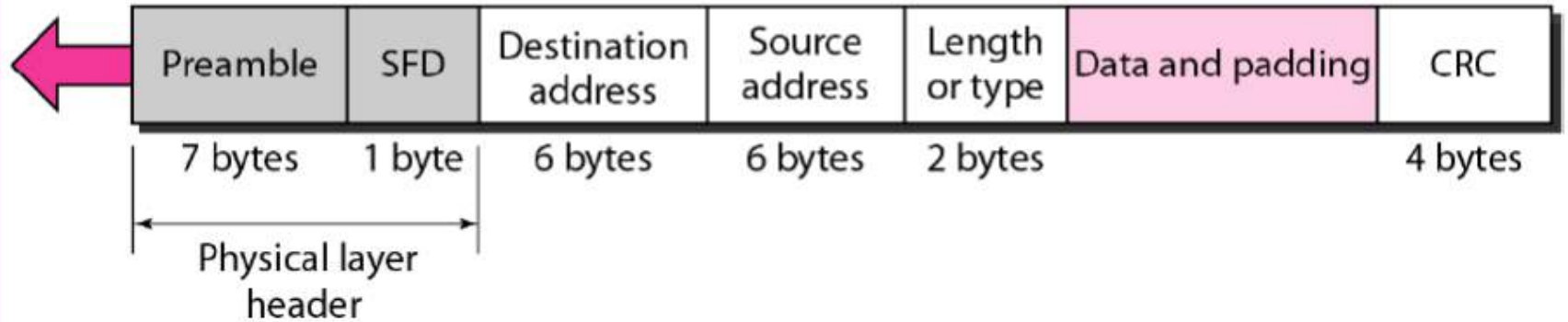
EVOLUTION OF ETHERNET



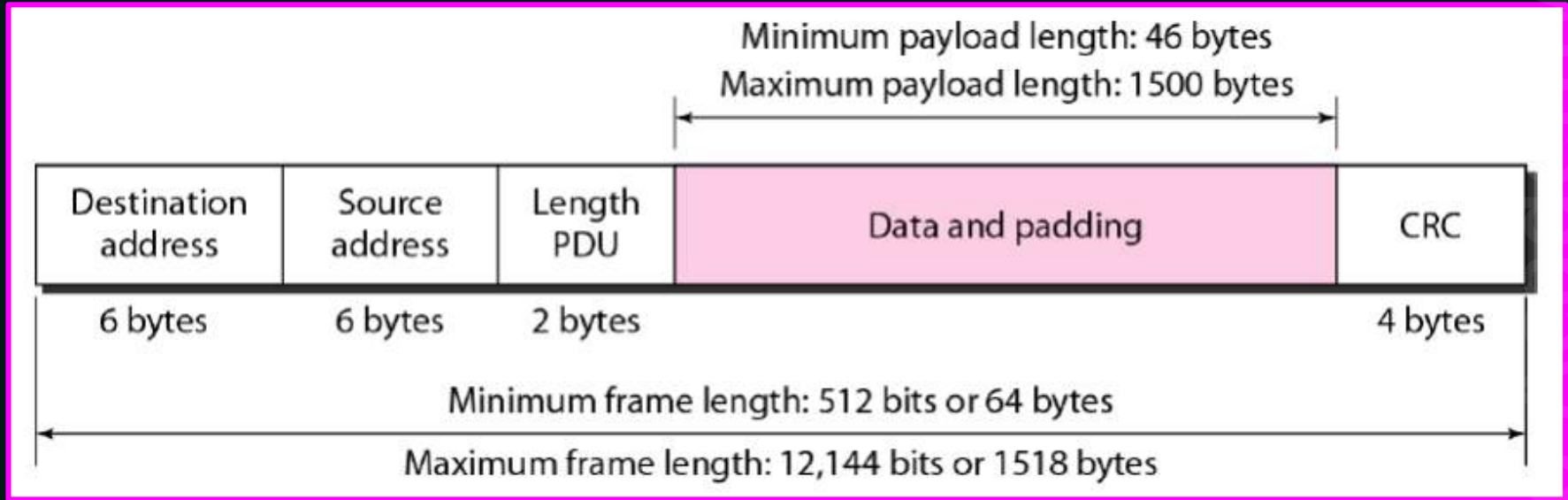
ETHERNET FRAME FORMAT

Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)



ETHERNET FRAME – MIN AND MAX LENGTH



Frame length:

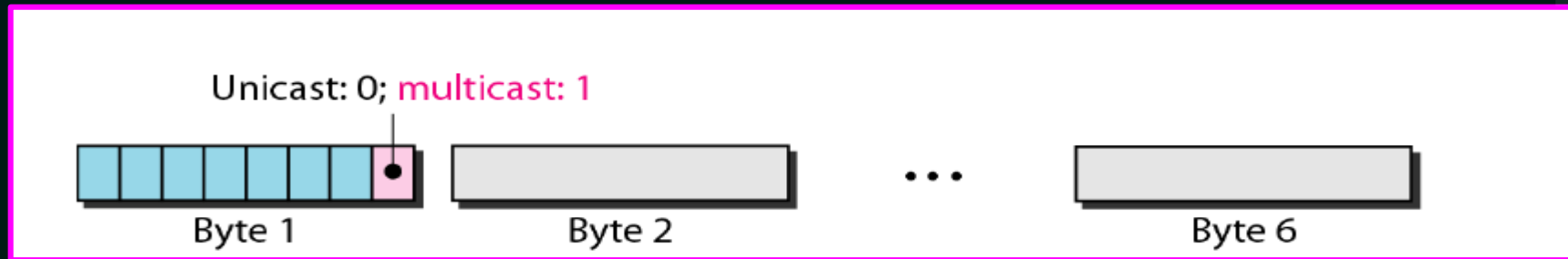
Minimum: 64 bytes (512 bits)

Maximum: 1518 bytes (12,144 bits)

ETHERNET ADDRESS

Example – 06:01:02:01:2C:4B

06:01:02:01:2C:4B $\Leftrightarrow$ 6 bytes $\Leftrightarrow$ 12 hex digits $\Leftrightarrow$ 48 bits



The least significant bit of the first byte defines the type of address.

If the bit is 0, the address is unicast; otherwise, it is multicast.

If all bits are 1, then it is broadcast address

HEXADECIMAL TO BINARY

1 Hexadecimal digit = 4 binary bits.

Hex: 0-9,A,B,C,D,E,F

Binary: 0,1

Hexadecimal to Binary

Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Conversion:

8	4	2	1
2^3	2^2	2^1	2^0

HEXADECIMAL TO BINARY – EXAMPLE

Convert $5B_{16}$ to binary.

Hexadecimal to Binary

Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

$$5B_{16} = 01011011_2$$

	8	4	2	1
	2^3	2^2	2^1	2^0
5	0	1	0	1
B	1	0	1	1

QUESTION 1

What is the hexadecimal equivalent of the following Ethernet address?

01011010 00010001 01010101 00011000 10101010 00001111

SOLUTION

01011010 = 5A

00010001 = 11

01010101 = 55

00011000 = 18

10101010 = AA

00001111 = 0F

Hexadecimal to Binary

Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

8	4	2	1	
0	0	0	0	0
1	1	1	1	F

QUESTION 1

What is the hexadecimal equivalent of the following Ethernet address?

01011010 00010001 01010101 00011000 10101010 00001111

SOLUTION

01011010 = 5A

00010001 = 11

01010101 = 55

00011000 = 18

10101010 = AA

00001111 = 0F

Hexadecimal to Binary

Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

The Answer is: 5A 11 55 18 AA 0F

QUESTION 2

Define the type of the following destination addresses:

(A) 4A:30:10:21:10:1A (B) 47:20:1B:2E:08:EE (C) FF:FF:FF:FF:FF:FF

SOLUTION

(A) 4A

QUESTION 2

Define the type of the following destination addresses:

(A) 4A:30:10:21:10:1A (B) 47:20:1B:2E:08:EE (C) FF:FF:FF:FF:FF:FF

SOLUTION

(A) 4A $\Leftrightarrow$ 0100 1010 $\Leftrightarrow$ 0100 1010 $\Leftrightarrow$ Unicast Address.

(B) 47 $\Leftrightarrow$ 0100 0111 $\Leftrightarrow$ 0100 0111 $\Leftrightarrow$ Multicast Address.

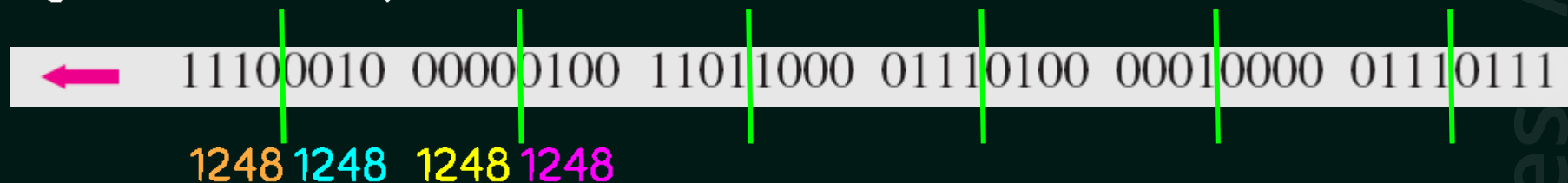
(C) All digits are 'F' so it is a Broadcast Address.

QUESTION 3

Show how the address 47:20:1B:2E:08:EE is sent out on line.

SOLUTION

The address is sent left-to-right, byte by byte; for each byte, it is sent right-to-left, bit by bit, as shown below:



QUESTION 4

The address 43:7B:6C:DE: 10:00 has been shown as the source address in an Ethernet frame. The receiver has discarded the frame. Why?

ANSWER:

The first byte in binary is 01000001.

The least significant bit is 1.

This means that the pattern defines a multicast address.

A multicast address can be a destination address, but not a source address.

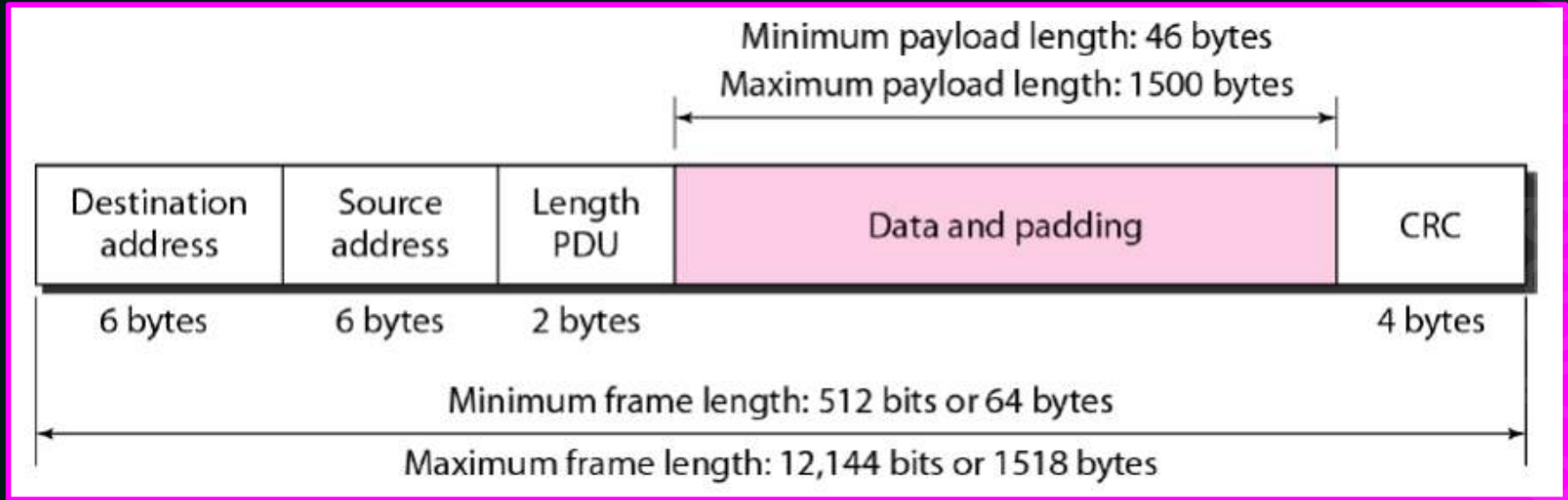
Therefore, the receiver knows that there is an error, and discards the packet.

QUESTION 1

An Ethernet MAC sublayer receives 42 bytes of data from the upper layer.
How many bytes of padding must be added to the data?

ANSWER:

ETHERNET FRAME – MIN AND MAX LENGTH



Frame length:

Minimum: 64 bytes (512 bits)

Maximum: 1518 bytes (12,144 bits)

QUESTION 1

An Ethernet MAC sublayer receives 42 bytes of data from the upper layer.
How many bytes of padding must be added to the data?

ANSWER:

The minimum data size (Payload) in Ethernet is 46 bytes.

Therefore, we need to add 4 bytes of padding to the data so as to make the minimum data size of 46 bytes.

QUESTION 2

An Ethernet MAC sublayer receives 1510 bytes of data from the upper layer. Can the data be encapsulated in one frame? If not, how many frames need to be sent? What is the size of the data in each frame?

ANSWER:

The maximum data size in the Standard Ethernet is 1500 bytes. The data of 1510 bytes, therefore, must be split between two frames. The standard dictates that the first frame must carry the maximum possible number of bytes (1500); the second frame then needs to carry only 10 bytes of data (it requires padding). The following shows the breakdown:

Data size for the first frame: 1500 bytes

Data size for the second frame: 10 bytes (with padding)

QUESTION 3

What is the ratio of useful data to the entire packet for the smallest Ethernet frame? What is the ratio for the largest frame?

ANSWER:

The smallest Ethernet frame is 64 bytes and carries 46 bytes of data (and possible padding).

The largest Ethernet frame is 1518 bytes and carries 1500 bytes of data. The ratio is (data size) / (frame size) in percent.

We can then answer the question as follows:

Smallest Frame: Frame size = 64 & Data size = 46 Ratio $\leq 71.875\%$

Largest Frame: Frame size = 1518 & Data size = 1500 Ratio = 98.814%

QUESTION 4

The data rate of 10Base5 is 10 Mbps. How long does it take to create the smallest frame? Show the calculation.

ANSWER:

The smallest frame is 64 bytes or 512 bits. With a data rate of 10 Mbps, we have

$$\text{Transmission Delay} = \frac{\text{Message Size}}{\text{Bandwidth}}$$

$$\text{Transmission Delay} = \frac{512 \text{ bits}}{10 \text{ Mbps}}$$

$$\text{Transmission Delay} = \frac{512 \text{ bits}}{10 \times 1000 \times 1000 \text{ bps}}$$

$$\text{Transmission Delay} = 51.2 \mu\text{s}$$

HOMEWORK!

Choose the correct pair from the following <address-size> pairs.

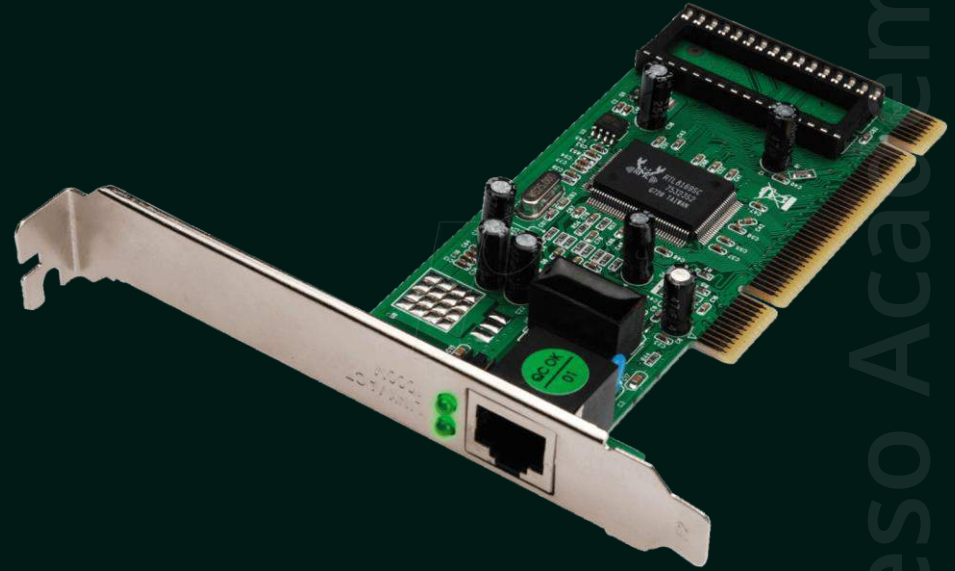
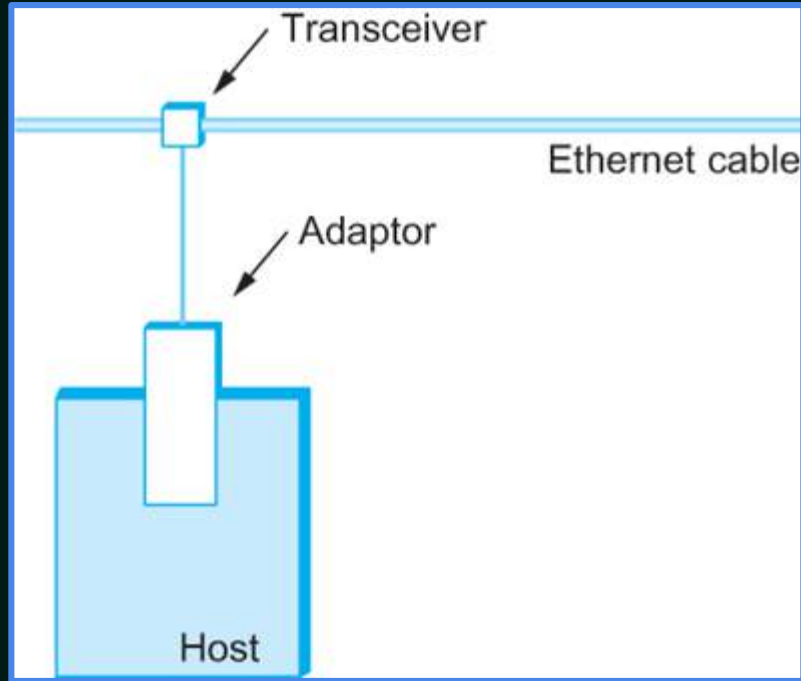
IPv4 – 32 bits	IPv4 – 32 bits	IPv4 – 32 bits	IPv4 – 32 bits
IPv6 – 128 bits	IPv6 – 128 bits	IPv6 – 128 bits	IPv6 – 128 bits
MAC – 40 bits	MAC – 48 bits	MAC – 48 bits	MAC – 46 bits
Port – 16 bits	Port – 32 bits	Port – 16 bits	Port – 16 bits
A	B	C	D

OUTCOMES

Upon the completion of this session, the learner will be able to

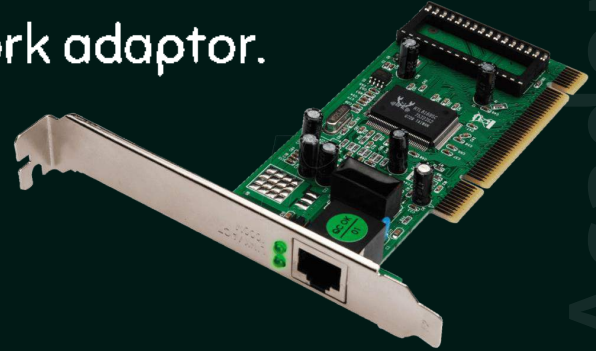
- ★ Understand the Ethernet Transmitter Algorithm.
- ★ Understand Runt Frames.
- ★ Know about Exponential backoff.

ETHERNET ADAPTOR



ACCESS PROTOCOL FOR ETHERNET

The algorithm is commonly called Ethernet's Media Access Control (MAC) which is implemented in Hardware on the network adaptor.



Access Method of Ethernet: CSMA/CD

Encoding method: Manchester Encoding Technique for converting data bits into signals.

ETHERNET TRANSMITTER ALGORITHM

- ★ When the adaptor has a frame to send and the line is idle, it transmits the frame immediately.
- ★ The upper bound of 1500 bytes in the message means that the adaptor can occupy the line for a fixed length of time.
- ★ When the adaptor has a frame to send and the line is busy, it waits for the line to go idle and then transmits immediately.
- ★ The Ethernet is said to be CSMA 1-persistent protocol because an adaptor with a frame to send transmits with probability 1 whenever a busy line goes idle.

ETHERNET TRANSMITTER ALGORITHM

- ★ Since there is no centralized control it is possible for two (or more) adaptors to begin transmitting at the same time, either because both found the line to be idle, or, both had been waiting for a busy line to become idle.
- ★ When this happens, the two (or more) frames are said to be collide on the network.

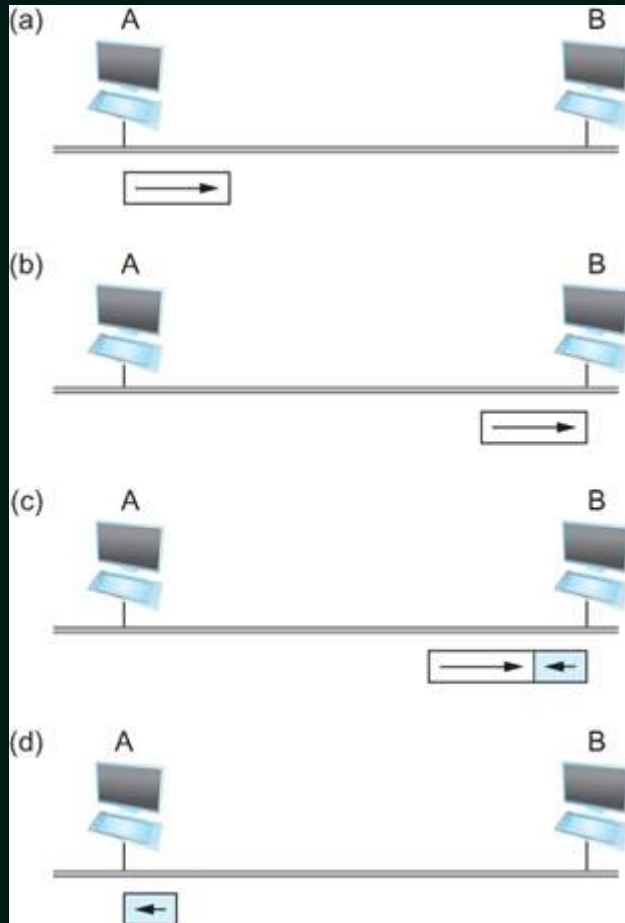
ETHERNET TRANSMITTER ALGORITHM

- ★ Since Ethernet supports collision detection, each sender is able to determine that a collision is in progress.
- ★ At the moment an adaptor detects that its frame is colliding with another, it first makes sure to transmit a 32-bit jamming sequence and then stops transmission.
- ★ Thus, a transmitter will minimally send 96 bits in the case of collision
64-bit preamble + 32-bit jamming sequence.

RUNT FRAMES

- ★ A runt frame is an Ethernet frame that is less than the IEEE 802.3's minimum length of 64 bytes.
- ★ Runt frames are most commonly caused by collisions.
- ★ Other possible causes are a malfunctioning network card, buffer underrun, duplex mismatch or software issues.

ETHERNET TRANSMITTER ALGORITHM



Worst-case scenario:

- (a) A sends a frame at time t .
- (b) A's frame arrives at B at time $t + d$.
- (c) B begins transmitting at time $t + d$ and collides with A's frame.
- (d) B's runt (32-bit) frame arrives at A at time $t + 2d$.

EXPONENTIAL BACKOFF

- ★ Once an adaptor has detected a collision, and stopped its transmission, it waits a certain amount of time and tries again.
- ★ Each time the adaptor tries to transmit but fails, it doubles the amount of time it waits before trying again.
- ★ This strategy of doubling the delay interval between each retransmission attempt is known as Exponential Backoff.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the pros of Ethernet.
- ★ Know the cons of Ethernet.
- ★ Know various categories of Ethernet cables.

ETHERNET – PROS

- ★ Ethernet is the most widely used wired LAN technology.
- ★ Ethernet is relatively inexpensive.
- ★ In Ethernet, all the node have the same privileges. It does not follow client-server architecture.
- ★ Maintenance and administration are simple.
- ★ The cable used to connect systems in ethernet is robust to noise.
- ★ As it is robust to the noise, the quality of the data transfer does not degrade. The data transfer quality is good.
- ★ With latest versions such as gigabit ethernet, the transfer speeds in Gbps have become possible.

ETHERNET – CONS

- ★ Under heavy loads, too much of the network's capacity is wasted by collisions.
- ★ It does not hold good for real-time applications and interactive applications.
- ★ As the network cannot set priority for the packets, it is not suitable for a client-server architecture.
- ★ For interactive applications, dummy data have to be fed to make the frame size 46B which is mandatory.
- ★ After receiving a packet, the receiver doesn't send any acknowledgement.

ETHERNET CATS

CATEGORY	SHIELDING	MAX TRANSMISSION SPEED (AT 100 METERS)
Cat 3	Unshielded	10 Mbps
Cat 5	Unshielded	10/100 Mbps
Cat 5e	Unshielded	1000 Mbps / 1 Gbps
Cat 6	Shielded or Unshielded	1000 Mbps / 1 Gbps
Cat 6a	Shielded	10000 Mbps / 10 Gbps
Cat 7	Shielded	10000 Mbps / 10 Gbps
Cat 8	Shielded	Up to 40 Gbps

QUESTION

In an Ethernet local area network, which one of the following statements is TRUE? [GATE CS 2016]

- (A) A station stops to sense the channel once it starts transmitting a frame. ✖
- (B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size. ✖
- (C) A station continues to transmit the packet even after the collision is detected. ✖
- (D) The exponential backoff mechanism reduces the probability of collision on retransmissions. ✔

QUESTION

Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

[GATE CS 2013]

- (A) 1
- (B) 2
- (C) 2.5
- (D) 5

SOLUTION

Given Data:

Rate of transmission = 500 Mbps

Size = 10^4 bits

Signal speed = 2×10^5 km/sec

Solution:

Transmission time $\geq$ round trip time of 1 bit

Transmission time $\geq 2 \times$ propagation time

SOLUTION

Transmission time $\geq 2 \times$ propagation time

$$\text{Transmission time} = \frac{\text{Message Size}}{\text{Bandwidth}}$$

$$\text{Propagation time} = \frac{\text{Length}}{\text{Propagation Speed}}$$

$$\frac{\text{Message Size}}{\text{Bandwidth}} = \frac{2 \times \text{Length}}{\text{Propagation Speed}}$$

$$\frac{10^4 \text{ bits}}{500 \text{ Mbps}} \geq \frac{2 \times \text{Length}}{2 \times 10^5 \text{ Km/sec}}$$

SOLUTION

$$\frac{10^4 \text{ bits}}{500 \text{ Mbps}} \geq \frac{2 \times L}{2 \times 10^5 \text{ Km/sec}}$$

$$\frac{10^4 \text{ bits}}{500 \text{ Mbps}} \geq \frac{L}{10^5 \text{ Km/sec}}$$

$$\frac{10^4 \text{ bits} \times 10^5 \text{ Km/sec}}{500 \text{ Mbps}} \geq L$$

$$\frac{10^4 \text{ bits} \times 10^5 \text{ Km/sec}}{500 \text{ Mbps}} \geq L$$

$$\frac{10^4 \text{ bits} \times 10^5 \text{ Km/sec}}{500 \times 10^6 \text{ bps}} \geq L$$

$$\frac{10^4 \text{ bits} \times 10^5 \text{ Km/sec}}{500 \times 10^6 \text{ bps}} \geq L$$

$$\frac{10^4 \text{ Km/sec}}{500 \times 10} \geq L$$

$$2 \text{ Km} \geq L$$

QUESTION

Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

[GATE CS 2013]

- (A) 1
- (B) 2 ✓
- (C) 2.5
- (D) 5

QUESTION

A network with CSMA/CD protocol in the MAC layer is running at 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be

[GATE CS 2005]

- (A) 10000 bits
- (B) 10000 bytes
- (C) 5000 bits
- (D) 5000 bytes

SOLUTION

Given Data:

Rate of transmission = 1 Gbps

Length = 1 Km

Signal speed = 2×10^8 m/sec

Solution:

Transmission time $\geq$ round trip time of 1 bit

Transmission time $\geq 2 \times$ propagation time

SOLUTION

Transmission time $\geq 2 \times$ propagation time

$$\text{Transmission time} = \frac{\text{Message Size}}{\text{Bandwidth}}$$

$$\text{Propagation time} = \frac{\text{Length}}{\text{Propagation Speed}}$$

$$\frac{\text{Message Size}}{\text{Bandwidth}} \geq \frac{2 \times \text{Length}}{\text{Propagation Speed}}$$

$$\frac{\text{Message Size}}{1 \text{ Gbps}} \geq \frac{2 \times 1 \text{ Km}}{2 \times 10^8 \text{ m/sec}}$$

SOLUTION

$$\text{Message Size} \geq \frac{2 \times 1 \text{ Km} \times 10^9 \text{ bps}}{2 \times 10^8 \text{ m/sec}}$$

$$\text{Message Size} \geq \frac{2 \times 1 \times 10^3 \text{ m} \times 10^9 \text{ bps}}{2 \times 10^8 \text{ m/sec}}$$

$$\text{Message Size} \geq \frac{2 \times 1 \times 10^3 \times 10^9 \text{ bits}}{2 \times 10^8}$$

$$\text{Message Size} \geq 10^4 \text{ bits}$$

$$\text{Message Size} \geq 10000 \text{ bits}$$

QUESTION

A network with CSMA/CD protocol in the MAC layer is running at 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be

[GATE CS 2005]

(A) 10000 bits ✓

(B) 10000 bytes

(C) 5000 bits

(D) 5000 bytes

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know about IEEE 802.11 Wi-Fi.
- ★ Know the access method of Wi-Fi.
- ★ See Wi-Fi adaptors.
- ★ Understand the modes of Wi-Fi.
- ★ Know different Wi-Fi protocols.

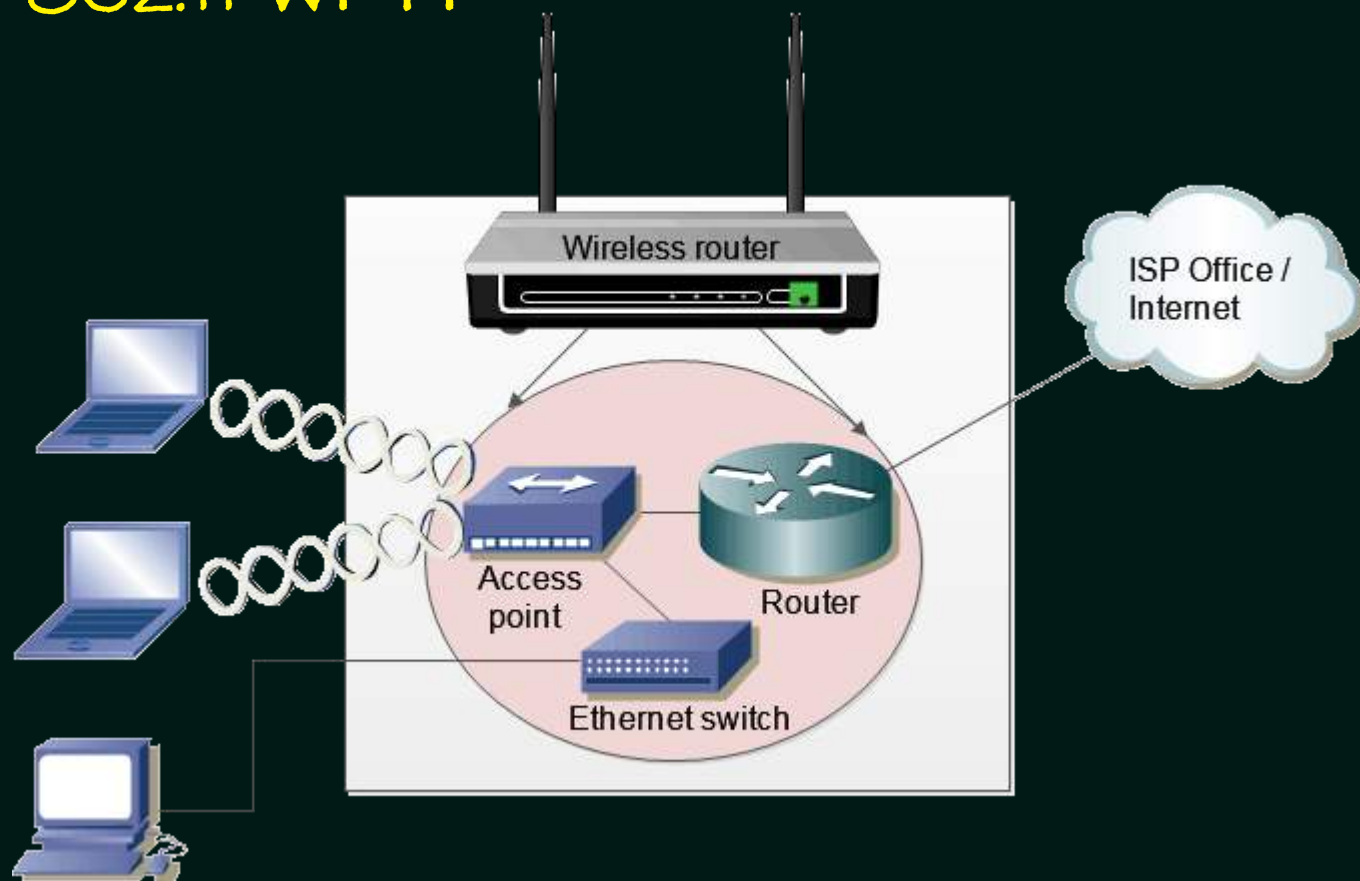
ANALOGY



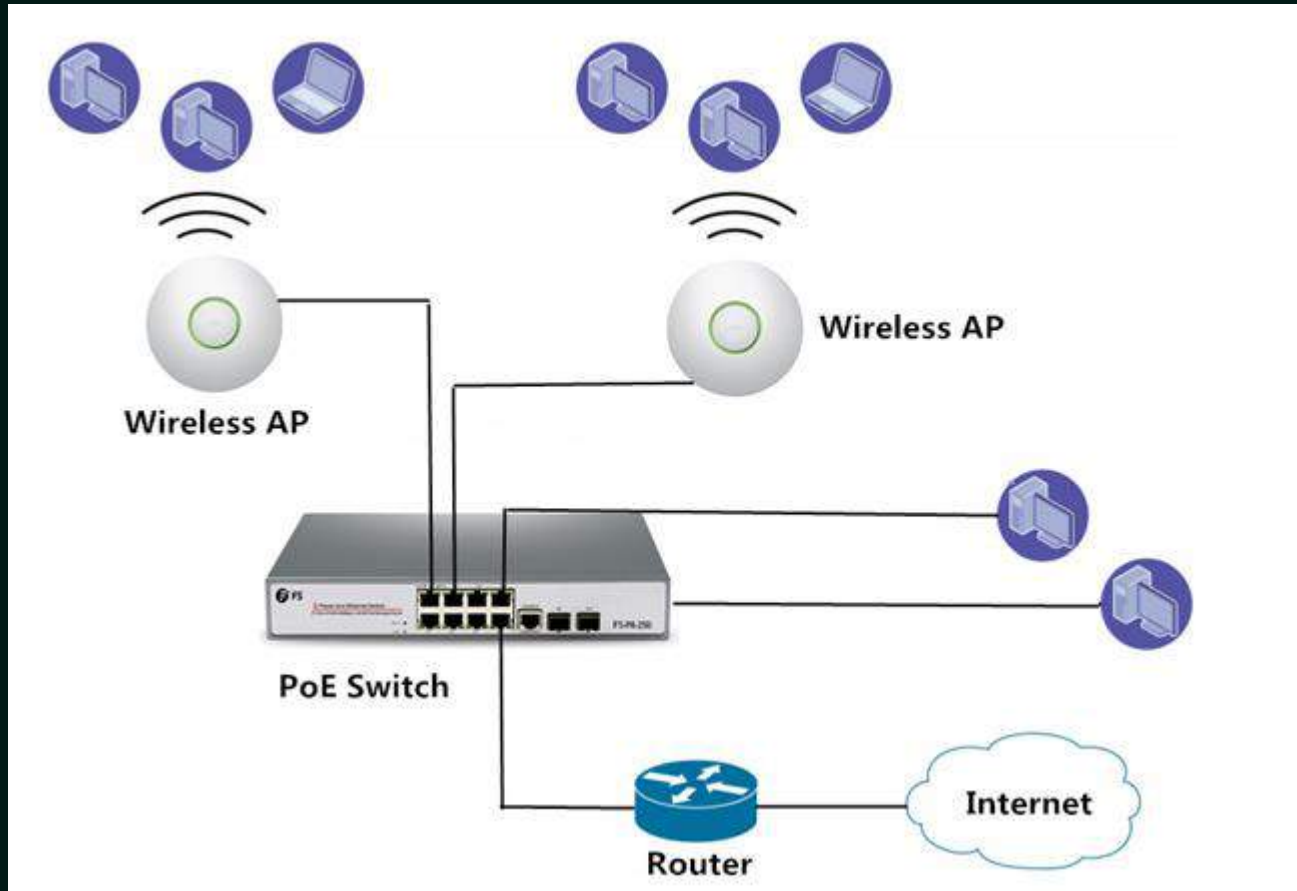
IEEE 802.11 Wi-Fi

- ★ Also known as Wireless Fidelity (Wi-Fi).
- ★ Like its Ethernet and token ring siblings, 802.11 is designed for use in a limited geographical area (homes, office buildings, campuses).
- ★ Primary challenge is to mediate access to a shared communication medium – in this case, signals propagating through space.
- ★ 802.11 supports additional features:
 - power management and
 - security mechanisms

IEEE 802.11 Wi-Fi



IEEE 802.11 Wi-Fi



IEEE 802.11 Wi-Fi

- ★ 802.11 uses 5 GHz Radio Band (High Frequency) which has 23 overlapping channels rather than the 2.4 GHz frequency band which has only three non-overlapping channels.

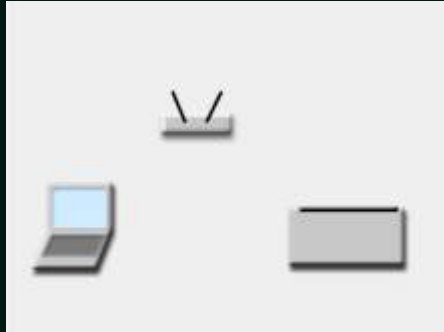
Access Method of IEEE 802.11 Wi-Fi: CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

WI-FI ADAPTOR



MODES OF Wi-Fi

★ Infrastructure mode



★ Ad hoc and Wi-Fi Direct

This wireless ad hoc network mode has proven popular with multiplayer handheld game consoles, such as the Nintendo DS, PlayStation Portable, digital cameras, and other consumer electronics devices.

DIFFERENT Wi-Fi PROTOCOLS

Protocol	Frequency	Channel Width	Maximum data rate (theoretical)
802.11 ax	2.4 or 5GHz	20, 40, 80, 160MHz	2.4 Gbps
802.11 ac wave2	5 GHz	20, 40, 80, 160MHz	1.73 Gbps
802.11 ac wave1	5 GHz	20, 40, 80MHz	866.7 Mbps
802.11n	2.4 or 5 GHz	20, 40MHz	450 Mbps
802.11g	2.4 GHz	20 MHz	54 Mbps
802.11a	5 GHz	20 MHz	54 Mbps
802.11b	2.4 GHz	20 MHz	11 Mbps
Legacy 802.11	2.4 GHz	20 MHz	2 Mbps

b - high rate wifi

n - multiple inputs multiple outputs

e - quality of service

i - security

g - short distance at up to 54 Mbps in 2.4GHz



11be (2023-2024) - Next Generation (Wi-Fi 7)

802.11be, branded as "Wi-Fi 7," operates in the same three bands as Wi-Fi 6E. It is expected to quadruple speed up to 40 Gbps. See Wi-Fi 7.

11ax (2021) - Highest Speed (Wi-Fi 6)

802.11ax, branded as "Wi-Fi 6," operates in the 2.4 GHz and 5 GHz bands with data rates in the multi-gigabit range. More multiuser modes support higher user density. Wi-Fi 6E (Extended) adds the 6 GHz band. See Wi-Fi 6.

11ac (2012) - High Speed (Wi-Fi 5)

802.11ac operates in the 5 GHz band with data rates into the gigabit range (see 802.11ac). Very valuable when multiple users are online at the same time.

11n (2009) - High Speed (Wi-Fi 4)

802.11n uses multiple antennas for speeds up to 450 Mbps. 11n operates in both 2.4 GHz and 5 GHz spectrum bands and is compatible with previous 11b/g and 11a standards (see 802.11n). More than enough speed for the casual user.

11g (2003) - Medium Speed (Wi-Fi 3)

Using orthogonal FDM (OFDM) transmission, 802.11g increased speed in the 2.4 GHz band to 54 Mbps. Both 11b and 11g are compatible, and equipment is often designated as 802.11b/g. See OFDM.

11a (1999) - Medium Speed (Wi-Fi 2)

Using orthogonal FDM (OFDM), 802.11a transmits up to 54 Mbps. It uses the 5 GHz band and is not compatible with 11b.

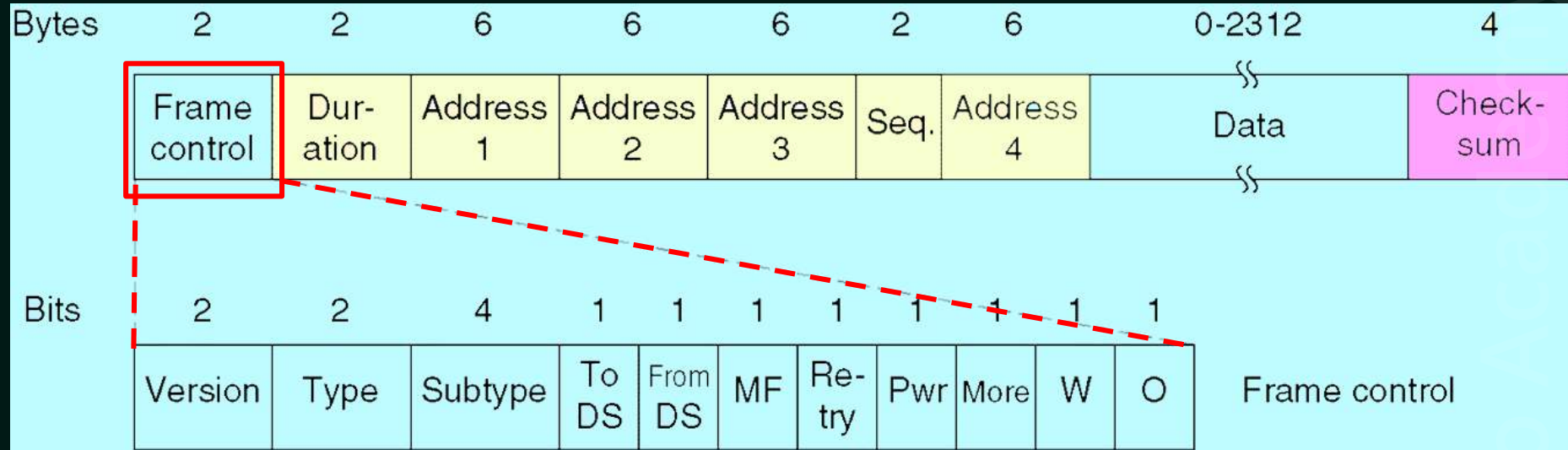
11b (1999) - Slow Speed (Wi-Fi 1)

Using DSSS and the 2.4 GHz band, 802.11b boosted speed to 11 Mbps while retaining the slower DSSS modes to accommodate weak signals. It was the first major wireless local network standard, and many laptops were retrofitted with 11b network adapters. Later, 11b was built into the laptop motherboard.

Original Spec (1997)

The original 802.11 specification included two spread spectrum methods in the unlicensed 2.4 GHz band: 1 Mbps frequency hopping (FHSS) and 1 and 2 Mbps direct sequence (DSSS). It also included an infrared method. Both FHSS and infrared were dropped by the Wi-Fi Alliance, but 1 Mbps DSSS method is still used by access points to advertise themselves (see beaconing).

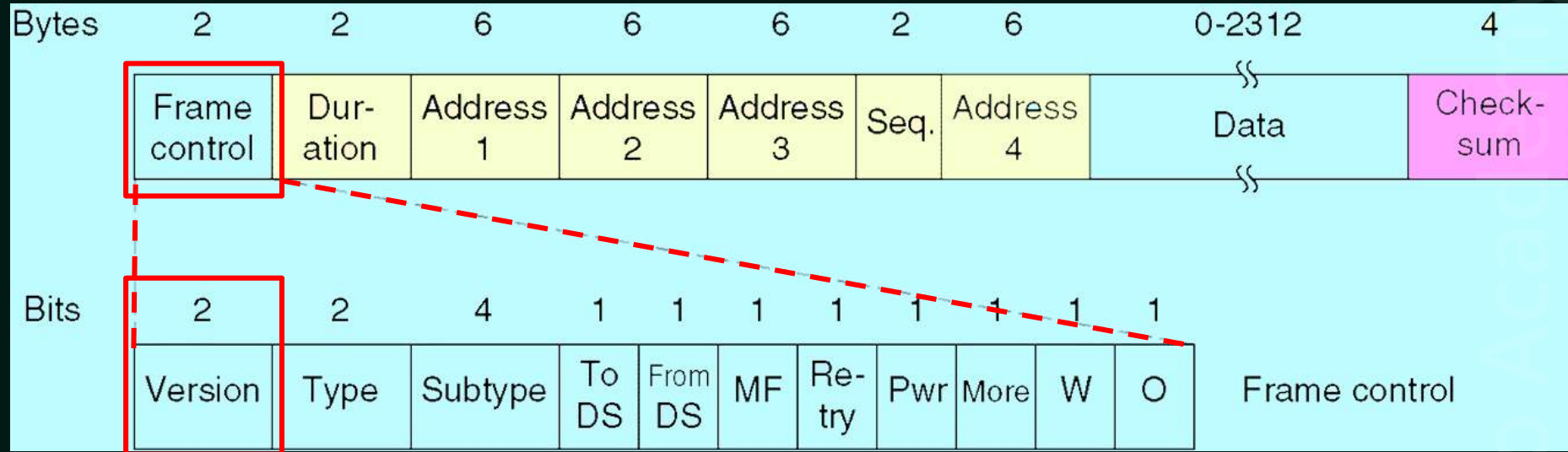
IEEE 802.11 Wi-Fi Frame Format



Frame Control:

- ★ It is a 2 bytes starting field composed of 11 subfields. It contains control information of the frame.
- ★ It has 11 subfields.

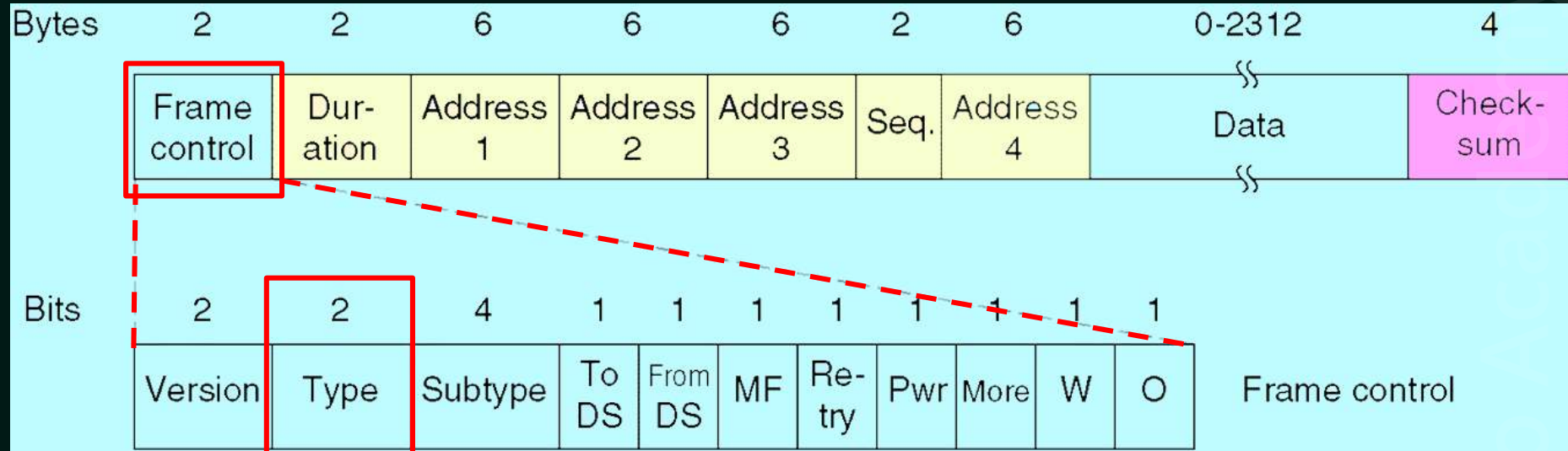
IEEE 802.11 WI-FI FRAME FORMAT



Protocol Version:

- ★ The first sub-field is a two – bit field set to 00. It has been included to allow future versions of IEEE 802.11 to operate simultaneously.

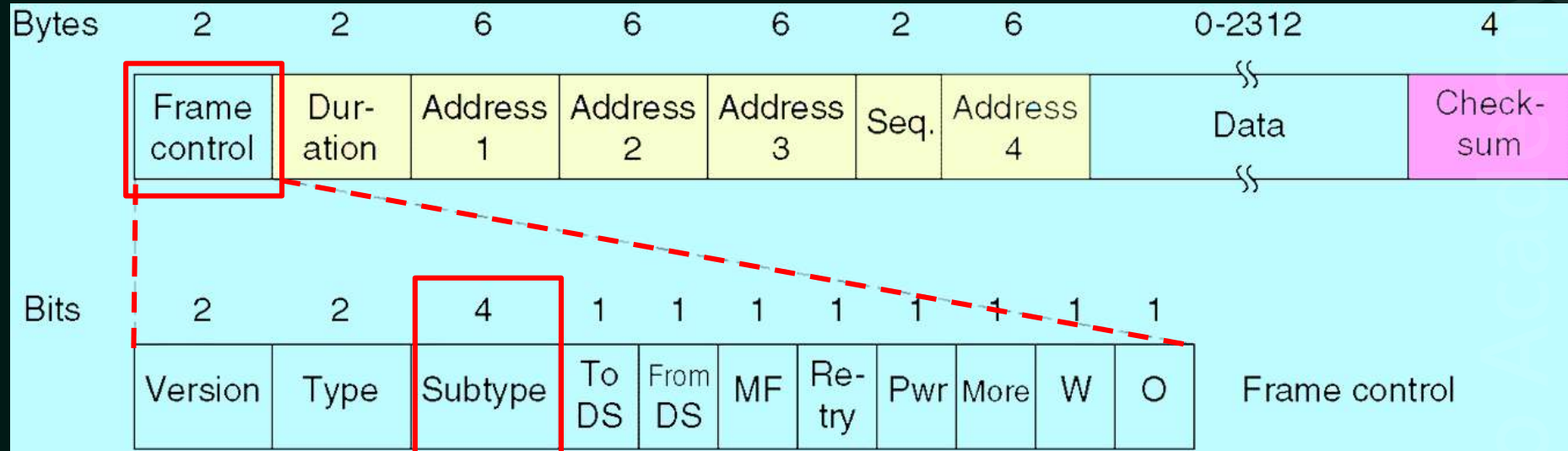
IEEE 802.11 WI-FI FRAME FORMAT



Type:

- ★ It is a two-bit subfield that specifies whether the frame is a data frame, control frame or a management frame.

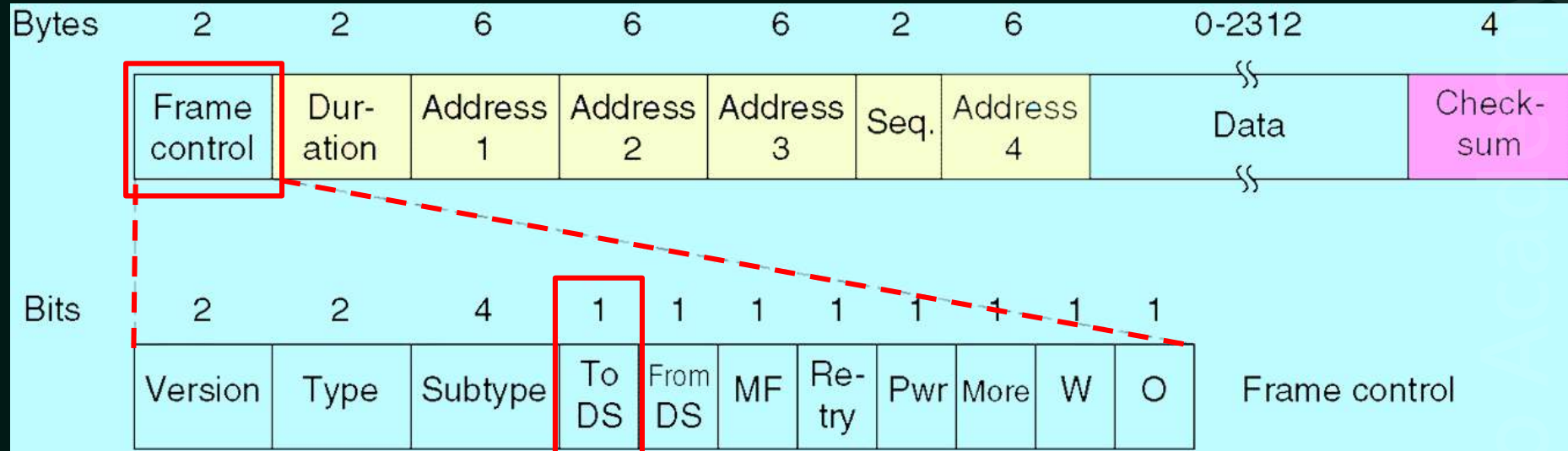
IEEE 802.11 Wi-Fi Frame Format



Subtype:

- ★ It is a four – bit subfield states whether the field is a Request to Send (RTS) or a Clear to Send (CTS) control frame. For a regular data frame, the value is set to 0000.

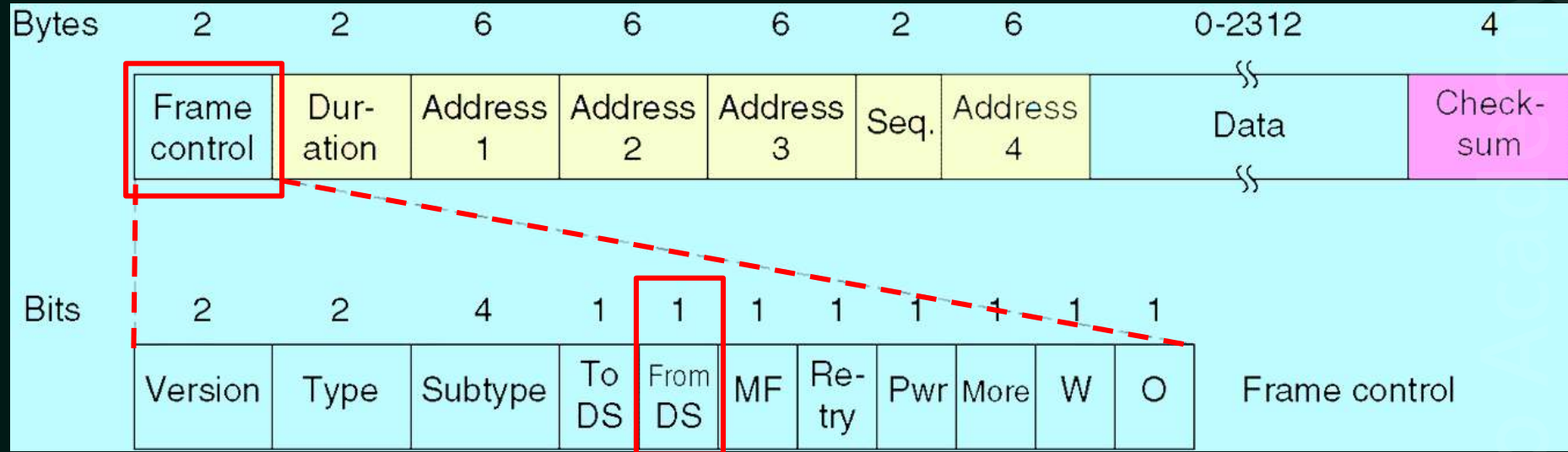
IEEE 802.11 Wi-Fi Frame Format



To DS:

- ★ A single bit subfield indicating whether the frame is going to the access point (AC), which coordinates the communications in centralised wireless systems.

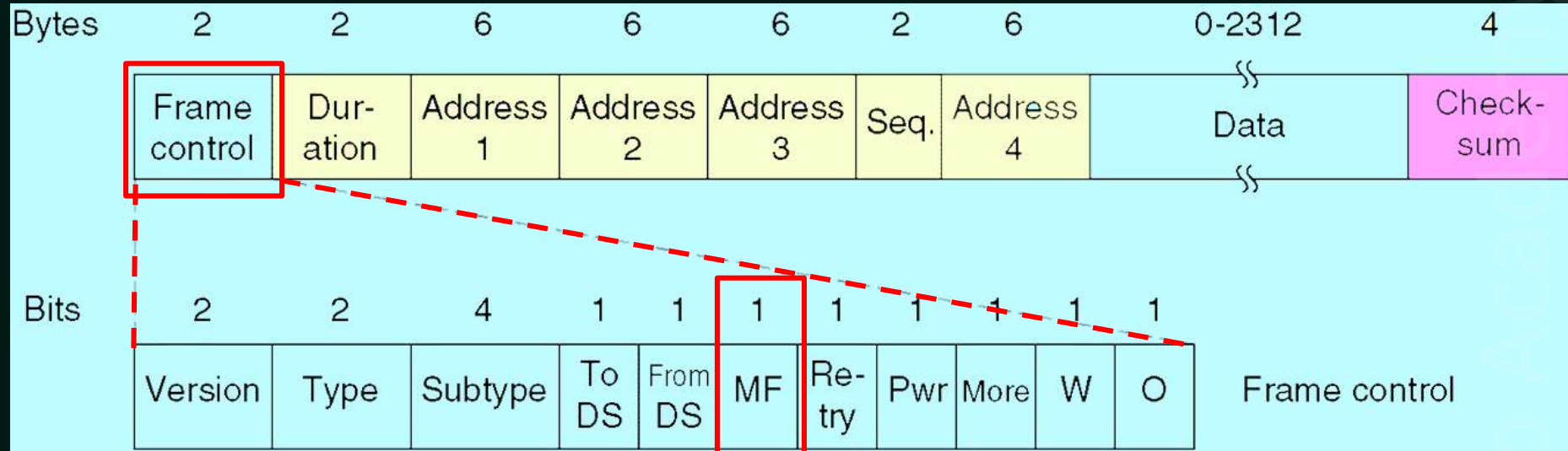
IEEE 802.11 Wi-Fi Frame Format



From DS:

- ★ A single bit subfield indicating whether the frame is coming from the Access point.

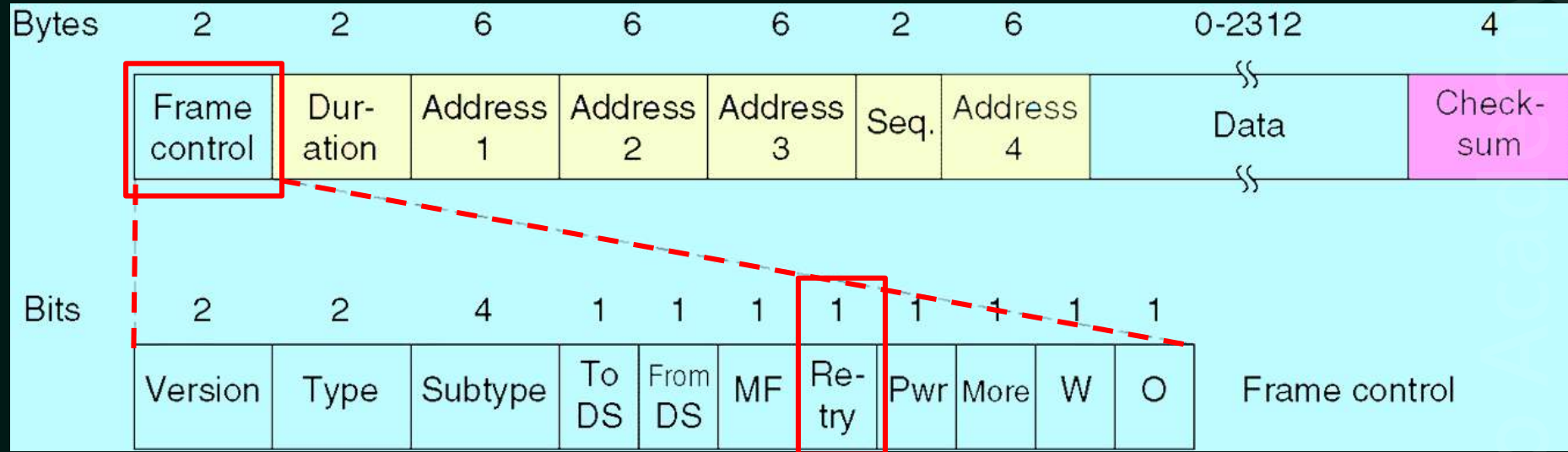
IEEE 802.11 Wi-Fi Frame Format



More Fragments:

- ★ A single bit subfield which when set to 1 indicates that more fragments would follow.

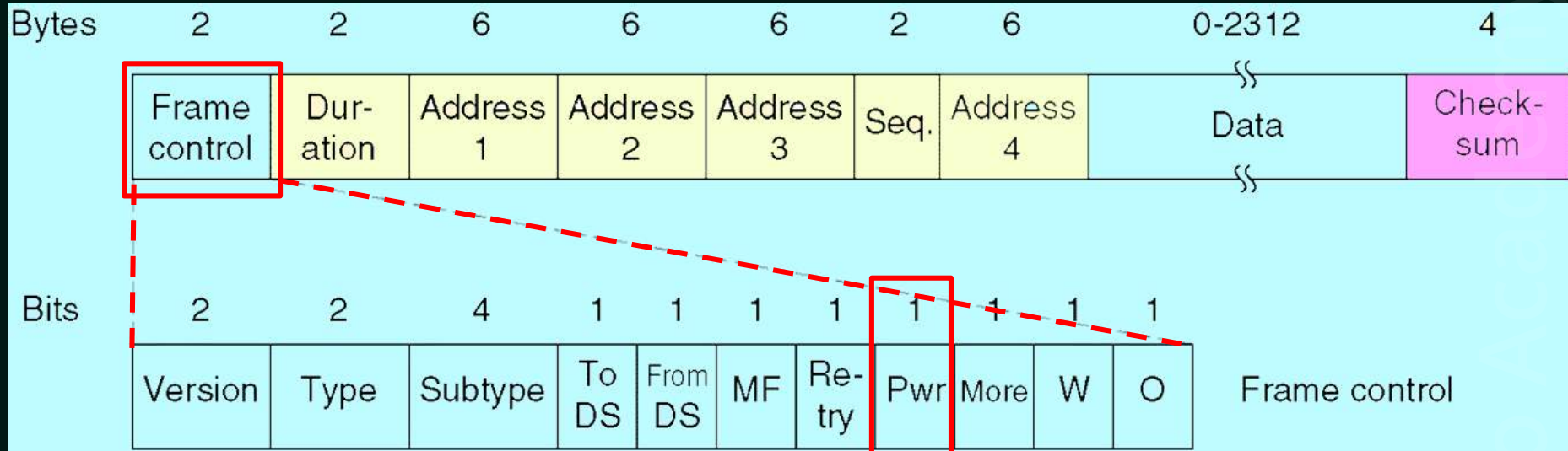
IEEE 802.11 Wi-Fi Frame Format



Retry:

- ★ A single bit subfield which when set to 1 specifies a retransmission of a previous frame.

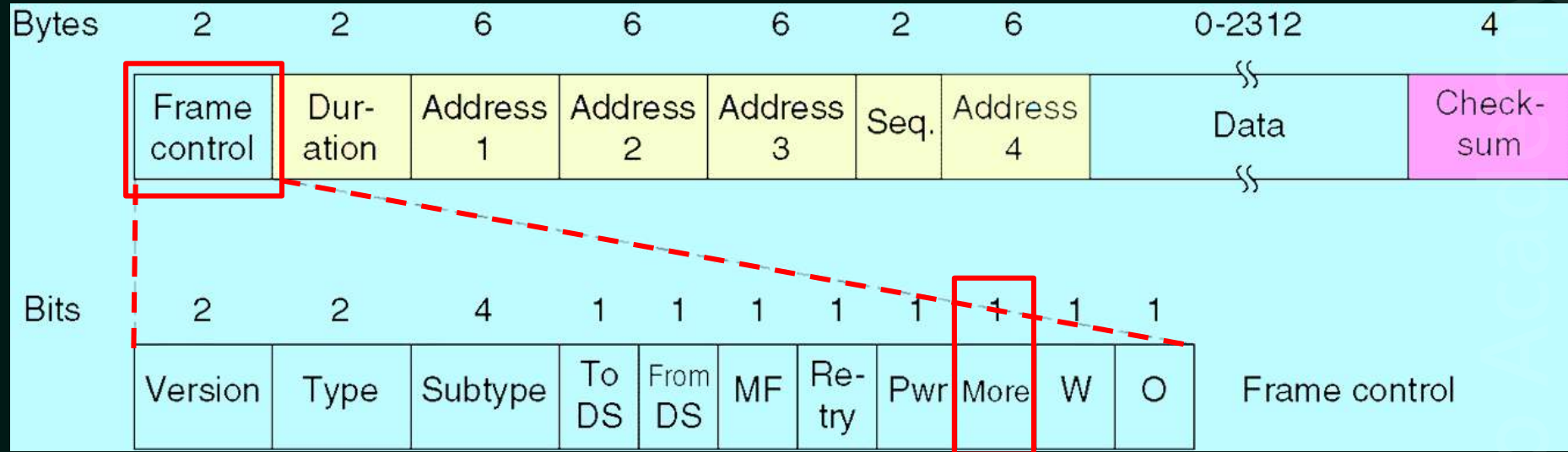
IEEE 802.11 Wi-Fi Frame Format



Power Management:

- ★ A single bit subfield indicating that the sender is adopting power-save mode.

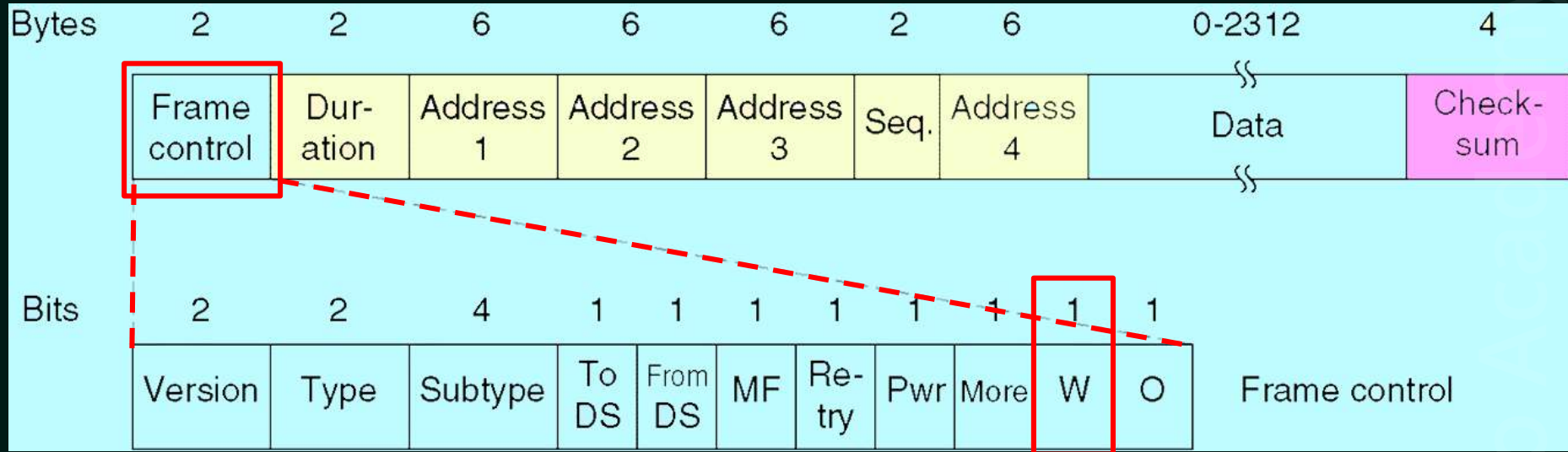
IEEE 802.11 Wi-Fi Frame Format



More Data:

- ★ A single bit subfield showing that sender has further data frames for the receiver.

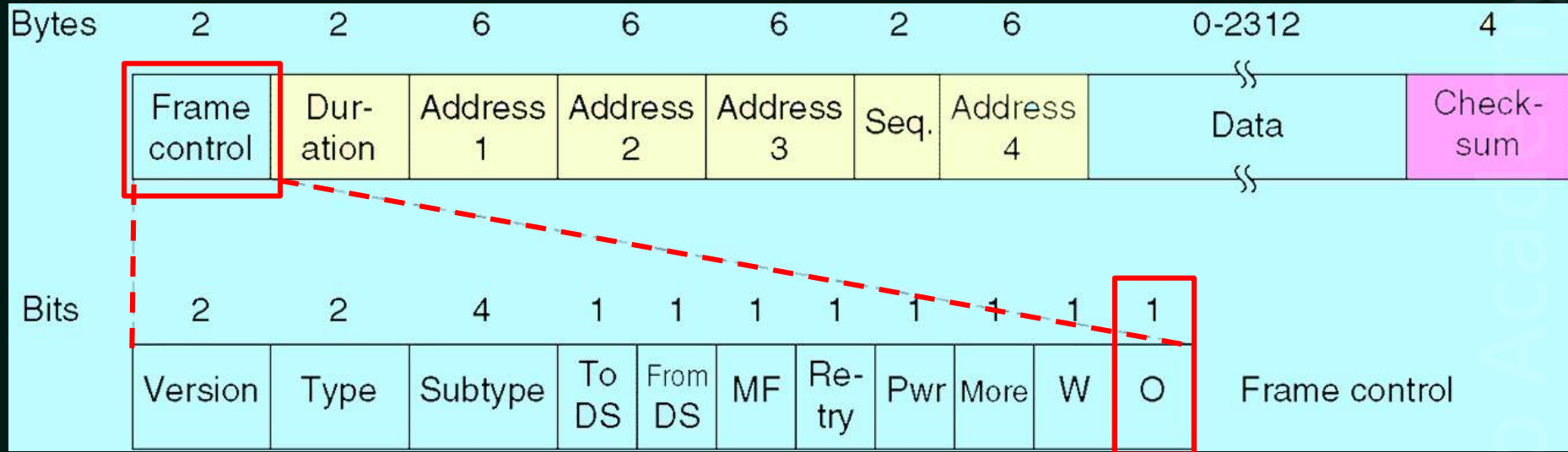
IEEE 802.11 Wi-Fi Frame Format



WEP:

- ★ A single bit subfield indicating that this is an encrypted frame.

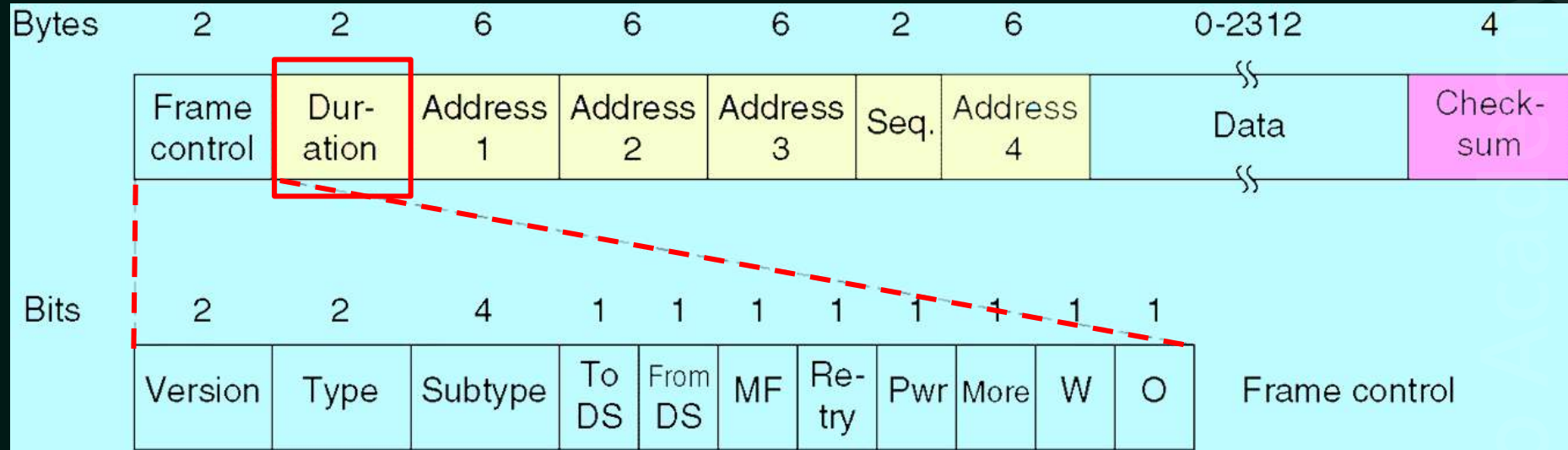
IEEE 802.11 Wi-Fi Frame Format



Order:

- ★ The last subfield, of one – bit, informs the receiver that to the higher layers the frames should be in an ordered sequence.

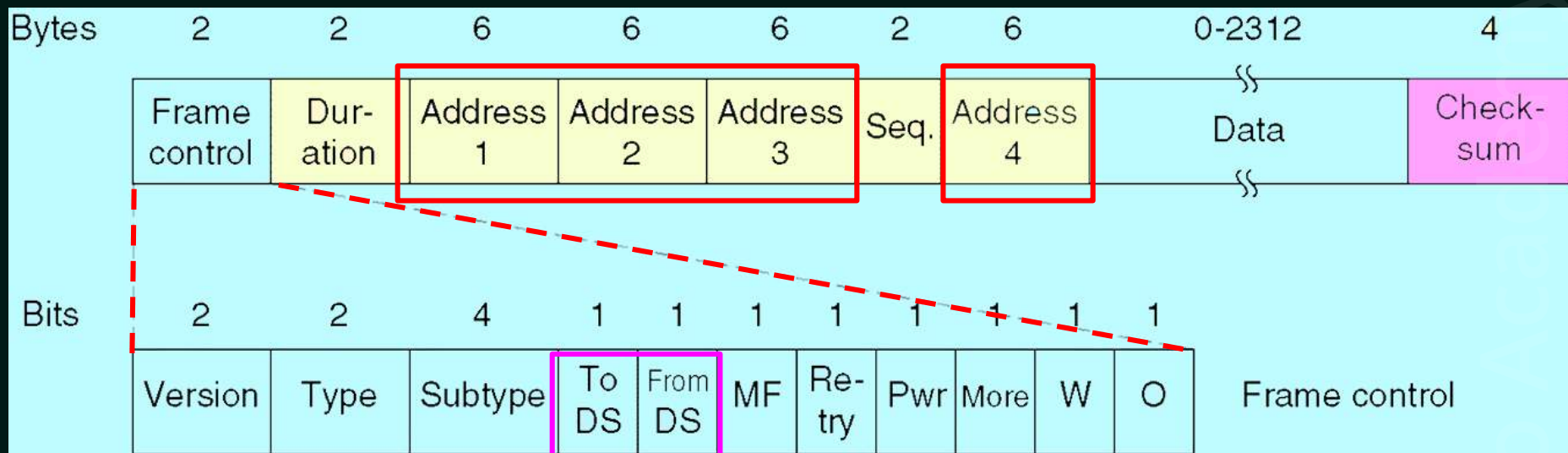
IEEE 802.11 WI-FI FRAME FORMAT



Duration:

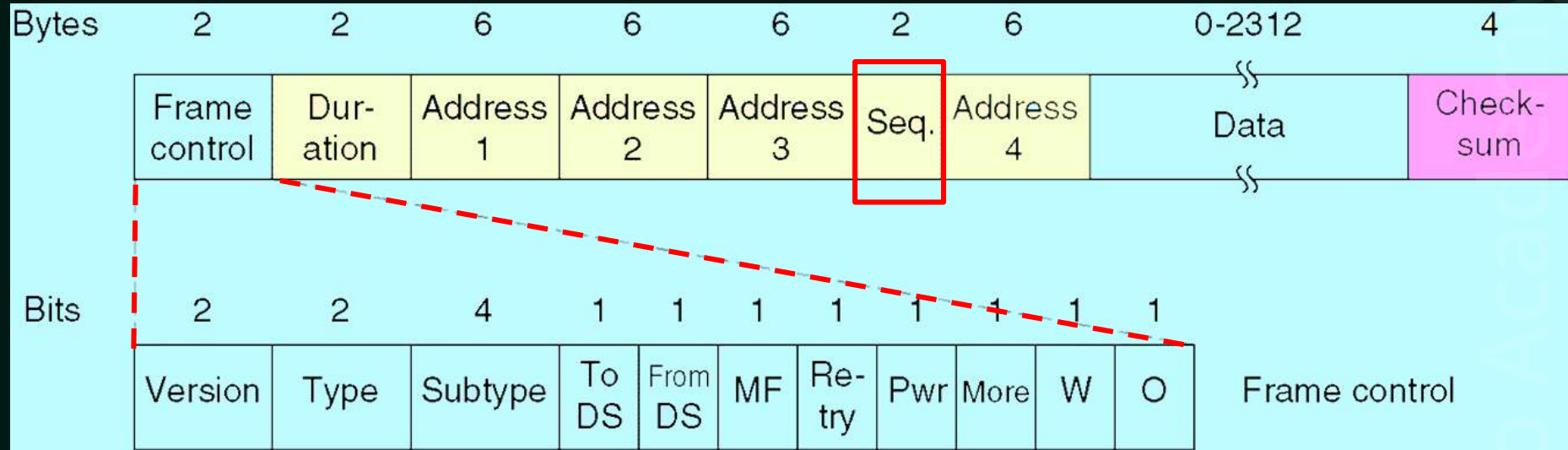
- ★ It is a 2-byte field that specifies the time period for which the frame and its acknowledgement occupy the channel.

IEEE 802.11 WI-FI FRAME FORMAT



To DS	From DS	Address 1	Address 2	Address 3	Address 4
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	SendingAP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	SendingAP	Destination	Source

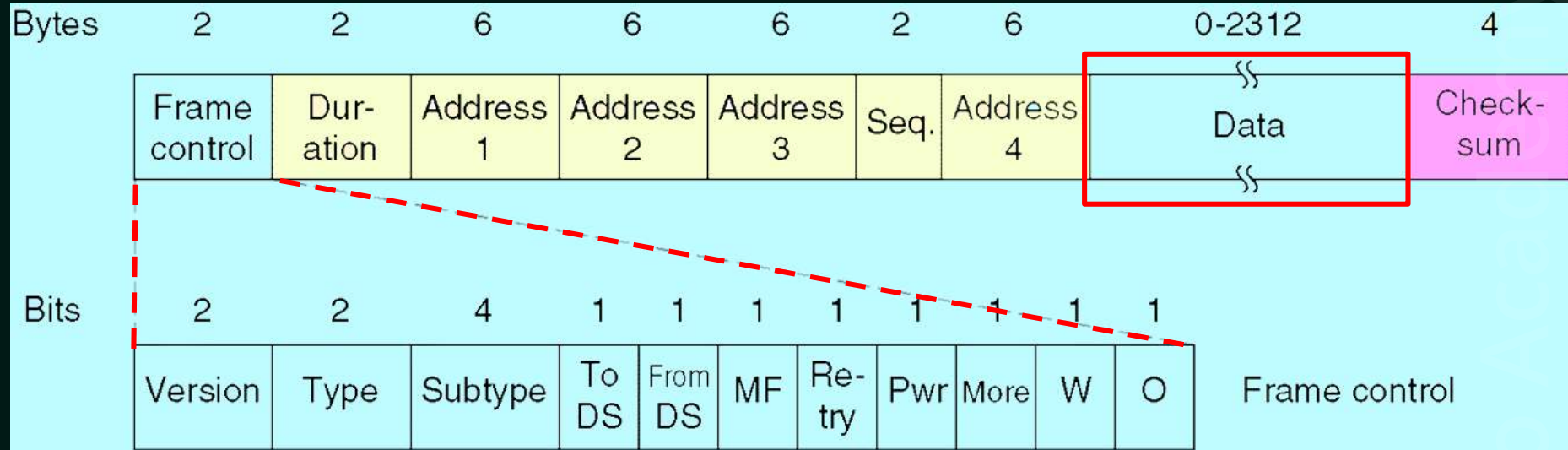
IEEE 802.11 WI-FI FRAME FORMAT



Sequence:

- ★ It is a 2 bytes field that stores the frame numbers. It detects duplicate frames and determines the order of frames for higher layers. Among the 16 bits, the first 4 bits provide identification to the fragment and the rest 12 bits contain the sequence number that increments with each transmission.

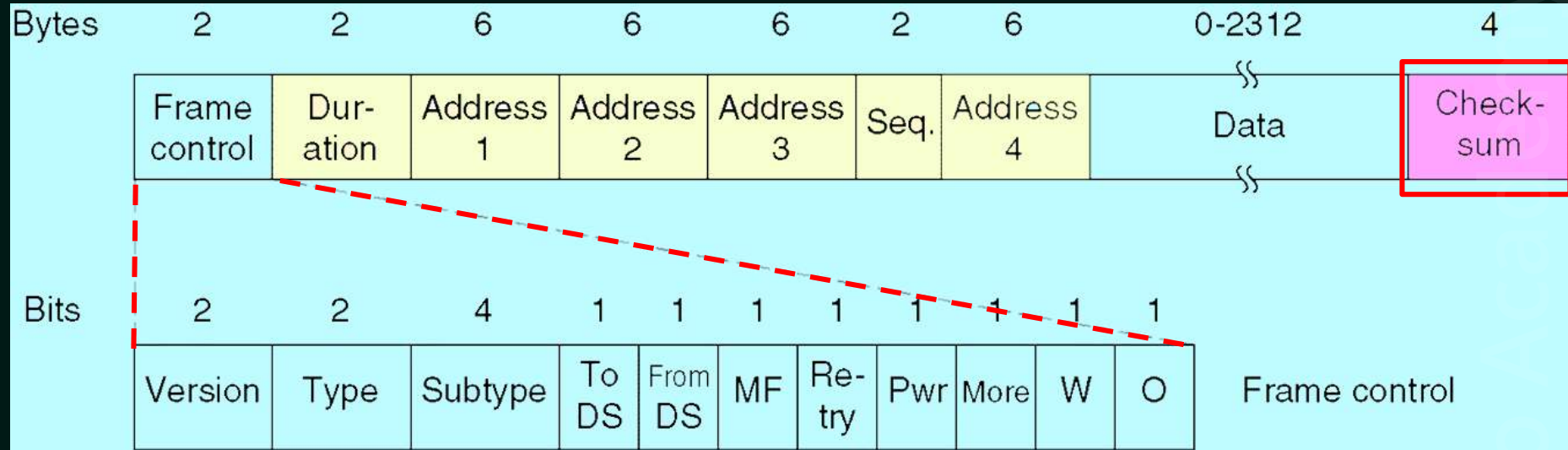
IEEE 802.11 Wi-Fi Frame Format



Data:

- ★ This is a variable sized field that carries the payload from the upper layers. The maximum size of data field is 2312 bytes.

IEEE 802.11 Wi-Fi Frame Format



Checksum:

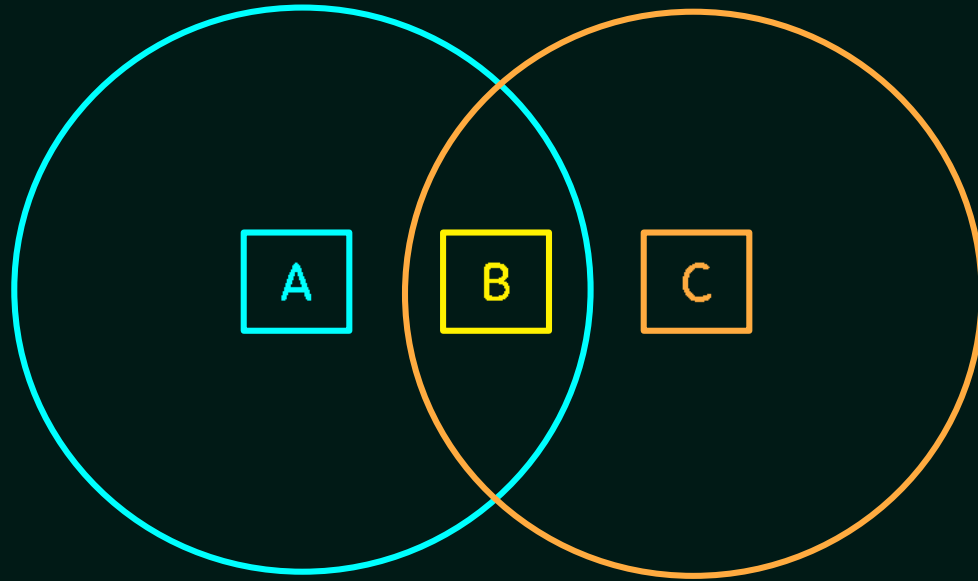
- ★ It is a 4-byte field for error detection purpose.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the hidden terminal problem.
- ★ Know the solution for hidden terminal problem.

HIDDEN TERMINAL PROBLEM

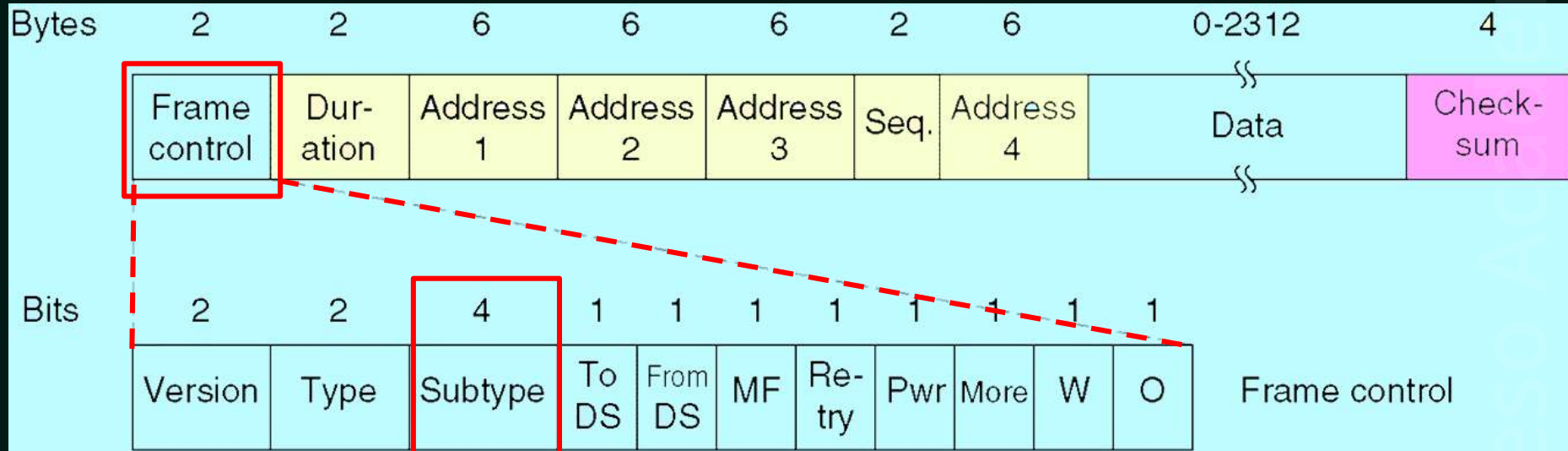


Suppose both A and C want to communicate with B and so they each send it a frame.

- ★ A and C are unaware of each other since their signals do not carry that far.
- ★ These two frames collide with each other at B (But unlike an Ethernet, neither A nor C is aware of this collision).
- ★ A and C are said to be hidden nodes with respect to each other.

HIDDEN TERMINAL PROBLEM – SOLUTION

Multiple Access Collision Avoidance (MACA) Algorithm – RTS and CTS frames



Subtype:

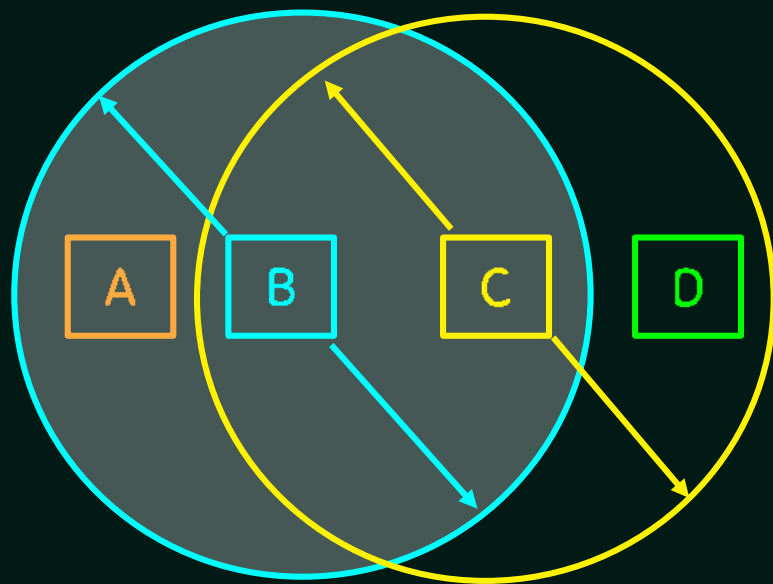
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OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Understand the exposed terminal problem.
- ★ Know the solution for exposed terminal problem.

EXPOSED TERMINAL PROBLEM

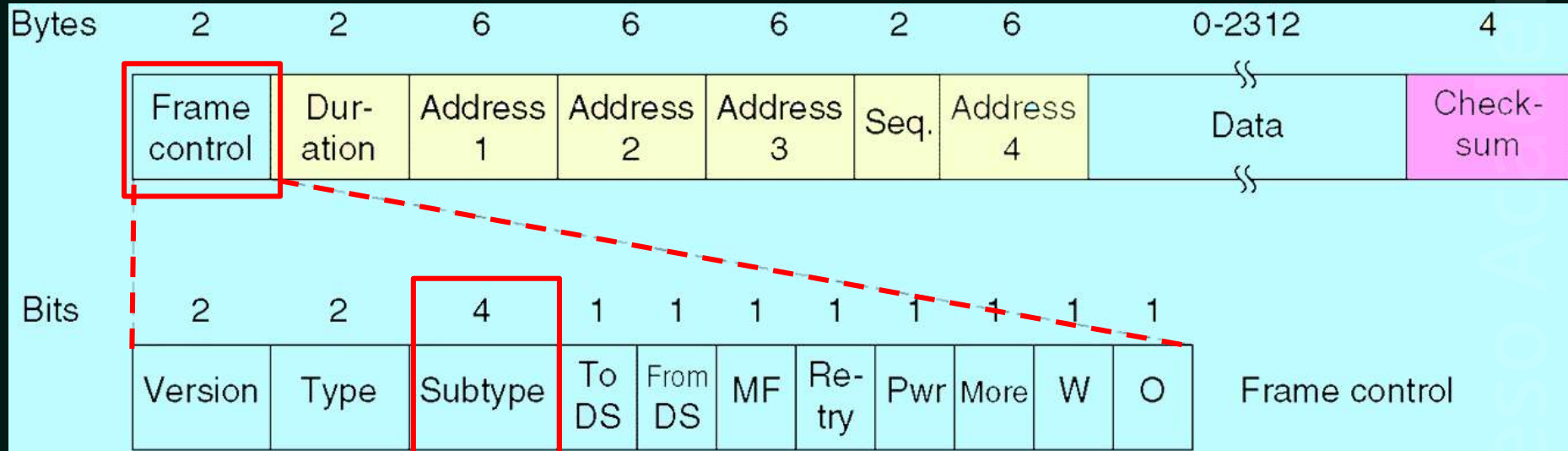


Suppose B is sending to A. Node C is aware of this communication because it hears B's transmission.

- ★ It would be a mistake for C to conclude that it cannot transmit to anyone just because it can hear B's transmission.
- ★ Suppose C wants to transmit to node D.
- ★ This is not a problem since C's transmission to D will not interfere with A's ability to receive from B.

EXPOSED TERMINAL PROBLEM – SOLUTION

Multiple Access Collision Avoidance (MACA) Algorithm – RTS and CTS frames



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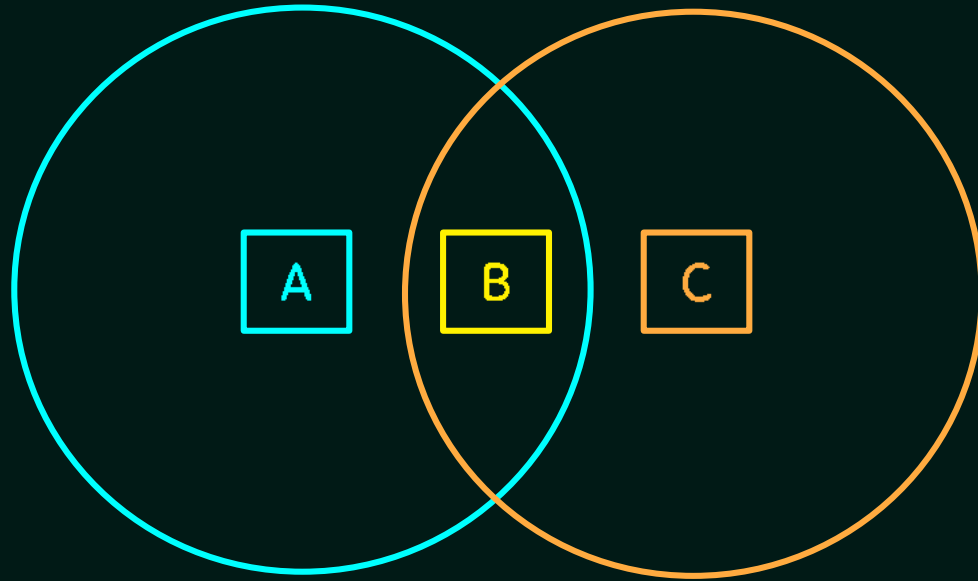
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OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Recall hidden terminal and exposed terminal problem.
- ★ Understand the MACA algorithm.

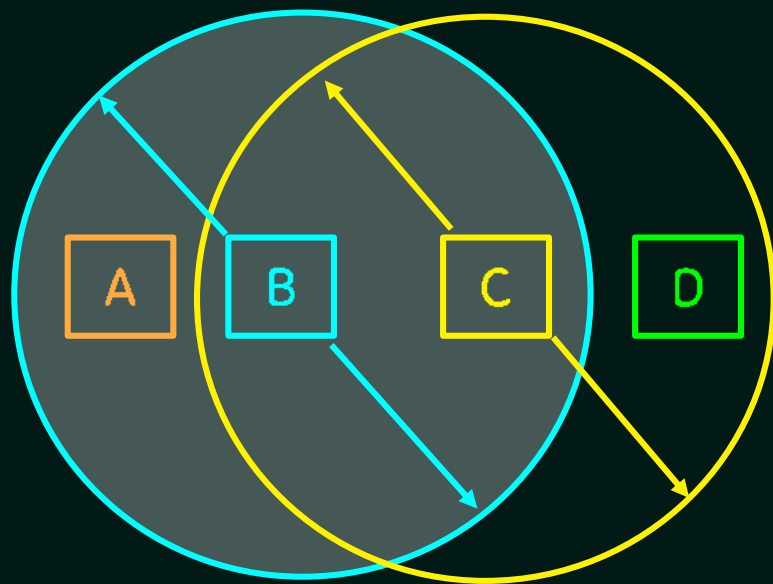
HIDDEN TERMINAL PROBLEM



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Suppose B is sending to A. Node C is aware of this communication because it hears B's transmission.

- ★ It would be a mistake for C to conclude that it cannot transmit to anyone just because it can hear B's transmission.
- ★ Suppose C wants to transmit to node D.
- ★ This is not a problem since C's transmission to D will not interfere with A's ability to receive from B.

IEEE 802.11 – COLLISION AVOIDANCE

- ★ 802.11 addresses these two problems with an algorithm called Multiple Access with Collision Avoidance (MACA).

Key Idea

- ★ Sender and receiver exchange control frames with each other before the sender actually transmits any data.
- ★ This exchange informs all nearby nodes that a transmission is about to begin.
- ★ Sender transmits a Request to Send (RTS) frame to the receiver.
 - The RTS frame includes a field that indicates how long the sender wants to hold the medium. Length of the data frame to be transmitted.
- ★ Receiver replies with a Clear to Send (CTS) frame
 - This frame echoes this length field back to the sender

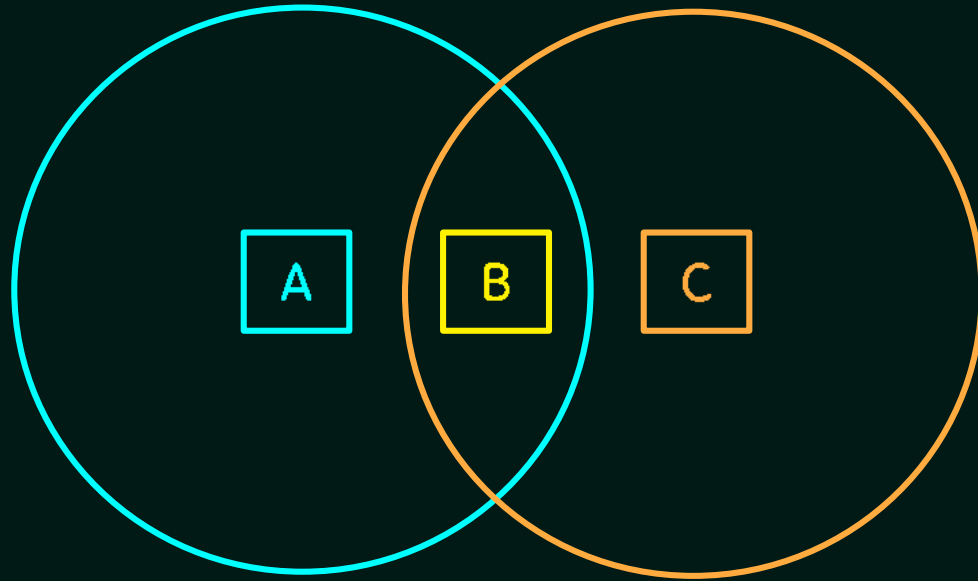
IEEE 802.11 – COLLISION AVOIDANCE

- ★ Any node that sees the CTS frame knows that
 - it is close to the receiver, therefore
 - cannot transmit for the period of time it takes to send a frame of the specified length
- ★ Any node that sees the RTS frame but not the CTS frame
 - is not close enough to the receiver to interfere with it, and
 - so is free to transmit

IEEE 802.11 – COLLISION AVOIDANCE

- ★ The idea of using ACK in MACA is Proposed in **MACAW: MACA for Wireless LANs**.
- ★ Receiver sends an ACK to the sender after successfully receiving a frame.
- ★ All nodes must wait for this ACK before trying to transmit.
- ★ If two or more nodes detect an idle link and try to transmit an RTS frame at the same time
 - ★ Their RTS frame will collide with each other
- ★ **802.11 does not support collision detection**
 - ★ So the senders realize the collision has happened when they do not receive the CTS frame after a period of time
 - ★ In this case, they each wait a random amount of time before trying again.
 - ★ The amount of time a given node delays is defined by the same **exponential backoff algorithm** used on the Ethernet.

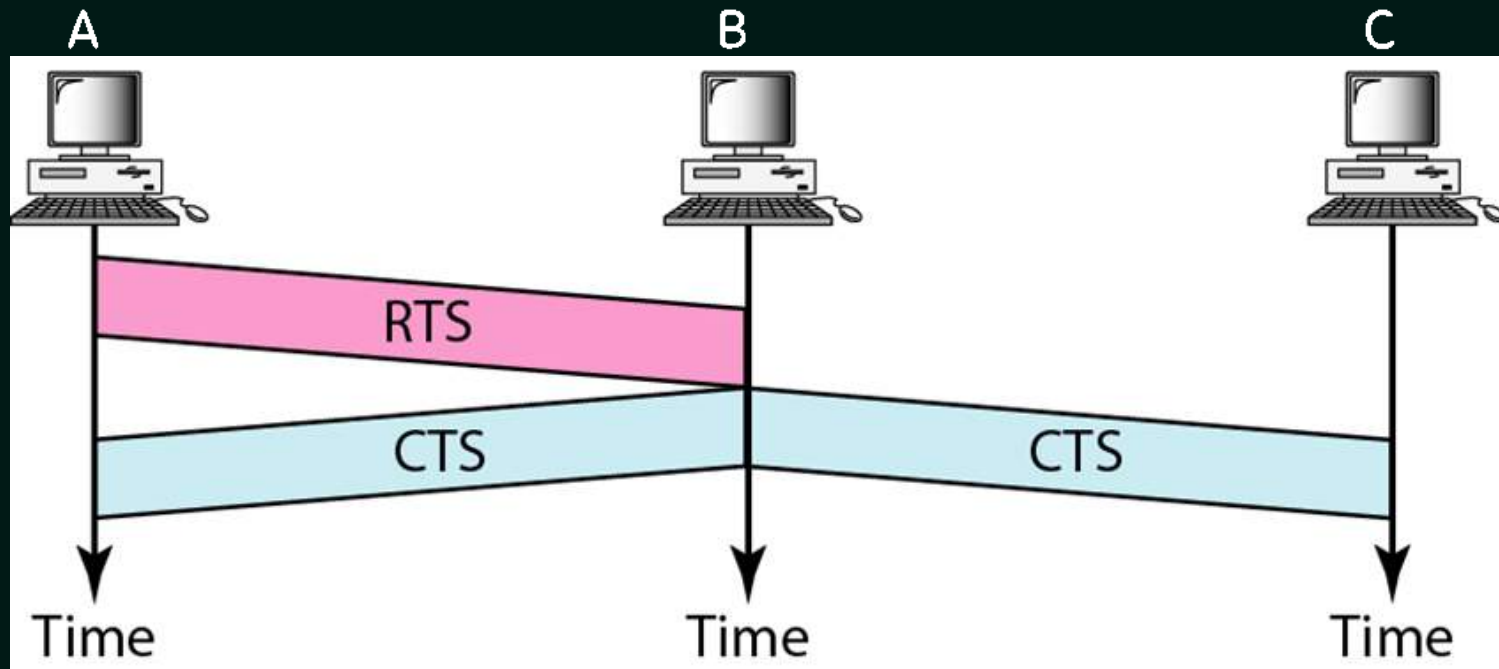
HIDDEN TERMINAL PROBLEM



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USE OF RTS/CTS HANDSHAKING



Station C doesn't hear RTS from A, but it does hear CTS from B, so it knows something is up.

QUESTION

For the IEEE 802.11 MAC protocol for wireless communication, which of the following statements is/are TRUE?

- I. At least three non-overlapping channels are available for transmissions.
- II. The RTS-CTS mechanism is used for collision detection.
- III. Unicast frames are ACKed.

[GATE CS 2016]

- (A) All I, II, and III
- (B) I and III only ✓
- (C) II and III only
- (D) II only

HOMEWORK

Which of the following statements is TRUE about CSMA/CD.

- (A) IEEE 802.11 wireless LAN runs CSMA/CD protocol
- (B) Ethernet is not based on CSMA/CD protocol
- (C) CSMA/CD is not suitable for a high propagation delay network like satellite network
- (D) There is no contention in a CSMA/CD network

[GATE CS 2005]

QUESTION

Which of the following statements is TRUE about CSMA/CD.

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[GATE CS 2005]

ANSWER

Which of the following statements is TRUE about CSMA/CD.

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- (C) CSMA/CD is not suitable for a high propagation delay network like satellite network.
- (D) There is no contention in a CSMA/CD network.

[GATE CS 2016]

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the basics of bluetooth.
- ★ Understand piconet and scatternet.
- ★ See the bluetooth adaptors.
- ★ Understand the pros and cons of bluetooth.
- ★ Know about bluesnarfing.



Neso Academy

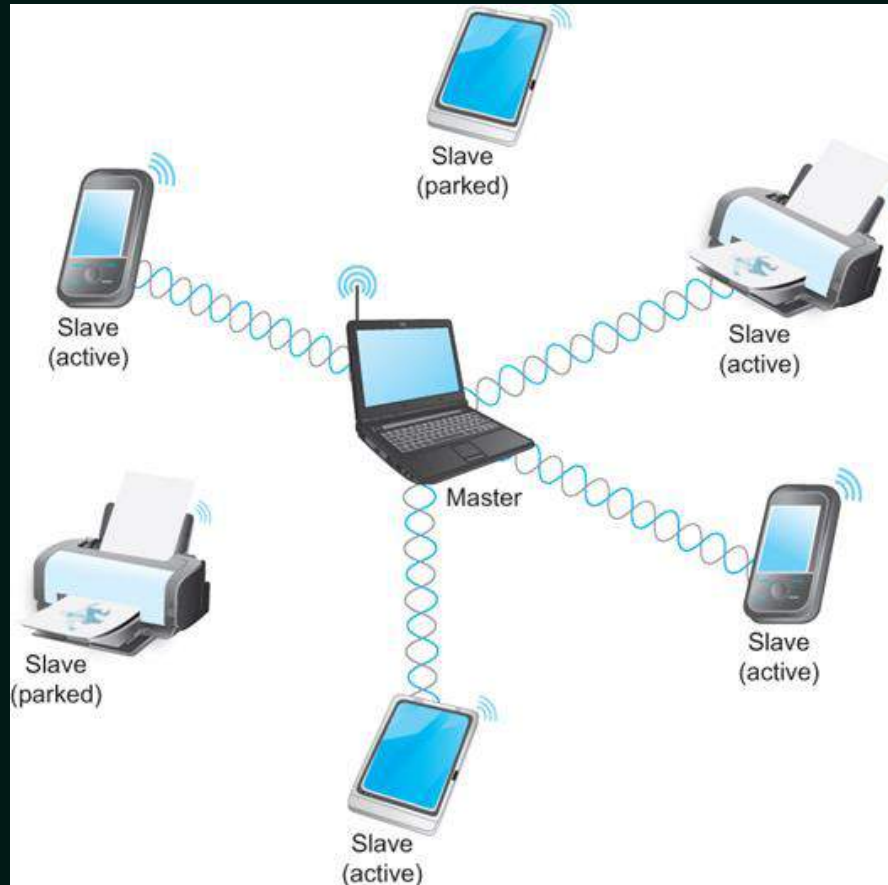
BLUETOOTH

- ★ Bluetooth is a wireless technology standard used for exchanging data between fixed and mobile devices over short distances using short-wavelength UHF radio waves in the ISM radio bands, from 2.400 to 2.485 GHz, and building personal area networks (PANs).
- ★ It was originally conceived as a wireless alternative to RS-232 data cables.
- ★ Bluetooth is a wireless technology named after Harald Bluetooth, based on an analogy that Bluetooth technology would unite devices the way Harald Bluetooth united the tribes of Denmark into a single kingdom.

BLUETOOTH

- ★ Bluetooth is specified by an industry consortium called the Bluetooth Special Interest Group.
- ★ It specifies an entire suite of protocols, going beyond the link layer to define application protocols, which it calls profiles, for a range of applications.
 - There is a profile for synchronizing a PDA with personal computer.
 - Another profile gives a mobile computer access to a wired LAN.
- ★ The basic Bluetooth network configuration is called a **piconet**.

AN EXAMPLE BLUETOOTH PICONET



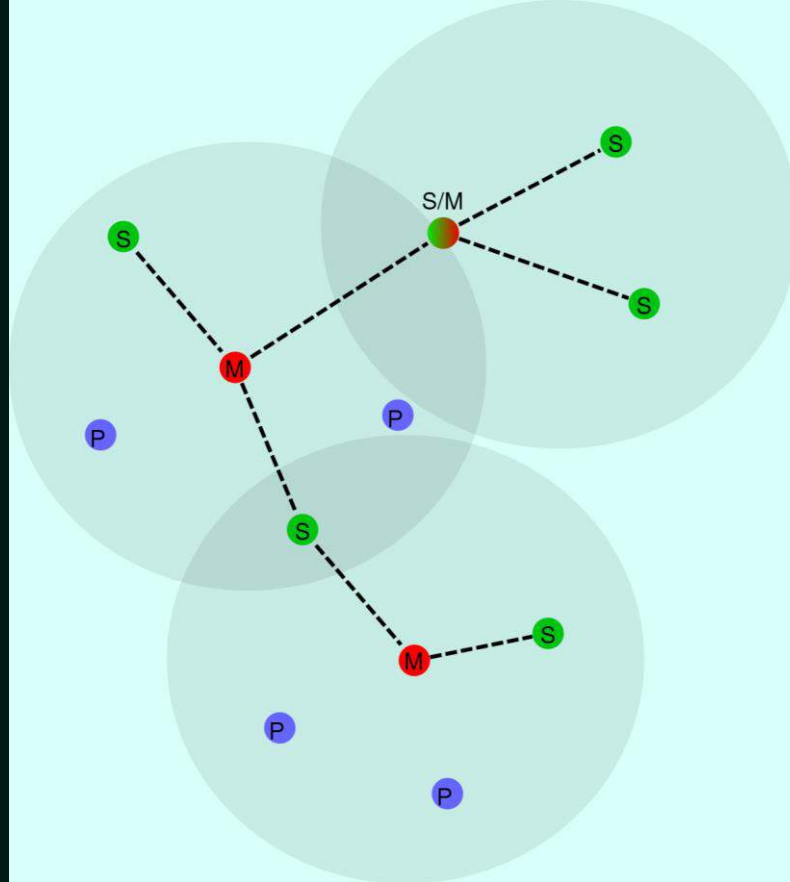
BLUETOOTH

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 - Another profile gives a mobile computer access to a wired LAN.
- ★ The basic Bluetooth network configuration is called a **piconet**.
 - Consists of a master device and up to seven slave devices.
 - Any communication is between the master and a slave.
 - The slaves do not communicate directly with each other.

BLUETOOTH

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- ★ The basic Bluetooth network configuration is called a **piconet**.
 - Consists of a master device and up to seven slave devices.
 - Any communication is between the master and a slave.
 - The slaves do not communicate directly with each other.
 - A slave can be parked: set to an inactive, low-power state.

PICONET AND SCATTERNET



BLUETOOTH DEVICES



PROS

- ★ Low cost.
- ★ Easy to use.
- ★ It can also penetrate through walls.
- ★ It creates an ad-hoc connection immediately without any wires.
- ★ It is used for voice and data transfer.

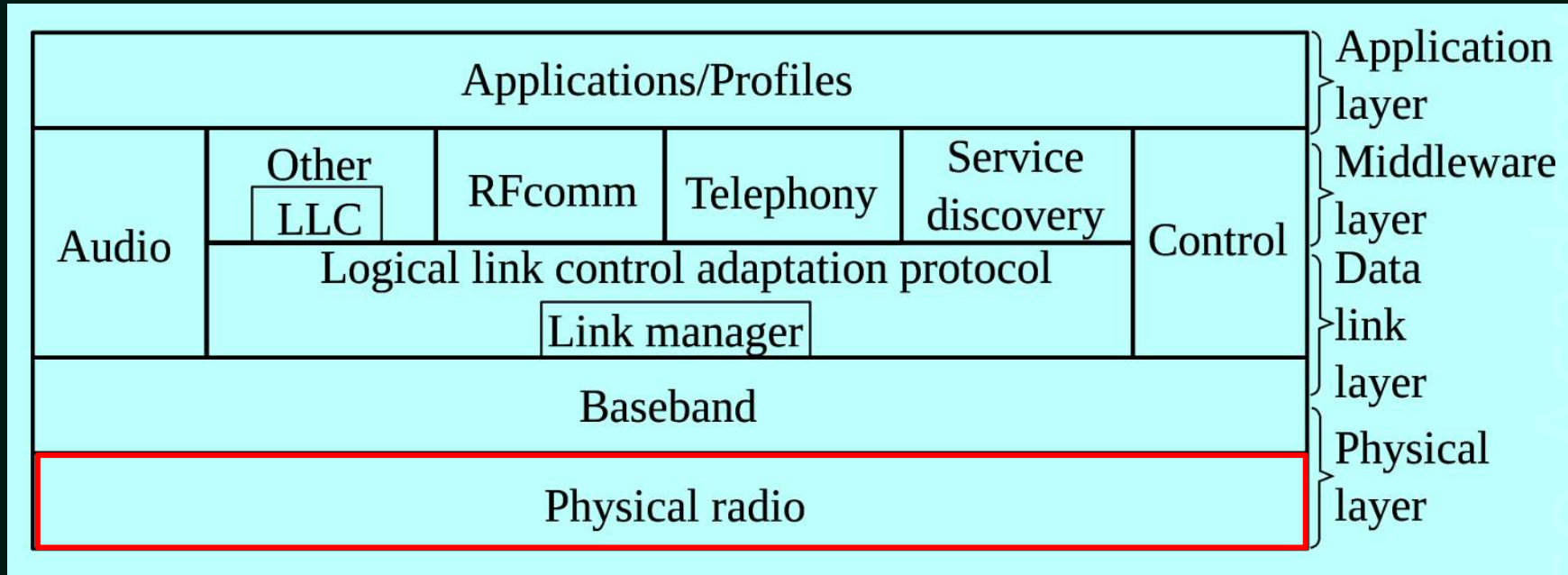
CONS

- ★ It can be hacked and hence, less secure.
- ★ It has slow data transfer rate.
- ★ It has small range: 10 meters.

BLUESNARFING

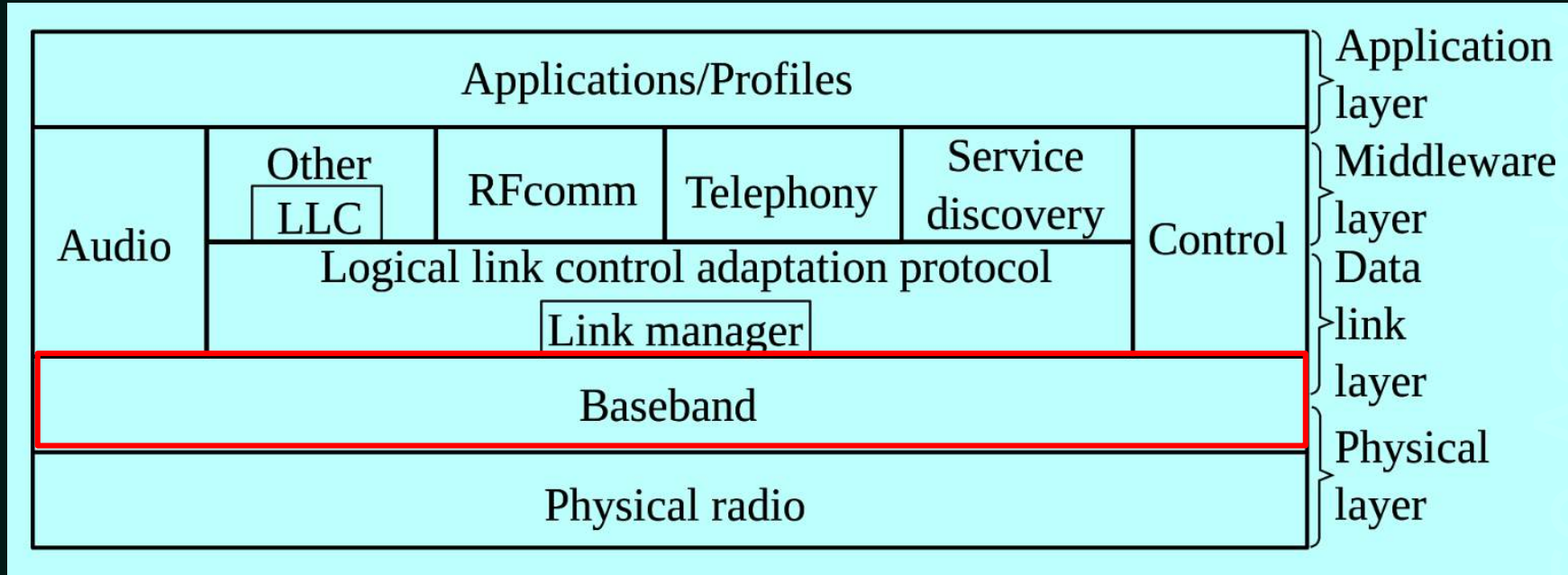
- ★ Bluesnarfing is the unauthorized access of information from a wireless device through a Bluetooth connection, often between phones, desktops, laptops, and PDAs.
- ★ This allows access to calendars, contact lists, emails and text messages, and on some phones, users can copy pictures and private videos.
- ★ Bluesnarfing is the theft of information from the target device.

BLUETOOTH PROTOCOL STACK



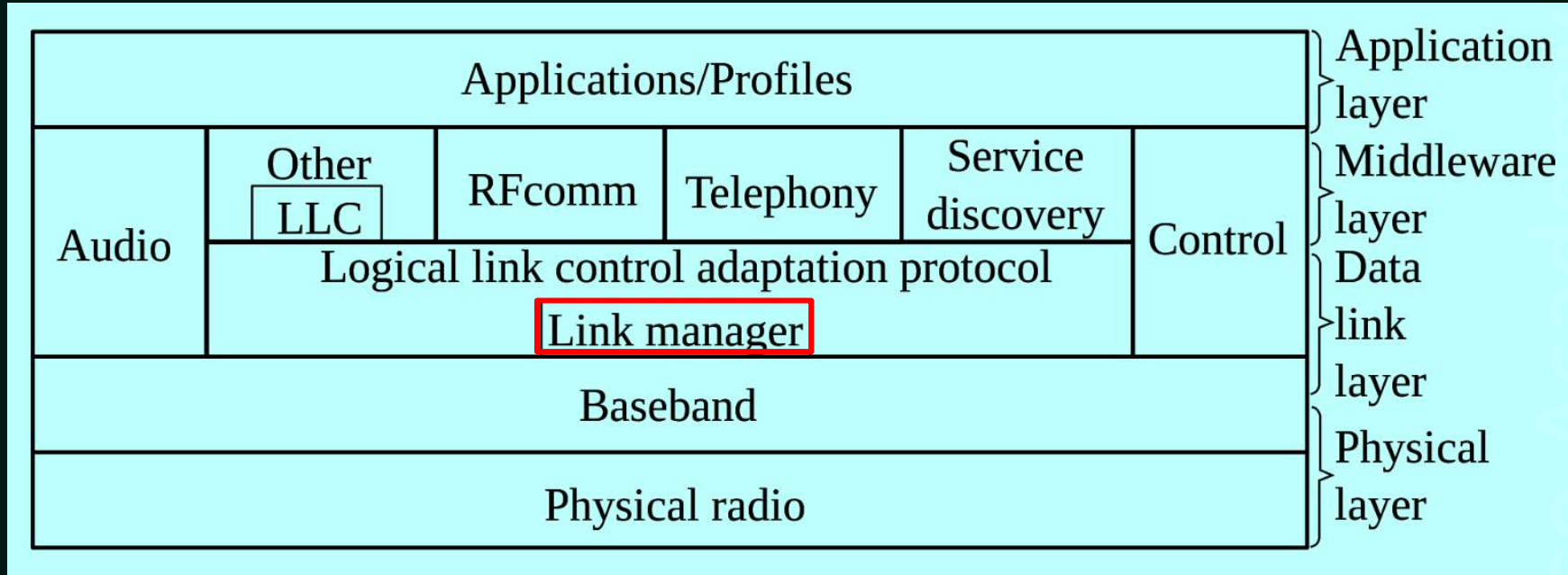
Physical Radio (RF) layer: It performs modulation/demodulation of the data into RF signals. It defines the physical characteristics of bluetooth transceiver. It defines two types of physical link: connection-less and connection-oriented.

BLUETOOTH PROTOCOL STACK



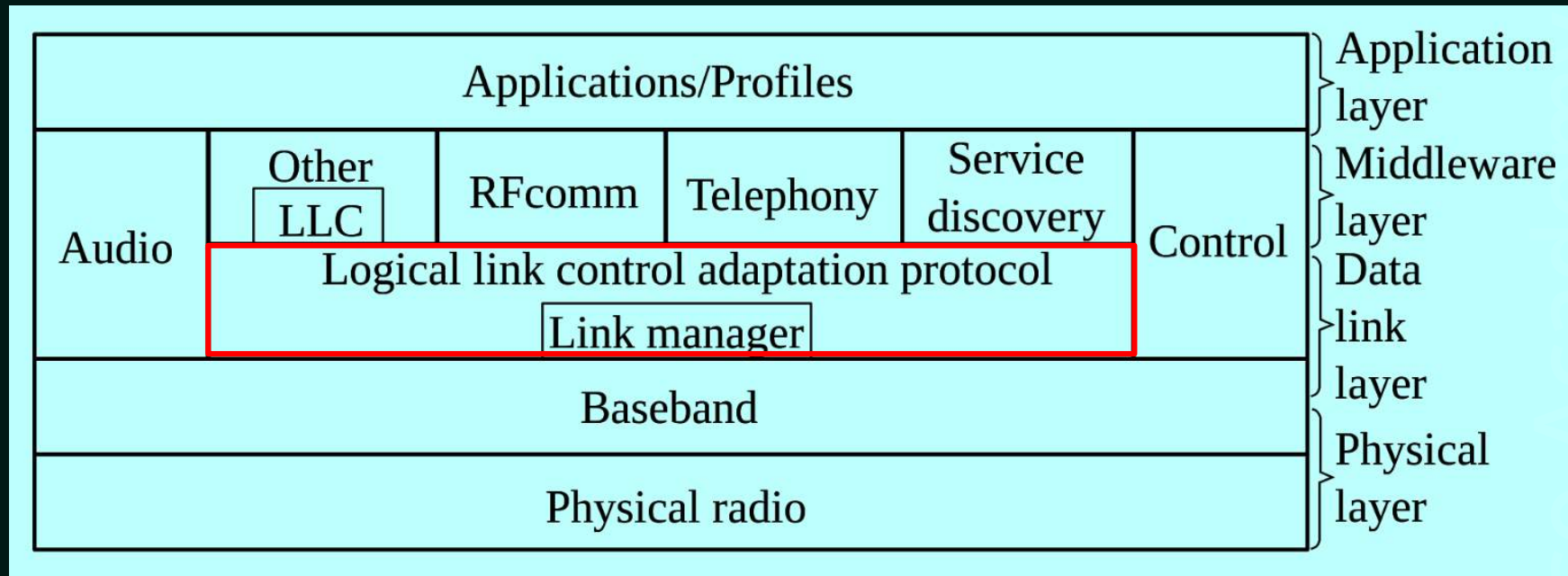
Baseband Link layer: It performs the connection establishment within a piconet.

BLUETOOTH PROTOCOL STACK



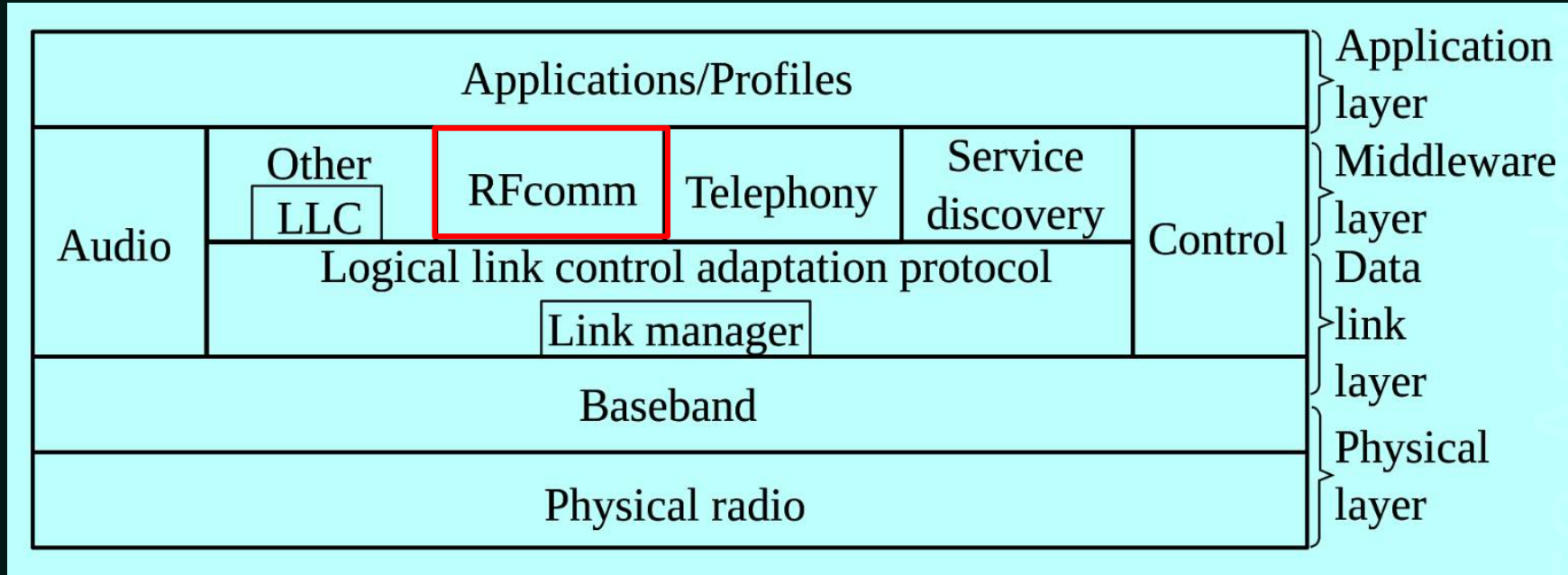
Link Manager protocol layer: It performs the management of the already established links. It also includes authentication and encryption processes.

BLUETOOTH PROTOCOL STACK



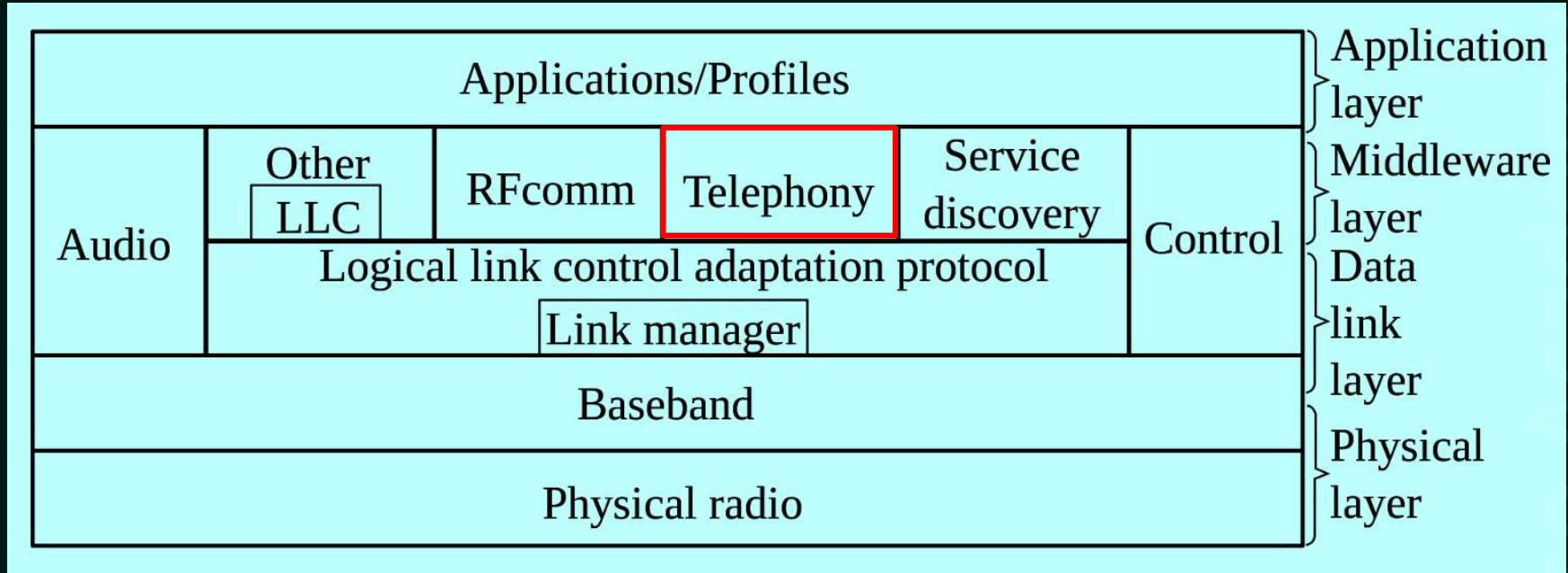
Logical Link Control and Adaptation protocol layer: It is also known as the heart of the bluetooth protocol stack. It allows the communication between upper and lower layers of the bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs the segmentation and multiplexing.

BLUETOOTH PROTOCOL STACK



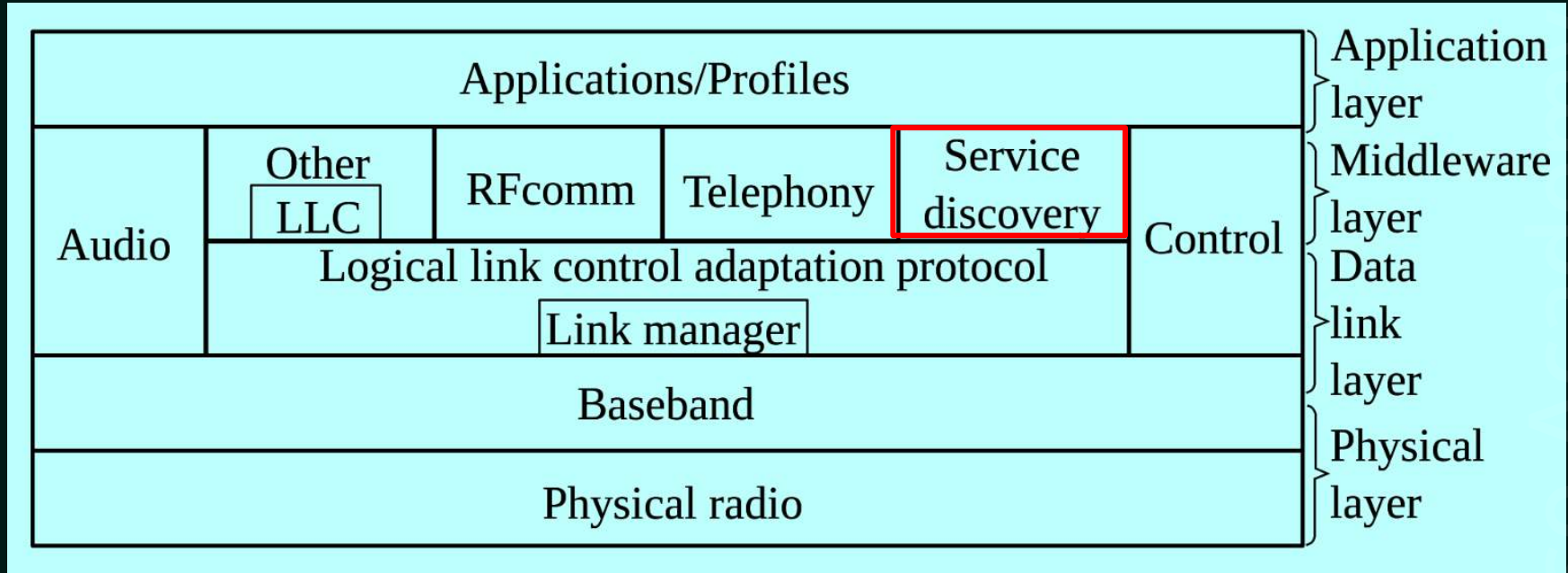
RF comm layer: It is short for Radio Frontend Component. It provides serial interface with WAP and OBEX.

BLUETOOTH PROTOCOL STACK



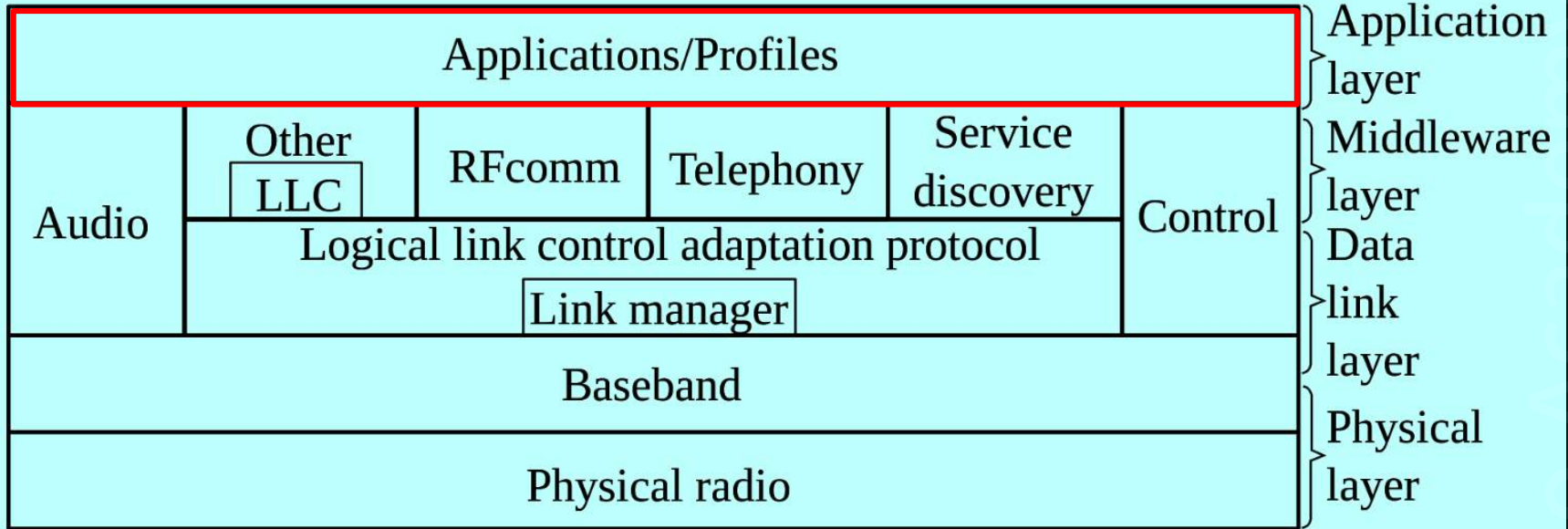
TCS: It is short for Telephony Control Protocol. It provides telephony service.

BLUETOOTH PROTOCOL STACK



SDP layer: It is short for Service Discovery Protocol. It allows to discover the services available on another bluetooth enabled device.

BLUETOOTH PROTOCOL STACK



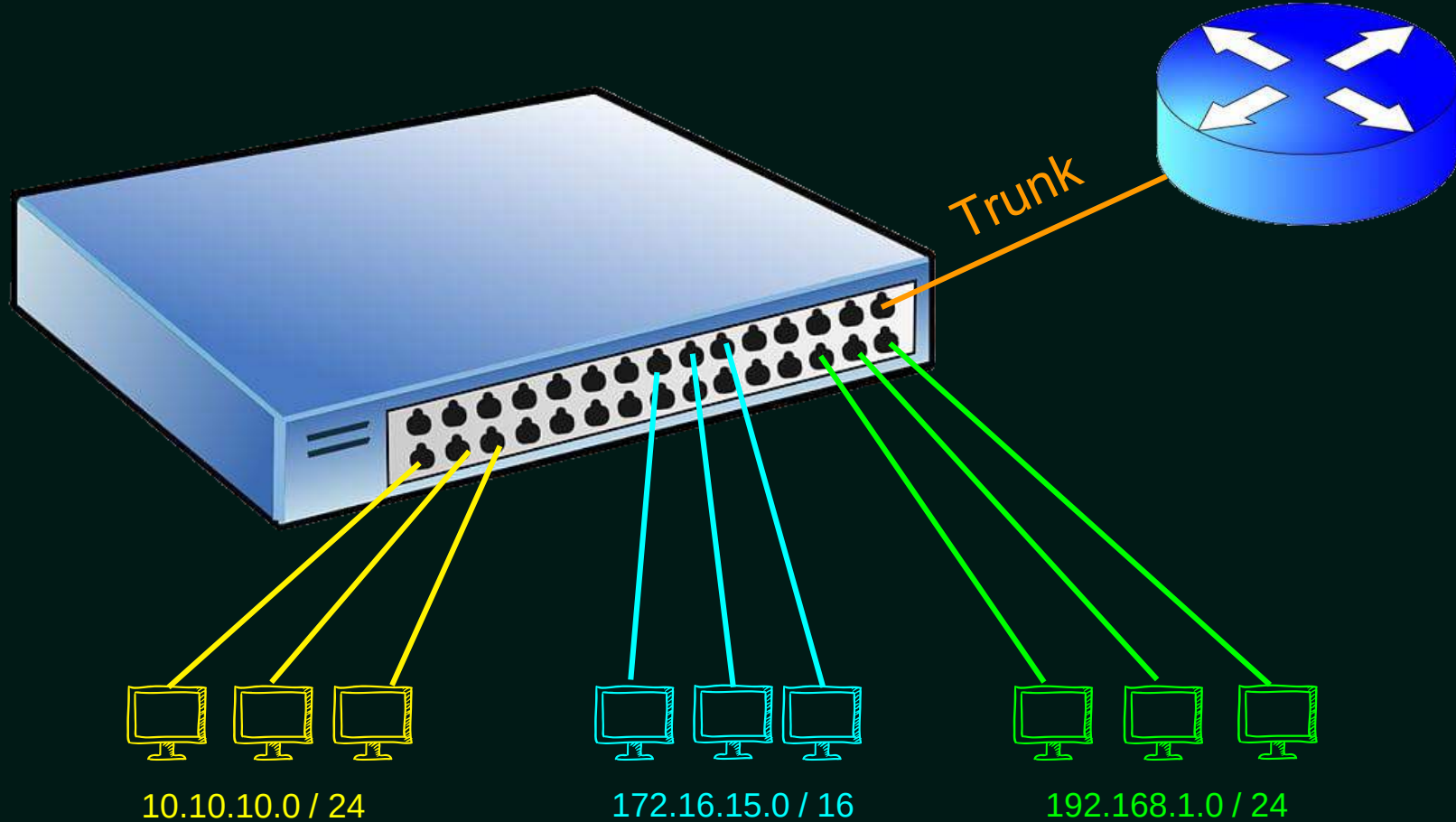
Application layer: It enables the user to interact with the application.

OUTCOMES

Upon the completion of this session, the learner will be able to

- ★ Know the need for VLAN.
- ★ Working of VLAN.
- ★ Benefits of VLAN.
- ★ Types of VLAN.
- ★ VLAN Frame Tagging.

VLAN



VLAN

- ★ A VLAN is a logical partition of a Layer 2 network.
- ★ Multiple partitions can be created, allowing for multiple VLANs to co-exist.
- ★ Each VLAN is a broadcast domain, usually with its own IP network.
- ★ VLANs are mutually isolated and packets can only pass between them via a router.
- ★ The partitioning of the Layer 2 network takes place inside a Layer 2 device, usually via a switch.
- ★ The hosts grouped within a VLAN are unaware of the VLAN's existence.

VLAN



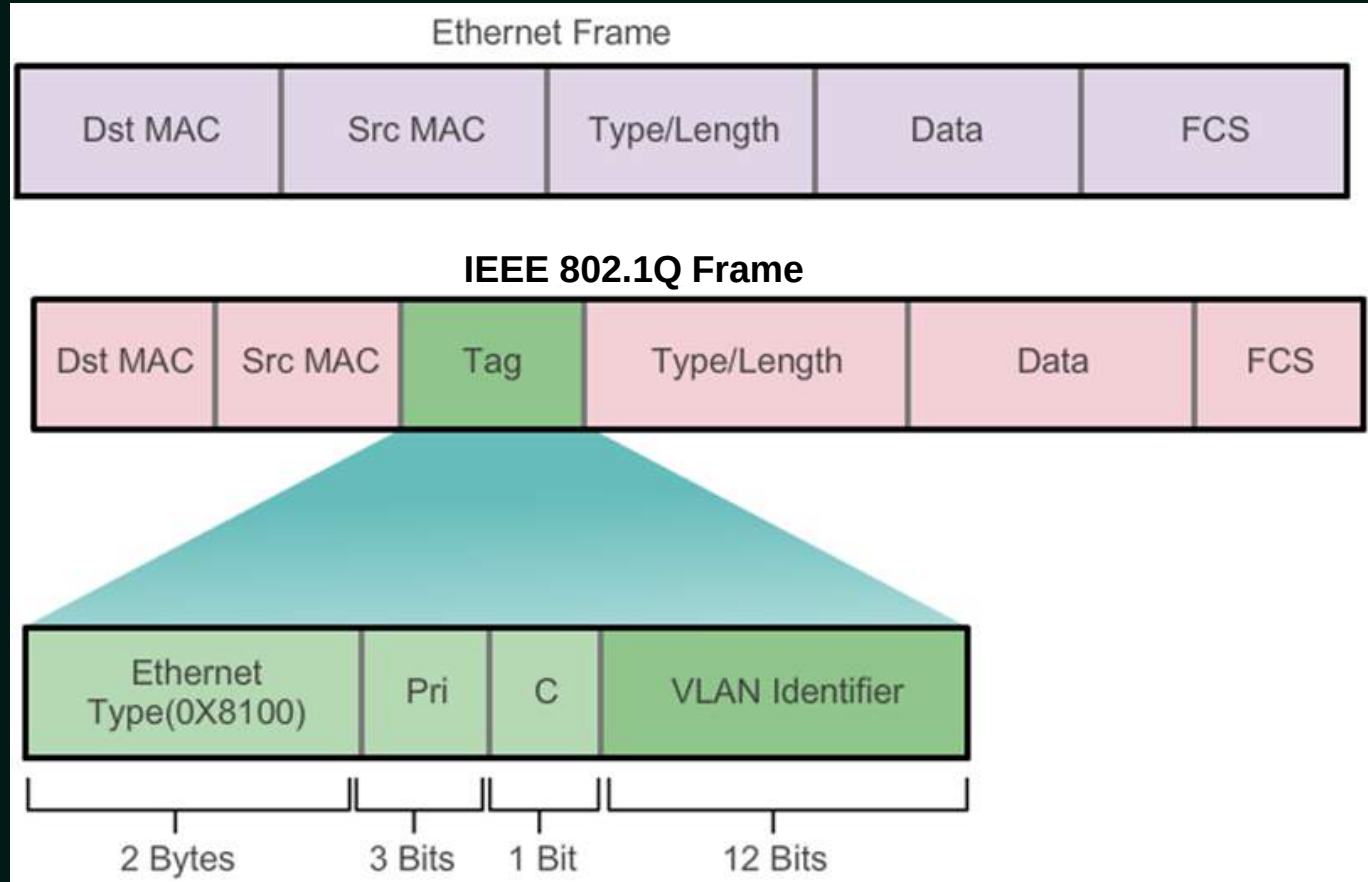
BENEFITS OF VLAN

- ★ Security.
- ★ Cost reduction.
- ★ Better performance.
- ★ Shrink broadcast domains.
- ★ Improved IT staff efficiency.
- ★ Simpler project and application management.

TYPES OF VLAN

- ★ Data VLAN
- ★ Default VLAN
- ★ Native VLAN
- ★ Management VLAN
- ★ Voice VLAN

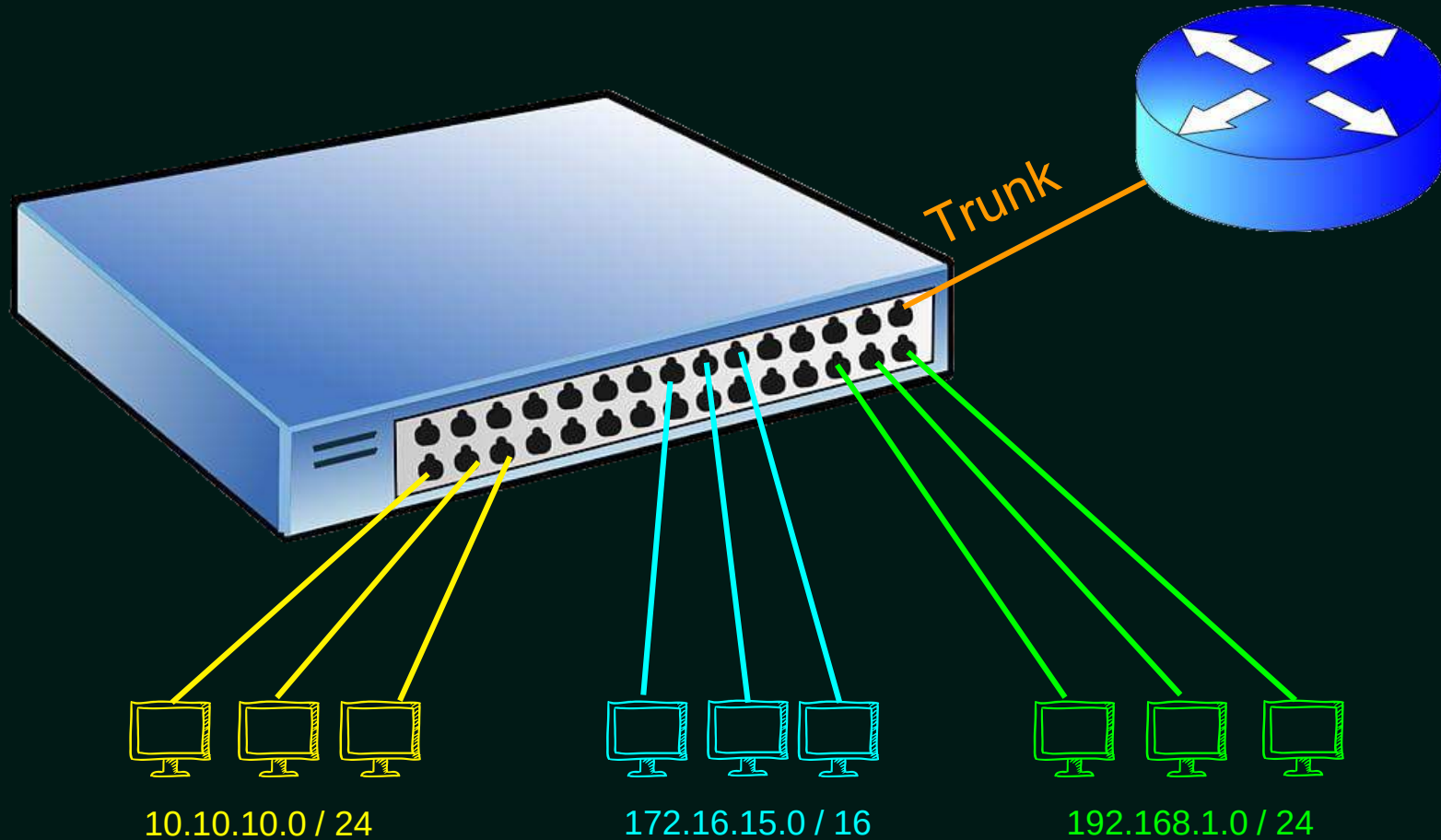
VLAN FRAME TAGGING



VLAN FRAME TAGGING

- ★ Frame tagging is the process of adding a VLAN identification header to the frame.
- ★ It is used to properly transmit multiple VLAN frames through a trunk link.
- ★ Switches tag frames to identify the VLAN to that they belong. Different tagging protocols exist; IEEE 802.1Q is a very popular example.
- ★ The protocol defines the structure of the tagging header added to the frame.
- ★ Switches add VLAN tags to the frames before placing them into trunk links and remove the tags before forwarding frames through non-trunk ports.
- ★ When properly tagged, the frames can transverse any number of switches via trunk links and still be forwarded within the correct VLAN at the destination.

VLAN



OUTCOMES

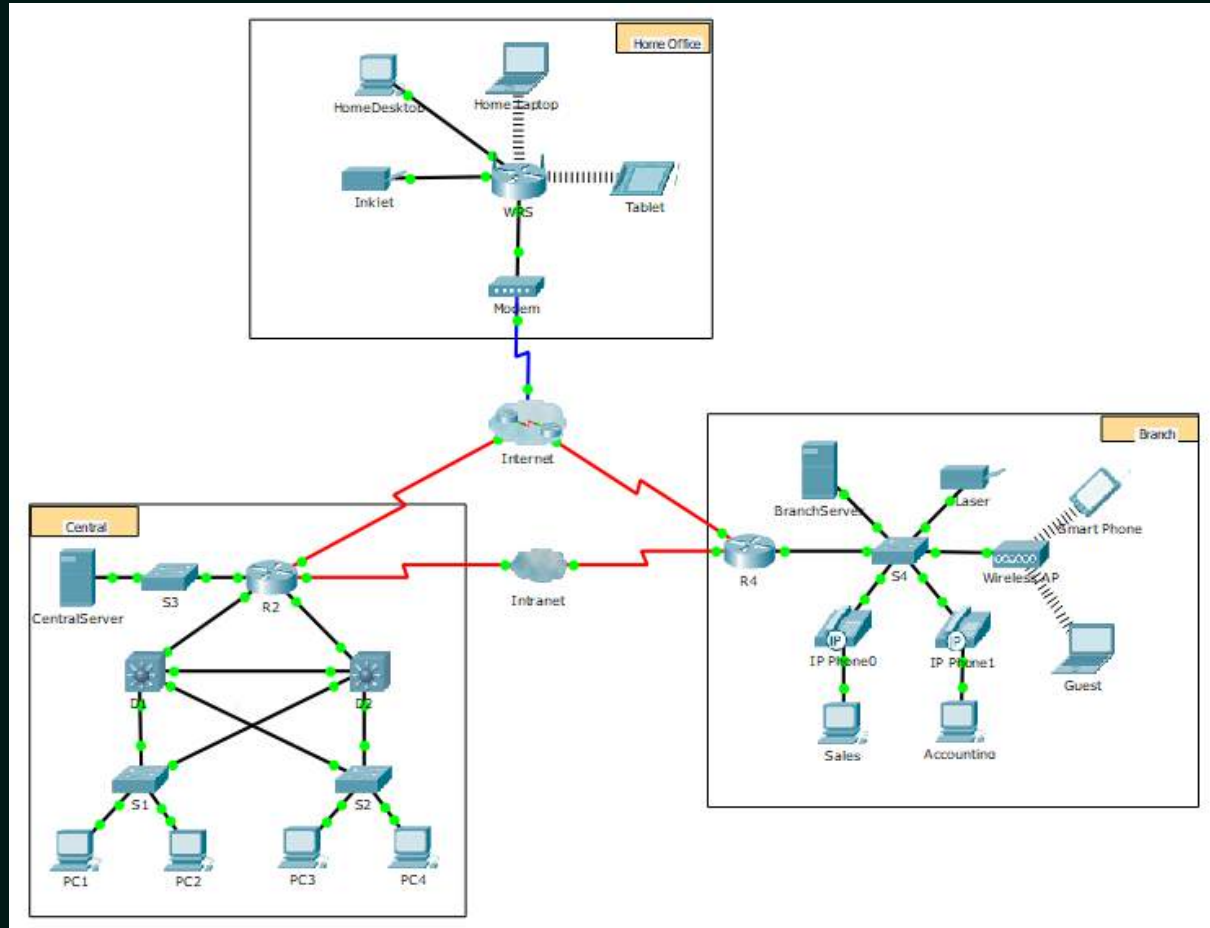
Upon the completion of this session, the learner will be able to

- ★ Understand the need for redundancy and how failure is handled.
- ★ Know about broadcast storm.
- ★ Understand Spanning Tree Protocol.
- ★ Understand various key concepts and terms in STP.

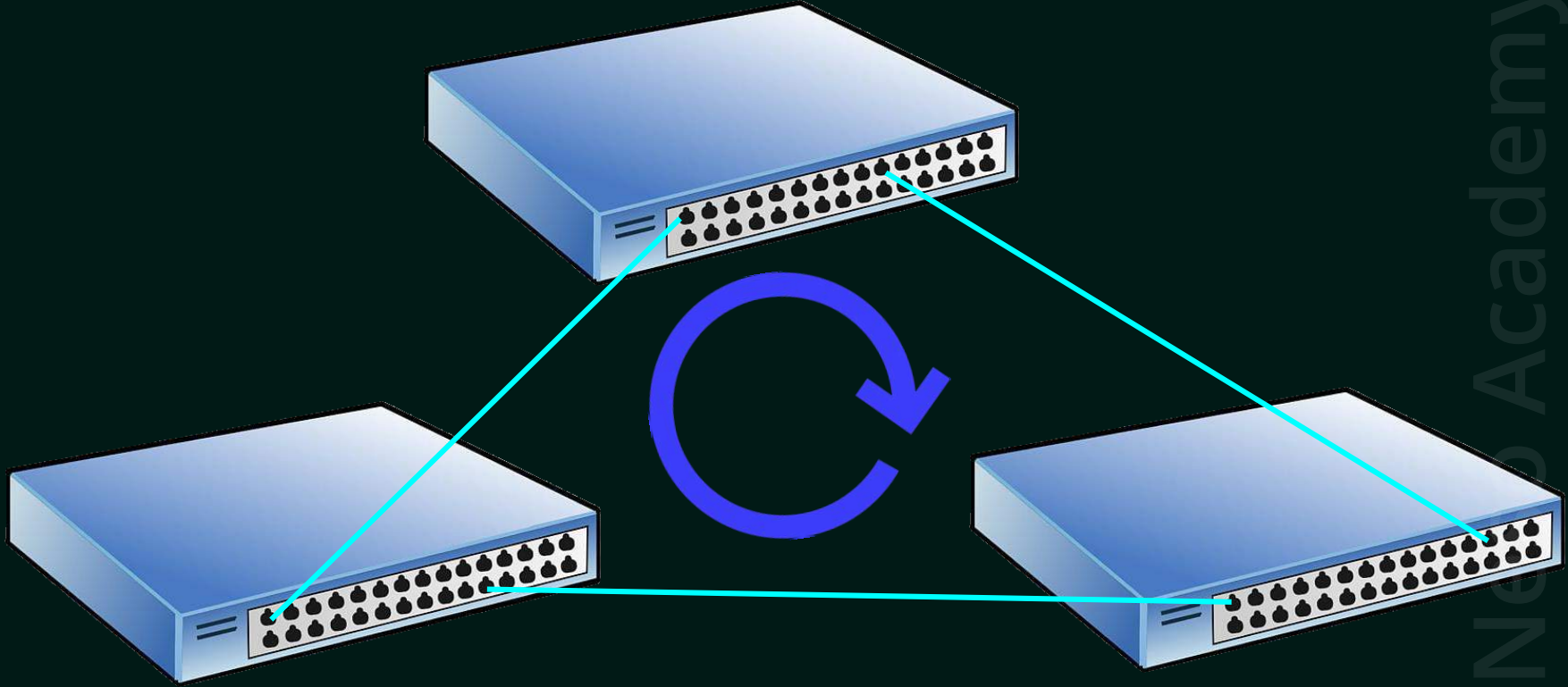
REDUNDANCY IS GOOD!

- ★ Enables users to access network resources, despite path disruption.
 - Improves reliability.
 - Improves availability.
- ★ In Technology, 2 is 1 and 1 is none. Single connection mean single point of failure.
 - Creating redundant links is very simple and is advisable.

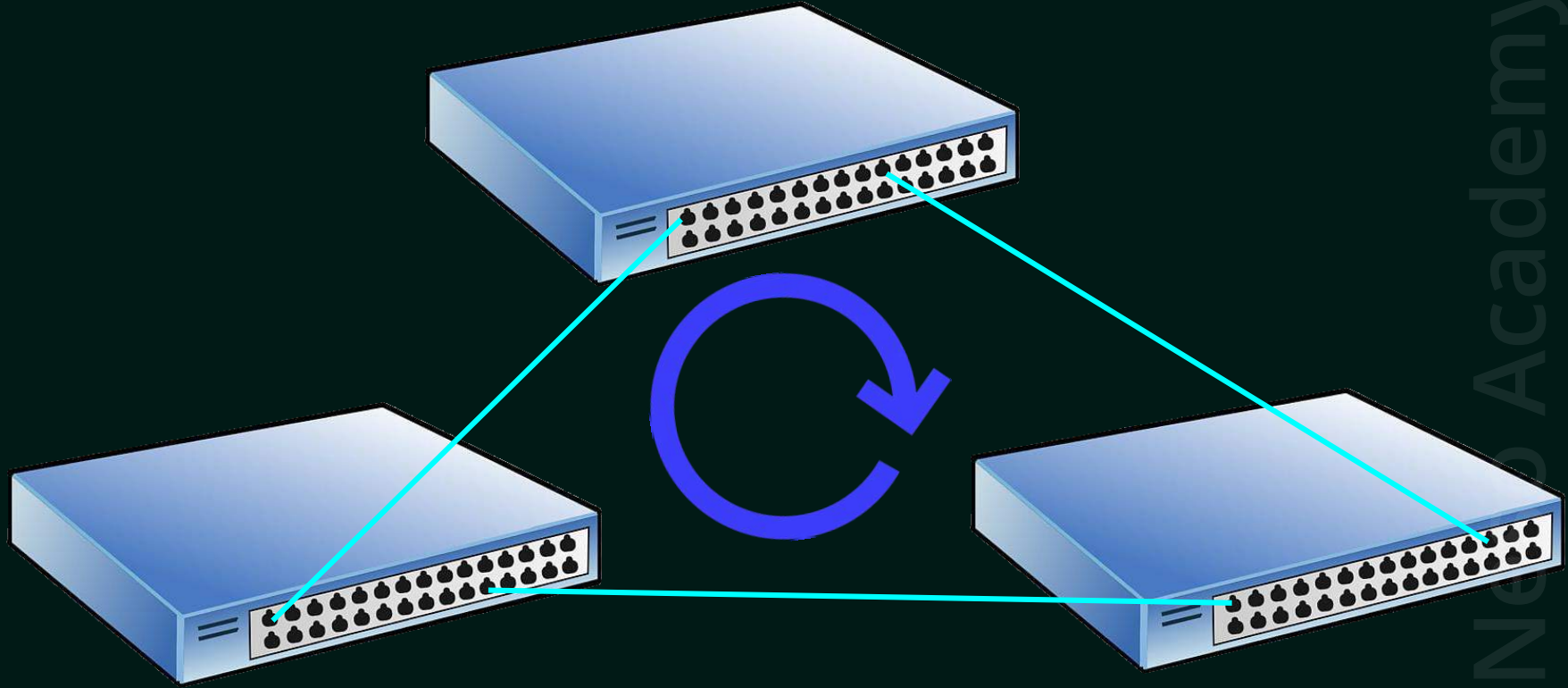
REDUNDANCY



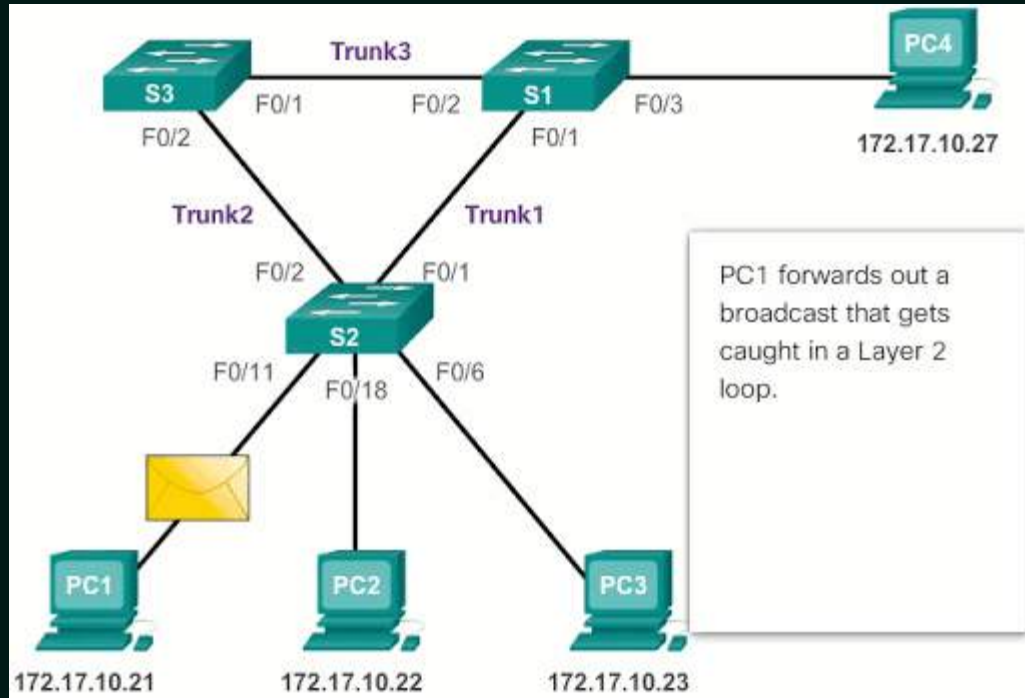
REDUNDANCY IS GOOD!



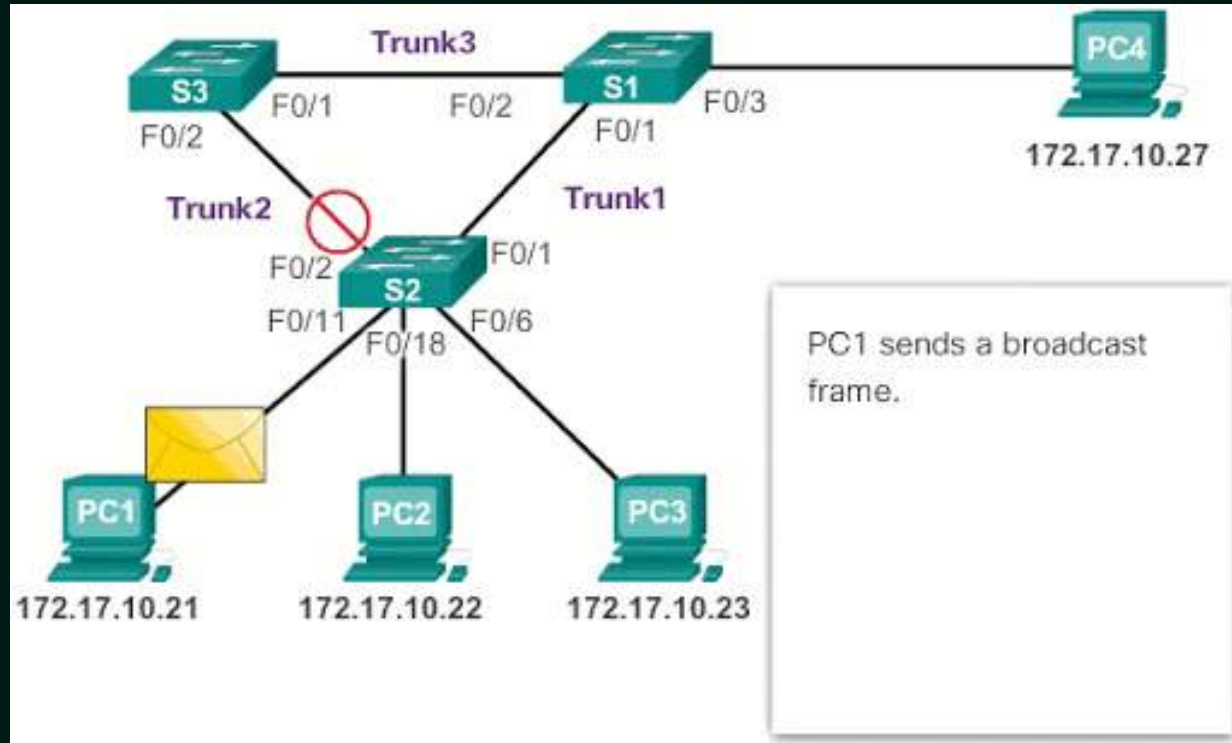
BROADCAST STORM



BROADCAST STORM

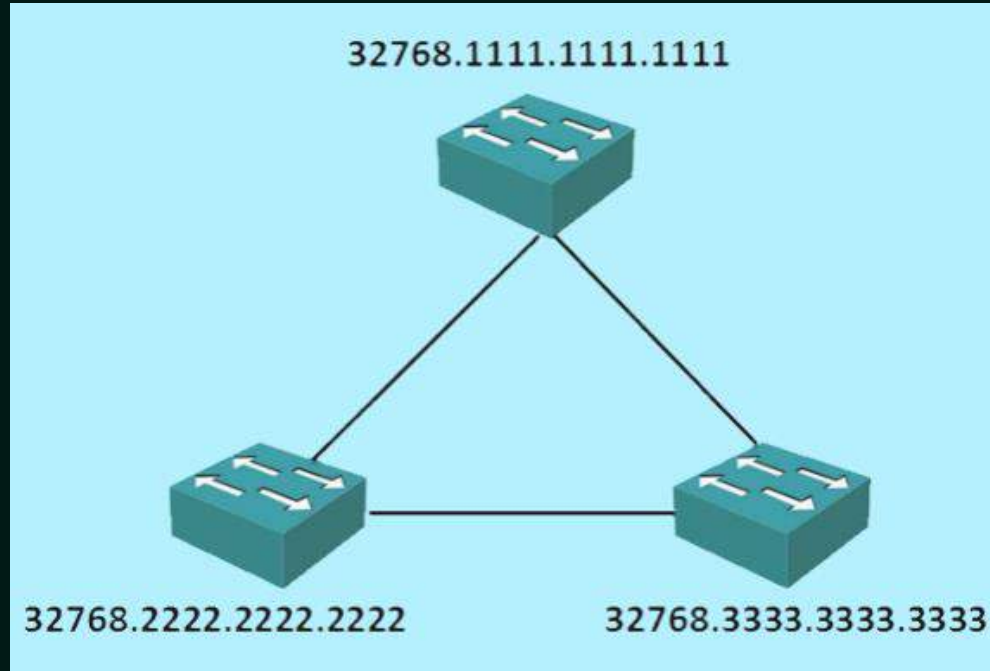


STP PREVENTS LOOPS



KEY FACTS – STP

- ★ Original STP (802.1D) was created to prevent loops.
- ★ Switches send probes into the network to discover loops.

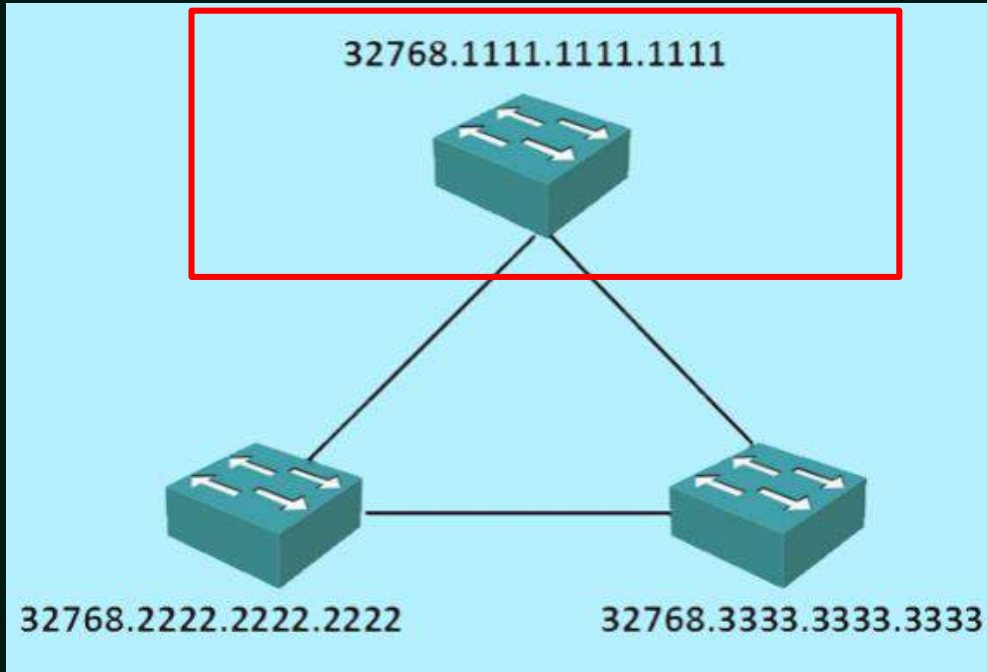


KEY FACTS – STP

- ★ Original STP (802.1D) was created to prevent loops.
- ★ Switches send probes into the network to discover loops.
- ★ These probes are called as BPDU.
- ★ BPDU = Bridge Protocol Data Unit.
- ★ BPDU will have specific information about the switch.
- ★ Switch multicasts BPDU probes (every 2 seconds) and if it receives its own BPDU back, it means there is a loop in the network.
- ★ Also the BPDU probes helps to elect the root bridge.
- ★ All switches will find the best way to reach the root bridge and the redundant links will be blocked. (Port cost)
- ★ This redundant links will be active only if the existing links or ports goes down.

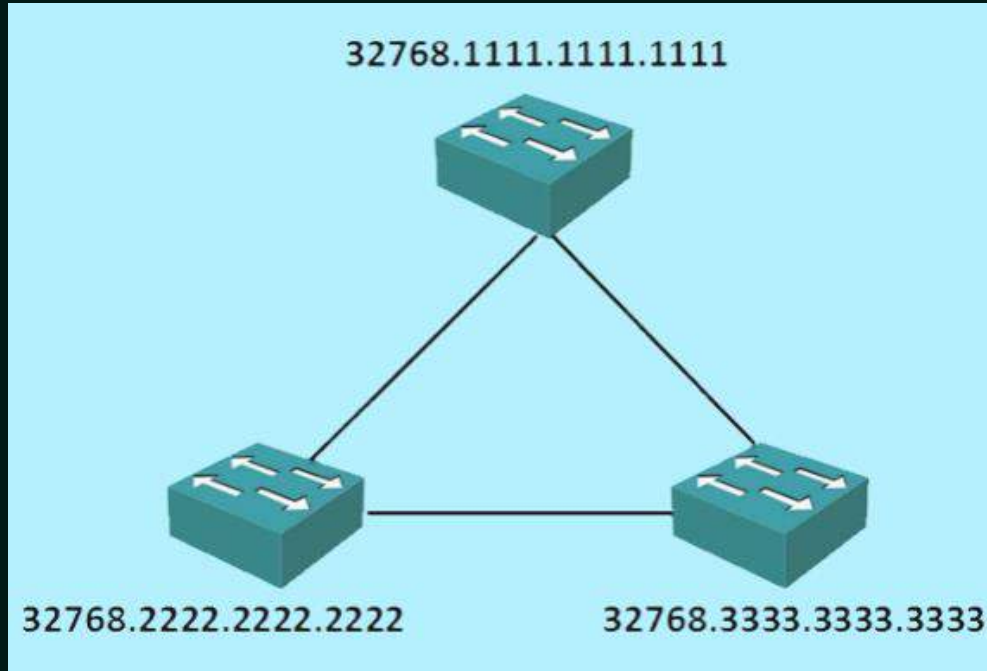
ELECTION – ROOT BRIDGE

BPDU (Bridge ID)	
Bridge Priority	MAC Address of the switch



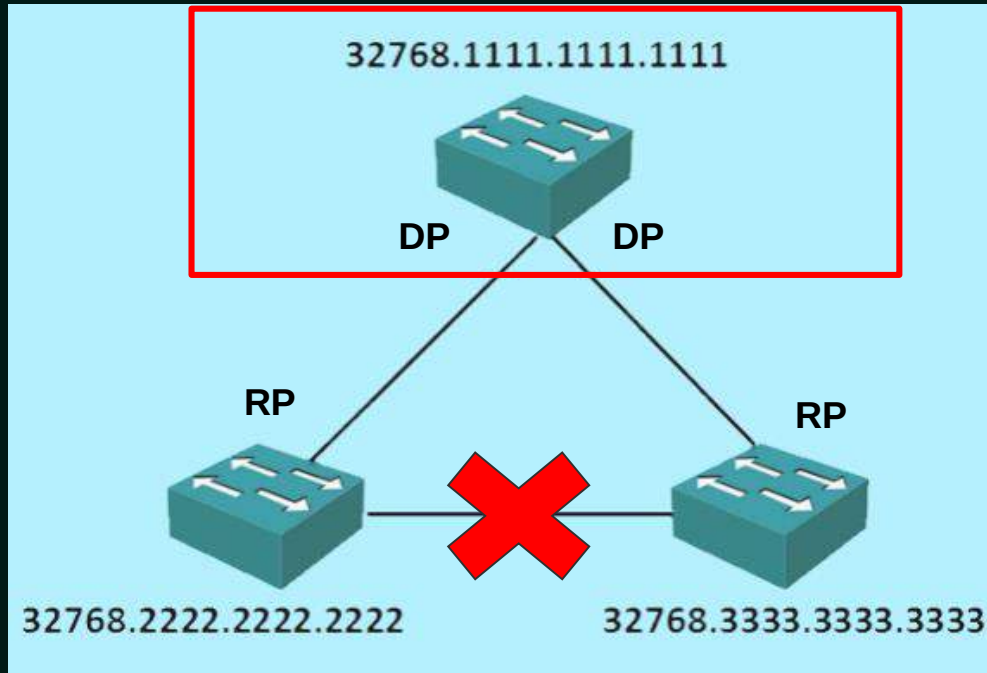
ROOT BRIDGE

- ★ The Root Bridge will have the lowest bridge id.
- ★ If Tie, The bridge with the lowest MAC address will be the Root bridge.

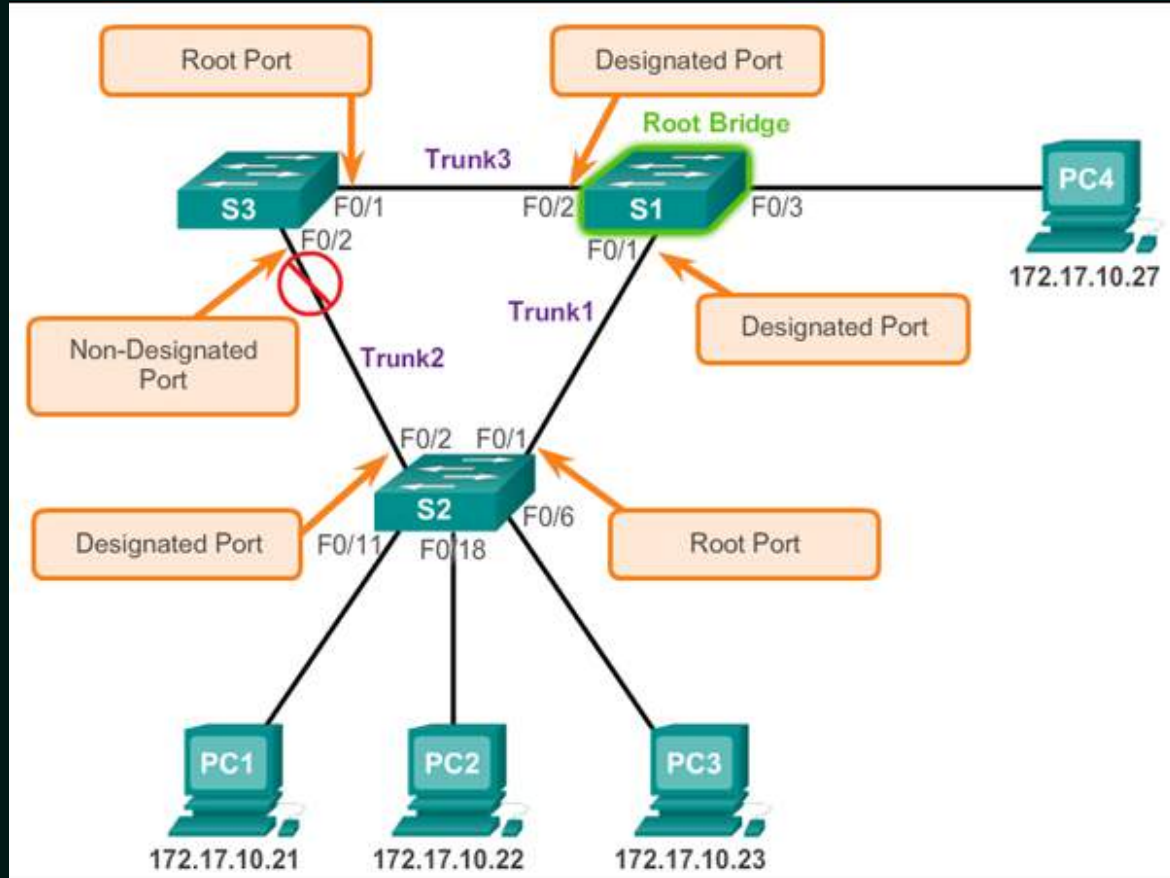


PORT ROLES

- ★ Root Port (Used to reach the root bridge)
- ★ Designated Port (Forwarding port; One per link)
- ★ Blocking/Non Designated Port. (Loops)



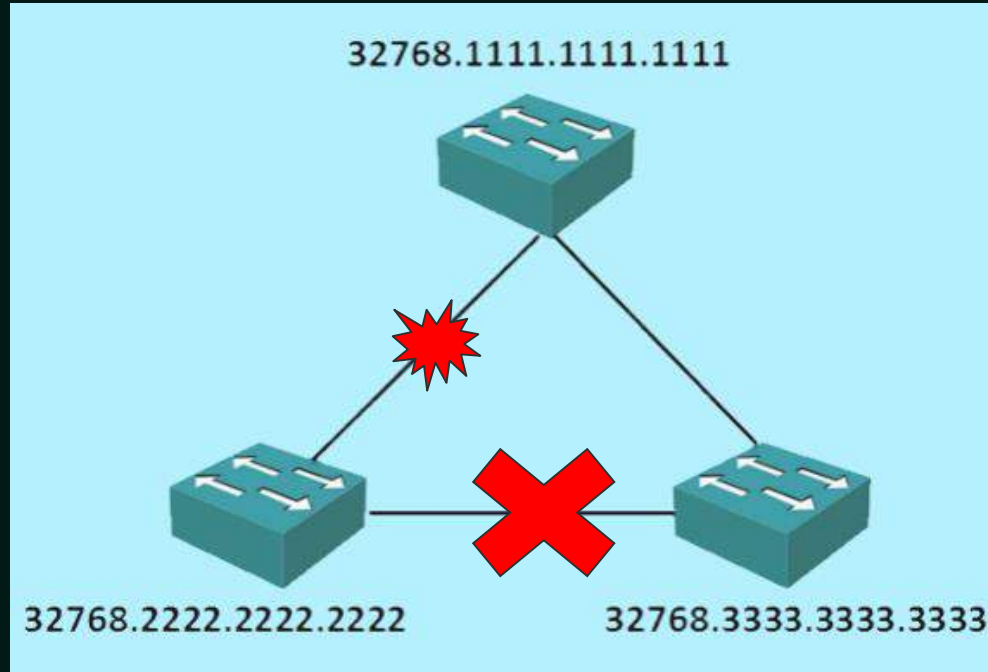
PORT ROLES AND PORT COST



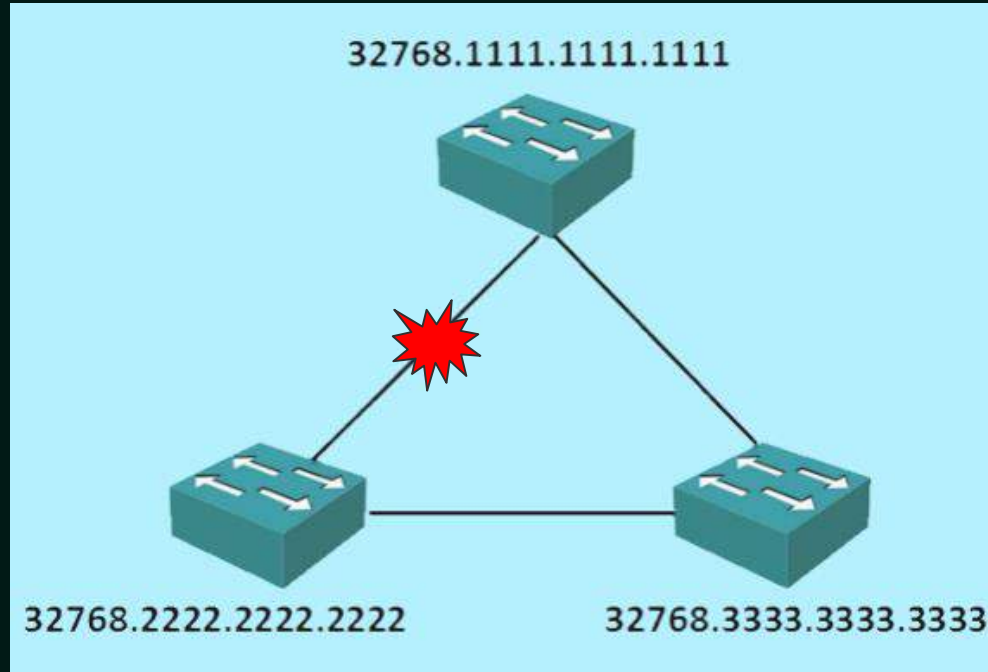
SPANNING TREE PROTOCOL

- ★ STP ensures that there is only one logical path between all destinations on the network by intentionally blocking redundant paths that could cause a loop.
- ★ A port is considered blocked when user data is prevented from entering or leaving that port. This does not include bridge protocol data unit (BPDU) frames that are used by STP to prevent loops.
- ★ The physical paths still exist to provide redundancy, but these paths are disabled to prevent the loops from occurring.
- ★ If the path is ever needed to compensate for a network cable or switch failure, STP recalculates the paths and unblocks the necessary ports to allow the redundant path to become active.

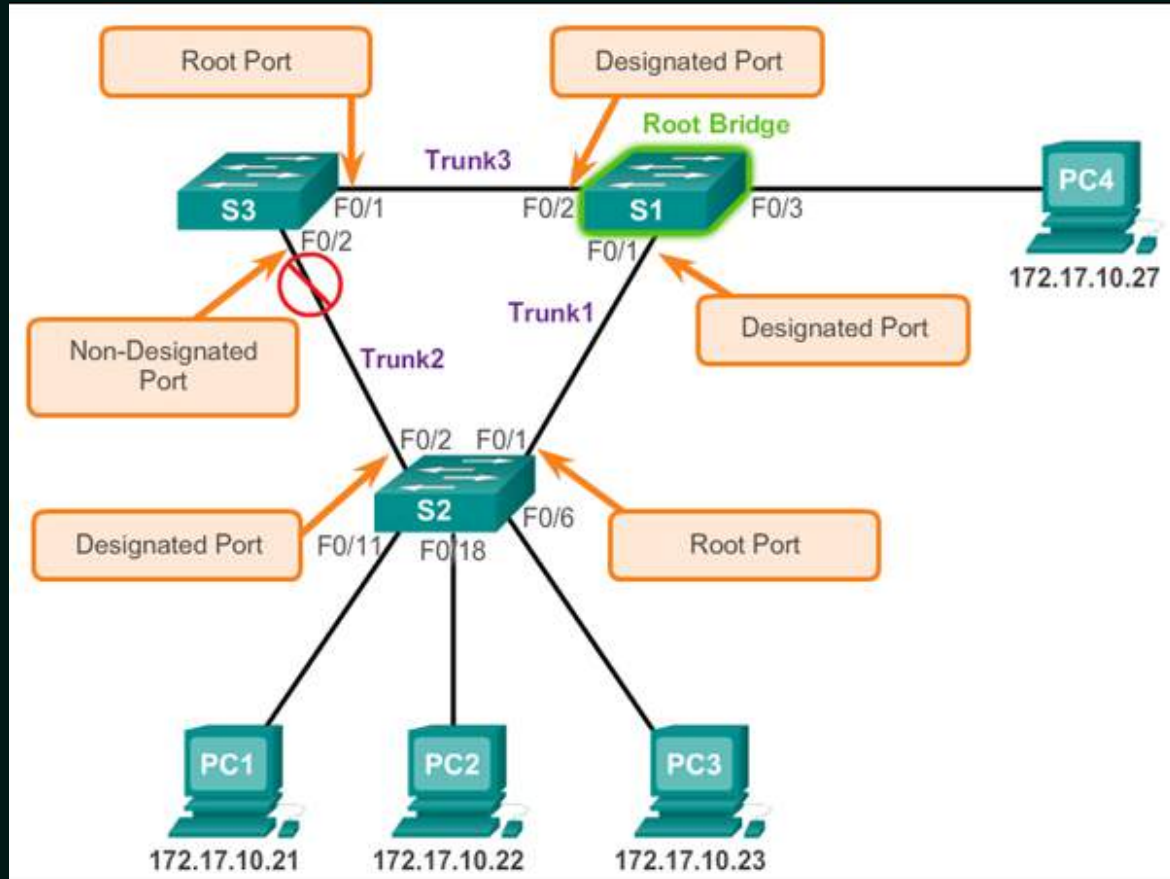
SPANNING TREE PROTOCOL



SPANNING TREE PROTOCOL



HOMEWORK!



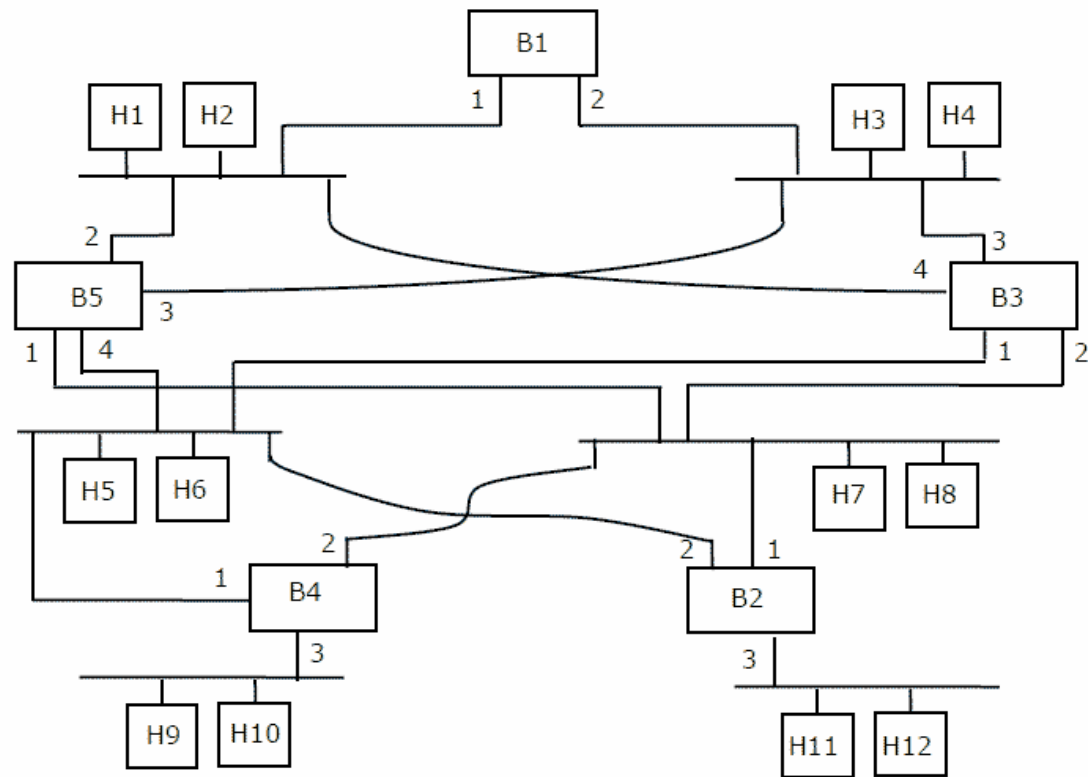
QUESTION

Consider the diagram shown below where a number of LANs are connected by (transparent) bridges. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge.

Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred.

[GATE CS 2006]

QUESTION



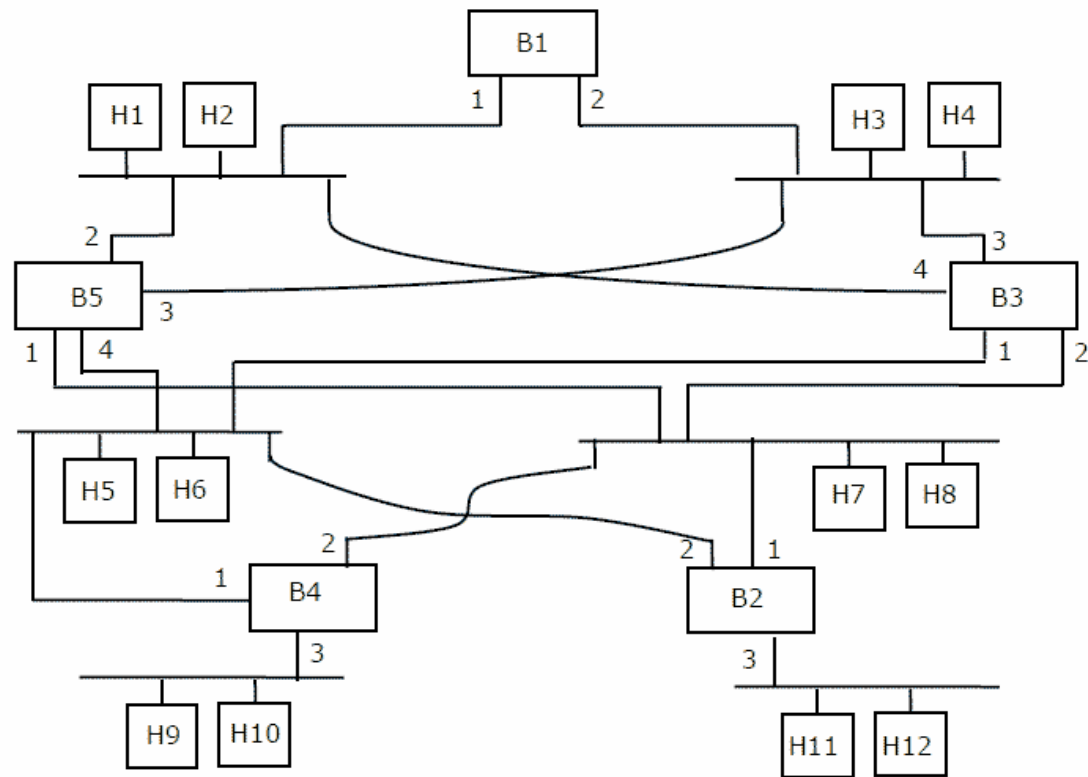
QUESTION

For the given connection of LANs by bridges, which one of the following choices represents the depth first traversal of the spanning tree of bridges?

[GATE CS 2006]

- (A) B1, B5, B3, B4, B2
- (B) B1, B3, B5, B2, B4
- (C) B1, B5, B2, B3, B4
- (D) B1, B3, B4, B5, B2

QUESTION



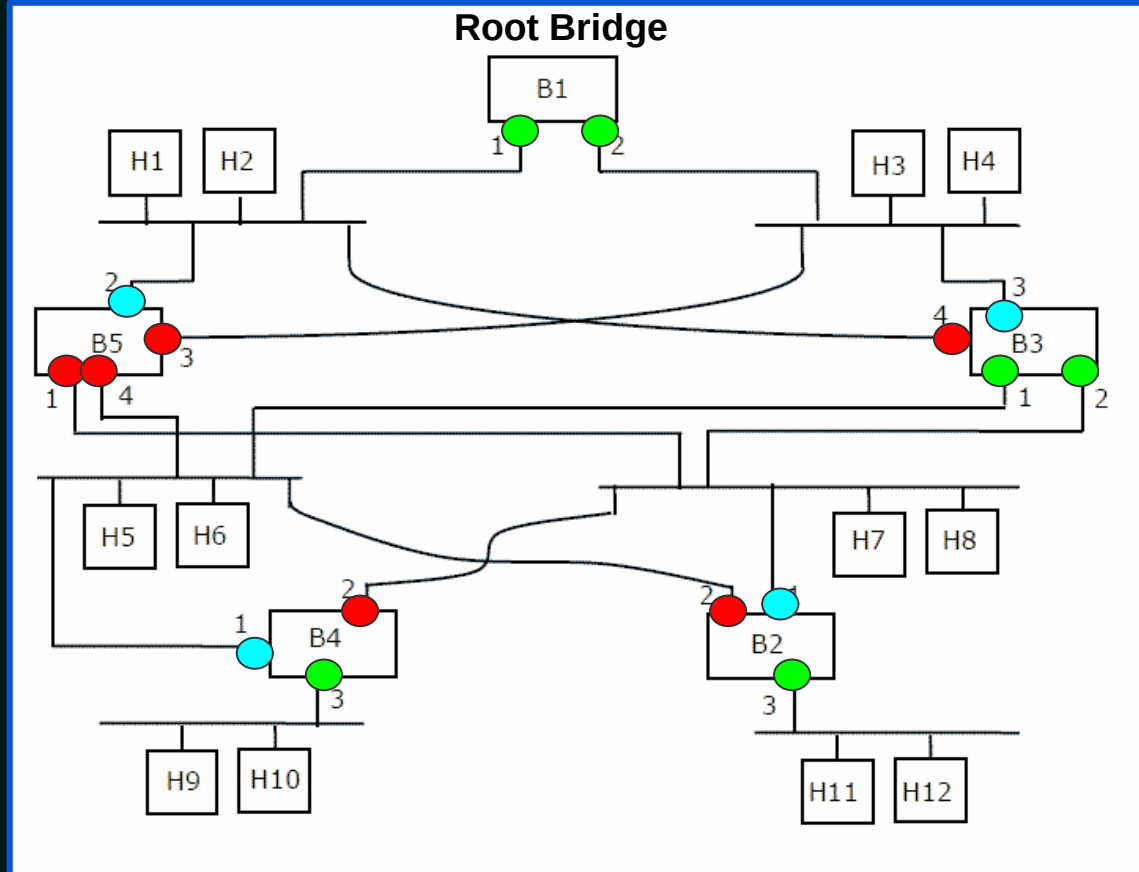
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[GATE CS 2006]

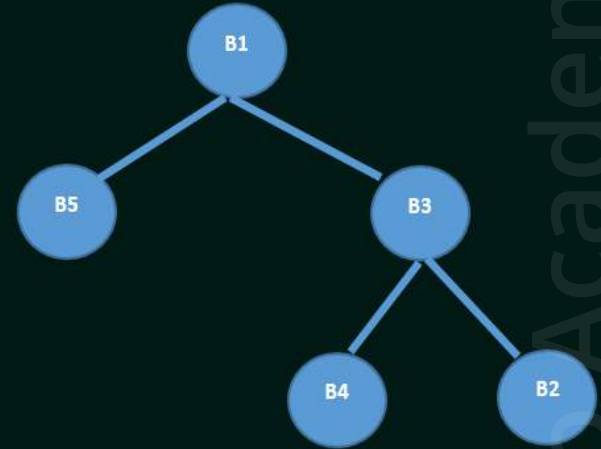
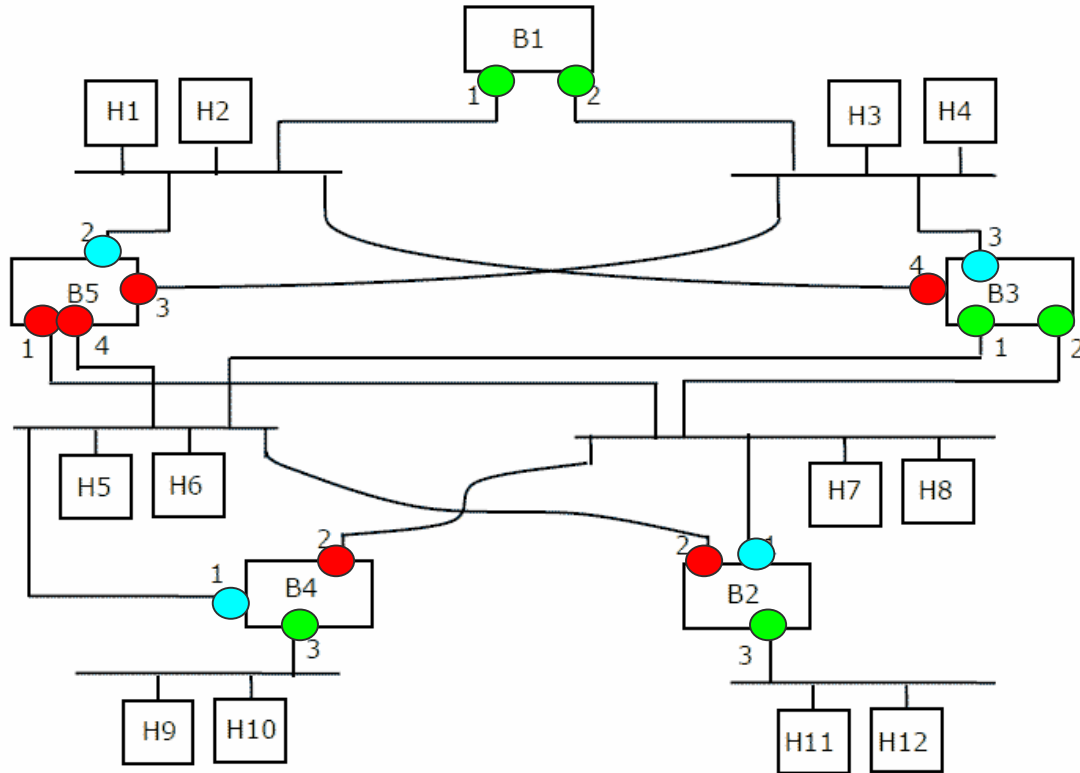
QUESTION



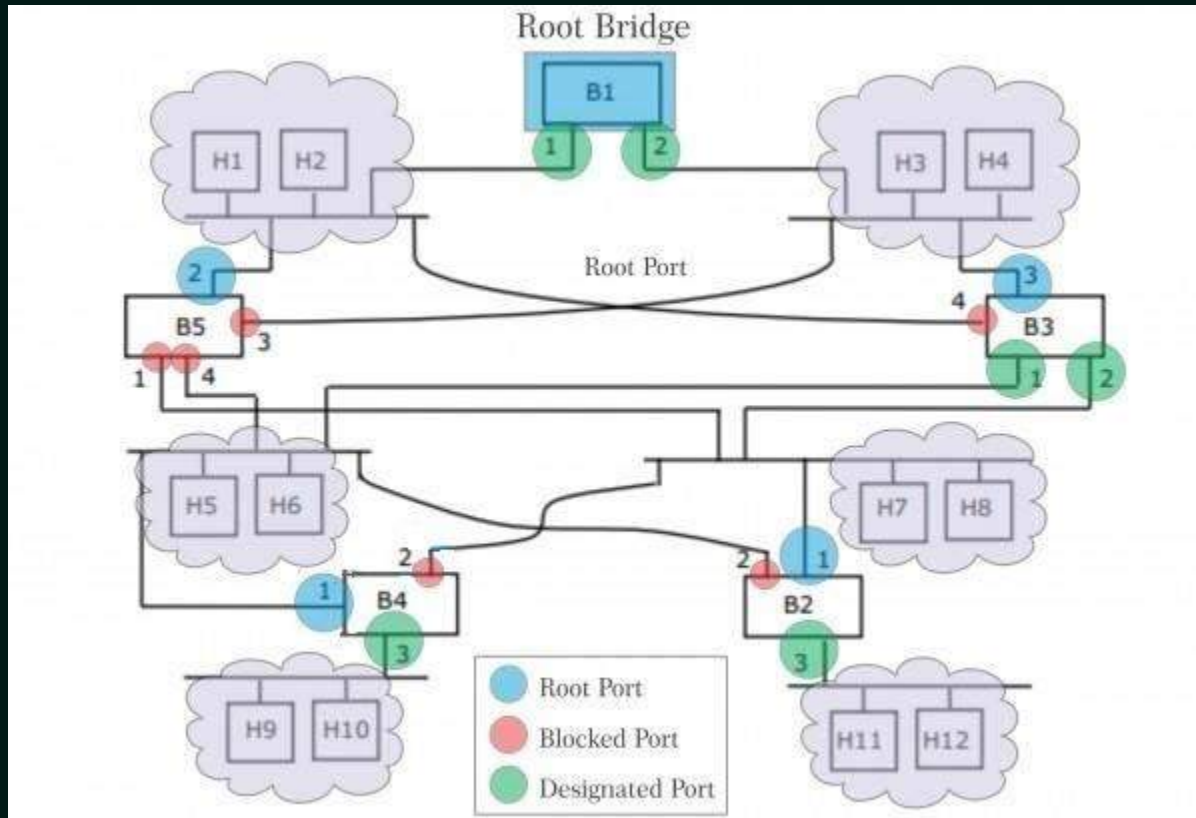
- Root Port
- Blocked Port
- Designated Port

QUESTION

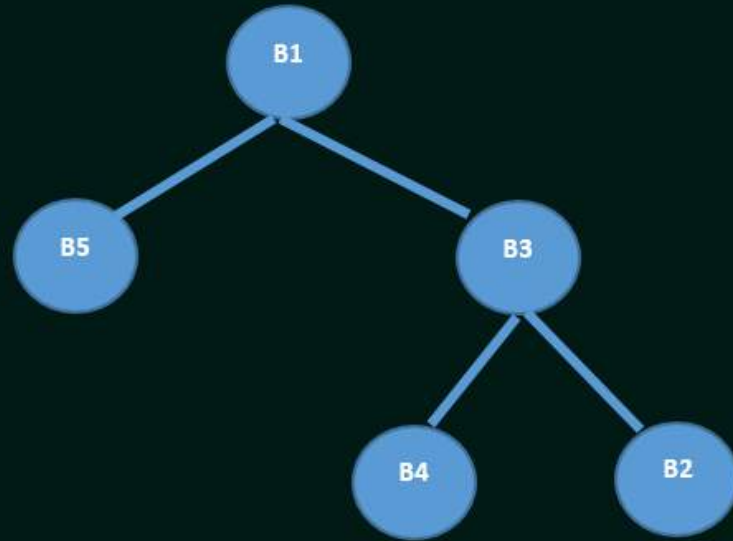
Root Bridge



SOLUTION



SOLUTION



B4 and B2 are connected through B3 (Not B5) because B3 has lower serial number than B5.

One DFS traversal of tree is: B1 – B5 – B3 – B4 – B2

SOLUTION

For the given connection of LANs by bridges, which one of the following choices represents the depth first traversal of the spanning tree of bridges?

[GATE CS 2006]

(A) B1, B5, B3, B4, B2 ✓

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(C) B1, B5, B2, B3, B4

(D) B1, B3, B4, B5, B2



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